



Virtual
**INTERNATIONAL
CONFERENCE**

PROCEEDINGS ON

**HIGHER EDUCATION 360°: INTEGRATING
BUSINESS STRATEGY, TECHNOLOGY, AND
INNOVATION IN THE DIGITAL AGE**

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FROM THE EDITOR'S DESK



It gives me immense pleasure to present this edited volume of conference proceedings, a scholarly outcome of the International Virtual Conference on “Higher Education 360°: Integrating Business Strategy, Technology, and Innovation in the Digital Age,” held on 23rd and 24th May 2025. This conference brought together a diverse community of educators, researchers, policy thinkers, and innovators from across the globe who are committed to reimagining higher education. The dynamic discussions, forward-thinking presentations, and insightful paper contributions included here reflect the urgency and creativity needed to navigate the evolving educational landscape.

In a time marked by rapid technological advancement and a global shift toward strategic, skill-based learning, the integration of business intelligence and innovation into academic frameworks is no longer optional—it is essential. These proceedings offer multidisciplinary perspectives and practical models that showcase how education can adapt to remain relevant, inclusive, and impactful. Each paper selected for publication represents rigorous thought and a shared vision of educational transformation. The diversity of contributions—spanning countries, academic disciplines, and institutional roles—demonstrates the global commitment to change and collaboration.

I extend heartfelt thanks to all the paper presenters, keynote speakers, session chairs, and reviewers who enriched this volume with their knowledge and critical engagement. I am also deeply grateful to the organizing committee for their tireless coordination and to the participating institutions for their support.

May this volume serve not only as a record of scholarly dialogue but also as a springboard for future research, partnerships, and policy innovation in higher education. Let us continue this journey—of learning, evolving, and leading education into the future.

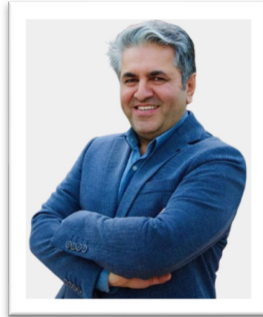
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MESSAGE FROM

Dr. HAMED BARJESTEH

Associate Professor, Institution Islamic Azad university, Iran



It has been a privilege to serve as a Co-Editor for this conference proceedings volume, a scholarly culmination of the International Virtual Conference on “*Higher Education 360°: Integrating Business Strategy, Technology, and Innovation in the Digital Age.*” As an academician dedicated to curriculum development, digital transformation, and policy in higher education, I was honored to deliver the Inaugural Address and engage with thought leaders from diverse cultural and disciplinary backgrounds.

In my address titled “*The Digital Renaissance: Transforming Higher Education through Strategic and Sustainable Practice,*” I emphasized the necessity for educational institutions to move beyond temporary digital adjustments and instead embed long-term strategies for sustainable growth. This renaissance is not simply about technological upgrades; it is about a paradigm shift in how we view learning, knowledge dissemination, and institutional relevance in a fast-changing world.

This conference acted as a true global dialogue, bringing together educators, policymakers, technologists, and researchers to explore the challenges and opportunities of transforming higher education. It was heartening to witness the enthusiastic exchange of models and solutions—from the use of artificial intelligence in learning environments to cross-border educational partnerships and digital inclusion initiatives.

The articles and insights documented in this proceedings volume reflect a collective commitment to moving forward with vision and responsibility. As Co-Editor, I take pride in supporting a platform that not only elevates academic discourse but also encourages real-world action. Each paper included in this compilation adds a unique voice to the larger mission of global academic transformation.

I thank the organizing team for their unwavering commitment, my fellow editors for their intellectual guidance, and all the presenters for their inspiring contributions. May this publication serve as both a reference and a rallying point for institutions striving to meet the demands of a digital, interconnected, and purpose-driven educational future.

MESSAGE FROM

Dr. MEHDI MOOCHEHRZSDEH

Founder, Institution Zerodale Center for Research in Entrepreneurship Education
and Development, Canada



It is with immense honor and academic enthusiasm that I extend my reflections as Co-Editor of this significant volume stemming from the International Virtual Conference on “*Higher Education 360°: Integrating Business Strategy, Technology, and Innovation in the Digital Age.*” As a keynote speaker and research advocate in the domain of entrepreneurship and innovation in education, I am deeply inspired by the powerful insights, cross-border knowledge exchange, and actionable frameworks presented throughout this two-day conference.

In my keynote address, I focused on “*Reimagining Higher Education through Digital Innovation: Driving Change through Technology, Academia, and Industry for a Sustainable Educational Ecosystem.*” This theme resonates with my mission at the Zerodale Center, where we bridge the gap between educational institutions and entrepreneurial ecosystems through digital strategies and research-led innovation. The conference, in many ways, mirrored this mission—placing innovation at the core of strategic educational transformation.

One of the most impactful takeaways from this gathering was the common commitment among international scholars to not only adapt but to lead change in higher education. From integrating artificial intelligence in pedagogy to aligning curricula with real-world industry needs, the presented works reaffirmed that education is no longer confined to the classroom. It is evolving into a dynamic, interconnected, and entrepreneurial force for global good.

As a Co-Editor, I am proud to see this proceedings volume capture the vibrant diversity of thought and depth of scholarship that the conference offered. These contributions form more than just a record—they provide a roadmap for universities and institutions across the world to implement visionary, sustainable, and student-centric models of education.

I thank my fellow editors, the organizing committee, and all contributors for their dedication to excellence. May this publication serve as a lasting academic beacon and inspire continued collaboration in transforming education for the digital future.

MESSAGE FROM

Dr. ABD GHOFUR

Founder, Share – To – Care Volunteers, Indonesia



It is with a heart full of gratitude and a mind enriched by dialogue that I pen this message as Co-Editor of the proceedings of the International Virtual Conference on “*Higher Education 360°: Integrating Business Strategy, Technology, and Innovation in the Digital Age.*” As the moderator of this prestigious conference and founder of the Share-To-Care Volunteers initiative, I was honored to serve not only as a facilitator but also as a learner in a truly global exchange of knowledge and vision.

This conference brought together academicians, researchers, and thought leaders from more than ten countries, each contributing unique perspectives on the transformation of higher education. As I moderated the sessions and witnessed the diverse paper presentations, it became evident that education in the 21st century must go far beyond traditional models. We are called to develop institutions that are not only technologically advanced but also human-centered, values-driven, and socially responsible.

In my opening remarks, I highlighted the importance of community, collaboration, and compassion in education—values that we at Share-To-Care Volunteers deeply cherish. The conference echoed those values in its inclusive structure and shared commitment to advancing knowledge that empowers, not just informs.

What we see in these proceedings is more than a collection of academic papers. It is a testament to the power of global academic unity. Each contribution reflects a sincere effort to integrate innovation with pedagogy, technology with values, and research with real-world impact.

I extend my heartfelt thanks to the organizing team, the fellow co-editors, and each presenter whose voice now resonates through these pages. Let this volume inspire not only future research but also a renewed commitment to education that serves all—with strategy, innovation, and heart.

SPEAKER'S MESSAGE

Prof. Luis Miguel Cardoso – Portugal



“Participating in this esteemed international conference has been both a privilege and a profound learning experience. The theme—‘Higher Education 360°: Integrating Business Strategy, Technology, and Innovation in the Digital Age’—perfectly resonates with the challenges and opportunities facing European higher education today. As an educator and researcher from Portugal, I view education as the foundation upon which we must build resilient societies capable of adapting to disruption and change.

The discussions held during this conference have emphasized a necessary shift from traditional models to forward-looking approaches that embrace interdisciplinary thinking, entrepreneurial spirit, and digital transformation. In Portugal, initiatives for digital literacy and industry-academia linkages are gaining momentum, but global collaboration is crucial to scale these efforts.

What made this conference exceptional was its inclusivity—scholars from Asia, the Middle East, Europe, and beyond came together with a shared vision. The diversity of ideas, challenges, and innovations presented reaffirmed that although our contexts may differ, our goals converge. This unified approach to education as a strategic pillar for national development is vital.

I commend the organizers for facilitating such a seamless and intellectually stimulating virtual experience. Their efforts have fostered not just knowledge exchange but deep reflection on how higher education must evolve to remain relevant in the 21st century.

Let this conference be the start of lasting academic bridges between nations and institutions. I look forward to future collaborations, research partnerships, and collective efforts that empower educators and learners alike across the globe.”

SPEAKER'S MESSAGE

Dr. Neda Fatehi Rad – Iran



“It is with great enthusiasm that I extend my heartfelt appreciation to the organizing committee of the Higher Education 360° International Conference. Participating in this forum has been an enriching experience, offering a unique platform for international dialogue on how business strategy, technology, and innovation can be integrated into the future of academia.

From the perspective of Iranian institutions, we are undergoing a pivotal transition—seeking to preserve cultural and academic heritage while embracing digital tools, AI-powered learning, and entrepreneurial education. This conference illuminated the global relevance of such efforts and offered diverse frameworks that can guide institutional reform.

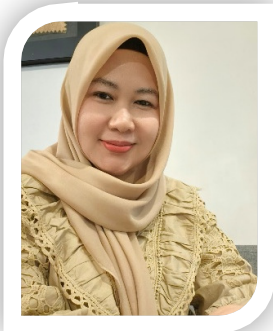
One of the most powerful elements of this gathering was its inclusivity. Whether discussing policy innovation, digital pedagogy, or strategic academic partnerships, each speaker contributed a vital voice to this collective conversation. I was particularly inspired by the emphasis on collaboration and the redefinition of educators' roles in a digitally driven world.

As we envision the classrooms of tomorrow, we must remember that technology is a tool—but vision, empathy, and purpose are the drivers. The discussions here challenged us to think not just about what we teach, but how and why we teach it.

My sincerest gratitude goes to the organizers for hosting this visionary and well-curated event. It provided a space not just for academic discourse, but also for intellectual growth and mutual respect. May the seeds planted here bloom into global educational ecosystems that are equitable, innovative, and sustainable.”

SPEAKER'S MESSAGE

Dr. Nurul Fadhillah – Indonesia



“It is with sincere gratitude that I participated in the International Virtual Conference on ‘Higher Education 360°.’ The focus on integrating business strategy, technology, and innovation in education is both timely and visionary. In Indonesia, our academic institutions are navigating a dynamic shift—balancing traditional classroom settings with the necessity of digital transformation. This conference gave us the rare opportunity to connect globally and examine how similar challenges are being addressed in different contexts.

As an educator, I believe we are standing at a critical juncture in history. The need to equip students with not only theoretical knowledge but also technological and entrepreneurial skills is now imperative. Our role is transforming—from instructors to mentors, innovators, and global collaborators.

The diverse presentations and expert insights shared during this event were enlightening. I was especially impressed by the way this platform facilitated cultural exchange and deep reflection on the future of higher education. It was not just about tools and strategies—it was about creating a community of educators who are future-ready.

The virtual format also demonstrated how education itself must become more flexible, inclusive, and accessible. The very structure of this conference echoed its theme, showcasing the power of digital tools to unite minds across borders.

I express my heartfelt appreciation to the organizing team for their exceptional coordination and vision. May the insights gained here inspire global partnerships and collective action in transforming education into a more equitable, strategic, and technologically empowered force for good.”

SPEAKER'S MESSAGE

Dr. Mohd Zuber – India



“I extend my warmest thanks to the organizing committee for the opportunity to participate in this vital conference. The theme—‘Higher Education 360°’—touches the core of what academic leadership needs to embody in the 21st century. As someone engaged with both classroom teaching and institutional development in India, I found the discussions incredibly relevant and forward-thinking.

What resonated most with me was the call to embrace strategic alignment between academia and industry. We must move from isolated academic silos to ecosystems where education fuels entrepreneurship, technological literacy, and social responsibility. In India, the emphasis on National Education Policy 2020 aligns well with this global vision of integrated, interdisciplinary education.

The international diversity of this conference elevated the experience. It was fascinating to learn how different countries are navigating similar issues: adapting to AI, redefining pedagogy, leveraging edtech, and ensuring that learning outcomes are meaningful and measurable.

This conference was not merely academic—it was transformative. It allowed for introspection, collaboration, and the co-creation of ideas that transcend geographical boundaries. I return to my institution with renewed vision and concrete strategies to implement.

I commend the organizing team for curating a balanced blend of scholarly discussion and practical reflection. This event is a testament to how virtual platforms can bridge global academia, inspiring collective growth. I hope this is only the beginning of a larger, sustained dialogue toward transforming higher education worldwide.”

SPEAKER'S MESSAGE

Dr. Andi Asrifan – Indonesia



“My sincere thanks to the organizing committee for inviting me to contribute to this prestigious international conference on ‘Higher Education 360°.’ As a scholar from Indonesia, I found the theme deeply relevant and inspiring—especially as we navigate the post-pandemic educational landscape where digital learning, academic entrepreneurship, and global connectivity have become central.

The focus on integrating business strategy and technological innovation into academic models is a vital shift for developing nations like ours. We are witnessing a growing awareness that traditional instruction is no longer sufficient. Students today must be prepared for dynamic global careers, and that preparation starts with a reimagined curriculum and digitally fluent teaching methodologies.

This conference has created a vital space for collaborative exchange among educators who are not only aware of this global transition but are also actively working toward it. I was particularly moved by the discussions around education as a driver of inclusive progress, where institutions are not just centers of learning but also agents of economic and social development.

The intellectual diversity represented here—from different continents, disciplines, and cultures—was truly enriching. It reminded us that innovation in higher education must be inclusive, adaptable, and culturally grounded.

To the organizing team: thank you for your meticulous planning and warm hospitality throughout the virtual sessions. This experience has not only expanded my professional insights but has also opened doors for future cross-border collaborations. I look forward to seeing how the ideas seeded here grow into transformative action across our institutions.”

SPEAKER'S MESSAGE

Dr. Gul Erkol Bayram – Turkey



“It has been an absolute honor to be part of the International Virtual Conference on ‘Higher Education 360°.’ As a participant from Turkey, I commend the initiative to bring together a global academic community to explore the intersections of strategy, technology, and innovation in shaping the future of higher education.

The conference’s thematic focus reflects an urgent global need—to move beyond the conventional classroom and redefine how we impart knowledge, develop competencies, and prepare future citizens. In Turkey, we are increasingly aligning with global trends by incorporating digital learning platforms, interdisciplinary courses, and strategic university-industry linkages. This conference provided valuable international context to our national efforts.

The wide-ranging perspectives shared—from artificial intelligence in pedagogy to entrepreneurial thinking in university governance—offered me a renewed sense of purpose as an educator. I was particularly inspired by the commitment shown by every speaker to inclusivity, student empowerment, and ethical technology use.

This was more than just an academic gathering; it was a convergence of minds with a shared mission—to make education more responsive to the world it serves. The virtual nature of this event itself underscored how technology can bridge gaps and create equal platforms for knowledge sharing.

I sincerely thank the organizers for their vision, dedication, and seamless execution of this high-caliber academic event. The takeaways are many, and I am certain that the relationships and ideas built here will resonate long into the future. Let us continue this journey of educational transformation together.”

SPEAKER'S MESSAGE

Dr. Muthmainnah – Indonesia



“Participating in this remarkable global conference has reaffirmed my belief in the power of international academic collaboration. The theme ‘Higher Education 360°’ is more than timely—it is transformative. In Indonesia, like in many developing nations, we are striving to build educational systems that not only impart knowledge but also instill adaptability, creativity, and a global mindset. This conference helped spotlight that journey.

One of the most important insights I gained was the emphasis on strategic thinking in curriculum design. Our institutions must prepare students not just for employment, but for leadership, innovation, and ethical decision-making in an unpredictable world. Technology, when guided by vision and purpose, can enhance this goal.

Each session I attended was rich in thought and global in relevance. From data-driven learning environments to the fusion of humanities and digital skills, the ideas shared here were both intellectually stimulating and practically useful. As an academic leader, I have gained valuable strategies to bring back to my institution, especially around integrating local wisdom into global learning models.

The camaraderie among participants from diverse backgrounds made this event even more impactful. It highlighted how educators worldwide are united by a common cause: to elevate the learning experience and ensure its relevance in the digital age.

I offer heartfelt thanks to the organizing committee for providing this outstanding platform. May this initiative continue to inspire academic excellence and innovation across borders. Together, we are planting the seeds for a more connected, inclusive, and forward-thinking future in higher education.”

SPEAKER'S MESSAGE

Ana Mae Monteza – Philippines



“Joining this international conference on ‘Higher Education 360°’ has been an inspiring milestone in my academic journey. As an educator and researcher from the Philippines, I am deeply committed to integrating meaningful innovation and inclusive practices into higher education. This conference amplified that mission by bringing together brilliant minds from across the globe to explore how strategy, technology, and creativity can reshape learning systems.

In the Philippines, we face distinct challenges—limited resources, digital access gaps, and shifting student needs. Yet, we also see opportunity: to create solutions tailored to our cultural and educational context, while aligning with global standards. The dialogues I witnessed during this conference highlighted that educational reform is not a one-size-fits-all process. It requires empathy, adaptability, and the courage to experiment.

One of the most striking elements of this event was its emphasis on collaboration—not just between institutions, but between disciplines and cultures. We are stronger when we work together, when we learn from each other’s successes and failures. I take back with me not just ideas, but a renewed commitment to mentor and guide my students to become innovative and globally aware citizens.

To the organizing committee: your vision, coordination, and hospitality were outstanding. You have built a truly global platform where we could reflect, reimagine, and reconnect as educators. Thank you for giving voice to so many unique perspectives. May the outcomes of this event ripple far beyond its schedule, shaping the classrooms of today and the leaders of tomorrow.”

SPEAKER'S MESSAGE

Dr. Khorshid Mousavi – Iraq



“It is with immense respect and gratitude that I acknowledge my participation in the International Virtual Conference on ‘Higher Education 360°.’ From my academic experience in Iraq, I see firsthand the urgency to rebuild and reimagine higher education—not only as a path to knowledge but as a bridge to peace, empowerment, and progress.

This conference provided a rare and meaningful platform where educators from vastly different contexts came together to discuss common goals: strategic transformation, technological integration, and innovation in pedagogy. It was particularly encouraging to witness conversations about equity and access, which are deeply relevant in regions affected by instability and inequality.

The sessions I attended offered not only theoretical insights but practical models that can be adapted in conflict-impacted environments like ours. As we strive to revitalize our institutions, the inspiration and guidance drawn from global best practices are invaluable. Equally, the empathy and understanding shared among speakers reflected the human side of education—one that prioritizes dignity, dialogue, and resilience.

For me, this conference was more than an academic exchange—it was a call to action. It reminded me that education is not just an intellectual endeavor, but a transformative force that can rebuild communities, heal divisions, and shape hopeful futures.

I extend my heartfelt appreciation to the organizers for their dedication, professionalism, and inclusiveness. This event stands as a testament to what can be achieved when minds and hearts unite beyond borders. I look forward to future collaborations that continue to uplift and empower educators and students around the world.”

SPEAKER'S MESSAGE

Dr. Madya Tuslin Binti – Indonesia



“Being part of the Higher Education 360° International Conference has been a profound and meaningful experience. The initiative to explore the synergies between business strategy, educational innovation, and technology is not only timely—it is transformative. As an academician from Indonesia, I view this as an essential dialogue that empowers institutions to meet the needs of a rapidly evolving global society.

The insights presented throughout the conference reaffirmed that education must become more than content delivery—it must be a system of growth, adaptability, and collaboration. I was particularly impressed by the discussions on curriculum redesign, digital learning tools, and the importance of aligning academic programs with market realities and societal challenges.

In Southeast Asia, we are working toward bridging the gap between academic knowledge and industry demands. The perspectives shared during this event offered valuable frameworks and case studies that I am eager to contextualize within my own institution’s practices.

This conference was also a celebration of diversity—of thought, language, and learning methods. The way it brought together professionals from across continents speaks to the power of virtual connectivity in enhancing academic diplomacy. The sense of unity, purpose, and passion was palpable in every session.

I commend the organizers for their vision and meticulous planning. Their effort has not only created an academic event but a network of collaboration that I hope will continue to grow. I return to my campus with new knowledge, meaningful connections, and a deeper commitment to educational excellence and innovation.”

SPEAKER'S MESSAGE

Dr. K. J. Varghese – India



“It was a great privilege to participate in the International Virtual Conference on ‘Higher Education 360°.’ This theme could not be more relevant at this critical juncture in global education. As a professional from India, I see higher education at the crossroads—striving to harmonize traditional academic excellence with the urgency for digital transformation and strategic relevance.

What this conference offered was not merely a platform for presentation, but a catalyst for rethinking the core purpose of higher education. The rich discussions around industry-academic collaboration, leadership in digital learning, and policy-level reform gave me insights I believe will influence how institutions in India shape their futures.

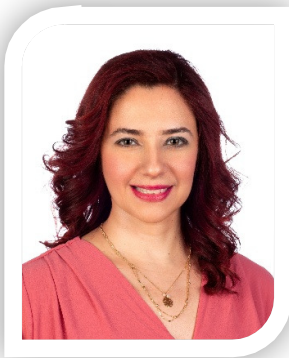
I was particularly inspired by the emphasis on interdisciplinary learning and the convergence of business thinking with academic governance. In today’s fast-paced world, universities must not only be centers of knowledge creation but also hubs for entrepreneurship, innovation, and social impact.

What stood out to me was the truly international character of this conference. Each speaker brought a unique cultural and academic lens, yet we all found common ground in our pursuit of excellence and equity in education. This alignment of global values with local aspirations is the essence of what ‘360°’ represents.

I express my deep appreciation to the organizing team for crafting such a well-structured, inclusive, and intellectually rich event. The clarity of purpose and execution reflected a deep commitment to educational reform. I return to my academic environment with renewed energy, practical strategies, and a deep sense of global belonging. May the impact of this conference continue to echo across classrooms and campuses for years to come.”

SPEAKER'S MESSAGE

Dr. Rania Lampou – Greece



“Participating in the Higher Education 360° International Conference has been both an honor and an enlightening experience. As a science educator and innovation advocate from Greece, I am passionate about bringing transformative thinking into classrooms—and this event perfectly aligned with that vision. The focus on integrating business strategy, technology, and innovation into higher education reflects the global shift toward more holistic, adaptable, and learner-centric systems.

The conference was not only about sharing knowledge—it was about reimagining the very structure and mission of higher education. I was especially impressed by the conversations that went beyond tools and techniques to emphasize creativity, emotional intelligence, and lifelong learning as core educational values.

One key takeaway for me was the growing importance of transdisciplinary education. We must move beyond rigid academic boundaries and equip our students with the skills to think critically, act ethically, and adapt quickly. Technology is a means—but the end must be human development and societal progress.

The international nature of this event added immense depth. Hearing perspectives from Asia, the Middle East, Africa, and Europe created a mosaic of ideas that showed just how interconnected our challenges—and solutions—are. Despite cultural and systemic differences, our shared commitment to educational transformation united us.

I deeply thank the organizing committee for their meticulous planning and inclusive approach. You have succeeded in creating not just a conference, but a global dialogue. I leave this experience more inspired than ever to continue contributing to education that is innovative, inclusive, and truly global in spirit. Let us continue working together to turn vision into impact.”

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REVISITING THE PUZZLING METHOD: A COGNITIVE APPROACH TO BUILDING RESILIENCE THROUGH CRITICAL THINKING

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Abstract

Puzzling Method Rooted in cognitive linguistics, constructivist pedagogy, and contrastive analysis, the Puzzling Method promotes grammatical mastery through native-to-target language mapping, structured puzzles, and scaffolded sentence-building activities. This paper explores the theoretical foundation and instructional implications of *puzzle-based pedagogy* as a complementary approach to traditional content-driven learning models. A puzzle-based pedagogy creates a broader contextual framework and fosters critical thinking as well as logical reasoning skills that can then be used to improve a student's performance on content-specific assessments. By engaging learners in domain-independent problem-solving activities, this method encourages metacognitive awareness and cognitive flexibility (skills that are transferable across disciplines). The study also compares puzzle-based learning with problem- and project-based approaches, highlighting how puzzles can serve as both an entry point into complex reasoning and a reinforcement tool for conceptual understanding. Practical applications, assessment strategies, and the pedagogical value of integrating puzzles into mainstream curricula are also discussed.

Keywords: Critical Thinking; Emotional Intelligence; Puzzling Method; Resilience

Introduction

Education has always been a cornerstone of human development, serving as the primary mechanism for transmitting knowledge, culture, and skills across generations (Kucuksuleymanoglu, 2025). From oral traditions around tribal fires to today's AI-enhanced classrooms, teaching methods have continually evolved in response to shifting philosophies, technologies, and societal needs (Barjesteh & Isaei, 2024). In early human societies, education was informal and experiential, grounded in observation, imitation, and storytelling. The invention of writing catalyzed a major transformation, enabling the formal transmission of complex knowledge in early civilizations such as Mesopotamia, Egypt, the Indus Valley, and China. Ancient Greece introduced dialogic methods through philosophers like Socrates, Plato, and Aristotle, laying the foundations for critical thinking in Western education. In parallel, Confucian teachings in China and the gurukul system in India emphasized moral development and holistic learning. Islamic education thrived through madrassas, preserving and expanding classical knowledge, particularly in science and mathematics (Wijayatunga, 2024). Medieval Europe's educational systems were predominantly religious, centered on memorization and rote learning. However, intellectual advancements continued in the Islamic world. The Renaissance reignited interest in classical learning, integrating the arts and sciences into broader curricula. Enlightenment thinkers like Rousseau, Pestalozzi, and Froebel introduced child-centered approaches, focusing on natural learning, emotional development, and play-based instruction (Setyadi et al., 2025). The Industrial Revolution ushered in mass public education systems aimed at producing a literate, efficient workforce. Standardized curricula and rigid classroom structures mirrored factory models. Progressive educators such as John

Dewey and Maria Montessori challenged this paradigm, advocating for experiential learning, creativity, and respect for children's developmental needs.

In the 20th century, educational psychology reshaped instructional practices. Behaviorism introduced structured reinforcement techniques, while cognitive theories from Piaget and Vygotsky emphasized active learning, language, and social interaction. Bloom's Taxonomy and Gardner's theory of multiple intelligences further expanded educators' understanding of how students learn and what they are capable of achieving. The digital age has radically transformed education, introducing computers, the internet, and AI-driven platforms that support personalized, interactive learning experiences. Pedagogical trends now emphasize 21st-century skills (critical thinking, creativity, communication, and collaboration) alongside emotional intelligence, inclusivity, and global citizenship. Amid this dynamic educational landscape, the Puzzling Method emerges as a contemporary, learner-centered approach with particular relevance for language instruction (Hazaymeh et al., 2025). Rooted in cognitive linguistics, constructivist pedagogy, and contrastive analysis, this method integrates guided discovery, template-based learning, and scaffolded instruction. It reframes grammar not as a set of abstract rules but as a structured system that can be internalized through puzzles, patterns, and language mapping between a learner's native and target languages. Through sentence-building activities and scaffolded progression, the Puzzling Method fosters grammatical mastery, metalinguistic awareness, and cognitive flexibility. It aligns with Vygotsky's Zone of Proximal Development by supporting learners as they move from basic to complex constructions. Its flexible, inclusive design makes it ideal for diverse, multilingual classrooms (Tarigan et al., 2025). The puzzling Method offers more than an instructional technique—it represents a paradigm shift. It connects the wisdom of past educational traditions with the adaptability and personalization of modern pedagogy. In doing so, it empowers learners not just to absorb language but to use it as a tool for thinking, creating, and connecting.

Despite the extensive evolution of educational philosophies and instructional methodologies—from ancient oral traditions to AI-powered personalized learning environments—many language classrooms today still rely heavily on rigid, rule-based grammar instruction that lacks contextual relevance and learner engagement. Traditional methods often fail to foster metalinguistic awareness, critical thinking, and active language use, particularly in multilingual and diverse educational settings. As a result, learners struggle to internalize grammatical structures and transfer them into meaningful communication. In response to this challenge, the Puzzling Method has been introduced as a learner-centered approach grounded in cognitive linguistics and constructivist pedagogy. However, empirical research on its efficacy, applicability across varied learner contexts, and impact on grammatical competence and cognitive flexibility remains limited. There is a pressing need to investigate whether and how this method can bridge the gap between abstract grammar instruction and functional language use in contemporary classrooms. While historical and modern pedagogies have each contributed to our understanding of how students learn, there remains a notable gap in integrating these perspectives into a unified, practical framework that effectively supports grammar acquisition in multilingual, cognitively diverse classrooms. Existing studies often focus on traditional or digital-enhanced approaches but overlook innovative, puzzle-based methodologies that align with constructivist and sociocultural learning theories. Specifically, there is a lack of rigorous academic inquiry into: (a) The cognitive and linguistic outcomes of implementing the Puzzling Method; (b) Its alignment with 21st-century educational goals such as critical thinking, problem-solving, and learner autonomy; (c) Its comparative effectiveness versus conventional grammar teaching strategies. Addressing this gap is essential for reimagining grammar instruction in a way that is not only pedagogically sound but also adaptable to the evolving needs of global language learners.

Critical thinking

Critical thinking is rooted in Socrates's theory. According to Socrates, one's justification could not rest on confused meaning, inadequate evidence, and self-contradictory beliefs (Fahim & Eslamdoost, 2014; Davies & Barnett, 2015; Caceres, Nussbaum, & Ortiz, 2020; Fatehi Rad et al., 2024). Sound thinking, citing Kanik (2010), needs "approaching issues with critical scrutiny and does not allow human beings to commit themselves to beliefs they do not know to be absolutely true because knowledge they acquire is subject to change under conditions in life" (p. 13). In a recent publication, Dewey (1933) defines critical thinking as a kind of "reflective thinking" which incorporate in "an active, persistent, and careful consideration of a belief or supposed form of knowledge in the light of the grounds which support it and the further conclusions to which it tends" (p. 9). According to Reed and Kromely (2001), critical thinkers apply sound justification in reaching a conclusion. To Menkes (2005), critical thinking is a "cognitive skill that determines how well someone gathers, processes, and applies information in order to identify the best way to reach a particular goal or navigate a complex situation" (p. 3). For Paul (1995), critical thinking contains rational aspect of human thinking. In Paul's words "thinking about your thinking while you're thinking to make your thinking better" (p. 91). Paul (1995, p. 21) provided the most comprehensive definition, incorporating most features as follows:

Components of Critical Thinking

According to Kanik (2010), critical thinking skill encompasses: (a) a set of information and belief-generating and processing skills, (b) the disposition to use those skills to guide behavior (p. 19). Following Kanik, critical thinking is not the acquisition and storage of information but seeking and processing information. Moreover, it contradicts with simple possession of skills. Critical thinking theorists stress on continuous employment of those skills (Mozafari & Barjesteh, 2016; El Soufi & See, 2019; Caceres et al, 2020; Leach et al., 2020). Critical thinkers apply their knowledge and information to obtain broader perceptions and deeper insights. To cite Kanik, critical thinkers focus on "identifying central issues and assumptions in an argument, recognizing important relationships, making correct inferences from data, deducing conclusions from information or data provided, interpreting whether conclusions are warranted based on the data given" (p. 20). Likewise, Halpern (1997) provides the taxonomy of critical thinking as verbal-reasoning skills, argument-analysis skills, thinking skills, for example, hypothesis testing, reasoning in terms of probability, decision making, and problem-solving skills. On the other hand, most researchers highlight the successful application of skills. One can have critical thinking skills, but he doesn't use them, signifying that he has a low level of critical thinking ability. Given that, personality and traits of mind are necessary to be considered in critical thinkers. For Ennis (1991), critical thinking incorporates "exercising open-mindedness, considering the total situation, staying relevant to the main point, looking for alternatives, making judgment based on evidence, using credible sources, to name but a few" (cited in Kanik, 2010, p. 20). Similarly, Paul (1995) argues that considering others' ideas and opinions in order to be open-minded goes beyond egocentric and socio-centric thinking, which is what good critical thinkers do. The following figure summarizes the components of CT.

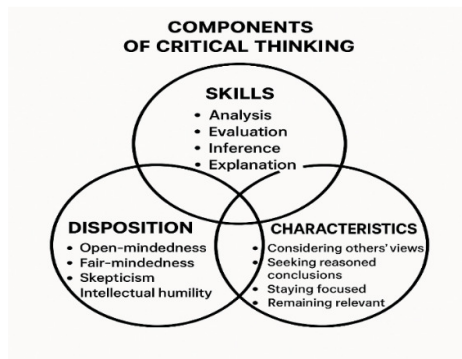


Figure 1. Components of Critical Thinking

Resilience: What is it and why does it matter for students?

Resilience, broadly defined as the capacity to adapt and thrive in the face of adversity, is increasingly recognized as a crucial competence in both academic and real-world settings (Masten, 2014). In educational contexts, resilience is not only about emotional strength but also about cognitive flexibility—the ability to face intellectual challenges without disengaging or succumbing to defeat. The Puzzling Method, by engaging students in structured problem-solving tasks that emphasize critical thinking and logical reasoning, provides a low-stakes environment for practicing persistence, mental agility, and reflective judgment. Each puzzle becomes a microcosm of real-life uncertainty, allowing learners to confront ambiguity, tolerate failure, and iteratively refine their strategies—hallmarks of resilient behavior (Duckworth et al., 2007). Moreover, by decoupling learning from domain-specific content and focusing on transferable reasoning skills, The Puzzling Method promotes self-efficacy and intrinsic motivation—two psychological factors closely linked to resilience (Bandura, 1997). As students learn to navigate novel, ill-structured problems, they are encouraged to question their assumptions, revise their hypotheses, and embrace uncertainty as part of the learning process. This iterative, inquiry-based approach fosters not only cognitive endurance but also emotional regulation and adaptability, equipping learners with the mindset required to approach future challenges with confidence and creativity. Thus, the Puzzling Method serves as a pedagogical model that not only enhances academic skills but also cultivates the psychological resilience essential for lifelong learning and success in dynamic environments. Resilience refers to a set of essential abilities that help individuals navigate challenges effectively and recover from adversity. For students, developing resilience is crucial for both academic success and personal growth. Key components of resilience include: (a) *Emotional regulation* – the ability to stay calm and express emotions constructively, even in stressful situations; (b) *Impulse control* – the capacity to pause before acting, manage urges, delay gratification, and persist toward goals; (c) *Causal analysis* – the skill of accurately analyzing problems and identifying their root causes; (d) *Empathy* – understanding and being sensitive to the emotions and needs of others; (e) *Realistic optimism* – maintaining a hopeful outlook while staying grounded in reality; (f) *Self-efficacy* – believing in one’s ability to overcome challenges and manage stress; (g) *Opportunity-seeking* – actively pursuing new experiences and building supportive relationships. Overall, students who demonstrate higher levels of resilience tend to experience greater well-being, achieve better academic outcomes, and engage in fewer behavioral issues. This is largely because resilient individuals are motivated and courageous in the face of difficulties—they acknowledge problems realistically and respond with confidence, persistence, and a positive mindset (Booth & Neill, 2017).

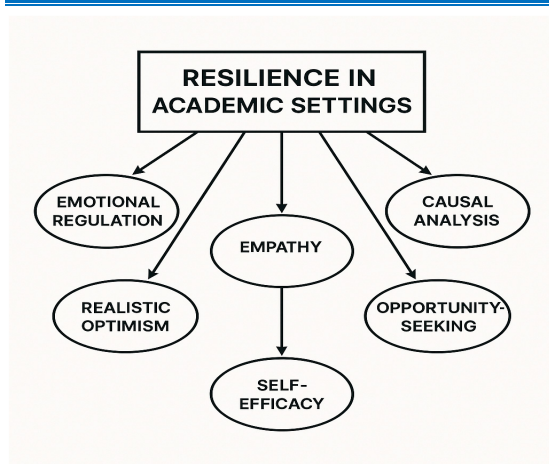


Figure 2. Graphic Conceptualization of Resilience

Toward a Dynamic Future and the Emergence of Puzzle-based Pedagogy

The evolution of teaching methods is a testament to humanity's ongoing pursuit of better ways to learn and grow. From oral traditions to online classrooms, each shift reflects deeper cultural, scientific, and technological transformations. Today, we stand at a crossroads where education is no longer confined to classrooms or textbooks. The most effective teaching methods are those that are adaptable, inclusive, and grounded in understanding how people truly learn (Hazaymeh et al., 2025). As we move into the future, the key to effective education lies in balancing tradition with innovation, honoring the wisdom of past methods while embracing new tools and ideas that empower learners for a rapidly changing world. Central to the Puzzling Method is the concept of **language mapping**, where grammatical structures in the target language are taught through direct parallels in the learner's first language. This draws on contrastive analysis and transfer theory, suggesting that known structures in L1 can facilitate the acquisition of similar patterns in L2 when taught explicitly. The Puzzling Method aligns with **constructivist theories** of learning (Piaget, Vygotsky), which posit that learners build knowledge actively through experience, reflection, and contextual application. Problem-based learning tasks foster deeper cognitive engagement, transforming grammar exercises into meaningful, student-driven discovery processes. Drawing from Vygotsky's **Zone of Proximal Development**, the method emphasizes scaffolded progression—from short answers to extended responses, from simple structures to complex constructions—thus supporting learners at every stage.

Puzzle-based learning approach: Concept, Principles, Theoretical Assumptions

Puzzle-based instruction

The puzzling method in education, also known as the puzzle-based learning approach, is an instructional framework that involves presenting students with challenging problems to work together to solve complex, real-world problems that require creative thinking, critical reasoning, and problem-solving skills (Hazaymeh et al., 2025). Focuses on cognitive (critical thinking) and emotional (psycho-emotional) development. PBI can also be used within the revised framework of Bloom's Taxonomy to include all six categories of cognitive processes by which thinkers encounter and work with knowledge, comprising "remember",

“understand”, “apply”, “analyze”, “evaluate”, and “create” (Figure 3). Involving learners in higher-level cognitive activities can positively impact their levels of engagement and knowledge retention.



Figure 3. Bloom's Taxonomy for the cognitive processes

The primary aim of Puzzle-based Learning is to equip students with the foundational skills necessary for effective real-world problem-solving. At its most advanced stage, real-world problem-solving involves three core skill sets: managing unpredictability and evolving situations, utilizing specialized knowledge and techniques, and engaging in critical thinking through the use of broad problem-solving strategies. These essential competencies are reflected in the three interconnected learning approaches illustrated below: project-based, problem-based, and puzzle-based learning.

Puzzle-based Learning and Problem-Solving Framework

The overarching goal of **Puzzle-Based Learning** is to establish a foundational framework that enables learners to become proficient and adaptive problem solvers in real-world contexts. At the most advanced level, effective problem-solving entails a multidimensional skill set that encompasses: (1) the ability to operate under uncertain and evolving conditions, (2) the application of domain-specific knowledge and disciplinary methodologies, and (3) the development and use of general, transferable problem-solving strategies grounded in critical thinking. These skill domains are conceptually represented through a triangular model that connects three complementary instructional paradigms: **Project-Based Learning**, **Problem-Based Learning**, and **Puzzle-Based Learning**. Each approach targets a specific facet of cognitive and applied competence. Project-Based Learning emphasizes managing ambiguity, collaboration, and decision-making in complex, often unpredictable scenarios. Problem-Based Learning focuses on structured inquiry and the systematic application of disciplinary knowledge to defined challenges. In contrast, Puzzle-Based Learning fosters abstract reasoning, logic-based inquiry, and the formulation of generalizable strategies across varying contexts.

Together, these interconnected modalities provide a robust pedagogical framework that cultivates not only subject-matter expertise but also the cognitive agility required to transfer skills across disciplines and settings. This triangulated model highlights the synergy between context-specific expertise and abstract reasoning, fostering a more comprehensive preparation for real-world problem-solving (Michalewicz & Michalewicz, 2008).

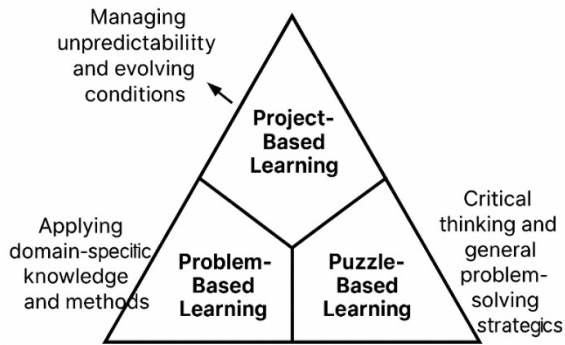


Figure 4. Conceptual Distinctions Among Project-, Problem-, and Puzzle-Based Learning

In the layered continuum of learning models, each pedagogical approach builds progressively upon the cognitive skills fostered by its predecessors. **Project-based learning** and **problem-based learning** are both well-established educational methodologies that support complex reasoning and collaborative inquiry (Thomas, 2000; Barrows, 1986). Project-based learning is particularly suited for navigating ambiguous, real-world challenges that do not have a single correct solution. For instance, a question such as *“How can we improve treatment adherence among patients with cystic fibrosis?”* often entails multifaceted considerations, stakeholder perspectives, and iterative testing of strategies. The pedagogical aims of project-based learning include developing tolerance for ambiguity, synthesizing diverse approaches, validating outcomes through practical testing, and fostering interdisciplinary teamwork (Thomas, 2000). In contrast, **problem-based learning** is closely tied to the acquisition of deep domain-specific knowledge. It is typically emphasized in discipline-specific undergraduate education, such as in courses on electromagnetism, data structures, or circuit theory, where students are expected to apply known theories to structured problems (Barrows, 1986). In both models, the **problem serves as the central driver of learning**: students must assess their current understanding, identify what further knowledge is needed, and devise strategies to close the knowledge gap. **Puzzle-based learning**, however, diverges by focusing on **domain-neutral cognitive skills**, such as abstract reasoning, logical deduction, and metacognitive reflection. Learners are encouraged to analyze not only the puzzle itself but also their internal thought processes—asking questions like *“What assumptions did I make?”*, *“Why didn’t I arrive at the solution earlier?”*, and *“What reasoning steps led me to the answer?”* (Michalewicz & Michalewicz, 2008). This introspective aspect differentiates puzzle-based learning from more content-dependent frameworks. One key distinction between a **puzzle** and a **problem** lies in the **degree of domain knowledge required**. While problem-solving often necessitates specialized training, puzzles are generally solvable through universal reasoning skills. Ideally, puzzles should be accessible to individuals from varied backgrounds—a scientist, a musician, or an artist should all be capable of solving the same puzzle using critical thinking rather than field-specific expertise. The cognitive demands of problem- and puzzle-based learning can be likened to two investigative roles: the former reflects the practice of a field investigator relying on tools and disciplinary insight, while the latter resembles an armchair detective who relies primarily on reason and logic. Nonetheless, the distinction is not always absolute. A puzzle may require domain knowledge, and a problem may incorporate an inherently puzzling dimension. Thus, the terms “problem” and “puzzle” are often used interchangeably throughout

the literature and even within this framework. What remains central is the goal of fostering ***broad reasoning abilities*** that extend beyond methodical execution and engage learners in reflective, cross-disciplinary thinking (Michalewicz & Michalewicz, 2008).

Principles

PBI employs *active learning strategies* through *naturalistic educational programs* to engage the mind, motivate learners, and transform learning into real-life situations. It connects knowledge and skills under two main theoretical underpinnings: loving pedagogy and critical thinking. PBI proposed seven-step learning strategies (Energizing, Navigating, Generating, Applying, Gauging, and Extending) that involve learners actively performing activities that reinforce their learning. A traditional lecture-based pedagogy conveys information and content without developing sufficient critical thinking skills or problem-solving abilities. A puzzle-based pedagogy creates a broader contextual framework and fosters critical thinking and logical reasoning skills that can then be used to improve a student's performance on content-specific assessments. Hodge et al. (2009) presented that “self-authorship enables learners to evaluate information critically, form their judgments, and collaborate with others to act wisely”. To fill this gap in the literature, PBI was proposed to explore how PBI can help develop *critical thinking skills* and enhance *emotional intelligence* in modern classrooms. The key feature of puzzle-based learning that distinguishes it from problem-based or project-based learning is the emphasis on domain-independent critical thinking and abstract reasoning (Falkner et al., 2010; Merrick, 2010; Michalewicz, 2008). This is not to say that a puzzle-based learning course is about presenting and discussing a variety of puzzles, but instead that it is about presenting and understanding problem-solving principles through puzzles that serve both as an illustration of broad concepts and, of course, specific material (Falkner et al., 2010; Michalewicz, 2008).

Principle	Theoretical Aspect	Practical Example
1. Critical Thinking	Encourages analysis, synthesis, and evaluation rather than rote memorization.	Students deduce grammar rules by solving a language puzzle with intentional gaps.
2. Engagement	Active participation enhances cognitive processing and retention.	Learners solve a riddle-based task to unlock the next stage of a lesson module.
3. Challenge	Tasks should be moderately challenging to stimulate problem-solving without frustration.	A logic-based word arrangement task that requires testing different combinations.
4. Discovery Learning	The learner constructs knowledge through exploration and inference.	Students infer a scientific principle by solving a visual or data-based puzzle.
5. Feedback Integration	Immediate and implicit feedback promotes learning without explicit correction.	A language app that visually reacts when the learner makes a logical syntax error.

6. Contextualization	Learning is situated in meaningful, realistic contexts for deeper understanding.	A social studies puzzle that reconstructs a historical timeline with missing events.
7. Iterative Reasoning	Promotes hypothesis testing, trial-and-error, and gradual refinement of solutions.	A math puzzle where incorrect answers lead to hints rather than direct solutions.

Conclusion

Assessment in the Puzzling Method is not merely a tool for assigning grades; it is an **integrated, dynamic, and student-centered process** that supports growth, fosters confidence, and develops both linguistic and cognitive skills. Formative assessment provides real-time guidance, summative tasks consolidate learning through meaningful output, and peer and self-assessment promote autonomy and reflective thinking. When students are actively involved in evaluating their progress and sharing learning strategies, they become more invested, resilient, and successful in their language learning journey. While highly effective in guided instruction, the Puzzling Method may initially require significant teacher preparation, particularly in designing language-aligned templates and puzzles. Furthermore, the approach may not be equally practical across all language pairs, especially where syntactic structures diverge significantly. Training for instructors is essential to ensure linguistic accuracy and pedagogical alignment. The Puzzling Method repositions grammar from a static subject to a dynamic, interactive learning process. By aligning native and target language structures through puzzles and templates, learners engage more deeply and retain more effectively. The method blends cognitive development, language awareness, and learner agency into a holistic instructional framework. Future research could explore its impact across age groups, its adaptability to different languages, and its integration with AI-driven learning tools.

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THE IMPACT OF USING ONLINE BRAINSTORMING TECHNIQUES ON IMPROVEMENT OF READING ACTIVITIES: INNOVATION IN EFL LEARNERS' INTERACTION VIA VIRTUAL CONTEXT

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Abstract

Today online brainstorming techniques which needs unique kinds of interaction have an important role among learning methods. The method reveals a significant shift away from teacher-centered method containing students' intellectual effort, or student and teacher together. The present research study aimed at discovering the effect of using online brainstorming techniques in electronic classes and discussed the effectiveness of this method on beginners' reading comprehension ability. Besides, uncovering the attitudes of EFL learners towards the using online brainstorming techniques in E-classes was the other goal of the study. In order to achieve the aims, a mixed-methods study was used. The data gathering tools included a homogeneity test, pretests and posttests in reading comprehension as well as the students' interviews. The sample of the study consisted of 38 elementary students of both genders, divided in two groups of experimental and control. The experimental group (N=19) received online brainstorming techniques in reading activities while the control group (N=19) practiced without the mentioned instruction. The results of independent samples t-tests and paired t-tests indicated that the members in the treatment group outperformed those in control. Moreover, findings of the interviews revealed both positive and negative views of the students on using brainstorming techniques in virtual context. The research study offered some pedagogical implications for EFL teachers and learners in using online brainstorming techniques in virtual context.

Keywords: Online Brainstorming Techniques, Reading Comprehension, Interaction, Virtual Context

Introduction

Nowadays English learning is essential because English is the international communication medium. The language is necessary for different activities, including education, politics, and socio-economics (Mckay, 2012). Reading skill, in particular, is the most important skill in second or foreign language learning. Previously reading has been a passive act and reader identified the meaning of small text unit. Over the time the meaning of reading has changed. Reader plays a very important role in reading and seen as an active participant who brings his/her stored knowledge in understanding the text without just heavily

relying on the reading material. In the late century, reading has been an interactive process where reader finds out the meaning of a text by both relying on the printed text and on the previously stored knowledge (Carrell, 2008).

Chatwirote (2013) suggested that the teachers could provide reading promoting online activities, such as the activities that interest the learners. The activities should include the reading objectives that suit the learners and teacher's interest. Reading is considered as a significant life skill and the lack of reading comprehension influence the learner's academic advances. Reading comprehension, based on Razi (2010), is significant for learners who consider English as a foreign language, therefore, there is a little chance to speak English in their daily lives. Many teaching techniques have been used to improve students' comprehension level. One of such techniques is believed to be brainstorming. Brainstorming is a technique which causes achievement and capability in reading comprehension. We cannot just rely on conventional techniques of reading instruction that involve reading text, explaining its meaning, and answering its questions. Kang (2014) believed that reading includes so many mental activities that cause students to understand the main idea of the text thus learners should employ tools which improve text comprehension. Students need techniques that enable them to have full comprehension of what they have already read.

Brainstorming is a combination of a relaxed, informal approach to problem-solving and lateral thinking. People are asked to find ideas and thoughts that can at first seem to be a bit irrelevant. It is a powerful practice that creates new ideas, and motivates students to carry out reading comprehension tasks. The idea is to use some of these ideas to form original, creative solutions to problems. Even some seemingly useless ideas can spark still more ideas. The goal of brainstorming is to direct people to new ways of thinking and break from the usual way of reasoning. Brainstorming contributes to the generation of creative solutions to a problem. It teaches students to break away from old patterns of reasoning to new unexplored paths of thinking.

Today, the need to develop reading skill among English language learners is more than before. To achieve this goal and to give a more engaged and active role to the learners, teachers and researchers have encouraged to look for strategies to promote more interactional roles to the learners. One challenge that teachers often face is that not all students can learn at the same rate, and as a result not all of them can participate during a discussion session. For this reason, they need strategies and methods that might foster their language skills. Even though brainstorming has been a valuable learning tool for learners, it has not been used enough in EFL classrooms. This lack of usage maybe attributed to the lack of research on this topic. Due to little attention that has been given to the mentioned method and its implications especially in the language classes in Iran, it is not known if the learners really think positively about the effectiveness and practicality of this method. Therefore, the present research has been designed to study the effectiveness of brainstorming on the improvement of reading activities as they relate to learning outcomes at the language institutes. In other words, the current paper investigated whether the online brainstorming activities help the EFL learners to get better comprehension or not, and examined the EFL learners' attitudes on the use of it. In order to meet the research objectives, the following questions have been designed:

1. To what extent does using online brainstorming techniques affect the improvement of reading activities of EFL learners' reading comprehension?
2. What are the EFL learners' views on the use of online brainstorming techniques in terms of improving reading comprehension ability?

Literature Review

Theoretical Framework

Brainstorming has a great importance in the learning process. Referred to its importance for students in (Widiastuti et al., 2022,) as follows:

1. Helps students to solve problems, an innovative solution.
2. Helps students to benefit from the ideas of others through the development and build on them.
3. Helps the cohesion of the students and build relationships among them and assess the views of others.

The following criteria are essential to the idea-generation phase of a brainstorming session (Adams, 2009):

1. There is no criticism, evaluation, judgment, or defense of ideas. During the brainstorming session. The purpose of brainstorming is to generate as many ideas related to the topic as possible in the time allowed. Evaluation, judgment, and selection of ideas are the purposes of subsequent sessions.
2. Freewheeling and free association is encouraged. Group members are asked to voice any solutions they can think of, no matter how outrageous or impractical they seem. There is no limit on “wild” or “far-fetched” ideas. Every idea is to be expressed. It is easier to tone down an idea and to select out later than it is to think up new and creative possibilities.
3. Quantity is more desired than quality. Group members are encouraged to contribute as many ideas as they think of. The greater the number of ideas generated, the more likely it is that there will be several useful ideas.
4. Building on ideas is encouraged. Combining, adding to, and “piggybacking” on ideas is part of the creative process. Members can suggest improvements, variations, or combinations of previous ideas.

According to Humaidan (2005, cited in Al-Khatib, 2012), the major purpose of brainstorming as a teaching strategy is to foster and enhance communication skill, help to promote thinking and decision-making skill as well as foster different viewpoints and opinions. It may equally be used in all key areas of learning. The teacher must equally be able to guide and give aid as necessary considering the class environment as such considerations often determine the outcomes. In brainstorming techniques, the instructor carefully plans the lesson to reach the desired learning outcomes. The group interacts in response to questions, and the instructor refrains from entering the discussion as an active participant. Students are encouraged to learn about the subject by actively sharing information, experiences, and opinions. The flow of communication is a transaction among all the students rather than recitation and response between individual students and the instructor.

Stages in brainstorming technique are presented by Al-Blwi (2010);

1. Performing warm up: To catch the attention of the students and incline their interests towards the topic.
2. Getting ideas: To know what the students think about the topic by bringing out different ideas from the students.

3. Writing/displaying the four principles of brainstorming on board: To involve the students in the systematic way by focusing the process as displayed for brainstorming and channelizing their knowledge output.
4. Writing and presenting ideas (Proposed): To write the stimulated information and display it.
5. Frustration must be avoided: To include all the students of class by gathering all sorts of points.
6. Boring must be avoided: Linking all the points and information and integrating the knowledge in a way that the session is made more and more interesting.

Regarding the above discussion, the present study was conducted within the concept of brainstorming techniques, considering Al-Blwi's procedures.

Related Studies

Al-abadi (2010) conducted a study to explore the effect of an educational program in developing creative thinking skills among gifted disabled students. Students were distributed equally into an experimental group and a control group, 11 students on each group. The findings showed that there were statistically significant differences between the means of both groups on creative thinking test in the favor of the experimental group attributed to the educational program. In another study, Al-qarni (2011) studied the effectiveness of brainstorming strategy in developing creative thinking among 115 students. The findings showed that there are statistical differences between the means of both groups on the test in the favor of the experimental group studied through brainstorming. Moreover, Al-khatib (2012) carried out a study to investigate the influence of applying brainstorming method on promoting creative problem-solving skills through female learners in university. The results indicated that there were statistical important differences between the experimental group and control group in total score.

In Iranian context, Shokrpour and Fotovatian (2013) conducted an experimental study to determine the effect of consciousness metacognitive strategies on a group of Iranian EFL students' reading comprehension. The results revealed that compared to the control group the experimental group showed an important improvement in reading comprehension at the end of treatment period. In addition, Maleki and Dabaghi (2014) examined the influence of concept maps on improving reading comprehension of students at some high schools. Findings showed that students who received treatment performed better in reading comprehension. In the same year, Ghabanchi and Behrooznia (2014) aimed at showing the effects of brainstorming as a pre-reading technique on reading comprehension and also the ability of EFL learners in critical thinking. Findings indicated that students' reading comprehension ability as well as their critical thinking ability promote after employing brainstorming strategy. In the same vein, Maghsoudi and Hariran (2015) conducted a study to investigate the influence of brainstorming method on EFL students' writing performance. According to data analysis the experimental group performed better in post-test comparing to their performance in pre-test.

Method

Research Design

The purpose of this study is to find out the effectiveness of online brainstorming techniques on improvement of reading activities of Iranian EFL learners. Quantitative research starts with an experimental design in which a hypothesis is followed by the quantification of data and some sort of numerical analysis is carried out. In the present study, the effectiveness of online brainstorming techniques was analyzed through scores on students' performance which was shown in terms of numerical data.

Participants and Sample

The participants of this study were pre-intermediate EFL learners who were studying English in a private Language Institute of Iran, Kerman. 40 EFL learners were selected based on convenience sampling method and to check the homogeneity of the sample, all were participated in Oxford placement test. Based on the results, 38 participants whose scores on the language proficiency test fell within ± 1 standard deviation of the mean score, participated in the present research. Then, they were randomly assigned in to two groups as experimental and control. All of the participants received English for 4.5 hours a week and there were no opportunities of actual English use outside the classroom. The whole research was done in 12 sessions.

Data Collection and Analysis Procedures

A pre- and post-test of reading comprehension were administered to all students for analyzing their reading performance. Both tests were extracted from their English textbook, American English File 2 (pre-intermediate level). The tests were administered to measure the students' reading comprehension ability twice, before and after the treatment, once as pre-test and the other as post-test. In the experimental group, the students benefited from online learning through Big Blue Bottom (BBB). It is a web-based tool that facilitates communication among students and teachers and suggests rich resources of peer feedback and teacher feedback simultaneously. As mentioned earlier, the present study was conducted within the concept of brainstorming technique, considering Al-Blwi's (2010) procedures. The researchers modeled the method (using online brainstorming techniques) and gave learners enough practice on using it. On the other hand, the control group received conventional method of teaching. They were taught and learnt under the supervision of the teacher (one of the researchers) in classroom condition. The teacher just relied on the conventional techniques of reading instruction which involved reading text, explaining the meaning, and answering the questions. In fact, the students of this group were taught and learnt under the supervision of the teacher (researcher) in classroom condition. Fortunately, none of the students were absent during the treatment, and this would increase the reliability of the results.

After the treatment on the experimental group, a post test of 20 multiple-choice items including three reading comprehension texts was applied to achieve the purpose of the research. All of the participants in control and experimental groups participated in post-test and they had to finish the test within 40 minutes. In addition to the tests of reading comprehension, a related questionnaire developed by Unin and Bearing (2018) were addressed to the experimental group to investigate the students' attitudes on the online brainstorming techniques. The Participants had to read the instructions carefully and responded the questions without any time limitation. Finally, all the data were gathered for analyzing and interpreting. The scores of the tests and questionnaires were taken for analyzing and all data were inserted into SPSS software for analyzing and interpreting. Descriptive statistics was used in this study and the data from the questionnaires and tests were analyzed quantitatively. To compare the performance of the participants on the post-tests, the mean score after the treatment was compared through an independent sample t-test to highlight the possible difference between the students' achievement in pre and posttest. Moreover, descriptive statistics including frequency, percentage, and mean were included to find out the effect of the mentioned techniques on reading comprehension skill.

Results

Quantitative method was applied in conducting this research. In fact, the data were collected through pre and post-tests' scores and the related questionnaires and the following tables present the result.

Data Analysis of Pre and Posttest' Scores

After gathering the scores, sample t-tests were run to analyze the quantitative data including pre-test and post-test results. Then, the differences between the mean scores in the first examination and second examination were measured. The null hypothesis for testing two independent samples, indicated the equality of means, and the opposite assumptions inferred the inequality of the means. The significance level was considered to be 0.05. If p- value is less than 0.05, it means there is a significance difference between the mean scores of two tests in control and experimental groups. Table 1 displayed the numbers, mean, and standard deviation of all the scores of two groups, control and experimental. According to careful analysis, in both groups the mean scores has increased in the post- test. However, the difference between the means of the pre- and posttest in experimental group is remarkable. In fact, the mean of the post- test in experimental group has increased more than the mean score of control group.

Table 1- Descriptive Statistics of T-Tests

Paired Samples Statistics					
		Mean	N	Std. Deviation	Std. Error Mean
Control Group	Pre-test	13.50	19	2.188	0.489
	Post-test	13.90	19	2.198	0.491
Experimental Group	Pre-test	13.10	19	2.245	0.502
	Post-test	15.35	19	2.476	0.553

Table 2- Paired Sample T-test

Paired Samples Test									
		Paired Differences					t	df	Sig. (2-tailed)
		Mean	Std. Deviation	Std. Error Mean	95% Interval	Confidence of the			
					Difference				
					Lower	Upper			
Con. Group	Pretest								
	- Posttest	-0.40	1.46	0.32	-1.08	0.28	-1.22	18	0.23
Exp. Group	Pretest								
	- Posttest	-2.25	1.83	0.40	-3.10	-1.39	-5.49	18	0.00

After implementing online brainstorming techniques in the experimental group, the scores of the tests in two groups demonstrated the different means. In other words, the means of pre-test and post-test in control group are 13.50 and 13.90 respectively. The difference between the two means is 0.40, which indicates a slight difference between two means. However, the difference between mean score of the experimental group is observable. Based on the careful analysis, the mean of the pre-test was 13.10, and the mean of the post-test was 15.35 in experimental group. In other words, EFL learners had better performance in reading comprehension post-test ($13.10 < 15.35$). Thus, a remarkable difference between them is clear, which is 2.25. Furthermore, p-value in control group is more than 0.05, but in experimental group is 0.00 that illustrates the mean scores of the experimental group has changed. Findings of the first question proved that there were statistically significant differences between the mean scores of the experimental group in the pre and post-test.

Results of the Questionnaire

In respect to the second research question, a related questionnaire was published to explore the attitudes of the participants on the effectiveness of online brainstorming techniques on reading comprehension. The results are presented by the table 3.

Table 3- Descriptive Statistics of the Questionnaire

N	Items	N	M	SD
1	This method makes me understand the text better.	19	3.8	3.24
2	This method gives me the opportunity to practice thinking.	19	3.9	3.42
3	This method creates a positive environment for reading skills.	19	3.25	3.16
4	I can work with anyone I like.	19	4.0	4.52
5	I feel more relaxed using this method.	19	3.75	2.73

6	This method is fun and interesting.	19	4.15	4.30
7	Many interesting topics are given to us using this method.	19	3.95	3.08
8	My vocabulary is enriched.	19	4.03	4.30
9	I am more active and not shy to speak.	19	3.15	2.22
10	I can give out more ideas faster and easily.	19	3.95	3.53
11	I am given more chances to cooperate with my classmates.	19	3.45	2.87
12	I can share my ideas with my group members.	19	3.65	2.82
13	I am motivated to work by my group members.	19	3.45	3.16
14	My mistakes are tolerated by my group members.	19	2.09	1.41
15	I am more confident when working in groups.	19	3.98	3.39
16	I can communicate better using this method.	19	3.65	3.08
17	I can work more independently from my teacher.	19	3.35	2.23

Discussion

According to Alam Khan (2013), brainstorming is an activity used to generate ideas in small groups, it focuses on the participation. It can help students to learn to take risks. The purpose is to generate as many ideas as possible within a specified time-period. These ideas are not evaluated until the end and a wide range of ideas is often produced. As seen above, there is a statistically significant difference between the means of the performance of the group that received training and the control that did not receive any training even in the total score of the test. This may be attributed to the nature of brainstorming strategy as a collective discussion strategy that encourage students to generate the highest number of ideas that are varied and creative in a spontaneous and free open climate that is not critical and does not limit the freedom of launching ideas. Moreover, its nature based on phases allows students to move from one step to another freely after completing the previous step. The effect of this strategy in developing creative thinking as a whole and in its sub-skills may be attributed to the advantages of this strategy that are accepted among students. Some of those advantages are the preparing element and making students ready to participate in the sessions as well as joy environment that provide students with a free climate that does not contain any critics and interference. The results of the current research support the study conducted by Widiastuti et al., (2022) who investigates the brainstorming strategy and indicated the importance of conducting brainstorming strategies in learning activities. The findings of this study are also consistent with Maleki and Dabaghi, 2014; Salehi, Jahandar, and Khodabandelou, 2013; Maghsoudi & Hariran, 2015. They conducted some research to investigate the effectiveness of brainstorming in developing English language skills as reading, speaking, and writing. The results revealed that compared to the control group the experimental group showed an important improvement in the considered language skill at the end of treatment period.

The second research question of the study highlighted the EFL learners' attitudes toward the online brainstorming techniques. Descriptive statistics revealed the students' attitudes by the related statements on the effectiveness of these techniques. Based on the results of table 3, most of the students of the experimental group understood the text better by this method. Also, they stated that using brainstorming method gave them the opportunity to practice thinking and the method created a positive environment for reading skills. Moreover, the students claimed that they can work with anyone and felt more relaxed using this method. In their views, this method is fun and interesting. They thought that interesting topics were given to them and their vocabulary is enriched. Most of them were active and not shy to speak, and they were able to give more ideas faster and easily. Furthermore, they were given more chances to cooperate with their classmates, and they could share their ideas with group members. Besides, the EFL learners were motivated to work by group members and their mistakes were tolerated by group members. Their mistakes were also tolerated by using these techniques. In their views, they can communicate better using this method and they could work more independently through using online brainstorming techniques. The result supported findings of previous studies. Unin, (2014), for example, maintained brainstorming strategy to address students' reluctance to speak English more voluntarily. Besides, brainstorming strategy, can be used as a way that reflects active involvement and participation (Brandes & Ginnis, 2006). Also, the outcomes of the project support the existing literature by Slavin (2005), as well as, Hamzah and Lu (2010), who found that students are more confident and motivated to express their ideas when brainstorming in small groups.

Conclusions

By carrying out this study, it has been found that brainstorming is an effective teaching strategy, and it can be easily introduced into language classes. The research studies previously discussed, confirmed that brainstorming technique helps students to become better learners. In addition, it is a fun activity which students enjoy and well worth trying out in language classes. Moreover, brainstorming had a strong positive effect on the classroom environment and the learning behavior of the students. The students in the present study got involved in the brainstorming enthusiastically. The EFL students participated constructively and enthusiastically in the classroom interventions and indicated their positive perceptions toward the brainstorming techniques. The results also suggest the majority of students participating in the study felt positive about the effectiveness of the brainstorming techniques used in an EFL class. Most of the students agreed to some extent that the techniques had helped them prepare more effectively for their language learning tasks. The success of "brainstorming" as a teaching-learning strategy is further supported by the observation and experience of the teacher-researcher. This is quite important to bear the fact that brainstorming

technique can be used in many ways in educational setting in general. It can be intentionally planned as a two-way learning process involving both the students as learners and the teacher as facilitator.

Since the research has carried out in reading classes in which employing the brainstorming techniques has not been previously used, the results may provide information to compare teacher-centered and learner-centered techniques, as learner centeredness is a key element in learning effectively. Additionally, the teachers who have not used this method in their reading classes, may be encouraged to use it after seeing that using brainstorming technique affects the individuals learning. If there is a positive response to this method and the teachers find out an influence of it on learners' attitudes to reading courses, they may want to design it in their classes. The obtained results can be useful and effective not only for teachers and learners, but also improve teaching methods, curriculum development, syllabus design, and pedagogical implications. Since using this method in the language classes is based on new approaches, the results would really reveal the attitudes of the learners and teachers in the effectiveness of such technique. Besides that, this research is expected to give valuable information for English teachers about brainstorming as their technique used to generate a long list of diverse responses without making judgments about students' ideas.

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STRATEGIC SYNERGY IN THE DIGITAL AGE: INTEGRATING BUSINESS STRATEGY, TECHNOLOGY, AND INNOVATION IN HIGHER EDUCATION

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Abstract

The digital age has compelled higher education institutions (HEIs) to reevaluate and transform their traditional models by integrating business strategy, technology, and innovation. This literature review explores the intersection of these domains, highlighting the challenges and opportunities that arise from such integration. Drawing on recent studies and theoretical frameworks, the review examines how HEIs can adapt to the rapidly evolving digital landscape to enhance their competitiveness, relevance, and societal impact. Key themes include digital transformation, innovation ecosystems, strategic alignment, and the role of emerging technologies in reshaping higher education. The review identifies both systemic barriers and enabling strategies, emphasizing the importance of agile leadership and data-informed decision-making in fostering sustainable change.

Introduction

Higher education is undergoing a paradigm shift driven by rapid technological advancements, changing societal needs, and increasing competition (Barjesteh & Isaee, 2024; Butcher et al., 2024; Zaki et al., 2025). The demands of the digital economy are challenging traditional models of teaching, research, and community engagement, necessitating a strategic integration of business principles, technological innovation, and adaptive leadership (Ahn, Song, & Lee, 2024; Isaee & Barjesteh, 2023). This transformation involves more than just adopting new digital tools—it demands a fundamental rethinking of institutional missions, value propositions, and operational models.

Business strategy in HEIs now incorporates market-oriented thinking, entrepreneurial frameworks, and efficiency metrics traditionally associated with private enterprise (Al-Emran, Mezhyuev, & Kamaludin, 2024). At the same time, technology is enabling expanded access, personalized learning, and streamlined processes. Together, these forces demand innovation at both the pedagogical and institutional levels.

This literature review synthesizes current research on the integration of business strategy, technology, and innovation in higher education. It identifies best practices and frameworks that institutions can leverage to thrive in the digital age.

Digital Transformation in Higher Education

Digital transformation (DT) refers to the comprehensive application of digital technologies to all facets of institutional operations, reshaping both academic and administrative functions. DT differs from digitization in that it involves strategic, cultural, and systemic change (Ahn et al., 2024). Challenges include insufficient infrastructure, digital skill gaps, and resistance to change.

When effectively implemented, DT fosters personalized learning, boosts operational efficiency, and promotes innovation (Becker & Eube, 2018). Nevertheless, these benefits must

be weighed against ethical concerns like data privacy and algorithmic bias (Zhang et al., 2024). Institutions are increasingly using AI to analyze student engagement, predict dropout risks, and automate administrative processes, underscoring the need for ethical AI frameworks.

HEIs are also leveraging cloud-based learning management systems, digital credentialing through blockchain, and hybrid learning environments. These innovations enable new pedagogical approaches while reducing operational bottlenecks. However, such transformation requires ongoing investment in infrastructure, faculty training, and change management to ensure long-term sustainability.

Strategic Alignment and Business Models

The integration of business strategy into higher education involves aligning academic goals with market dynamics, operational excellence, and financial sustainability. This often entails Business Model Innovation (BMI), where institutions redesign value propositions, delivery mechanisms, and revenue streams (Al-Emran et al., 2024). However, entrenched academic traditions, regulatory hurdles, and internal inertia can impede innovation.

Strategic partnerships with industry, government, and global networks can mitigate these challenges. Collaborations support resource sharing, co-designed curricula, and real-world relevance, aligning HEI offerings with labor market needs (Etzkowitz & Leydesdorff, 2000). Many HEIs are also adopting lean and agile methodologies in administrative units to enhance service delivery and reduce inefficiencies. Increasingly, universities are exploring entrepreneurial education, innovation hubs, and incubators as new revenue models. These business strategies not only diversify income streams but also cultivate a culture of creativity and problem-solving.

Innovation Ecosystems and Collaborative Frameworks

HEIs are integral components of regional innovation ecosystems, facilitating knowledge creation, technology transfer, and entrepreneurial activity. The Triple Helix model (university–industry–government) explains how dynamic collaboration fosters systemic innovation (Etzkowitz & Leydesdorff, 2000). Expansions into the Quadruple and Quintuple Helix frameworks include civil society and environmental sustainability, respectively (Carayannis & Campbell, 2010).

Through these models, HEIs can evolve from siloed institutions into innovation hubs that address complex societal challenges. Cross-sector alliances and interdisciplinary centers of excellence exemplify how collaboration can fuel discovery and application. International partnerships further amplify impact by encouraging cross-border research and global talent pipelines. These ecosystems also nurture student entrepreneurship, technology commercialization, and knowledge-intensive startups, contributing to regional economic development and societal progress.

Emerging Technologies and Pedagogical Innovation

Emerging technologies, especially AI, machine learning, and blockchain, are redefining pedagogical methods. AI-powered platforms enable adaptive learning, automate grading, and generate insights for early intervention (Zhang et al., 2024). However, without proper data

governance, such tools may inadvertently reinforce inequities or compromise privacy. Pedagogical innovation must accompany technological adoption. Active learning, flipped classrooms, and project-based learning models are increasingly facilitated by technology, yet require thoughtful instructional design and faculty development to be effective. Gamification and immersive technologies like AR/VR also present new opportunities to engage digital-native learners in transformative ways (Zaki et al., 2025). Crucially, faculty must be empowered as co-designers in the innovation process, supported by professional learning communities, instructional designers, and ongoing digital literacy development.

Leadership and Organizational Change

Leadership plays a crucial role in enabling transformation. Institutional leaders must articulate a compelling vision, build digital capacity, and foster a culture that supports experimentation. Agile governance and distributed decision-making are becoming vital as HEIs manage increasingly complex innovation portfolios (Ahn et al., 2024). Change management must address not only systems and structures, but also the mindsets of faculty and staff. Continuous professional development, transparent communication, and inclusive planning processes are key enablers of sustained transformation. Leadership frameworks that embrace uncertainty, systems thinking, and stakeholder co-creation are especially valuable in guiding institutions through periods of disruption and reinvention.

Conclusion

The integration of business strategy, technology, and innovation is no longer optional for higher education, as it is imperative. HEIs must develop adaptive strategies that embrace complexity, foster cross-sector collaboration, and deploy technology with purpose and care. The literature reveals a roadmap for transformation grounded in alignment, agility, and accountability. As higher education continues to navigate the uncertainties of the digital era, success will depend on the capacity to innovate with integrity and lead with foresight.

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INVESTIGATING THE EFFECTIVENESS OF DIGITAL ANNOTATION TOOLS IN VOCABULARY LEARNING THROUGH LAUFER'S FIVE 'I' MODEL

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Abstract

This study explores the role of digital annotation tools in vocabulary learning, using Laufer's Five 'i's Model—Input, Instruction, Involvement, Interaction, and Interpretation—as a conceptual framework. Digital platforms such as Hypothesis, Kami, and Diigo enhance the learning experience by offering increased opportunities for learner engagement and improved vocabulary acquisition. These tools not only enable learners to engage more meaningfully with texts but also promote active learning through note-taking, highlighting, and collaborative discussions. A total of 60 intermediate EFL learners participated in the study over eight weeks, divided into two groups: one group used digital annotation tools during online textbook reading, while the other followed a traditional vocabulary learning approach. Data were collected through vocabulary tests, learner engagement questionnaires, and student interviews. The findings revealed that the experimental group demonstrated significantly better word recall and higher cognitive engagement. The use of digital annotation tools provided richer input, clearer instructional support, increased involvement, more interaction, and deeper interpretative opportunities. Qualitative data further indicated that learners using these tools exhibited greater autonomy, active participation, and reflective learning, making vocabulary acquisition a more individualized and collaborative process. The study concludes that well-implemented digital annotation strategies can enrich pedagogical practices and support more purposeful vocabulary instruction in EFL contexts.

Keywords: Digital annotation tools, Laufer's five 'i' model, Learner engagement, Technology-enhanced language learning, Vocabulary learning

1. Introduction

Over the last few years, the integration of digital technology in language learning has revolutionized how learners interact with texts and acquire new vocabulary (Sun & Chen, 2019; Yang, 2020). Among all the developments in technology, digital annotation platforms have been a promising innovation that can encourage deeper reading engagement (Liu & Lan, 2016). Tools such as Hypothesis, Kami, and Diigo not only allow students to highlight and annotate texts but also collaborate, share meaning, and personalize reading activities (Novak, Caruso, & Brown, 2012; Zheng & Warschauer, 2015).

The current study adopts Laufer's five "i" model—Input, Instruction, Involvement, Interaction, and Interpretation—as the theoretical framework to examine the effectiveness of computer-

based annotation tools in the acquisition of vocabulary by English as a Foreign Language (EFL) learners (Laufer, 2005; Laufer & Nation, 2012). While earlier studies revealed the strength of computer-based tools over traditional comprehension strategies in both comprehension and student engagement (Fredricks, Blumenfeld, & Paris, 2004; Barkaoui, 2016), fewer researchers have explicitly examined how those tools translate into cognitive and pedagogical models of vocabulary acquisition.

By using a mixed-methods approach, this research investigates how digital annotation tools aid in vocabulary learning and cognitive engagement compared to traditional teaching procedures. Findings are anticipated to offer language educators applied guidelines for integrating digital annotation tools into teaching designs guided by Laufer's model.

“To achieve the objectives of this study, the following research questions were posed:”

1. How do digital annotation tools affect the vocabulary acquisition of intermediate EFL learners based on Laufer's five 'i' model?

2. What are the differences in cognitive and affective engagement between learners using digital annotation tools and those using traditional vocabulary instruction methods?

3. How do learners perceive their experience using digital annotation tools in vocabulary learning in terms of motivation, interaction, and meaning interpretation?

2. Literature Review

2.1 Vocabulary Acquisition in EFL Contexts

Vocabulary acquisition is a cornerstone of second language learning, as it directly impacts learners' ability to comprehend texts, express ideas, and participate in communication. In EFL contexts, where authentic exposure is often limited, effective vocabulary instruction requires structured input, learner engagement, and active processing strategies (Nation, 2001; Schmitt, 2010).

2.2 Laufer's Three 'i' Model and Its Extension

Laufer (2005) proposed a pedagogical model for vocabulary acquisition that relies on three key elements:

- *Input*: Target vocabulary exposure in real contexts
- *Instruction*: Direct teaching and explanation of word use and meaning
- *Involvement*: Cognitive and affective participation in the learning task by the learners

The three elements are the foundation for effective vocabulary acquisition. However, the model is further extended in this study to include two additional dimensions:

- *Interaction*: Peer or teacher-learner conversation and joint negotiation of meaning
- *Interpretation*: Learners' personalization and recognition of meaning based on context and existing knowledge

This five-part model is suggested by the researcher as a more encompassing lens through which digital annotation tools can be assessed within language learning environments, specifically in how learners can be assisted in developing vocabulary in more dynamic and meaningful senses.

2.3 Digital Annotation Tools in Language Learning

Digital annotation software such as Hypothesis, Kami, and Diigo has revolutionized students' interaction with reading content. Students can highlight, annotate, and exchange interpretations in electronic reading content using these software tools, promoting engagement and reflective reading. Previous studies (Novak et al., 2012; Liu & Lan, 2016; Yang, 2020) have shown that digital annotation facilitates more engagement, collaborative learning, and deeper processing—all of which positively influence vocabulary retention.

3. Methodology

3.1 Research Design

A convergent mixed-methods design was employed for this study in order to explore the impact of digital annotation tools on vocabulary learning. The quantitative strand was responsible for the measurement of vocabulary gain via pre-tests and post-tests, and the qualitative strand investigated learner motivation and learner perceptions via interviews and open-ended questionnaire items. The convergence of the two strands provided greater insight into how and why digital annotation might affect vocabulary learning.

3.2 Participants

The study involved 60 intermediate-level EFL students of a private language institute in Tehran, aged between 18 and 24 years. All the participants received upper-intermediate reading and vocabulary classes and were randomly assigned to either the control group (n = 30) or the experimental group (n = 30). They were put on the basis of a standard Oxford Placement Test to ensure homogeneity in linguistic competence.

3.3 Instruments

- *Vocabulary Tests*: The researcher constructed a test of 40 target words, which was administered before and after the intervention in order to measure vocabulary gains.
- *Engagement Questionnaire*: A 20-item Likert-scale questionnaire with modifications from Fredricks et al. (2004) measured behavior, emotional, and cognitive engagement.
- *Semi-Structured Interviews*: Conducted among 10 participants in each group in order to better understand their experience and perception.
- *Digital Tools*: The experimental group used Hypothesis, an open-source web annotation platform that allows users to highlight, comment, and share notes on digital texts collaboratively.

3.4 Procedure

The study spanned eight weeks. Both groups of participants read semantically similar online materials during each session. The experimental group annotated texts with Hypothesis, engaged in peer discussions, and completed vocabulary-focused activities. The control group received traditional vocabulary lessons via printed worksheets and teacher discussion.

Vocabulary pre-tests were done at the start of the first week, and post-tests were done at the end of the eighth week. Engagement questionnaires were distributed in week 4 and repeated in week 8 to monitor change. Interviews in the last week were transcribed for thematic analysis.

3.5 Data Analysis

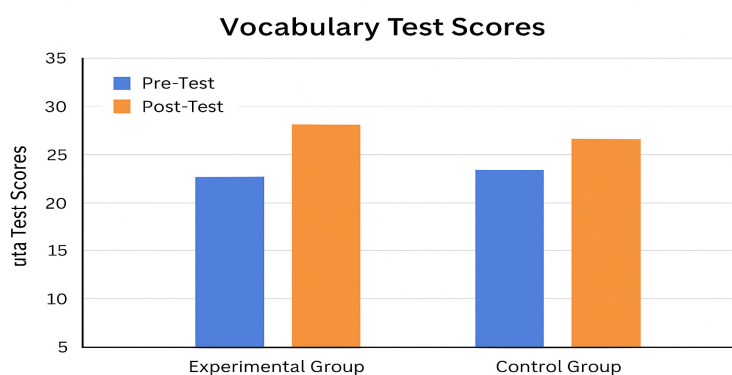
Quantitative data from questionnaires and vocabulary tests were contrasted in paired and independent t-tests in SPSS. Qualitative data from interviews were thematically coded using open and axial coding to enable the recognition of frequent themes in learner participation, annotation tool use, and perceived advantages.

4. Results

4.1 Quantitative Findings

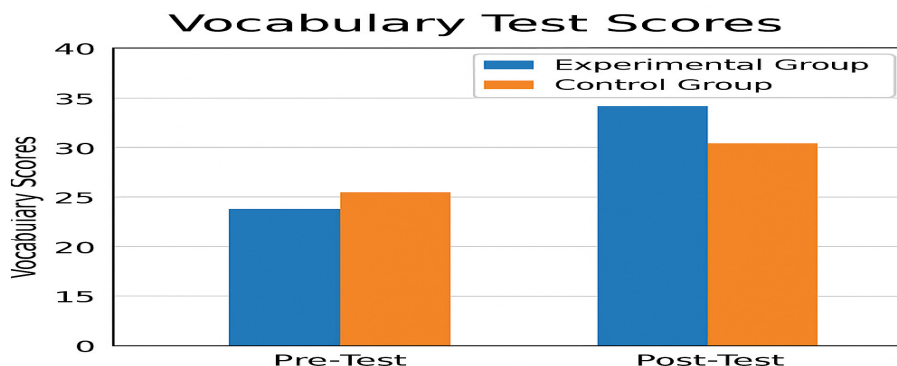
Both groups' vocabulary test results showed improvement after the intervention. But the experimental group with annotation tools using digital devices had a higher rise in vocabulary retention. Pre-test mean of the experimental group was 18.5 (SD = 3.2) and rose to 31.4 (SD = 2.9) during the post-test. In contrast, the control group mean pre-test was 19.1 (SD = 3.5), increasing to 26.2 (SD = 3.7). Independent t-tests showed that post-test score difference between groups was statistically significant ($t(58) = 5.87, p < .001$).

Figure 1. Comparison of vocabulary test scores between experimental and control groups (pre- and post-test).



Similarly, engagement questionnaire results showed higher cognitive and affective engagement among the experimental group in comparison to the control group, with a focus on motivation and active engagement in vocabulary practice. Paired sample t-tests revealed improvement in engagement scores over time in the experimental group ($p < .01$).

Figure 2. Comparison of engagement dimensions (cognitive, emotional, behavioral) in week 4 and week 8 for both groups



4.2 Qualitative Findings

Thematic analysis of interview responses revealed several significant themes supporting the quantitative findings:

- *Rich Input Access Enhanced*: Group members in the experimental group appreciated the ease of accessing definition, synonym, and contextual example through annotations linked directly to the text.
- *Enhanced Instruction and Guidance*: The availability of teachers' immediate feedback and peer scaffolding through digital annotation made students understand word usage better.
- *Increased Engagement and Motivation*: The students were more motivated and engaged in active learning of vocabulary with the use of annotation tools.
- *Interactive Learning Environment*: The value of interacting with others through exchange of comments and discussion for better understanding was highlighted by participants.
- *Meaning Construction through Interpretation*: The strength to adapt meaning by relating words to one's own experiences and contexts was highly valued.

5. Discussion

The findings of this study demonstrate that digital annotation tools can significantly enhance vocabulary learning by providing learners with enriched input, clearer instruction, and opportunities for active involvement, interaction, and interpretation. These findings support and extend Laufer's original three 'i' model by emphasizing the importance of interaction and interpretation as critical dimensions in digital learning environments. By merging the two 'I's, this research highlights the multileveled nature of vocabulary acquisition in technology-aided situations. The experimental group's increased gains in vocabulary and higher degrees of engagement suggest that computer-supported annotation facilitates a more dynamic and intersubjective learning process than existing methods.

The increased interaction that digital annotation technologies afford prompts learners to socially negotiate meaning, a trait that generally is lacking in conventional vocabulary instruction. Moreover, interpretation as a dimension encourages students to engage actively to personalise and construct meaning, in line with constructivist models of second-language acquisition.

The research is also in line with cognitive theory emphasizing the operation of deep processing in word memorization. Electronic annotation software, in combining multiple sensory and social inputs, encourages deeper mental processing and stronger memory traces.

6. Conclusion

This study set out to examine the effectiveness of digital annotation tools in vocabulary learning through the lens of Laufer's five 'i' model—involvement, interaction, interpretation, inferencing, and individual accountability. Using a mixed-methods design, the research provided valuable insights into both the measurable outcomes of vocabulary retention and the lived experiences of learners in digitally mediated environments.

Findings showed digital annotation software, specifically websites like Hypothesis, significantly boosted vocabulary acquisition via active text processing. Not only did students acquire more words, but they also exhibited greater levels of processing, critical thinking, and

reflection learning. These findings were highly correlated with the components of Laufer's model, which verify that optimal vocabulary acquisition occurs when learners are both socially and cognitively active.

The study showed that learners were able to derive meaning from context, acquire new vocabulary collaboratively, and take ownership of their learning through persistently working with digital activities. The peer-to-peer nature of the annotation tools supported peer interaction and peer feedback, with the traceability of the individuals' comments ensuring accountability. All these combined to provide a more substantial and enduring process of vocabulary acquisition.

Furthermore, the research also points to the pedagogical advantage of the integration of digital tools in learning vocabulary. Rather than applying typical, often decontextualized vocabulary exercises, digital annotation places learning into actual, interactive reading situations. This phenomenon signifies a direction toward student-centered teaching, where technology enhances autonomy, cooperation, and higher cognitive involvement.

Despite its strengths, the study also identifies limitations, including the short duration of the intervention and varying levels of participants' digital literacy. These limitations suggest possible lines of future research, including longitudinal studies and studies in more diverse school environments.

In conclusion, digital annotation tools, when guided by a robust theoretical model like Laufer's five 'i' framework, hold significant potential for enhancing vocabulary learning in EFL contexts. They not only improve lexical retention but also cultivate critical educational skills necessary for modern learners.

7.Implications

The findings of this study offer several pedagogical and practical implications for language educators, curriculum designers, and educational technology developers:

For Language Teachers: The inclusion of digital annotation tools in reading lessons can potentially enhance vocabulary acquisition since it makes reading interactive and collaborative. It would be crucial for teachers to consider teaching students proper usage of the tools and guiding them to critically engage with vocabulary presented in texts.

For Curriculum Designers: Vocabulary instruction should move beyond rote memorization and incorporate frameworks like Laufer's five 'i' model. Annotated reading tasks aligned with this model can be embedded into EFL curricula to foster deeper learning.

For Educational Technology Developers: There is an opportunity to design and refine annotation platforms that support vocabulary learning explicitly, with features that facilitate inferencing, personal reflection, and social interaction.

For Teacher Training Programs: Professional development programs should include training on how to implement digital tools like Hypothesis effectively in the classroom, both in terms of technical ability and pedagogical fit with models like Laufer's.

For Future Research: The experiment lends itself to further study of the long-term implications of annotation on vocabulary, as well as to its application in other proficiency levels and age groups. Comparative research on digital and paper-and-pen annotation would also be valuable.

Overall, the study shows that when technology is well integrated into established pedagogical frameworks, it can turn vocabulary learning into a richer, more productive, and more contextual process.

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EXPLORING THE ROLE OF CRITICAL PEDAGOGY IN SHAPING IRANIAN EFL TEACHERS' CLASSROOM TECHNOLOGY DECISION-MAKING IN MARGINALIZED AREAS

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Abstract

In contemporary times, technology is incorporated into language teaching and learning classes. This is a commonly encountered phenomenon. It is imperative to mention, however, that this considers the marginalized context's sociopolitical intricacies. Guided by critical pedagogy's lens (Freire, 1970; Giroux, 1988), this research looks at how Iranian EFL teachers operating from the bottom levels of the socio-economic ladder consider technology use—its use or lack thereof—in their classrooms. The dominant discourse shifts towards digitalization: national policies advocate for it. However, social inequity, infrastructural inequity, and limited teacher autonomy create local realities that are often ignored. This qualitative study analyzes the purposes of instruction and the degree of social activism undertaken by 20 EFL teachers in rural and neglected regions of Iran through semi-structured interviews. One of the research outcomes suggests that the choices made by the educators were not solely based on requirements or needs, rather, decisions were strongly influenced by idealistic values surrounding equity, empowerment, and the active role of learners within the education system. It is argued, in this case, that for meaningful technology, contextualization frameworks of integration should be guided by educators' critical changes aims, not imposed from above. Adaptations are considered for teacher training, policy formulation, and inclusive digital pedagogy in English language teaching and learning.

Keywords: Critical Pedagogy, EFL Teacher Decision-Making, Digital Inequity, Marginalized Educational Contexts

Introduction

Background

In the past decades, there has been a growing incorporation of technological innovations into pedagogical practice, and in particular, in language instruction (Warschauer, 2000). Emerging technologies enhance learners' motivation, their communicative competence, and grant access to authentic linguistic materials (Kern, 2006). Yet, this global transformation is not uniformly possible, especially in marginalized societies where some of the greatest obstacles include inadequate infrastructure, socio-economic disparities, and limited pedagogical autonomy for teachers (Warschauer & Matuchniak, 2010). In Iran, though there is growing national attention to inserting digital technologies into the pedagogical system, rural and economically disadvantaged regions pose unique obstacles that discourage successful implementation of such technologies. Critical pedagogy provides a theoretical foundation for confronting such asymmetries. Drawing from Freire (1970) and Giroux (1988), it emphasizes equity, empowerment, and social justice, urging educators to overturn exploitative power dynamics and empower students' critical awareness. Critical pedagogy, in language teaching, urges teachers to attend to the ways that technology can reinscribe or contest social inequalities,

particularly in situations of marginalization. Despite its relevance, there has been limited inquiry into the ways critical pedagogy has shaped technology-related decisions teachers make in these settings. This study explores Iranian English as a Foreign Language (EFL) teachers' use of technology in impoverished communities through the lens of critical pedagogy. The study aims to find out how their instructional decisions reflect a commitment to equity and learner agency in techno-poor settings.

Statement of the Problem

The Iranian educational system promotes classroom technology usage through official policies although schools in marginalized regions face implementation constraints. The combination of insufficient infrastructure along with scarce digital resources and centralized decision-making processes creates obstacles for teachers who want to integrate technology into their classrooms (Rahimi & Yadollahi, 2011). The current gap between educational policy goals and actual classroom activities requires an investigation into how English as a Foreign Language (EFL) teachers from disadvantaged areas choose to use technology in their classrooms. A thorough examination of how critical pedagogy influences their teaching methodology and social justice activities represents a necessary step to resolve these educational disparities.

Research Questions

This study explores the following questions:

RQ1: How do Iranian EFL teachers in marginalized areas perceive the role of technology in their teaching practices?

RQ2: What factors influence their decisions regarding the integration of technology in the classroom?

RQ3: How does critical pedagogy shape their approach to technology use in teaching?

The study seeks to understand how critical pedagogy together with social activism and teaching goals affect teachers' choices related to technology implementation.

Significance of the Study

This research expands the field of inquiry research by examining the dynamics of technology implementation in resource-constrained settings. Also, it emphasizes the role of critical pedagogy as a means of empowering instructors and learners alike, while simultaneously demonstrating the degree to which digital technology promotes social equity and equal treatment. Furthermore, this study emphasizes the degree to which educational objectives overlap with social activism in order to expand our understanding of radical pedagogical technology. Ultimately, the results should inform EFL teacher education and policy-making and inclusive teaching strategies to facilitate technological integration with equity.

Literature Review

Critical Pedagogy in Education

The field of education is based on critical pedagogy, which originated from Freire's work in 1970 and Giroux's work in 1988. Through education, Freire (1970) developed the term "conscientization" which describes the development of learner awareness regarding oppressive structures using dialogue and reflection. Giroux (1988) further developed the concept by presenting teachers as intellectuals who help students challenge established power structures. Through critical pedagogy in language instruction, students develop the ability to use language for both social criticism and social transformation (Shor, 1992). EFL education benefits greatly from this approach because language learning helps students overcome social and cultural boundaries (Canagarajah, 2005). The implementation of critical pedagogy together with technology remains an area that researchers need to explore further in language teaching.

Technology Integration in Language Instruction

Technology has transformed language learning by enhancing learner motivation, enabling communicative competence, and providing access to authentic materials (Warschauer, 2000; Kern, 2006). Studies indicate that technology, such as computer-assisted language learning (CALL), improves vocabulary and interaction (Ghorbani & Marzban, 2013). Successful integration, however, depends on teachers' digital literacy, pedagogical perspectives, and availability of resources (Ertmer, 2005). Too much access to technology can actually work to increase education disparities, specifically in marginalized settings (Warschauer & Matuchniak, 2010). A critical pedagogy model of technology implementation can work against these disparities by emphasizing empowerment and equity; this intersection is in need of more research.

Teaching EFL in Iran

In Iran, the integration of technology in EFL settings is supported by national policies but faces significant challenges. Research indicates that teachers are interested in technology use, but identifies a shortage of infrastructure, lack of training, and inflexible curricula as widespread challenges (Rahimi & Yadollahi, 2011; Ghorbani & Marzban, 2013). These challenges are even more pronounced in marginal regions, where socio-economic and logistical issues undermine the utilization of technology (Akbari & Dadvand, 2014). Although Iranian EFL teachers have tried to make use of technology for cultural and learning advantages, the influence of critical pedagogy on their decision-making has not yet been widely researched.

Challenges in Marginalized Educational Contexts

Teachers encounter unique difficulties when working in schools that lack resources and suffer from socio-economic inequalities. Teachers in such environments experience multiple challenges due to insufficient professional training, as well as cultural obstacles and outdated physical facilities (Johnson & Golombek, 2011). Technology implementation within these environments requires teachers to create flexible systems that take into account their specific local circumstances (Warschauer & Matuchniak, 2010). Research shows that critical pedagogy treats technology as a resource for students to gain power and not just as a teaching medium; however, no studies have analyzed its practical implementation in disadvantaged EFL contexts. The investigation of how technology integration impacts EFL teacher choices in Iranian marginalized contexts through critical pedagogy methods requires further research because this

area remains underdeveloped. The research paper explores the impact of critical pedagogy on EFL teachers in Iran who work in low-resource schools to select their technological tools.

Method

Design of the Study

The qualitative approach that the researcher used for this study was phenomenological methodology for exploring and describing Iranian EFL teachers' lived experiences who teach in under-resourced areas. More specifically, this study employed hermeneutic phenomenology which is based on the philosophy of Heidegger (1972). Its primary focus is on interpreting the lived experiences rather than mere description.

Participants

The research analyzed 20 EFL teachers who taught in economically deprived rural locations across Iran. The researcher utilized purposive sampling to choose teachers who had firsthand understanding of integrating technology into educational environments that lacked sufficient resources. The study population consisted of both male and female instructors who taught English as a Foreign Language for periods between 5 and 20 years. All of the participants were Persian native speakers with a working knowledge of English, making them effective teachers of EFL. Informed consent was obtained from all the participants prior to their participation in the study, in accordance with ethical research standards.

Instruments

Data were collected using semi-structured interviews, which were aimed at gaining a deep insight into the experiences of teachers and their decision-making in terms of bringing technology into the classroom. The interview schedule was informed by seminal works in critical pedagogy (Freire, 1970; Giroux, 1988) and was particularly developed to probe issues of equity, empowerment, and learner agency. Some of the sample questions included, "How do you assess the introduction of technology in your EFL lessons?" The interview questions were, "What contextual factors guide your technology use decisions?" To allow for complete and truthful answers, interviews were conducted in Persian, lasting between 45 and 60 minutes. Interviews were tape-recorded with the participants' agreement.

Procedure

The study was conducted over the course of six months. Potential participants were first located through contacts with local teaching agencies and professional teaching networks in rural Iran. They were then emailed or phoned to introduce the aims of the research and invite their participation. Agreeable participants were arranged to be interviewed at locations and times convenient to them in order to optimize comfort and convenience. Audio recordings of the interviews were taken with open consent, and the recordings were subsequently accurately transcribed. To ensure participant confidentiality, all identifying features were stripped from the transcripts, in line with ethical requirements.

Data Analysis

The interview data was analyzed thematically, using the approach outlined by Braun and Clarke in 2006. The researchers began by reading the transcripts multiple times to better familiarize themselves with the data. From these, initial codes were developed based on the highlights of the most prominent patterns and ideas. These preliminary codes were further refined and reviewed to ensure they presented meaningful insights into the phenomena being studied and grouped into overarching themes. Once the most salient themes had been developed, clearer operational definitions were established along with the justification for the findings using representative quotes. The analysis was conducted in Persian, and only relevant extracts were translated for reporting purposes.

Results

The thematic examination of semi-structured interviews with Iranian EFL teachers from marginalized regions identified three primary themes that directly corresponded to the study's research objectives: (1) Technology Perception in Teaching, (2) Determinants of Technology Usage Decisions, and (3) Critical Pedagogy Influence on Technology Implementation. The study based its conclusions on interview coding and quotations from participants to determine three major themes and their corresponding sub-themes. The main findings appear in the next section with a summary table showing the themes and sub-themes which leads to a visual depiction of their relationships using a concept map.

Theme 1: Perceptions of Technology in Teaching

Educators held diverse opinions towards technological tools, which demonstrated the benefits and obstacles these tools presented in their particular educational settings.

Sub-theme 1.1: Technology as a Motivational Tool

Educators found technology to be beneficial for increasing student motivation. Teacher 7 explained that students become more interested in English learning when videos and apps replace traditional textbooks. The results of this study confirm previous research that shows technology boosts student motivation (Kern, 2006).

Sub-theme 1.2: Technology as a Cultural Bridge

Educators appreciated technology and its value in providing access to authentic English settings. Teacher 12 made the comment that “the internet enables my students to observe the different ways English is utilized outside of Iran.” This mirrors the findings of Warschauer (2000) concerning the use of technology and exposure to culture.

Sub-theme 1.3: Skepticism Toward Technology

Some educators expressed doubts due to practical issues. Teacher 3 said, “In my school, we have slow internet and not enough devices. There is a lot of frustration.” This deepens the understanding of infrastructural difficulties aligned with Rahimi and Yadollahi (2011).

Theme 2: Factors Influencing Technology Integration Decisions

The teachers' choice to incorporate technology into their practices was influenced by both external and internal factors.

Sub-theme 2.1: Infrastructural Constraints

Limited resources represented a significant barrier, as Teacher 9 stated, "I would use online quizzes, but half of the time the connection doesn't work." This is also an indication of the larger issue of digital inequity (Warschauer and Matuchniak, 2010).

Sub-theme 2.2: Pedagogical Beliefs

Beliefs about pedagogy also had a bearing on technology use. Teacher 15 identified her motive as: "I believe in student-centered learning. I want to provide students with chances to explore, so I use technology." This corresponds with the findings of Ertmer (2005) regarding the significance of beliefs in technology use.

Sub-theme 2.3: Administrative Support

The absence of support was also a common challenge. Teacher 18 identified that "Our school is not providing training or support. We just have to figure it out on our own." This is consistent with Akbari and Dadvand's (2014) findings documenting limited support in the Iranian context.

Theme 3: The Role of Critical Pedagogy in Shaping Technology Use

Initially, the critical pedagogy framework was one-way teachers approached technology use. The emphasis/focus was on empowerment and equity.

Sub-theme 3.1: Technology for Empowerment

Teachers discussed facilitating student agency through technology. For example, teacher 5 said, "I urge students to create their own blogs. This gives them a voice." This represents Freire's (1970) idea of education as empowerment.

Sub-theme 3.2: Addressing Inequity Through Technology

Some participants viewed technology as a way to address inequity. Teacher 11 said, "In rural settings, technology helps give students access to opportunities they wouldn't be able to access otherwise." This aligns with Giroux's (1988) concept of education for social justice.

Sub-theme 3.3: Resistance to Top-Down Policies

Teachers resisted changes to their practice from external mandated policies. For example, teacher 14 said, "The ministry is calling for us to move to digital textbooks, but they don't understand what we face." This relates to Shor (1992) in emphasizing the need for context-sensitive approaches to pedagogy.

Discussion

This research investigated Iranian EFL teachers in marginalized areas' perceptions of technology, the factors affecting their decisions in integrating it into their pedagogy, and the impact of critical pedagogy on their instructional practices. The results demonstrate that while teachers recognize technology as a motivating factor and an agent of cultural membership, they

are pessimistic due to infrastructural problems, including irregular internet services and lack of access to devices. These conceptions capture the dual aspects of possibility provided by technology and adversity present in environments where resources are minimal. In addition, teachers' technology use decisions are also determined by infrastructural constraints, their pedagogical beliefs, and the amount of support they have from administration. Critical pedagogy has also been a lens through which teachers employ technology to empower students and confront inequities, frequently in opposition to prevailing educational policy.

Teachers' positive attitudes towards technology as a stimulus for motivation and cross-cultural exchange are consistent with previous studies stressing its role in increasing students' engagement and exposure to authentic language use (Kern, 2006; Warschauer, 2000). Nevertheless, their reluctance, rooted in pragmatic issues, is consistent with the digital divide observed in underprivileged educational settings (Warschauer & Matuchniak, 2010). This result demonstrates incongruence between the national technology initiatives and the local context of Iran, where infrastructural constraints hinder its effective implementation (Rahimi & Yadollahi, 2011).

The interplay of influences on technology integration decisions emerges as a dense decision-making landscape. Educators who cherished student-centered pedagogy took up technology despite challenges, vindicating Ertmer's (2005) argument that pedagogical philosophies are central in taking up technology. Conversely, participants who were struggling with lack of administrative support and resources did not use it in an optimal manner, just as institutional research into barriers in EFL environments (Akbari & Dadvand, 2014) has established. This implies that effective integration demands not only availability but also value alignment with educators and high degrees of supporting mechanisms.

Critical pedagogy offers teachers a critical approach to using technology beyond the simplistic instructional purposes, facilitating student autonomy and reducing inequalities. Some teachers, for example, utilized online platforms to facilitate student-generated content, which is couched in Freire's (1970) liberation model of education.

Their efforts to close opportunity gaps also echo Giroux's (1988) vision for education as an instrument of social justice. Another unexpected finding was teacher resistance to policies ignoring local constraints, e.g., mandated technology use without a corresponding infrastructure. This resistance, carried out through selective or adaptive technology use, indicates a form of critical praxis not adequately covered in previous EFL research (Shor, 1992). The unique strength of this research is that it portrays technology integration as both a sociopolitical and pedagogical act. That instructors use technology for conforming to or resisting policies indicates their agency and adds a new dimension of complexity to the discussion on technology in language learning in marginalized environments.

Conclusion

This study highlights the multifaceted relationship between technology, pedagogy, and context for Iranian EFL teachers in marginalized settings. It illustrates that while teachers recognize the potential of technology to inspire students and reduce cultural distance, their practice is limited by infrastructure and administrative issues. When implementing principles of critical pedagogy, teachers engage with technology to give voice to their students and

ultimately create opportunities for agency and equity, despite interacting with misaligned policies that produce disillusionment and perhaps even resistance. These results illustrate the need for contextually appropriate productive use of technology, which relies on teacher agency and beliefs about education. This study contributes to a broader understanding of how digital materials can be used as tools to promote social change in resource-constrained EFL contexts and lends support to further policy and practice in national contexts that promote local needs and realities.

Implications of the Study

Teacher Training: Low-resource settings require professional development programs to teach educators effective technology implementation and digital literacy, while focusing on adaptive problem-solving methods (Warschauer & Matuchniak, 2010). Training needs a specific approach to deal with real-world problems, rather than using one-size-fits-all solutions.

Policy Development: Teachers need to collaborate with policymakers during technology planning to create realistic and appropriate integration solutions (Rahimi & Yadollahi, 2011). The digital divide between people requires investments in equitable infrastructure to create educational equality.

Classroom Practices: The implementation of technology in classrooms should empower students through collaborative digital projects that parallel critical pedagogy's goal of student empowerment (Freire, 1970).

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NEW STRATEGIES FOR TEACHER TRAINING IN ONLINE ENGLISH LANGUAGE INSTRUCTION: ANALYZING EFFECTIVE TEACHING AND LEARNING METHODOLOGIES IN DIGITAL ENVIRONMENTS IN IRAN

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Abstract

Digital literacy has become increasingly important in today's world due to the widespread use of technology and the internet in various aspects of life, including educational settings. With the growing trend towards online teaching in educational institutions, efficient teacher training strategies for English language teachers have gained paramount importance. This study explores innovative methodologies aimed at equipping teachers for online English instruction, highlighting those approaches that cultivate meaningful teaching and learning experiences. Utilizing a mixed-methods design, the existing literature was examined, alongside qualitative case studies were conducted with 25 experienced EFL teachers. Data collection encompassed in-depth interviews, classroom observations, online surveys, and reflective journals, yielding valuable insights into pedagogical practices that facilitate language acquisition in virtual environments. The findings underscore the efficacy of modern teaching strategies such as interactive technologies, digital collaboration tools, virtual classrooms, and gamified platforms. Ongoing professional development and peer mentorship are identified as vital components contributing to the enhancement of teachers' digital competence and confidence. Training programs should include insights into digital teaching methods, effective online engagement techniques, and formative assessment strategies that provide timely feedback to learners. The implications of this study suggest pathways for educational institutions to refine their online English language teacher training programs by effectively integrating digital methodologies to promote meaningful engagement.

Keywords: Digital literacy, Interactive technologies, Online education, Teacher training.

Introduction

21st-century digital technology has become an ordinary aspect of everyday life, transforming not only communication and social relations but also education. Teachers' high levels of digital literacy skills are required by the rapid development of digital tools and platforms—a set of competences to effectively, critically, and responsibly use digital technologies for the sake of teaching (Ng, 2019; Ferrari, 2013). For English as a Foreign Language (EFL) teachers, specifically, planning for virtual settings is especially daunting as they need to integrate technology with language pedagogy in order to promote learner engagement and education goals (Kukulska-Hulme, 2021).

The COVID-19 pandemic has hastened the globe's transition to online and blended learning, reveals the enormous chasm that exists between teachers' preparation to teach online

(Bao, 2020; Trust & Whalen, 2020). The majority of teachers, even language teachers, reported difficulty in working on digital platforms, managing online classrooms, and being effective in engaging learners from a distance (Hodges et al., 2020; Zhang et al., 2021). These difficulties have been even more extreme in cases like in Iran, where the absence of nationwide professional development opportunities in digital literacy exacerbates teachers' challenges .

In addition to technical proficiency, digital literacy encompasses more general pedagogical skills such as designing interactive content, facilitating digital communication, and assessing learners in online settings (Tondeur et al., 2021). Hence, teacher education programs must evolve by integrating these multi-dimensional aspects of digital literacy, specifically tailored to the specific needs of EFL teachers in Iran and similar contexts.

Despite the growing concern over the importance of digital literacy, few studies have examined effective teacher training programs for enhancing technological competence and pedagogical adaptation in online English language teaching. This current study endeavors to fill the gap through research on how Iranian EFL instructors experience and interpret innovative methods of enhancing digital literacy, investigating their experiences and perceptions towards the training courses.

Through the mapping of the challenges and best practices in digital teacher education, the study intends to contribute to the development of better training models that can support EFL teachers in adapting to the challenges of the digital age. Ultimately, this study aims to improve the quality of online teaching of English and promote a more balanced education in Iran's evolving digital environment.

Based on the research objectives of this study, the following research questions inform the inquiry:

1. What are the most effective digital teaching approaches for online instruction, as perceived by experienced EFL teachers in Iran?
2. In what manner do these teachers implement interactive and learner-centered approaches in virtual classrooms?
3. What are the obstacles facing Iranian EFL teachers in embracing online learning environments?
4. What aspects should teacher training courses successfully include to promote digital competency and instructional quality?

Literature Review

This transition from traditional face-to-face instruction to e-learning environments necessitated a careful overhaul of teacher training approaches, especially in English language teaching. Various studies have identified digital literacy as an essential 21st-century competency for teachers, extending well beyond mere technical expertise to include pedagogical knowledge and critical thinking competencies that enable teachers to use technology effectively in their instructional practices (Ng, 2012; Spante et al., 2018; Vuorikari et al., 2022).

In distance English language instruction, research points out teachers' necessity to acclimatize themselves in order to effectively deploy interactive technologies such as video conferencing tools, virtual whiteboards, learning management systems (LMS), and gamified

learning software, which all serve to enhance learners' motivation and engagement (Kessler, 2018; Zhang & Zou, 2020; Wang & Chen, 2023). However, the effective utilization of such technologies is heavily dependent upon teachers' ability to create productive learning activities and monitor active virtual discussions (Huang et al., 2023).

Continuous professional development models that ensure ongoing teacher growth outline some key attributes that result in effectiveness, including aggressive content emphasis, active involvement, orientation towards instructional goals, longer duration, and collaborative involvement (Desimone, 2009; Darling-Hammond et al., 2017). These same fundamentals are particularly critical in the online classroom context, where peer-to-peer collaboration, mentoring, and safe space for experimentation with new digital tools can significantly contribute to teachers' confidence levels and competency (Trust & Horrocks, 2017; Johnson & Aragon, 2003).

In the Iranian EFL context, existing studies yield directions toward serious problems concerning teachers' preparedness for online instruction. Research by Ahmadi (2018), Razavi et al. (2021), and recent analysis (Karimi, Marashi, & Mall-Amiri, 2024) suggest that the majority of Iranian EFL teachers are offered little training in web-based pedagogies and receive no institutional facilitation. Institutional challenges such as inadequate infrastructure, technological reluctance, and absence of ongoing professional development further complicate the effective implementation of virtual teaching methodologies.

To address these voids, scholars advocate for specialist teacher training programs with integrated blended learning frameworks, microteaching in online classrooms, and the application of formative assessment to monitor and guide learner progress (Boettcher & Conrad, 2016; Chen & Zhang, 2022). Reflective practice, using peer feedback and teaching diaries, is also emphasized as fundamental elements for upholding professional improvement in virtual teaching environments (Farrell, 2019).

This literature review gives this study a solid theoretical and empirical basis by highlighting the central significance of digital competence, pedagogical adaptability, and institutional backing to the negotiation of optimal online English language teaching. It puts the necessity of creating innovative and contextually suitable training methods for preparing teachers for the evolving demands of virtual classrooms

Methodology

Research Design

This study employed a mixed-methods approach that combined qualitative and quantitative methods to obtain a rich understanding of effective training techniques for EFL instructors in e-learning. Mixed-methods were employed to gather both measurable trends as well as thick, descriptive data among the participants.

Participants

The study comprised 25 veteran EFL teachers from various language institutes and universities across Iran. The participants were selected through the purposive sampling technique in which participants with teaching experience of three years and above and prior involvement in online teaching were prioritized.

Data Collection

Data were collected using four primary instruments:

Online surveys: Likert-scale and open-ended questions were present to elicit demographic data, teaching experience, and views of EFL teachers towards digital teaching.

In-depth interviews: Semi-structured interviews with 10 participants were conducted for in-depth understanding of their experience, strategies, and challenges.

Classroom observations: Online lessons recorded were analyzed for observing how digital tools and interactive strategies were utilized by teachers.

Reflective teaching journals: Participants reflected on their weekly activities for a period of one month, detailing their teaching strategies and participation techniques.

Data Analysis

Quantitative data were tested using descriptive statistics (percentages, frequencies, means). The qualitative data obtained from interviews, observations, and journals were tested using thematic analysis, where recurring themes on teaching practices, challenges, and training needs were identified. The integration of both data sets offered a more in-depth interpretation of findings.

Results

Quantitative Findings

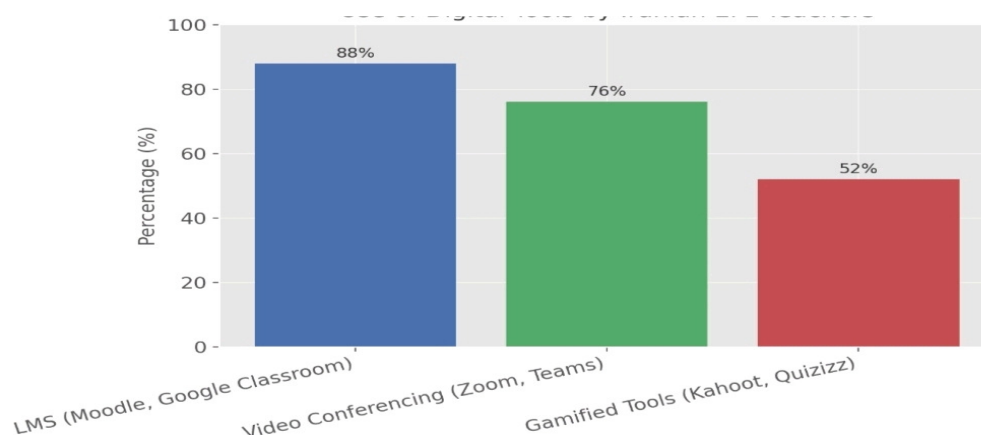
The quantitative information obtained with the help of surveys from Iranian EFL teachers showed fascinating patterns with regards to technology use, self-efficacy levels, and instructional methods in online teaching. Widespread usage of Learning Management Systems such as Moodle and Google Classroom—used on a daily basis by 88% of teachers—suggests a paradigm shift towards digital means for content delivery, assignment management, and communication. These tools enable instructors to organize their courses in a systematic way, share materials, collect assignments, and provide feedback efficiently, something that is really needed during the pandemic-induced shift towards online classes.

Similarly, video conferencing tools such as Microsoft Teams and Zoom are also vital in facilitating synchronous lessons. As 76% of teachers regularly hold live classes, the sites have now become the foundation for live teaching with the capability to facilitate instant interaction, voice communication, and facial expressions required for language learning. Teachers employ functions such as screen sharing, breakout rooms, polls, and chat for optimizing participation and a sense of community among students.

However, the uptake of more dynamic and creative tools—is not complete, with 52% only utilizing gamified software (e.g., Kahoot, Quizizz). The relatively low use rate is perhaps due to factors like unfamiliarity with tools, perceived complexity, or uncertainty about efficacy. It suggests that there is a need for targeted training to familiarize teachers with such interactive tools, which, as studies have shown, can significantly boost motivation and engagement (Hwang & Chen, 2019).

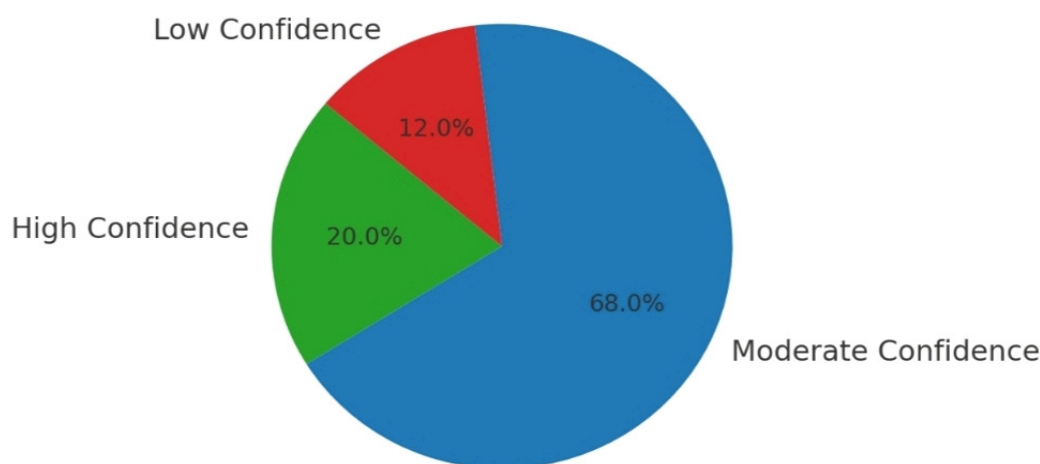
Of the teachers' self-confidence, 68% described themselves as having moderate confidence in their digital teaching abilities, 20% as having high confidence, and 12% with low confidence. This pattern indicates that while the majority of teachers are fairly confident about working with technology, a large percentage remain unsure, which negatively affects their capacity to apply new pedagogies successfully. Confidence is strongly linked with the desire to test and apply new devices and techniques, and therefore, continued professional development is vital.

Figure 1 _Use of Digital Tools by Iranian EFL Teachers



The teachers' survey revealed that 68% of them were moderately self-assured in digital teaching, 20% highly confident, and 12% low in confidence.

Figure 2 _Teacher Self-Confidence in Digital Teaching



Qualitative Findings

From reflective journals, classroom observations, and in-depth interviews, several underlying themes emerged that enhance our understanding of online teaching practice and concerns:

Student Interaction and Engagement: Teachers pointed out that it is harder to keep students engaged in online settings. To counter this, most use interactive strategies such as breakout room conversations, shared collaborative documents (e.g., Google Docs), and multimedia exercises. As a teacher noted, "Using small groups within breakout rooms encourages shy students to participate more actively, something that is more challenging in big, in-person classes."

Infrastructure and Technological Challenges: The common challenge that kept reappearing was the poor internet connectivity, unavailability of suitable devices, and lack of knowledge of some of the digital tools that at times caused disruption of the lessons or lowered participation. For example, teachers reported that poor internet connectivity caused them delays, disconnections, or audio-visual issues, which lowered the quality of teaching.

Pedagogical Adaptation and Innovation: The majority of instructors emphasized remodeling lesson plans to accommodate online formats, with a focus on student-led, interactive activities. They desired training on making online tests, multimedia application, and gamification strategies to activate learners.

Peer Support and Professional Development: Teachers stressed the need for collective learning communities, where they share resources, learn best practice, and troubleshoot technical issues together. A few mentioned mentorship schemes under which veteran teachers mentor first-year teachers to build confidence and continually improve.

Assessment and Feedback: Teachers value the limitation of traditional tools for assessment online and are experimenting with formative assessment strategies, e.g., tests, reflective journals, and peer assessment, in order to monitor progress and give immediate feedback. They believe that ongoing assessment detects areas of learning gaps and modifies teaching accordingly.

Summary of Main Results

In general, with high levels of adoption of technologies among Iranian EFL teachers, issues still exist in pedagogical preparedness, confidence levels, and infrastructure. The findings highlight an urgent need for comprehensive, long-term professional training that strikes a balance between technical expertise and innovative pedagogical strategies. Additionally, establishment of a social community of professionals and investment in the technological infrastructure are essential steps towards improvement in quality of online language teaching.

Discussion

"The conclusions of this study align with established research emphasizing that effective online language teaching goes beyond just providing technological tools; it demands pedagogical adaptability, active participation strategies, and ongoing support (Boettcher & Conrad, 2016). The widespread use of LMS and video conferencing software is a testament to their position as fundamental tools, but moderate levels of confidence reflect that most instructors are still in uncharted territory without proper training.

The employment of interactive tools like breakout rooms and collaborative documents similarly indicates consideration for learner-centered methodologies in web-based learning.

Such activities ensure active participation, speaking practice, and peer discussion, critical to language acquisition (Wang & Vásquez, 2012). The judicious use of gamified tools, however, opens the possibility for the incorporation of more engaging, motivation-inducing tasks.

Infrastructural technology remains a significant barrier. Repeated grievances of internet instability, limited access to devices, and lack of technical support indicate systemic issues in need of policy and institutional correction. Without stable infrastructure, even the best teaching techniques cannot be maximally applied.

The aspiration of teachers towards ongoing professional development underscores the imperative to create tailored training programs that are context-sensitive, practically oriented, and long-term. The training programs need to interact with pedagogical theory and its application on the ground, e.g., designing interactive lessons, tracking student progress, and applying technologies creatively.

Peer mentorship and peer communities offer promising areas for professional growth. Teachers benefit from expertise sharing, pooling of resources, and collaborative problem-solving that builds confidence and creativity. Institutionalizing mentorship programs, online communities, and professional learning networks can have a significant impact on teaching quality.

Finally, the evolution towards formative assessment practices in online learning contexts is indicative of adaptive teaching mindsets. Teachers are experimenting with various methods of monitoring student progress, providing instant feedback, and adjusting instruction accordingly, consistent with formative assessment best practices in education (Black & Wiliam, 2009).

Conclusion and Implications

This study highlights the multifaceted nature of Iran's online EFL education, which indicates both the benefits of embracing technology and shortcomings in pedagogical enactment, infrastructure, and instructor self-efficacy. To enhance online language learning experience, the stakeholders must hold the following all-inclusive strategies in mind

Develop Strong and Long-Term Professional Development Programs: These need to combine diverse technical training with pedagogical methodologies appropriate for online learning. Interactivity, evaluation, engagement, and multimedia use should be areas of workshops, webinars, and micro-credentials that can empower instructors.

Establish Collaborative Professional Communities: Develop online forums, mentoring activities, and peer observation systems to facilitate knowledge sharing, peer support, and continuous development. Such communities can be utilized as a safe space to experiment with new techniques and solve problems.

Invest in Resources and Infrastructure: Policymakers and schools must prioritize equal access to high-speed internet, devices, and digital materials by all, especially for lower-income communities. Partnerships with technology providers and government incentives can be beneficial.

Curricula Design: Student-Centered and Engagement-Focused: Leverage interactive activities, gamification, and authentic materials for engaging students. Make use of formative assessments on a periodic basis to monitor progress and readjust teaching.

Foster Reflective Practice and Feedback: Teachers need to be encouraged to develop reflective practices to assess their teaching effectiveness, modify strategies, and exchange knowledge with colleagues.

Policy and Institutional Support: School authorities should set policies to encourage digital literacy, offer incentives for innovative pedagogy, and earmark budgets for technology upgradation and professional development.

Future Research Directions:

Assessing Student Outcomes: Study the impact of such teaching strategies on language proficiency, motivation, and autonomy of learners.

Longitudinal Studies: Track teachers' confidence development, pedagogical practices, and student engagement over a duration as they are trained repeatedly.

Technology Integration: Study innovative technologies such as artificial intelligence, virtual reality, and adaptive learning systems implemented in online language instruction.

Cultural and Contextual Factors: Examine cultural orientations to technology and education that determine adoption and success of online instruction techniques.

Through their embrace of these recommendations, education stakeholders can significantly contribute to improving the quality of Iranian online EFL education so that teachers are adequately trained, motivated, and supported in delivering effective language instruction in electronic environments.

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A QUALITATIVE STUDY OF UNIVERSITY INSTRUCTORS' EXPERIENCES WITH AI-POWERED LEARNING ANALYTICS IN BUSINESS-ORIENTED CURRICULA

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Abstract

Implementing artificial intelligence (AI) technologies is profoundly changing teaching methods in business education as the use of AI within post-secondary education institutions increases. The purpose of this qualitative study is to explore how university instructors engage with AI Learning Analytics (AI LA) concerning curriculum development, student evaluation, and instructional choices relative to the different levels of LA integration. Using a phenomenological approach, faculty from different business disciplines were selected and participated in in-depth semi-structured interviews. Thematic analysis offered distinct themes such as deepened pedagogical understanding through customized instruction, ethical issues of data collection and usage, and preparedness of the institution to receive new technology. One of the striking themes was how the instructors-professionals' ideal images of AI-based analytic tool training in the context of advanced data examination partially diverged from the actual training they received. The research adds to the body of literature detailing the limited practical experiences, along with conceptual frameworks educators have designed, which guides them to adapt AIDMPs (AI-driven decision making processes) and instructional policy frameworks for evolving teaching policies.

Keywords: Artificial Intelligence, Learning Analytics, Business Education Curriculum

Introduction

The growing application of artificial intelligence (AI) in tertiary education is reshaping the dynamics of teaching, learning, and evaluation. The most promising use of AI is Learning Analytics (LA), which entails tracking and analyzing learner data to optimize teaching practices, enhance learning outcomes, and guide curriculum development (Ferguson, 2012; Ifenthaler & Yau, 2020). In business education—where skills such as flexibility, impromptu decision-making, and data analysis are paramount—AI-driven learning analytics becomes even more significant.

While promising, AI LA implementation is not all smooth sailing. University instructors struggle with understanding how to read and respond to AI-generated data, especially where training on the machines is lacking or not in sync with real-world needs (Daniel, 2015). Additionally, ethical considerations such as data privacy, algorithmic bias, and the transparency of AI decision-making processes complicate the rollout of such technologies (Slade & Prinsloo, 2013). Faculty are not only supposed to integrate these tools into practice but also to help develop curriculum and institutional policy based on such data. However, there is a lack of knowledge concerning how business program instructors really experience and manage this changing, AI-driven learning context.

The purpose of this research work is to investigate the experiences of those university instructors who use AI-supported learning analytics in business-related curricula. More

particularly, it attempts to find out how they interact with these technologies regarding pedagogical decision-making, the assessment of students, and the development of curricular materials. Furthermore, the research examines how the training provided to the instructors for AI-powered learning analytics corresponds to their actual professional demands in the workplace.

This research adds to the growing body of literature related to the application of artificial intelligence in learning environments by prioritizing the voice of educators and taking a grounded and practice-based stance. It offers insights into the readiness of institutions for AI adoption and addresses a key gap in understanding pedagogical and ethical complexities involved in integrating AI into higher education. The results of this research will be of paramount importance to education policymakers, curriculum developers, and training developers who aim to match technological innovations to the actual needs and expectations of teachers.

The study is guided by the following research questions:

1. How do university instructors in business disciplines experience the integration of AI-powered Learning Analytics in their teaching practices?
2. In what ways do instructors utilize AI Learning Analytics in curriculum development and student evaluation?
3. What challenges and ethical concerns do instructors face in engaging with AI-driven decision-making processes?
4. How do instructors perceive the adequacy and relevance of the training they receive on AI Learning Analytics tools?

Literature Review

AI in Education

Artificial intelligence (AI) is revolutionizing higher education through the provision of tools that support and enhance teaching and learning activities. AI facilitates personalized learning experiences, automates administrative tasks, and supports data-informed decision-making (Zawacki-Richter et al., 2019). For example, AI programs are able to tailor learning content to individual students' requirements, which improves learning and academic outcomes (Tuomi, 2018). In spite of these developments, there are still concerns, especially the requirement for educators' training to utilize AI technologies and data privacy concerns (Daniel, 2015). These challenges point to the dual nature of AI as both a potential and a challenge in education.

Learning Analytics in Higher Education

Learning analytics (LA), one of the primary applications of AI, refers to the collection and analysis of data about learners with the intent to enhance learning environments and educational practices (Ferguson, 2012). LA provides educators with feedback on student performance and conduct to enable them to make data-driven pedagogical decisions (Siemens & Baker, 2012). Post-secondary education has used it to identify at-risk students and enhance

retention (Sclater et al., 2016). However, its use is ethically challenging, particularly regarding data privacy and the possibilities of algorithmic bias, and thus requires careful consideration (Slade & Prinsloo, 2013).

AI and Learning Analytics in Business Education

AI together with learning analytics delivers special benefits to business education programs which emphasize essential abilities for data analysis and decision-making. Students experience practical simulations of real-world business cases through AI tools which utilize data analysis (Kumar & Singh, 2020). Through learning analytics, students pursuing online business courses receive customized feedback that enhances their engagement levels (Lee & Kim, 2019). The effective implementation of AI tools in business education programs needs instructors who possess advanced competencies in these systems yet their development gets blocked by insufficient training opportunities (Ifenthaler & Yau, 2020).

Curriculum Development and Student Evaluation with AI

Curriculum development and student assessment processes experience significant AI-based influence. AI technology allows educators to construct personalized learning routes which match student requirements through individual performance data assessment (Chen & Liu, 2021). AI assessment tools streamline grading processes to deliver instant feedback while simultaneously decreasing human bias and operational time requirements (Nguyen et al., 2020). AI-powered educational applications face ethical problems because they fail to provide transparent decision-making methods while struggling to avoid inequity replication (Baker & Hawn, 2021).

Ethical Concerns and Challenges

The coming of AI together with learning analytics to education systems creates numerous moral problems. The primary concern about student data is their privacy since educational institutions must obtain proper consent and maintain information security (Slade & Prinsloo, 2013). AI systems trained on biased datasets face the risk of producing unjust results through their decision making (Baker & Hawn, 2021). AI process opacity leads to trust problems between educators and students which highlights the requirement for explicit operational policies and responsible oversight (Daniel, 2015).

Instructor Training and Institutional Preparedness

The successful implementation of AI and learning analytics requires institutions together with instructors to be properly prepared. The lack of essential competencies among educators demonstrates the critical requirement for customized training programs according to Crompton's 2022 study. Organizations need to establish strong technical infrastructure while building an environment that promotes new ideas according to Macfadyen's 2014 research. The study focuses on revealing the difference between what instructors expect from training and what they actually receive in order to investigate this particular gap further (Mirzadigohari, ongoing research).

Method

Design of the Study

The research adopted a qualitative phenomenological framework to investigate how university instructors experience AI-powered Learning Analytics implementation in business-focused courses. The phenomenological research method enables the study of participants' detailed interpretations and reactions to phenomena according to Creswell and Poth (2018). The researchers selected this framework to gain insight into how instructors experience LA in actual educational contexts instead of focusing on theoretical knowledge.

Participants

The study included 12 full-time university instructors who teach business-related subjects including management, marketing, finance, information systems at recognized educational institutions in Iran. The research employed purposive sampling to choose instructors who had employed AI tools for at least two years in their teaching practices. The selection process focused on participants who directly participated in curriculum development and assessment to maintain research objectives alignment.

Instruments

Researchers used a semi-structured interview guide as their main data collection instrument. The guide consisted of open-ended questions which aimed to collect comprehensive information about how AI LA affected curriculum planning and student assessment and teaching decision-making for participants. The interview protocol underwent validation by experts and pilot testing through two university instructors who were not part of the final sample.

Data Collection Procedure

The researcher gathered information through in-depth semi-structured interviews conducted continuously for three months with each interview session lasting between 45 and 60 minutes. The interviews took place on secure video conferencing platforms because of distance between participants and institutional needs. Participants granted permission for audio recording of all interviews which were later transcribed word for word. The research received ethical approval from the university's research ethics board and participants received confirmation that their personal details would remain confidential during the study.

Results

Thematic analysis of interviews with twelve university lecturers revealed four broad themes regarding their experiences with AI-supported Learning Analytics (LA) in business education:

Theme 1: Enhanced Pedagogical Insight through Data-Driven Personalization

Nearly all instructors noted that the use of AI LA tools allowed them to gain deeper insights into student learning behaviors and performance patterns. They pointed out the importance of data in instruction customization for individual students.

Instructor 3 claimed, *"The dashboards allow monitoring of student activity which enables the modification of teaching methods in real-time."*

Theme 2: Gaps Between Training and Real World Requirements

A number of instructors cited an existing gap between their training and practical application of AI LA tools.

“I was not able to overcome the obstacles presented in my courses. The training sessions were too generic,” shared Instructor 2.

Theme 3: Ethics and Data Privacy

The ethics of the issue, particularly data collection and algorithmic opacity, emerged quite often.

“I have concerns about how much information, or data, is being captured. More importantly, do students understand the implications?” Instructor 4 shared.

Theme 4: Readiness and Support of the Institution

Responses regarding institutional support differed, as some reported strong support, while others pointed out policies and structural inadequacies.

Instructor 1 said, *“AI applications are within our institution’s scope of concern, and they trained and resourced us to utilize AI in our work.”*

Discussion

This research explored the experiences of Iranian business faculty with AI-driven Learning Analytics (LA) tools within Iranian higher education settings. The study found that although AI LA can play an essential role in pedagogical decision-making by facilitating real-time analysis and personalized learning, its effective adoption continues to be constrained by gaps in teacher training, ethical issues, and variable institutional support.

The enhanced understanding of pedagogical practices gained from AI learning analytics software is consistent with existing research demonstrating that these systems facilitate the early detection of vulnerable students and individualize learning trajectories (Ifenthaler & Yau, 2020; Chen & Liu, 2021). Yet, our research substantiates the broadly recognized difficulty that the majority of instructors are not in a good position to decipher multifaceted outputs generated by AI (Crompton et al., 2022). The lack of context-specific, practical experience encroaches upon their ability to maximize the tools’ potential (Macfadyen & Dawson, 2012).

Ethical issues surrounding the use of students’ data, discriminatory algorithms, and accountability of AI decision-making, since this research’s results, align with Slade & Prinsloo (2013) and Baker & Hawn (2021) advise caution, emphasizing the importance of thorough institutional policies. Institutions should not only adopt AI technologies but also educate faculty members and students about their ethical implications and promote open models of data governance.

Such reported discrepancies in institutional readiness also highlight the necessity to develop comprehensive strategic plans for AI integration (Zawacki-Richter et al., 2019). Interdisciplinary cooperation, e.g., IT assistance, faculty training, and leadership commitment, is essential for long-term integration.

Conclusion

This study adds to the expanding body of literature exploring the use of AI in higher education including its application in business courses. The four major themes: pedagogical enhancement, training gaps, ethical concerns, and institutional readiness reveal the promise and the sobering challenges AI-powered Learning Analytics presents. Meaningful adoption, however, goes beyond having the technological framework; there need to be policies and training that are geared towards human interaction.

With AI being an essential driver of educational change, it is crucial that faculty are adequately supported – equipped, and informed with ethical considerations. Further investigation is needed into the implications of the adoption of AI Learning Analytics on student performance and institutional culture over a longer time horizon, and in other educational settings.

Implications of the Study

This study has various implications:

1. With respect to policymakers: It is important to form national and institutional policies regarding the ethical and responsible use of AI technology in education at all levels.
2. With respect to curriculum innovators: Integrating data competency and AI awareness into relevant professional training is a necessity.
3. With respect to designing trainings: General training sessions should be deprioritized in favor of more sophisticated, precise training tailored to specific roles.
4. With respect to other researchers: There is a need for more longitudinal studies that examine the effect of continual integration of AI LA on teaching practices, equity pedagogy, and educational equity.

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INNOVATIVE TEACHING STRATEGIES IN EFL CLASSROOMS: ASSESSING HOW MODERN TECHNOLOGIES CAN ENHANCE THE EFFECTIVENESS OF ENGLISH LANGUAGE TEACHERS AT THE ELEMENTARY LEVEL IN IRAN

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Abstract

Effective English teaching at the elementary level demands competent and literate educators who possess the knowledge of modern technologies and the ability to implement them effectively in educational settings. As technology has revolutionized educational settings, the present study aimed to explore innovative teaching strategies in EFL classrooms, considering how modern tools can enrich teaching effectiveness at the elementary level in Iran. To this end, a mixed methods approach was selected, utilizing surveys and semi-structured interviews to gather insights from teachers and students at different language institutes. These methods helped to assess the impact of integrating technologies such as interactive whiteboards, language learning apps, and online platforms on students' motivation and engagement. The study sample included a total of 30 Iranian male and female EFL teachers and 50 male and female learners, selected via available sampling methods. The findings revealed that the incorporation of gamified learning experiences and multimedia resources significantly enhanced students' interest while accommodating different learning styles, making language acquisition both enjoyable and effective. Moreover, the study highlighted the importance of equipping teachers (through professional development programs on digital pedagogy) with the essential skills to implement modern technologies in their classrooms, so that they become capable of addressing their students' needs and fostering a more dynamic and interactive learning environment. The findings have several implications for various stakeholders, including EFL teachers, teacher educators, and curriculum developers in teacher education, emphasizing the need to prioritize technology integration and comprehensive teacher training to enhance teacher effectiveness at the elementary level in Iran.

Keywords: Effective elementary-level teaching, Innovative teaching strategies, Professional development, Technology integration.

Introduction

English Foreign Language (EFL) primary education is crucial to the formation of the early language attitudes and language capacity of students. Early English learning is an international issue due to globalization and increasing demand for English language competency at educational, working, and social levels (Butler, 2015; Cameron, 2001). In Iran, reforms in education have been introduced lately that have encouraged teaching English at the lower levels of education in order to equip learners with future competencies (Iran Ministry of Education, 2020). Teaching at such a level, however, requires more than the actual applicability of the curriculum; it requires professional teachers with sound pedagogical know-how and Information and Communication Technology (ICT) skills (Rahimi & Yadollahi, 2011; Reinders & White, 2011).

The accelerated pace of educational technology innovation has reshaped the traditional classroom environment, offering technology that supports interactive, student-centered pedagogy. Technologies such as interactive whiteboards, tablet PCs, mobile applications, and web-based collaboration tools support pedagogical practices that focus on student engagement, differentiated instruction, and immediate feedback (Kukulska-Hulme, 2012; Warschauer & Meskill, 2000). These technologies encourage a paradigm shift from teacher-centered to communicative and task-oriented learning approaches that are more appropriate for the facilitation of language acquisition (Stockwell, 2010; Godwin-Jones, 2011).

Despite their potential, Iranian EFL classrooms at the elementary level have substantial barriers in embracing technology-supported pedagogy. The majority of schools do not have good infrastructure, have weak internet connectivity, and lack digital hardware (Zarei & Aghaei, 2014; Rezaei & Meshkat, 2017). Moreover, educators lack appropriate digital literacy and professional training facilities to effectively integrate technology into instruction (Ertmer & Ottenbreit-Leftwich, 2010; Shahsavari, 2012). This is aggravated by unequal resource distribution between schools in rural and urban areas, resulting in disparity in access and education quality (Farhady, 2011).

Systemic and cultural factors also affect the teaching and learning environment. Iran's education culture of examinations is dominant, making grammar-based instruction and rote memorization more relevant than communicative competency and student autonomy (Sadeghi & Richards, 2016; Kiany et al., 2011). This cultural leaniness loathes the adoption of student-centered and adaptive approaches that are so vital for meaningful technology integration.

Against this context-specific set of challenges, the current study investigates how existing technology can be purposefully integrated into Iranian primary-level EFL classrooms to optimize teaching effectiveness and enhance learners' motivation. Employing a mixed-methods approach, the study considers not only the rate of technology adoption but also teachers' and learners' everyday lives, with the aim of accounting for enablers and barriers from an all-encompassing viewpoint. The results intend to inform EFL instructors, curriculum planners, and education policymakers about effective action plans for supporting the meaningful integration of modern technology in first language instruction in Iran (Tondeur et al., 2012; Koehler & Mishra, 2009).

Based on the research objectives of this study, the following research questions inform the inquiry:

1. In what ways do the available technologies impact the efficacy of English language teaching at the elementary level in Iran?
2. In what ways do computer-based materials influence students' motivation and engagement in elementary EFL courses?
3. What are the most important challenges for instructors to implement their teaching methods with the help of modern technologies?

Literature Review

Technology Integration in EFL Teaching

Information and communication technologies have transformed language learning and teaching worldwide. Networked technologies provide the potential to construct more communicative, collaborative, and interactive learning cultures, Warschauer and Kern (2000) say. Computer technology can provide actual language input and output, student-led learning

pathways, and immediate corrective feedback (Stockwell, 2010). For instance, mobile language learning applications such as Duolingo and Babbel offer game-based learning exercises where students can exercise vocabulary and grammar outside the classroom through interactive drills (Godwin-Jones, 2011).

Both issues related to technology integration in EFL classes have been cited in the Iranian environment and demand. Zarei and Aghaei (2014) reported that despite the potential, infrastructural issues and professional development gaps hinder effective utilization. Similarly, Rezaei and Meshkat (2017) detail that limited access to digital tools and unequal access to the internet pose huge barriers, especially in the more rural areas. Moreover, the teachers themselves are inadequately trained in digital pedagogy, and technology hence gets utilized in patches and on the surface.

The study indicates the necessity of context-sensitive solutions that are aware of local constraints and demands (Rahimi & Yadollahi, 2011). For instance, low-tech interactive equipment (e.g., projectors, audio players) or offline-compatible learning applications can be more easily accessible in resource-poor environments. Further, facilitating teacher-to-teacher collaboration and peer learning towards integrating technology has been found to result in more long-term adoption (Tondeur et al., 2012).

Innovative Teaching Strategies

Innovative pedagogy entails pedagogical approaches that promote active student learning, critical thinking, creativity, and collaboration. These approaches move away from the traditional lecture approach towards student-centered approaches where students feel a sense of ownership over their learning (Brown, 2007).

Flipped classrooms, project-based education, and gamification have been identified in language education as powerful innovations. Through the flipped classroom model, content is presented online outside of the classroom and class time is reserved for interactive instruction, discussions, and one-on-one support (Bergmann & Sams, 2012). In EFL classes, this approach could be particularly effective in that it maximizes communicative practice with face-to-face classes.

Gamification, the use of the elements of game design in non-game situations, has proved effective in terms of motivation and engagement. Deterding et al. (2011) enumerate some of the key gamification mechanics that include points, badges, leaderboards, and challenges that elicit feelings of accomplishment and competition. Gamified tasks and applications in EFL classes reduce learners' anxiety levels, increase perseverance, and improve vocabulary retention (Murray & Nallaya, 2020).

Multimedia-based instruction, for example, video, animation, and interactive simulation, is suitable for different learning styles (visual, auditory, kinesthetic), making classes more accessible and memorable (Moreno & Mayer, 2007). Teachers' multimedia usage results in higher levels of student attention and participation.

The role of teachers in the process of educational technology adoption

Effective technology integration in education depends much on the capabilities, dispositions, and infrastructure of support among educators. Reference is made by Ertmer and Ottenbreit-Leftwich (2010) regarding teachers' attitudes towards technology, self-efficacy, and professional development opportunities. Educators' self-belief and preference for technology enhance the likelihood of them trying things out and persisting with its use.

In Iran, prevalence of traditional exam culture pedagogies and limited programs of continuous training have constrained teachers' adoption of innovative pedagogies (Shahsavari, 2012). But recent initiation of professional development programs emphasizing digital pedagogy has begun to bring about this change. For instance, initiatives by Iran's Ministry of Education and non-governmental organizations have facilitated technology integration, instructional design, and learner-centered approaches workshops (Ministry of Education, 2023).

Teachers also undertake a fundamental role as technology facilitators who redesign technology to suit the context and needs of their students. Teachers' creativity and problem-solving ability dictate how much technology assists in learning outcomes (Koehler & Mishra, 2009). Teachers should be enabled through regular training, peer assistance, and technical support.

Gaps in the Literature

Whereas a number of studies in the literature document the benefits of technology for language learning, there are few that explore elementary-level EFL classes in Iran. Some of these studies target secondary or university settings, in which students are older and more mature. Younger learners have unique developmental, cognitive, and motivational needs that require unique approaches to teaching and instruments (Lightbown & Spada, 2013).

Also, previous research tends to use qualitative or quantitative data alone, and that limits conclusions to richness. A mixed-methods approach may provide richer detail through access to quantifiable outcomes as well as to personal experience.

The current research aims to fill these gaps in studies in the following manners: by examining technology integration in primary EFL classrooms in different Iranian cities, from teachers' and students' points of view, and employing a mixed-methods research approach. Implications will guide evidence-based policy and practice suggestions for early English.

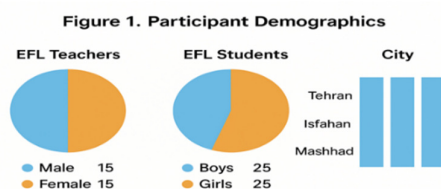
Methodology

Research Design

The study used a mixed-methods design, combining qualitative and quantitative data in a way that presented an inclusive picture of the impact of accessible technologies on the efficiency of EFL instruction at the elementary level level in Iran. The two datasets were gathered concurrently, examined separately, and only combined for analysis in the last phase using convergent parallel design.

Participants

Five language institutes in Tehran, Isfahan, and Mashhad comprised 30 Iranian EFL teachers (15 male and 15 female) and 50 elementary-level EFL students (25 male and 25 female). Purposive and convenience sampling helped the participants to be chosen to guarantee simple access and a range of background.



Data Collection Instruments

Quantitative data were collected by means of a Likert-scale questionnaire specially designed and validated using previous studies (e.g., Ertmer, 2010). It consisted of 20 statements assessing:

Frequency of use of technology

Attitudes toward digital tools

Perceived effects on motivation and engagement

Qualitative data were collected via semi-structured interviews with 10 teachers and 10 students (randomly recruited from the entire sample). Interview guides with open-ended questions on experience, issues, and attitude towards language learning with technology.

Sample Quantitative Item:

"Using multimedia tools (videos, animations) helps me better understand English lessons."

[Strongly disagree – Disagree – Neutral – Agree – Strongly agree]

Sample Qualitative Question:

"How does learning from apps or technology change your English classes?"

Analysis of Data

Quantitative data were examined using SPSS 26, including:

Descriptive statistics (mean, SD)

Independent sample t-tests

Pearson correlation analysis

Qualitative data were examined using thematic analysis, following Braun and Clarke's (2006) six-step process. Repeated themes and subthemes were established regarding motivation, engagement, teacher roles, and implementation barriers.

Ethical Considerations

Voluntary participation was sought, and informed consent documents were acquired from all participants. Parental or guardian consent was sought for students younger than 12 years of age. The study maintained anonymity and confidentiality during the research activity.

Results

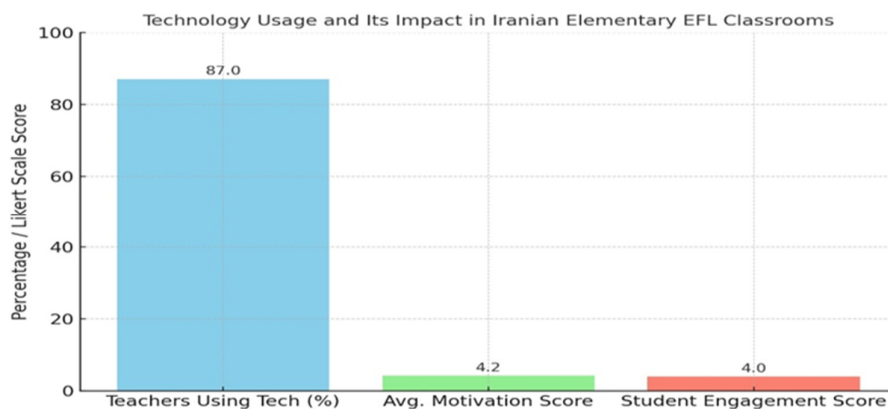
Quantitative Findings

The quantitative data analysis confirmed that 87% of teachers incorporated new technologies such as interactive whiteboards and language learning software into their normal daily routine. The average score, 4.2 (SD = 0.6), on a 5-point Likert scale for the perceived impact of technology on students' motivation showed a strongly positive attitude.

Independent sample t-tests showed there was no significant gender difference in the use of technology among female and male teachers ($t(28) = 1.15$, $p = .26$). Likewise, there was no significant gender difference in level of engagement among students ($t(48) = 0.95$, $p = .35$).

Pearson's correlation test also indicated that there is a positive relationship between frequency of use of technology and interest of students ($r = .68$, $p < .01$) and this also signifies

that greater usage of technological tools tends to lead to greater participation and interest among the students.

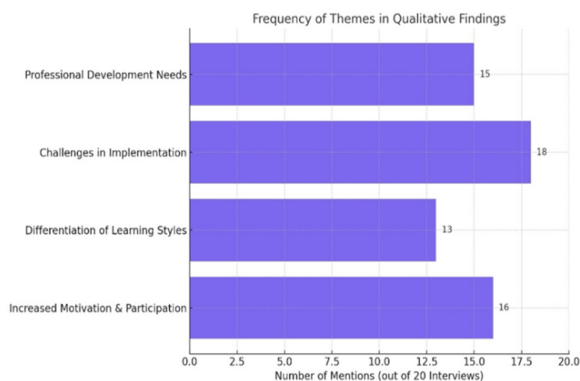


Increased Motivation and Participation: Students and teachers both mentioned that gamified and multimedia tools increased the enjoyment aspects of lessons and created a good mood for learning.

Differentiation of Learning Styles: Teachers mentioned that technology supported differentiating visual, auditory, and kinesthetic learners.

Challenges in Implementation: Participants mentioned challenges due to improper training, weak infrastructure, and sporadic technical issues.

Professional Development Needs: Staff members highly recommended the need for frequent workshop and training sessions to develop their digital literacy and pedagogy skills.



Discussion

The results confirm earlier studies suggesting that current technologies have the ability to significantly increase the efficiency of English language learning at the elementary level (Warschauer & Kern, 2000; Lee & McLoughlin, 2008). The empirically supported positive relationship between technology use and motivation of learners is confirmed by the studies dealing with gamification's motivational power (Deterding et al., 2011).

Nonetheless, the listed challenges reflect the listed challenges by Zarei and Aghaei (2014) and Rezaei and Meshkat (2017) of infrastructural and training deficiencies in Iranian EFL contexts. The necessity of extensive professional development is proposed as a key

component in successful technology integration, as hypothesized by Ertmer and Ottenbreit-Leftwich's (2010) study.

This research makes a contribution to the research by providing a combined quantitative and qualitative approach specifically tailored to the Iranian elementary EFL context and hence answering an important research gap.

Conclusion and Implications

The aim of the present study was to examine the role of modern technologies in enhancing the effectiveness of English language teaching at the primary level in Iran. The findings of the mixed-methods design were that the use of electronic tools such as gamified settings, multimedia resources, and online classrooms not only engages students and motivates student involvement but also supports teachers' facilitation of instruction differentiation for various learning styles. Quantitative results indicated a strong positive correlation between frequency of technology use and interest among students, but qualitative responses highlighted the motivational, pedagogical, and developmental benefits of technology-rich classrooms.

However, there are certain roadblocks to the implementation of these innovations, including weak infrastructure, poorly prepared teachers, and sporadic availability of digital resources. The teachers demanded immediate professional development programs on digital pedagogy and instructional design. The research is in line with previous findings emphasizing the significance of teacher training and infrastructural support in successful technology incorporation.

The study contributes to the literature by providing empirical evidence from Iranian primary EFL environments, a relatively under-developed arena. Based on the findings, the following implications follow:

For EFL teachers: Greater need to engage in ongoing professional growth in digital literacy and explore learner-centered, technology-enhanced pedagogies.

For teacher educators and policymakers: Curriculum needs to be restructured to encompass digital methodologies, and schools must invest in sustainable technological facilities.

For curriculum writers: Instruction materials must be updated in the context of current digital trends and made inclusive of technology for differentiated and interactive learning.

Overall, new technologies hold out promise to transform the horizon of elementary EFL education in Iran. Systemic change, capacity building, and policy alignment are nevertheless required to achieve this potential.

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ENHANCING CAREER READINESS THROUGH THE INTEGRATION OF INDUSTRY-ALIGNED DIGITAL TECHNOLOGIES IN HIGHER EDUCATION CURRICULA

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Abstract

This qualitative study examines the incorporation of modern industry digital technologies into courses offered in tertiary institutions and their effect on students' career preparation. The rapid evolution of technology, coupled with shifting market trends, has heightened the demand for effective integration of theoretical knowledge and real-life applications within higher education systems. The study sought instructors', curriculum developers', and final-year students' perspectives, experiences, and challenges with integrating relevant digital tools like data analytics, project management software, and AI-based applications through semi-structured interviews across three colleges. Thematic analysis of study data revealed that students' engagement with authentic industry-relevant technologies greatly enhances their practical skills, digital fluency, and self-efficacy within professional settings. However, educational institutional shortcomings concerning digital and training resources, infrastructure, and faculty professional development were revealed as barriers. The findings emphasize the urgent need for a comprehensive strategic approach for curriculum and professional development that enables effective and enduring alignment of digital technologies with labor market requirements.

Keywords: Career Readiness, Digital Integration, Higher Education

Introduction

Background

The advancing digital revolution reshaped the worldwide employment sector by establishing mandatory software tool proficiency requirements for graduates including data analytics software together with project management software and artificial intelligence tools (Brynjolfsson & McAfee, 2017). Student preparation for employment remains a fundamental responsibility for higher education institutions since they need to teach students the skills and knowledge and attitudes necessary to succeed in their careers (NACE, 2021). The majority of traditional educational programs lack real-world software integration which leads to a fundamental mismatch between academic training and professional workplace demands (Jackson, 2019). Developing countries such as Iran experience this problem most because educational institutions implement digital reforms at a pace that does not match industry needs (Farhadi & Rezaee, 2021).

The usage of computer applications including Tableau for statistical analysis and Trello for project management and AI such as Grammarly and chatbots result in practical skills development and enhanced employability (Dede et al., 2018). The practice of using these tools for hands-on learning enables students to develop competence with digital tools and build confidence in their ability to perform professional tasks (Bandura, 1997; Bennett et al., 2020).

Yet institutional advantages face substantial obstacles due to limited technology access and insufficient faculty development and outdated instructional materials (Selwyn, 2016; Alenezi, 2021).

Statement of the Problem

The failure of higher education institutions to align their academic programs with industry requirements creates a problem for students who seek careers in advanced technical fields. The growth of higher education institutions in Iran faces challenges in digital technology integration because of resource shortages and faculty members who lack experience (Farhadi & Rezaee, 2021). Students face diminished chances of securing professional positions that demand digital skills when educational programs fail to provide adequate preparation for these roles (Moore & Morton, 2017). The evaluation of curriculum-matching processes requires insights from faculty members and curriculum developers and student perspectives to identify existing obstacles and potential solutions for this alignment.

Research Questions

This study addresses the following research questions:

Q1. How does the integration of industry-aligned digital technologies into higher education curricula influence students' career readiness?

Q2. What are the perspectives and experiences of instructors, curriculum developers, and final-year students regarding the integration of digital technologies?

Q3. What barriers hinder the effective incorporation of digital technologies in higher education curricula?

Significance of the Study

The research addresses an existing research void through its exploration of digital technology implementation in Iranian higher education because this context lacks global attention. This research documents the viewpoints of stakeholders by producing practical recommendations that educators and policymakers can use to boost student career preparedness. The study exposes methods to overcome institutional challenges that serve as a pattern for other developing countries facing identical difficulties. The research provides a better understanding of the effects of digital technology on student work performance and digital ability development and self-assurance during work activities through an emphasis on qualitative data.

Literature Review

Incorporating industry-oriented digital technologies into the curricula of higher education has been widely supported in literature for its positive impacts on the employability, digital competence, and self-efficacy of the students as well as the general scholarly performance yielding to the demands of the labor market.

Brynjolfsson and McAfee (2017) term the "second machine age" as a period when digital technologies radically alter the labor market such that higher education institutions (HEIs) must train graduates well-versed in contemporary tools such as data analysis and artificial intelligence tools. The change emphasizes the necessity of curriculum development in line with such shifting technological imperatives (Jackson, 2019). However, as Farhadi and Rezaee (2021) argue, in settings such as Iran, integration remains always mismatched due to infrastructural as well as resource limitations.

A research work by Bennett, Knight, and Rowley (2020) points out that experiential working with digital tools commonly used in industries increases students' readiness for careers. Their findings supplement Bandura's (1997) self-efficacy theory that control of tasks technology-oriented makes one confident and prepared for working professional settings. Dede et al. (2018) also supplement the view that learning engineering and hands-on exposure to digital tools results in practical skills acquisition and digital fluency.

Despite these strengths, there are pitfalls. Selwyn (2016) refers to institutional obstacles like outdated materials, insufficient access to technology, and inadequate faculty professional development, which undermine effective digital transformation in education. Alenezi (2021) also explains that effective digital integration requires technological infrastructure as well as institutional readiness.

Several research studies suggest the need for systemic practice to enable closure between preparation in the academy and expectations in workplaces. Tynjälä, Välimaa, and Sarja (2003) call for pedagogical methods that connect theoretical know-how with authentic, real-world tasks. Billett (2015) and Oliver (2015) are both proponents of practice-based and work-integrated models of learning that directly enhance graduate employability. NACE (2021) and Moore & Morton (2017) further emphasize the importance of curricula developing not only digital skills but also more broadly conceived career readiness competences.

Institutional level is tackled by the triple helix model (Cai & Etzkowitz, 2020), proposing university-industry-government collaboration as a model for enhancing curriculum responsiveness to the needs of the industry. Johnson et al. (2016) document increasing trends in higher education enabling the adoption of such collaborative models, whereas Wang & Ellinger (2011) examine the organizational learning and innovation role in response to external environment signals.

In totality, literature highlights that successful integration of industry-specific digital technologies in higher education enhances the employability of students, working skills, and self-confidence. For this to be achieved, however, requires concerted actions in overcoming institutional, infrastructural, and pedagogical challenges through drawing on best practices and models of collaboration from across the globe among the key stakeholders.

Method

Participants

The research was conducted on 45 students of three faculties (business, engineering, and education) of one state university in Kerman city in Iran. The participants were 15 last-

year undergraduate students (8 males, 7 females, M age = 22.6, SD = 1.3), 15 teachers (9 males, 6 females, M teaching experience = 11.2 years, SD = 3.8), and 15 curriculum developers (10 males, 5 females, M experience = 8.5 years, SD = 2.9). The participants were recruited by purposive sampling for diverse perspectives on the embodiment of digital technology. Informed consent was obtained from all the respondents, and voluntary response were given without compensation.

Instruments

We collected data through semi-structured interviews intended to obtain the views and experiences of the participants. The guide for the interview included questions specifically worded:

- Students: "How have digital resources (e.g., data analysis or AI programs) prepared you for your profession?"
- Instructors: "What are the challenges you face when working with digital tools in instruction?"
- Curriculum Developers: How do you incorporate industry-related technologies in curriculum development?

Face-to-face interviews, each lasting 30–50 minutes, were performed in Persian, audio-recorded, and verbatim transcribed. Transcripts were then translated by a bilingual researcher into English and their accuracy verified through back-translation.

Data Collection Procedure

Data collection occurred from January through June 2024. Participants were students in pilot colleges that taught courses on digital tools, including Tableau, Trello, and artificial intelligence tools such as Grammarly. Interviews were conducted in privacy (privately or through secure online mediums) to ensure anonymity. Participants were informed of the reason why the study was being undertaken and that they could withdraw at any time. Transcripts were anonymized for confidentiality reasons.

Design of the Study

This qualitative study took a phenomenological stance in investigating the lived experience of participants in embedding digital technology.

Data Analysis

Thematic analysis, based on Braun and Clarke's (2006) six-step procedure, was used to code the data: (1) familiarization, (2) coding, (3) generating themes, (4) reviewing the themes, (5) defining the themes, and (6) reporting the result. Two researchers independently coded the transcripts, achieving an inter-rater reliability of 87%. Themes were generated through consensus, and the results were validated by member checking with 12 participants.

Results

Thematic analysis identified four broad themes: (1) Increased Practical Skills, (2) Improved Digital Fluency, (3) Greater Self-Efficacy, and (4) Institutional Barriers. Sub-themes and supporting quotes for these themes are provided in Table 1.

Table 1

Themes and Sub-Themes with Illustrative Quotes

Theme	Sub-Themes	Example Quote
Increased Practical Skills	Real-World Application, Mastery of Skills	“Tableau let me analyze data like a professional analyst.” (Student)
Improved Digital Fluency	Proficiency with Tools, Technological Adaptability	“I can now use Trello and other similar tools with ease.” (Student)
Increased Self-Efficacy	Professional Confidence, Career Preparedness	“Using AI tools made me feel confident about my work writing.” (Student)
Institutional Barriers	Resource Constraints, Lack of Training	“We lack the training or enough licenses to effectively use these tools.” (Instructor)

Theme 1: Improved Practical Skills

Participants showed that technology tools provided students with opportunities to develop skills in line with industry demand. Students reflected that Tableau and Trello facilitated activities such as data visualization and project planning, which are reflective of work-related activities (n = 13). Teachers reported increased problem-solving and critical thinking in students utilizing these tools (n = 11). Curriculum developers highlighted developing authentic tasks to bridge academic environments (n = 9).

Theme 2: Improved Digital Fluency

Increased Digital fluency emerged as a key top outcome, with students reporting increased capacity in industry software (n = 14). Teachers recognized that students were able to seamlessly integrate new software, making them more technologically flexible (n = 10). Curriculum developers suggested revision of curricula to include emerging technologies, such as AI tools, to remain contemporary (n = 8).

Theme 3: Increased Self-Efficacy

Students indicated improved confidence in their professional abilities collaborating with digital tools (n = 13). One student said, “Learning Grammarly and chatbots prepared me for workplace communications.” Teachers and curriculum developers agreed, indicating students’ self-efficacy improved as they learned the tools applicable in the industry (n = 18).

Theme 4: Institutional Barriers

Key barriers were unavailability of software licenses, outdated infrastructure, and poor preparation of teachers. Teachers stated lack of professional development as a leading barrier (n = 12), while curriculum developers stated budget as a leading barrier (n = 10). Students stated inconsistent tool availability in courses, hence causing learning interruptions (n = 11).

Discussion

The study demonstrates that students who learn industry-relevant digital technologies through university programs develop stronger professional skills which boost their readiness for employment. Previous work from Bennett et al. (2020) and Bandura (1997) has already established that students who actively use professional tools gain practical abilities and job task confidence.

The analysis of data found three important advantages which include better practical abilities and digital proficiency combined with stronger student self-confidence. Through practical application of Tableau, Trello and AI tools students acquire the ability to apply these technologies for real-world practice which leads to workplace skills development. Practice-based educational models from Tynjälä et al. (2003) and Billett (2015) support the enhancement of student skills through real-world problem solving and experiential learning approaches.

The implementation of digital transformation faces obstacles because institutions lack proper software tools along with having outdated technology infrastructure and incomplete teacher development programs. The problems institutions face matches the statements from Selwyn (2016) and Alenezi (2021) about requiring infrastructure investment and faculty development for successful technological implementation.

Conclusion

The research points to industry-specific digital technologies like data analysis and project management software and artificial intelligence as tools which boost practical skills and digital proficiency and student confidence thereby enhancing their employability prospects. Learning advantages exist but systemic challenges create major obstacles. The widespread use of technological tools faces barriers due to three main factors: insufficient financial resources and digital equipment quality and the absence of teacher training.

The complex nature of advanced challenges drives the need for educational policy to establish better alignment between curriculum and labor market requirements through faculty development while increasing institutional commitment toward digital capital development.

Implications of the Study

The given evidence has shown that educational policymakers need to formulate policies based on evidence strategies if they wish to enhance their universities' institutional digital transformation capabilities. For universities to acquire modern digital tools, there is a need for both government support and smart funding strategies. The research paper urges the educators to formulate learning plans that synchronize with current employment

opportunities across various sectors. The implementation of digital technologies in the course structure promises exceptional outcomes in students' active participation alongside their employment opportunities. This research offers a set of practical guidelines for the higher education institutions in developing countries for the incorporation of digital technologies into their education systems within the limitations of funding. The research provides strategies that any institution facing similar barriers can adopt to implement digital integration. Future research should focus on utilizing mixed-methods or longitudinal designs to explore the impact of digital integration on graduates' workplace success and job performance.

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EXPLORING THE IMPACT OF DIGITAL PEDAGOGICAL COMPETENCE ON THE SOCIAL-EMOTIONAL COMPETENCE AND RESILIENCE OF IRANIAN EFL TEACHERS

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Abstract

This study explores the impact of digital pedagogical competence on the social-emotional competence and resilience of Iranian EFL teachers. With the proliferation of digital technologies in educational frameworks, there is an increasing expectation for educators, in addition to having the requisite technological skills, to also possess complex classroom management skills and the related emotional and social skills. A mixed-methods approach was utilized to assess the relationship between pedagogical digital tool competence and the capacity to manage stress, foster meaningful relationships, and maintain emotional health amidst professional challenges. Employing validated questionnaires and semi-structured interviews, data was gathered from 120 Iranian EFL teachers. The results indicate that increased social-emotional skills and enhanced teacher resilience are linked to increased digital pedagogical competence, thereby underscoring the need for integrated holistic teacher training frameworks which address the psychosocial and technological demands of teaching.

Keywords: Digital Pedagogical Competence, Social-Emotional Competence, Teacher Resilience

Introduction

Background

The digital transformation of classroom learning has caused substantial changes in teaching methods thus making it essential for educators including EFL teachers to enhance their digital pedagogical skills. The required competencies for teachers involve two distinct areas: technical abilities to operate digital materials and educational skills to integrate these materials in language instruction (Zhang, 2023). The educational sector of Iran has experienced rapid digitalization, which has shown the urgency to equip EFL instructors with digital skills for better teaching outcomes and student participation (Shirazizadeh & Abbaszadeh, 2023).

Teaching as a profession demands from educators both a high level of emotional intelligence and the ability to handle challenges without showing weakness. Through their social-emotional abilities teachers learn to maintain their emotions, create positive connections and handle classroom complexities well (Derakhshan et al., 2021). For teachers to maintain their effectiveness and well-being they need resilience which enables them to keep working despite facing professional difficulties (Shirazizadeh & Abbaszadeh, 2023). The hybrid educational approach that integrates digital pedagogical elements with social-emotional

learning and resilience becomes most relevant in Iranian EFL classrooms because teachers encounter technology complications alongside emotional challenges.

Statement of the Problem

The educational field recognizes the importance of digital pedagogical competence, yet there is a research gap on its relationship to the social-emotional intelligence and resilience of English as a Foreign Language (EFL) teachers, specifically in Iran. While teacher resilience and emotional well-being have been investigated in research, the impact of digital competences on such psychological aspects is still under-investigated (Derakhshan et al., 2021). This interrelationship warrants investigation in light of the double-edged impact of digital technology on pedagogical practice, which can influence teachers' mental wellbeing and stress resilience. To safeguard EFL educators from the issues caused by digital advancement without compromising their social-emotional wellbeing, this knowledge gap should be filled in the design of comprehensive professional development approaches.

Research Questions

1. What is the relationship between digital pedagogical competence and social-emotional competence among Iranian EFL teachers?
2. What is the relationship between digital pedagogical competence and resilience among Iranian EFL teachers?
3. How do Iranian EFL teachers perceive the impact of digital pedagogical competence on their social-emotional competence and resilience?

Research Hypotheses

H1: There is a significant positive correlation between digital pedagogical competence and social-emotional competence among Iranian EFL teachers.

H2: There is a significant positive correlation between digital pedagogical competence and resilience among Iranian EFL teachers.

Significance of the Study

The current research has significance for various stakeholders within the educational community. Policymakers and educational administrators will benefit from the research outcomes to inform the development and rollout of strategic professional development programs to promote digital pedagogy capacities in tandem with social-emotional issues. For educators and teacher trainers, the findings of the study can be utilized in informing the incorporation of emotional intelligence and resilience training concepts into teaching digital literacy skills. Moreover, with a focus on digital competencies and the psychological, EFL teachers can choose to adopt new approaches to teaching that maximize professional engagement and psychosocial retention in the event of changes in the needs of education. The research contributes to the body of literature on teacher development in promoting an integrated approach that blends technological competence with emotional resilience.

Literature Review

Digital Pedagogical Competence

Digital pedagogical competence is the capacity of teachers to effectively integrate digital technologies into their pedagogy without compromising good pedagogical practice (Tondeur et al., 2020). This competence encompasses awareness of digital resources, comprehension of how to incorporate them within different learning contexts, and the ability to work under changing technological contexts. Spante et al. (2018) state that digital competence encompasses more than technical proficiency; it involves the ethical and critical utilization of digital tools, collaborative practice in digital settings, and pedagogical design for impact.

In the scenario of teaching English as a Foreign Language (EFL), the significance of digital pedagogical competence has grown, since language instruction is increasingly based on multimedia content, web tools, and online communication (Zhang, 2023). Based on research, educators who have higher levels of digital competence tend to implement student-centered instructional methods and establish active learning environments (Koehler et al., 2013).

Social-Emotional Competence in Teaching

In 2020, the Collaborative for Academic, Social, and Emotional Learning (CASEL) described social-emotional competence (SEC) as the capacity to comprehend emotions and regulate them alongside developing positive relationships and showing responsible decision-making. The ability to develop a positive classroom environment while handling stress effectively and meeting students' emotional requirements depends on a teacher's SEC abilities, according to Jennings & Greenberg (2009). According to Derakhshan et al. (2021), social-emotional competence serves as a fundamental quality for effective teachers in EFL classrooms because these environments demand high levels of communication, empathy, as well as patience.

Brackett et al. (2019) conducted recent research demonstrating that teachers who possess high SEC skills achieve better classroom management and develop stronger interpersonal bonds and experience improved personal wellness. These competencies are essential in helping teachers navigate the social complexities of diverse classrooms and mitigate burnout.

Teacher Resilience

In educational settings, teacher resilience denotes the ability to maintain both personal dedication and professional competence during challenging situations, according to Gu and Day (2013). In the EFL teaching environment, the combination of institutional demands with insufficient resources and cultural barriers creates the essential need for resilience, which secures career longevity for teachers, according to Shirazizadeh and Abbaszadeh (2023).

The study conducted by Mansfield et al. (2016) indicates that teachers who demonstrate resilience experience elevated job satisfaction and improved emotional stability, as well as better recovery from obstacles. The relationship between professional training programs and support systems and resilient behavior exists because these systems also affect the development of self-efficacy and emotional intelligence, as stated by Beltman et al. (2011).

Method

Design of the Study

This research utilized a design based on convergent mixed methods (Creswell & Plano Clark, 2018) that combined quantitative and qualitative data collection to examine digital pedagogical competence effects on Iranian EFL teacher social-emotional competence and resilience. The study analyzed quantitative and qualitative data through separate processes before combining them to form a complete understanding of the research topic. The combination of both data sources enabled researchers to confirm their findings through triangulation while producing more robust interpretations.

Participants

The study enrolled a group of 120 Iranian EFL teachers for the quantitative research, while a subgroup of 12 teachers joined the qualitative research.

Quantitative Phase

Researchers obtained quantitative participants through purposive sampling techniques. Three main requirements were established for selection: a minimum of two-year teaching background together with current English language teaching responsibilities alongside basic computer tool knowledge. The final participant group included 89 women and 31 men who served as teachers in different educational institutions throughout Iran. The participants demonstrated teaching experience between 2 and 20 years.

Qualitative Phase

Researchers selected 12 participants through purposeful selection for in-depth semi-structured interviews from the 120 EFL teacher participants using maximum variation sampling techniques. The selection process followed criteria which aimed to include different participant ages along with their gender variations and teaching settings and self-assessed digital competency. The research team conducted this process to gather a wide range of opinions and more detailed qualitative data. The 12 interviewees from the first phase agreed through formal consent to participate in the follow-up qualitative interviews.

Instrument

The study involved the use of two different instruments:

Quantitative Instruments

1. *Digital Pedagogical Competence Questionnaire*: This instrument which was derived from the work of Rubach and Lazarides (2021) evaluates how teachers view their capability to use digital tools in instructional settings.

2. *Social-Emotional Competence Scale*: This instrument measures teacher abilities based on the Jennings and Greenberg (2009) framework which evaluates their competencies in emotional regulation combined with empathy and social interaction skills.

3. *Teacher Resilience Scale*: The instrument by Mansfield et al. (2016) measures how teachers handle stress along with their ability to recover from failure and maintain professional dedication.

All questionnaires used a 5-point Likert scale which showed acceptable internal consistency (Cronbach's $\alpha > .80$).

Qualitative Instrument

The research team performed semi-structured interviews with 12 volunteers selected from the study population to investigate their teaching experiences with digital tools and their classroom emotional and social benefits. During the interviews a standardized thematic approach was used to maintain consistency yet allowed participants to provide detailed explanations.

Data Collection Procedure

Quantitative Data Collection

Throughout three weeks, quantitative data collection took place through Google Forms which provided an online survey to participants. The research instrument consisted of three standardized tools: Digital Pedagogical Competence Questionnaire and the Social-Emotional Competence Scale as well as the Teacher Resilience Scale. The survey process required participants to provide their data voluntarily after obtaining electronic consent at the survey's start. The survey link was shared across various teacher Telegram groups together with Instagram pages and EFL teaching association mailing lists in Iran to reach a broad sample of participants.

Qualitative Data Collection

The qualitative data collection process took place throughout the last three weeks of the survey period. The researchers selected 12 participants from the quantitative data pool for follow-up interviews by sending emails and phone calls. Zoom was used as the interview platform for sessions that lasted between 30 and 45 minutes. The researchers developed a semi-structured interview guide which they used to investigate participants' digital tool experiences and their emotional and interpersonal reactions to digital integration and their methods for handling work-related difficulties. All participants provided audio-recording permission which researchers used to transcribe the interviews for further examination.

Data Analysis

The SPSS software served to analyze the quantitative data. The analysis started with calculation of descriptive statistics including means and standard deviations then continued with Pearson correlation and multiple regression to assess the connections between digital pedagogical competence and social-emotional competence with resilience.

Researchers performed thematic analysis on qualitative data using the method developed by Braun and Clarke in 2006. The transcripts underwent inductive coding which afterward formed general themes regarding emotional, social and professional effects of digital

tool utilization. Two researchers performed the coding independently to maintain interrater reliability before discussing any discrepancies.

The interpretation process combined both data strands to detect points of agreement and disagreement as well as mutually beneficial perspectives within the research (Fetters, Curry, & Creswell, 2013).

Result

Quantitative Results

Descriptive statistics were employed to investigate the general levels of Iranian EFL teachers' digital pedagogical competence, social-emotional competence, and resilience. As Table 1 reports, the average scores on all constructs were considerably higher, revealing that the participants demonstrated positive levels across all three variables.

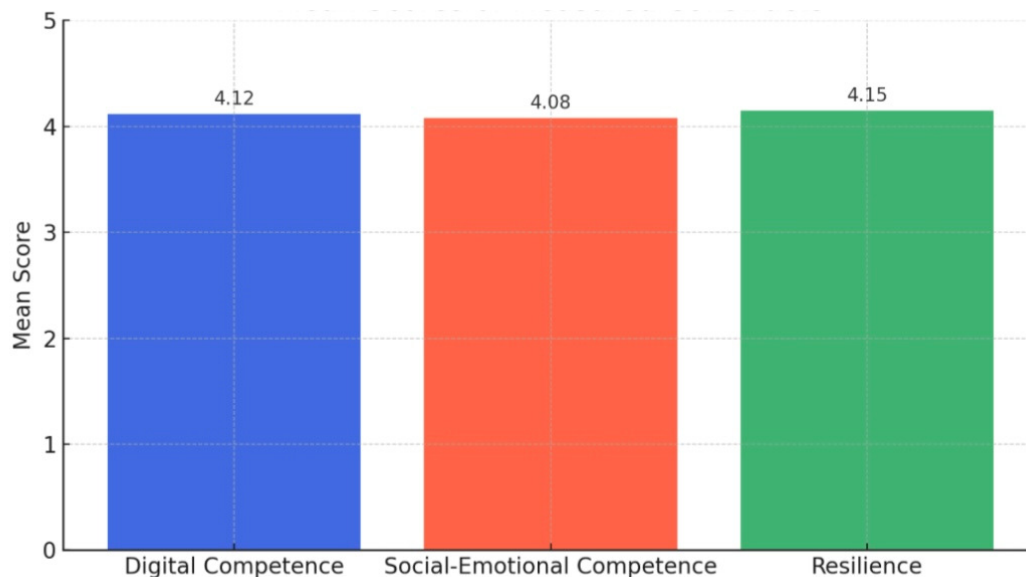
Table 1

Descriptive Statistics of Key Variables

Variable	Mean	Standard Deviation
Digital Competence	4.12	0.55
Social-Emotional Competence	4.08	0.60
Resilience	4.15	0.52

Figure 1

Mean Scores of Measured Constructs



Note: This figure presents participants' mean self-reported scores of digital pedagogical competence, social-emotional competence, and resilience.

In order to test the hypotheses, Pearson correlation coefficients were calculated. The results indicated that there were significant positive relationships among all three variables. Digital pedagogical competence was positively and significantly related to social-emotional competence ($r = .61$, $p < .01$) and resilience ($r = .65$, $p < .01$). Additionally, social-emotional competence and resilience were positively related ($r = .58$, $p < .01$).

Qualitative Results

Thematic analysis was used to code semi-structured interviews. Three main themes appeared:

1. *Digital Empowerment and Classroom Confidence*: Educators with greater digital literacy were more confident in conducting hybrid and online classes, as they reported increased engagement and lesson planning.

2. *Utilization of Digital Tools*: Most respondents indicated that the utilization of digital tools such as scheduling programs or mindfulness apps was beneficial in managing emotions and efficiently managing time.

3. *Resilience Formed by Peer Digital Networks*: Participants highlighted the value of online teacher forums and discussion boards for providing emotional support, exchanging resources, and buffering perceptions of isolation amidst professional stressors.

Integration of Findings

Both quantitative and qualitative findings converge to confirm the inference that pedagogical competence in digital technology enhances social-emotional capacity and resilience. This then suggests the implication for professional development programs, not only to prepare teachers in technology integration, but also to advance their emotional and psychological well-being.

Discussion

The results of the current research showed a statistically significant positive correlation between digital pedagogical capacity and social-emotional capacity and teacher resilience in Iranian EFL teachers. The results are consistent with previous studies proposing that digitally competent teachers not only practice more innovative teaching but also have greater emotional well-being and adaptability (Zhang, 2023; Derakhshan et al., 2021).

The correlation analysis revealed a strong correlation between digital competence and social-emotional competence ($r = .61$, $p < .01$), and resilience ($r = .65$, $p < .01$). The finding corroborates the work of Brackett et al. (2019), in which they highlighted that educators who have high emotional intelligence are more effective at tool integration and stress management. Qualitative findings also build on this further by demonstrating that digital empowerment boosts classroom confidence and facilitates peer collaboration—factors that play a significant role in resilience (Gu & Day, 2013).

In addition, thematic results indicated that educators use digital technologies not just for pedagogical practice but also for self-management and emotional self-regulation through digital peer networks and organizational structures. These practices are congruent with CASE's

social-emotional learning framework (Jennings & Greenberg, 2009) and underscore the imperative to have pedagogical innovations contextualized within consistent emotional support systems.

Conclusion and Implications of the Study

This study emphasizes the importance of digital pedagogical competence in developing social-emotional competence as well as the resilience of EFL teachers in Iran. The findings indicate that teacher training programs need to expand the scope of technical training to include emotional intelligence and resilience training directly tied to digital pedagogy.

Considering the combined challenges of the digital world and emotional strain in education, especially in high-stress language learning contexts, educational policymakers need to be more comprehensive. These findings support the construction of industry-guided, integrated, and comprehensive professional development frameworks that address the technological, emotional, and social dimensions of teaching (Tondeur et al., 2020).

In addition, institutions need to provide digital peer network and mindfulness resource access as part of standard active professional supervision support. This study strengthens the claim that resilience is not merely an individual trait but rather a product of organizational support, digital literacy, and emotional architecture.

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EXAMINING THE INTERPLAY BETWEEN DIGITAL PEDAGOGICAL RESILIENCE, TECHNOLOGICAL SELF-EFFICACY, AND PSYCHOLOGICAL CAPITAL AS PREDICTORS OF WORK ENGAGEMENT AMONG IRANIAN EFL INSTRUCTORS

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Abstract

The scope of this study covers the relationship between digital pedagogical resilience, self-efficacy of technology use, psychological capital, and their influence as predictors of work engagement within the context of Iranian EFL instructors. As digital tools in education increase, understanding how to maintain professional engagement among instructors is essential. Out of 180 Iranian EFL instructors from various universities and language institutes, a sample was drawn using stratified random sampling. Respondents completed validated questionnaires that assessed digital pedagogical resilience, self-efficacy of technology use, psychological capital, and work engagement. Structural equation modeling (SEM) analyses indicated that all three constructs had a significant predictive effect on one's work engagement, with psychological capital bearing the greatest impact. The findings illustrate that psychological resilience, confidence in technology, and positive psychological resources are vital for engaging instructors in active teaching and digital learning environments. This study impacts the design of professional training courses to improve the adaptational flexibility and psychological welfare of EFL instructors in relation to technology in education systems.

Keywords: Digital Pedagogical Resilience, Technological Self-Efficacy, Work Engagement

Introduction

Background

The use of modern digital tools has been implemented in teaching English as a Foreign Language which Sudhir and Pillai noted transformed practices globally. Online teaching platforms became widespread during the Covid-19 pandemic, which posed a challenge for EFL instructors who were unfamiliar with the technology and pedagogical strategies needed to cope with the challenges posed by digitization Schaufeli et al. (2006). While modern devices may have the potential to improve teaching efficacy, the degree of psychological resiliency, self-efficacy, and technological confidence required to implement these devices into classroom instruction, especially in the context of Iran's inconsistent digital education infrastructure, is concerning Sudhir and Pillai (2021). All of these aspects bring to light the gaps related to the technology and psychology of work that underpin EFL instructors' work engagement.

Work engagement is an important operationalization of working motivation and job satisfaction. It defines instructors' sustained energy, effort, and enthusiasm for work (Schaufeli

et al., 2006). Psychological factors have been found to emerge as consequential contributors to the formation of engagement, especially in cases involved with technological ambiguity (Luthans et al., 2007; Marsh et al., 2004). Digital pedagogical resilience refers to an educator's ability to cope and thrive when confronted with unforeseen technological challenges, while technological self-efficacy involves the educator's perception that he or she can utilize digital technology effectively (Compeau & Higgins, 1995). Psychological capital—hope, efficacy, resilience, and optimism—are positive psychological strengths that enable one to accumulate one's strength despite encountering challenges and remaining motivated (Luthans et al., 2007). Despite increased interest in the above constructs, little research has examined their interaction and predictive function for work engagement in the Iranian EFL teaching environment.

The unique challenges Iranian EFL teachers face, particularly in cities like Amol, demand further investigation into the sources of their professional commitment (Sudhir & Pillai, 2021). While earlier work has investigated individual dimensions of resilience, self-efficacy, or psychological capital in isolation, no integrated framework explaining their combined effect has been put forward. This literature shortfall demands exploring the interaction between these constructs in the prediction of work engagement which, in turn, can guide professional development interventions among EFL teachers.

Statement of the Problem

As pedagogical practice becomes increasingly penetrated by digital technology, Iranian EFL teachers are increasingly subject to the pressure of keeping pace with rapid rates of technological change while maintaining their professional interest and motivation. Institutional and urban disparities in access to technology and training in Amol amplify these pressures, with an added emphasis placed on psychological and adaptive ability (Sudhir & Pillai, 2021). Studies of organizational psychology have identified psychological capital, self-efficacy, and resilience as key determinants of work engagement and job satisfaction (Schaufeli et al., 2006; Luthans et al., 2007). However, a review of their joint effect in the virtual learning environment remains incomplete, particularly for non-Western countries such as Iran.

Compounding this issue are the specific stressors imposed upon Iranian EFL instructors, such as the need to reconcile pedagogical approaches with linguistic and cultural demands, as well as structural constraints such as inadequate education technology budgets (Brown, 2015). Although academic studies have determined work engagement to be key to productive pedagogical routines, motivation drivers responsible for engagement in virtual learning space—most notably on the part of EFL instructors—remains to be fully investigated (Schaufeli et al., 2006). These factors need to be identified and addressed for ensuring well-being among instructors and boosting their contribution to students' learning outcomes. In light of the problem stated above, this study seeks to address the following research questions:

Q1: What is the relationship between digital pedagogical resilience, technological self-efficacy, and psychological capital among Iranian EFL instructors in Amol?

Q2: How do digital pedagogical resilience, technological self-efficacy, and psychological capital individually and collectively predict work engagement?

Q3: Which construct has the strongest predictive effect on work engagement in the Iranian EFL teaching context?

Based on theoretical frameworks and prior empirical research (Compeau & Higgins, 1995; Luthans et al., 2007; Marsh et al., 2004), the following hypotheses are proposed:

H1: Digital pedagogical resilience significantly predicts work engagement among Iranian EFL instructors.

H2: Technological self-efficacy significantly predicts work engagement among Iranian EFL instructors.

H3: Psychological capital significantly predicts work engagement among Iranian EFL instructors.

H4: Psychological capital has the strongest predictive effect on work engagement compared to digital pedagogical resilience and technological self-efficacy.

Significance of the Study

This study is among the pioneers that integrate pedagogical resilience, technological self-efficacy, and psychological capital as predictors of work engagement in Iranian EFL instructors. Its contributions lie in both theory and practice. First, the study contributes to the body of literature on digital pedagogy by applying constructs that have been conventionally studied in organizational psychology—such as psychological capital—to the educational domain (Luthans et al., 2007; Compeau & Higgins, 1995). Second, the findings can be utilized to develop targeted professional development courses to establish EFL teachers' psychological resources and technological self-belief to thrive in evolving teaching environments (Sudhir & Pillai, 2021). These programs can include digital resilience workshops, technology-based self-efficacy training, and psychological capital building programs.

In addition, the study sheds light on how psychological and technological constructs interact to shape teachers' work engagement in a culturally constituted context. By way of analyzing EFL instruction in Amol, Iran, this research contributes to addressing a significant lacuna in the literature regarding understanding how local and institutional variables impact teachers' experience with computer-based learning tools (Schaufeli et al., 2006). The findings can also provide policymakers and institutions of developing countries with some lessons on how to improve the psychological well-being and online appropriateness of teachers.

Last but not least, encouraging work engagement among EFL teachers not only boosts their professional satisfaction but also boosts the learning experiences of the students. By encouraging resilience, confidence, and psychological resources, the study seeks to contribute to the broader agenda of sustaining good teaching practice in the age of technology.

Literature Review

Digital Pedagogical Resilience

Pedagogical resilience in the digital world refers to how well a teacher can learn and survive amidst technology disruptions and challenges in digital learning environments (Sudhir & Pillai, 2021). As digital tools such as learning management systems and virtual platforms become more integrated into EFL instruction, teachers must battle technical problems, shifting pedagogical demands, and student engagement issues. Sudhir and Pillai (2021) propose that pedagogical resilience in online environments entails problem-solving adaptability, emotional regulation, and persistence in the effective use of technology. Their conceptual framework portrays resilience as an energetic process through which teachers can maintain instruction quality regardless of technological ambiguity, a factor particularly relevant in Iran, where dependability in digital infrastructure may vary (Brown, 2015).

Technological Self-Efficacy

Technological self-efficacy, a derivative of Bandura's (1997) self-efficacy theory, is an indicator of how much a person believes they can successfully utilize technology to achieve tasks. Technological self-efficacy in EFL instruction is a prerequisite for teachers to embrace digital tools in instruction (Compeau & Higgins, 1995). Compeau and Higgins (1995) developed a scale of computer self-efficacy, showing that the higher the confidence in technology use, the more they will adopt digital tools and perform well on technology-dependent tasks. The research shows that technologically self-efficacious teachers are likely to utilize digital tools, enhancing instruction quality.

Psychological Capital

Psychological capital (PsyCap) according to Luthans et al. (2007) consists of four positive psychological resources: hope, efficacy, resilience, and optimism. PsyCap is a higher-order construct that enhances people's ability to cope with adversity in the workplace and maintain motivation (Luthans et al., 2007). In education, PsyCap has been linked to improved teacher performance and well-being. For example, Luthans et al. (2007) confirmed that PsyCap enables employees to maintain energy and productivity when coping with stress, something that can be applied to EFL teachers who have to endure the stress of teaching in a digital age.

Work Engagement

Work engagement, a fulfilling, positive work-related state characterized by vigor, dedication, and absorption, is a crucial outcome measure in educational research (Schaufeli et al., 2006). Schaufeli et al. (2006) developed the Utrecht Work Engagement Scale (UWES) that has been used intensively to measure engagement in various professions, such as teaching. Their cross-cultural study presented evidence that employed teachers are more job-satisfied and have better pupil outcomes, which highlights the importance of engagement in education.

Interplay of Constructs and Gaps in the Literature

The interaction between digital pedagogical resilience, psychological capital, and technology self-efficacy is not well studied, particularly in the context of EFL instruction in developing countries like Iran. While individual studies have tested these constructs, their aggregate effect on work engagement is not adequately examined (Compeau & Higgins, 1995; Luthans et al., 2007; Sudhir & Pillai, 2021). Bentler (1990) and Schermelleh-Engel et al. (2003), for example, document the strength of structural equation modeling (SEM) in examining intricate relationships between latent constructs, yet little research has employed SEM to examine these constructs in education.

Method

Design of the Study

To discuss the associations between digital pedagogical resilience, self-efficacy in technology, psychological capital, and work engagement, a correlational design was incorporated in the study. The dependent variable was work engagement, whereas the independent variables consisted of digital pedagogical resilience, self-efficacy in technology, and psychological capital. The relationships between variables, either direct or indirect, were analyzed using structural equation modeling (SEM). This type of study is cross-sectional, and the data was collected at one point in time only.

Participants

180 EFL teachers who taught at universities and private language institutes in Amol, Mazandaran Province, Iran participated in this research. Participants were sampled using stratified random sampling to represent institutional types (language institutes and universities) and teaching experience levels (novice, mid-career, and veteran teachers). The sample consisted of 102 female (56.7%) and 78 male (43.3%) teachers aged 24-55 years ($M=36.4$, $SD=7.2$). Teachers had 1-25 years of experience ($M=8.9$, $SD=5.6$). All the participants were teaching face-to-face or online EFL courses and had some experience using digital tools for teaching. The inclusion criteria necessitated the participation of at least one year of teaching experience and being actively working in an EFL teaching job in Amol. The participants were asked to sign an informed consent form, and ethical clearance was taken from the Institutional Review Board of a university in Amol.

Instruments

The following four instruments were utilized to gauge the associated variables for the study.

Digital Pedagogical Resilience Scale (DPRS): This is an 18-item questionnaire on a Likert scale regarding teachers' resilience in digital teaching which was adapted from Sudhir and Pillai (2021) and has a reliability score of $\alpha = .91$.

Technological Self-Efficacy Scale (TSES): This scale has 10 items and was adapted from Compeau and Higgins (1995). It measures teachers' perceived self-efficacy in technology use ($\alpha = .89$).

Psychological Capital Questionnaire (PCQ-24): This is a 24-item Hope, Efficacy, Resilience, and Optimism instrument developed by Luthans et al. (2007) with a reliability score of $\alpha = .93$.

Utrecht Work Engagement Scale (UWES-9): This is a 9-item measure of vigor, dedication, and absorption developed by Schaufeli et al (2006), reliability $\alpha = .92$.

The instruments were all culturally adapted and linguistically validated through forward and backward translation into Persian. They were tested on 30 EFL instructors for clarification.

Procedure

Data was gathered between March and June 2024. After obtaining the Ethics Board approval, a compilation of local institutions was prepared and permission was secured. Stratified random sampling was used to obtain representation from different institutions as well as from participants with varied experience. Through an email, participants received a consent form and questionnaire that they filled using a secured Google Forms link. Each participant took approximately 25-30 minutes to fill the survey. To enhance the response rates, reminders were sent. Out of the 192 responses received, 12 were incomplete and excluded, which gave a total of 180 participants.

Data Analysis

The data was analyzed in both SPSS (v26) and AMOS (v24) software. For analyzing normality, the Shapiro-Wilk Test along with skewness and kurtosis was employed. The minimally missing data (<2%) was addressed through mean imputation. For Confirmatory Factor Analysis (CFA), factors measurement validation was executed using Chi-square, CFI, TLI, RMSEA, and SRMR. The hypothetical models were evaluated using Structural Equation

Modeling (SEM) with maximum likelihood estimation that included bootstrapped (5,000 samples) mediation effects. Set standardized coefficients were used to determine the evidence's impact: small, medium, and large defined as ($\beta = .10 = \text{small}, .30 = \text{medium}, .50 = \text{large}$).

Results

This chapter presents the results derived from the study for testing the hypotheses concerning predictive relationships among Digital Pedagogical Resilience (DPR), Technology Self-Efficacy (TSE), Psychological Capital (PsyCap), and Work Engagement (WE) in a sample of Iranian EFL teachers. The findings were explained through the use of descriptive statistical procedures, correlational analysis, and Structural Equation Modeling (SEM), which tested direct and indirect relationships between the variables involved.

Descriptive Statistics

Table 1

Descriptive Statistics for Key Variables

Variable	Mean	SD
Digital Pedagogical Resilience	3.78	0.62
Technological Self-Efficacy	3.65	0.58
Psychological Capital	4.12	0.67
Work Engagement	4.02	0.69

Correlation Analysis

Table 2

Pearson Correlation Coefficients among Variables ($p < .01$)

Variable	DPR	TSE	PsyCap	WE
DPR	—	.48	.55	.52
TSE	.48	—	.50	.47
PsyCap	.55	.50	—	.61
WE	.52	.47	.61	—

Structural Equation Modeling (SEM)

Researcher analyzed the hypothetical structural equation model using AMOS software with maximum likelihood estimation methods. The following goodness-of-fit statistics indicate that the model met fit standards: $\chi^2 (84) = 172.45$, $p < .001$; CFI = .93; TLI = .91; RMSEA = .072; SRMR = .056.

All four hypotheses were confirmed and the standardized path coefficients were reported as follows:

- Digital Pedagogical Resilience → Work Engagement: $\beta = .28$, $p = .002$
- Technological Self-Efficacy → Work Engagement: $\beta = .22$, $p = .008$
- Psychological Capital → Work Engagement: $\beta = .41$, $p < .001$

Mediation analysis showed that there were partial mediation effects:

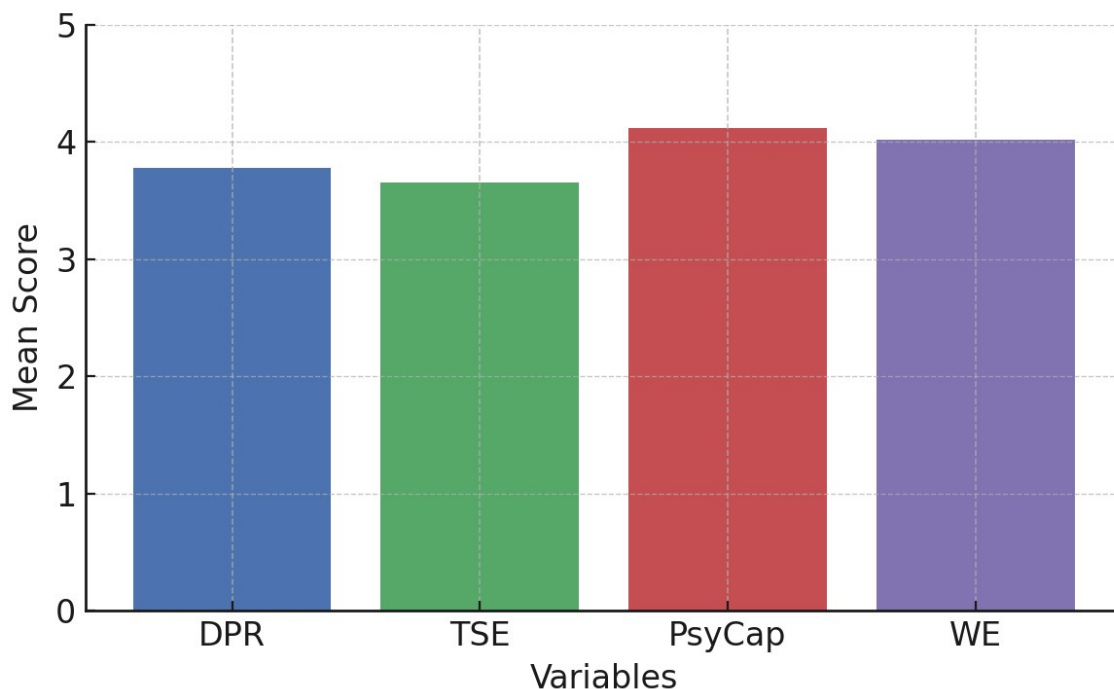
- DPR → PsyCap → WE: $\beta = .14$, $p = .003$
- TSE → PsyCap → WE: $\beta = .12$, $p = .006$

Findings indicate that psychological capital is a good predictor of work engagement with the further implication that it mediates the influence of digital pedagogical resilience and technology self-efficacy on work engagement.

Figure 1 illustrates the mean scores for the key study variables, and the indicator that Psychological Capital (PsyCap) had the highest average score among participants.

Figure 1

Mean scores for Digital Pedagogical Resilience (DPR), Technological Self-Efficacy (TSE), Psychological Capital (PsyCap), and Work Engagement (WE)



Discussion

The significance of psychological and technological factors in predicting work engagement of Iranian EFL instructors was particularly noteworthy in this study's findings. As expected from the model proposed by Luthans et al. (2007), psychological capital (PsyCap) stood out as the most prominent indicator of work engagement. The strong impact of PsyCap –hope, efficacy, resilience, and optimism– underscores the importance of positive psychological resources for high energy, dedication, and absorption in teaching.

Although DPR and TSE had lower effects than PsyCap, they still made significant contributions to predicting work engagement. Sudhir and Pillai (2021) observed that DPR indicates how instructors handle or cope with technological interruptions, suggesting that instructors possess considerable skill in managing fluctuating technologies. TSE showed that instructors who are more confident with digital tools tend to be more engaged, consistent with Bandura's (1997) self-efficacy theory.

The PsyCap effect is a type of partial mediation which describes its influence as a connector between adaptability to technology and professional motivation and well-being. This understanding illustrates the combined effect of these constructs in which technologic resilience and confidence facilitates positive psychological functioning which then leads to greater work engagement.

Conclusion

The study looked into how psychological capital, self-tech proficiency, and digital pedagogy resilience predict work engagement among EFL teachers in Amol, Iran. All three provided positive predictions, in the case of psychological capital it had the highest predictive power-strength, according to the structural equation modeling results. The findings assert that when educators need to actively participate in online and hybrid courses, the blend of psychological flexibility together with technological elasticity is the appropriate answer to motivate them.

As such, the findings validate the application of organizational psychology metrics for teaching research and propose that other studies need to further investigate the relations between these factors in other settings. Lastly, the construction of educators' psychological capital and self-efficacy needs to be prioritized to improve motivate and performance for distance education.

Implications of the Study

The results of this study impact policy implementors, leaders of education systems, and universities that focus on training educational professionals.

Self-Improvement: Training programs should strive to increase instructors' technological self-efficacy and adaptational pedagogical resilience through digital problem solving and virtual teaching. Teachers can greatly benefit from enhanced flexibility.

Organizational Support: Employers need to provide psychological support to teachers by having a favorable culture at work, making available psycho-digital resources, and safeguarding against undue workload. Such aid enables teachers to participate actively and improves their quality of teaching.

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THE IMPACT OF PEER-ASSISTED LEARNING-ORIENTED ASSESSMENT VIA MOBILE VOICE JOURNALS ON B2-LEVEL IRANIAN EFL LEARNERS' SPEAKING FLUENCY

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Abstract

This study evaluates the effect of peer-assisted learning-oriented assessment (LOA) through mobile voice journals on the speaking fluency of Iranian EFL learners at B2 proficiency level. The study is rooted in the principles of LOA, focusing on feedback-rich, formative assessment processes that actively involve learners in their self-monitoring and performance improvement. Sixty subjects were assigned into experimental and control groups. The control group was taught speaking using traditional methods; the experimental group participated in weekly mobile application sustained silent reading voice easy recording sessions followed by peer review. Fluency gain was measured with pre and post intervention speaking tasks performed with standard fluency criteria, and coupled with learner reflections and semi-structured interviews for qualitative data capturing perceptions and experience of learners. Quantitative data indicated considerable fluency improvement for learners in the experimental group, while qualitative data confirmed increased learner autonomy alongside enhanced engagement positively towards collaborative assessment. These insights illustrate mobile-assisted LOA coupled with peer interaction as a low-cost effective tool for cultivating speaking fluency.

Keywords: Learning-Oriented Assessment (LOA), Mobile Voice Journals, Speaking Fluency

Introduction

The rapid development of digital technologies, together with their integration into language learning environments, has reshaped the role of assessment in learning frameworks. One such innovation is Learning-Oriented Assessment (LOA), where formative practices with a lot of feedback aimed at enhancing learner autonomy and success are emphasized (Carless & Boud, 2018). At the same time, mobile-assisted language learning (MALL) has gained more popularity, offering flexible and context-sensitive approaches for learners to engage in language practice. Among these tools, mobile voice journals can provide the learner with a unique chance for continuous oral production and reflection, especially when coupled with peer feedback mechanisms. This is particularly opportune in EFL contexts like Iran, where classroom-based speaking instruction is often insufficient to provide learners with ample opportunities for spontaneous and interactive speaking practice.

Despite the recognized role of fluency in communicative competence, many Iranian EFL learners are not competent in this area due to a lack of opportunities for real practice. Traditional approaches to teaching speaking typically incline towards teacher-initiated feedback and occasional examination, which would demotivate active participation from learners themselves. Furthermore, integrating peer assessment into a controlled LOA model is less researched, particularly through mobile channels. This study aims to fill this gap with an

investigation into the effectiveness of peer-supported LOA using mobile voice diaries in improving speaking fluency in B2-level Iranian EFL learners. This study is guided by the following research questions:

RQ1: Does peer-assisted learning-oriented assessment through mobile voice journals significantly improve the speaking fluency of Iranian EFL learners at the B2 level?

RQ2: How do learners perceive the use of peer-assisted mobile voice journals in relation to their speaking fluency and engagement?

Based on the explanatory sequential mixed-methods design, the following hypothesis is proposed for the quantitative phase of the study:

H0: There is no significant difference in speaking fluency between learners who receive peer-assisted LOA via mobile voice journals and those who receive traditional instruction.

H1: Learners who receive peer-assisted LOA via mobile voice journals will demonstrate significantly greater improvement in speaking fluency compared to those receiving traditional instruction.

This study contributes to the body of research on technology-supported formative assessment in EFL settings. Also, it pinpoints the possibility of combining peer review and mobile applications in supporting learner autonomy, engagement, and speaking ability. Additionally, the findings of this research can be utilized to shape curriculum design and pedagogy in EFL settings, particularly where interaction with native speakers or immersive settings is limited.

Literature Review

Learning-Oriented Assessment (LOA)

Learning-Oriented Assessment (LOA) has been developed as a productive framework that integrates assessment into the process of learning, with a focus on feedback, self-regulation, and student engagement (Carless & Boud, 2018). Rather than viewing assessment as a discrete evaluative task, LOA promotes formative strategies that encourage continuous improvement. In English as a Foreign Language (EFL) contexts, LOA has shown promise in developing speaking and writing skills through peer feedback and scaffolded tasks (Evans, 2020). The use of peer-supported tasks within Learning-Oriented Assessment (LOA) frameworks has been associated with higher levels of learner autonomy and greater levels of motivation (Xu & Carless, 2022).

Mobile-Assisted Language Learning (MALL)

Mobile-Assisted Language Learning (MALL) provides learners with access to linguistic input and opportunities for practice beyond what is feasible in traditional classroom settings. Increasing smartphone penetration enables learners to participate in listening, speaking, and reflective tasks on the move (Burston, 2020). Recent research highlights the promise of mobile apps in facilitating the development of oral fluency, particularly if tasks are situated in authentic and purposeful activities (Kukulska-Hulme et al., 2021). Audio recording software programs particularly provide opportunities for repetitive practice, self-evaluation, and asynchronous peer feedback.

Speaking Fluency in EFL Contexts

Speaking fluency remains a key but challenging aspect of EFL learning. Speech rate, length of pause, and hesitation frequency are all indices of automaticity and communicative

proficiency (Tavakoli & Wright, 2020). It remains underprioritized in the classroom due to time constraints and problems of measurement. Hence, alternative approaches that offer regular, low-anxiety occasions to speak, such as mobile journaling and peer feedback, are being advocated more (Derwing et al., 2021).

Student Autonomy and Peer Assessment

Peer feedback is a central component of formative assessment that promotes collaboration and critical thinking. By evaluating each other's work, students are encouraged to adopt advanced levels of reflection and comparison that help them develop a deeper understanding of performance criteria (Lu & Zhang, 2020). Moreover, peer-to-peer communication in assessment settings has been demonstrated to reinforce learners' perception of agency and responsibility for their learning process, which are central components of autonomy. Within MALL environments, asynchronous peer review via audio journals provides an extra dose of flexibility and individualization.

Integration of MALL and LOA

Current research has started investigating the convergence of Learning-Oriented Assessment (LOA) and Mobile-Assisted Language Learning (MALL) for enhancing language learning experience. For example, Gao and Zhang (2023) showed that mobile-based LOA tasks, such as reflective voice diaries and peer feedback, resulted in noteworthy improvements in learners' speaking fluency and confidence levels. The use of mobile technology in formative, learner-centered assessment frameworks seems to break the constraints of classroom modes and facilitate ongoing language use and development.

Method

Design of the Study

This study utilized an explanatory sequential mixed-methods design. The first stage was quantitative wherein the outcomes related to speaking fluency were measured with pre- and post-testing. This was followed by a qualitative phase focused on explaining the quantitative results through interviews and learner reflections. With the aim of achieving an in-depth understanding of the influence of peer-assisted learning-oriented assessment (LOA) through mobile voice journals on Iranian EFL learners' speaking fluency, this study adopted a comprehensive mixed-methods approach.

Participants

Participants included 60 Iranian EFL learners at the B2 proficiency level. In the quantitative phase, participants were randomly assigned into an experimental group (n=30) and a control group (n=30). Twelve participants were purposefully chosen for the qualitative phase from the experimental group based on their improvement scores and availability to generate diverse responses to their learning process.

Instruments

The research utilized different tools including: (1) a speaking fluency assessment created by Lennon in 2020; (2) voice journaling mobile applications to track learner reflections; (3) feedback forms from peers; and (4) an interview protocol created by the researchers through the adaptation of Carless's 2019 LOA research.

Data Collection Procedure

Under the quantitative phase, pre-tests for baseline fluency scores were administered to both groups of participants. For the experimental group, weekly mobile voice journal recordings and peer ratings by peers through guided assessment forms were conducted. For the control group, however, conventional speaking instruction methods were maintained. Both groups completed a post-test after an eight-week period. In the qualitative phase, open-ended reflective questions and semi-structured interviews were used to examine learners' experiences with mobile-assisted peer assessment.

Data Analysis

The researcher examined quantitative data through paired-samples t-tests and ANCOVA to evaluate group differences in pre- and post-intervention fluency scores and evaluated the size of improvements through effect size measurements. The researcher also conducted a thematic analysis of qualitative data through NVivo software using the Braun and Clarke six-phase method for their analysis of interview and reflection data. The themes explored learner autonomy alongside engagement levels and students' views about peer assessment practicality.

Results

This chapter discusses the results that were gathered from both quantitative and qualitative phases of the study.

Quantitative Results

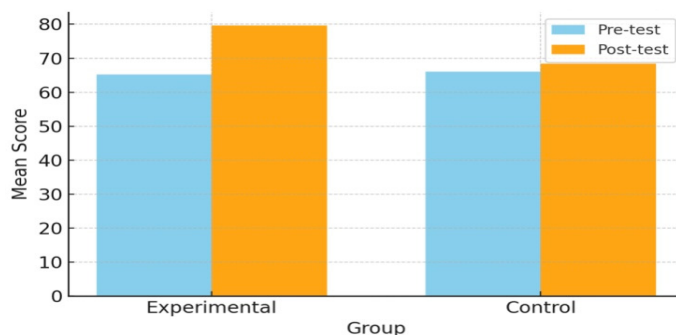
The quantitative data were designed to evaluate the efficacy of peer-assisted learning-oriented assessment (LOA) via mobile voice journals on the speaking fluency of B2-level Iranian EFL learners.

Table 1 illustrates pre-test and post-test mean scores in the experimental and control groups. The experimental group achieved a significant development in speaking fluency, as the mean score increased from 65.2 to 79.6. However, the control group under traditional teaching achieved a slight increase from 66.0 to 68.4.

Group	Pre-test Mean	Post-test Mean
Experimental	65.2	79.6
Control	66.0	68.4

Figure 1

Comparison of Mean Scores for Pre-test and Post-test Across Experimental and Control Groups



A paired-sample t-test within the experimental group showed a statistically significant enhancement in speaking fluency ($t(29) = 5.87, p < 0.001$), thus suggesting a strong effect of the peer-supported LOA intervention. No statistically significant gains were, however, recorded for the control group ($t(29) = 1.26, p = 0.216$). The findings provide evidence of the effectiveness of incorporating mobile-based peer feedback in language learning in an effort to promote speaking ability.

Qualitative Results

A qualitative analysis of learner reflection and semi-structured interview data revealed three broad themes: (1) Greater Learner Autonomy, (2) Developing Speaking Confidence, and (3) Positive Attitudes Towards Peer Feedback.

Most respondents indicated that the activity of recording their own voice performances and getting peer feedback assisted them in reflecting on speaking practices, enabled them to track their progress, and promoted confidence in communicating ideas. The following is an example of a prevalent perception:

“Recording myself weekly permitted a sense of my own progress. Feedback from my partner revealed aspects that I had not previously noticed.”

Discussion

The results of this study provide strong support for the assumption that peer-supported learning-oriented assessment (LOA) via mobile voice journals can effectively develop Iranian EFL learners' speaking fluency at the B2 level. Quantitative analysis revealed that the experimental group, who had performed weekly tasks using mobile voice journals complemented by peer feedback, developed their fluency scores on the post-test significantly compared to the control group. This supports previous studies arguing that technology-supported formative tests facilitate more in-depth learner engagement and language acquisition (Carless & Boud, 2018; Evans, 2020).

The qualitative results supported these findings, highlighting the development of learner autonomy, the increase in speaking confidence, and the development of positive attitudes towards peer feedback. These themes are consistent with current research that emphasizes the importance of mobile devices in creating flexible, learner-centered contexts that support self-monitoring and reflective practices (Gao & Zhang, 2023; Kukulska-Hulme et al., 2021). Peer feedback, when situated within a Learning Outcomes Assessment (LOA) context, promoted a culture of collaboration where students were actively involved in assessing and supporting each other's development—signature features of self-directed learning (Lu & Zhang, 2020).

These findings underline the pedagogical value of a mixture of LOA principles and mobile-supported practice in contexts where conventional assessment is unfeasible. Although problems of feedback quality and digital access can be predicted, general benefits to fluency development and learner agency seem considerable.

Conclusion and Implications of the Study

The existing literature receives an essential addition which supports using technology along with formative assessment in English as a foreign language teaching environments. A peer-supported LOA strategy through mobile voice journals enabled students to improve their speaking fluency alongside developing reflective competencies and self-directed learning

skills. The research confirms that mobile-supported peer feedback methods effectively facilitate student-directed oral language practice activities.

The research indicates the essential nature of implementing inexpensive advanced digital tools to support language development within educational settings. The development of mobile learning LOA strategies for promoting learner motivation and engagement in language programs would benefit from the study. The study recommends future research that explores how these programs affect learner proficiency development across different skill levels as well as how teacher involvement affects peer feedback effectiveness.

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HIGHER EDUCATION, STRATEGY, TECHNOLOGY (ROBOTS AND AI) & INNOVATION IN THE DIGITAL AGE.

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Abstract

With the development of digital technologies and its developments in various fields, educational systems have also changed. In this field, the curriculum, as one of the main pillars of the education system, requires revision and innovation to synchronize with the requirements of the digital age. This article examines the features, requirements, and challenges of innovative curriculum in this age and offers solutions to improve design and implementation. In the first part of the article, the basic concepts of innovative curriculum and the effects of modern technologies such as artificial intelligence, augmented reality, E-learning and learning management systems (LMS). Also, the basic principles of flexible, personalized and self-learning-based curriculum design are also mentioned. Below, the opportunities and challenges of implementing an innovative curriculum are discussed. The most important opportunities include promoting the quality of education, enhancing learners' interaction, developing critical thinking skills and creativity. Conversely, challenges such as lack of technology infrastructure, teachers' resistance to change, and digital inequality are among the obstacles to such programs. In addition, some successful global models and experiences of leading countries in innovative curriculum have been examined and compared to Iran. Finally, suggestions have been made to improve curriculum status in Iran that include developing support policies, upgrading teachers' digital skills, optimizing new technologies, and enhancing digital learning-related research. The results show that to achieve an efficient educational system in the digital age, the use of curriculum is a review of the curriculum. This requires the collaboration of educational policymakers, teachers, students and institutions to create a suitable platform for innovative curriculum.

Keywords

Digital technology, Critical thinking, Global model, Innovative curriculum

Introduction

Digital transformation today is recognized as one of the biggest challenges and at the same time the opportunities facing educational systems. In a world where information technology is progressing at an unprecedented speed. Traditional educational systems can no longer. Meet the needs and expectations of new generations. Innovative proper programs that use new technologies such as internet -pimples of augmented reality and virtual reality can respond to the rapid and dynamic changes of the digital age.

In this article, it is attempted to provide strategies for improving the quality of digital technologies in promoting the quality of education and successful domestic and international experiences by examining the theoretical and practical foundations of modern curriculum programs. The fundamental changes in the curriculum are not only in favor of promoting the scientific level of students. It also provides the growth of creativity and innovation in society.

Research Problem

The use of robotics and artificial intelligence technologies in the education and learning of children and students as a new approach has seen increased attention from education professionals and researchers. These two technologies are able to improve traditional teaching methods and enhance children's learning experience. Robotics, as an essential part of advanced technologies, allows children and students to learn scientific concepts and various skills in a practical and experimental way. This technology seems encouraging and effective and makes children want to learn. Programming and coding to control robots may help children strengthen important skills such as problem solving and reasoning. Artificial intelligence, as a key technology, allows for the collection and analysis of data to provide intelligent feedback to students. Using artificial intelligence algorithms, educational content can be provided in a personalized way according to the needs of each student. A deeper understanding of each individual's learning process is also obtained through artificial intelligence, which can be effective in improving their performance. The combination of robotics and artificial intelligence can improve the learning experience of children and students. By interacting with smart robots that use artificial intelligence algorithms, these technologies can help strengthen children's patience and desire to learn. This combination can help children understand more complex topics and develop deeper skills. Considering the importance and impact of robotics and artificial intelligence in the education and learning of children and students, the use of these technologies to improve the learning process and develop individual skills is of great importance. These technologies can help students acquire basic and advanced skills and become those who are ready to respond to future challenges. Therefore, the development and use of robotics and artificial intelligence in the education and learning of children and students seems essential and can play an effective role in the progress and excellence of individual and societal education.

Research Questions

- 1 What will educational management based on artificial intelligence look like?
- 2 According to research, how can artificial intelligence help improve the higher education process?
- 3 Based on current research, which AI-based educational tools are used in higher education?
- 4 Based on recent research, what opportunities has artificial intelligence provided in higher education?

2. Literature Review

The use of robotics and artificial intelligence in the education and learning of children and students is an area full of attractive and important possibilities and challenges that has received more attention in recent years. Much research has been conducted in this field, showing that these technologies can play a very effective role in improving the education and learning process of children and students. The effectiveness of using robotics in teaching science and technology skills Research has shown that the use of robots as educational tools can significantly improve the learning of science and technology concepts. Robots can encourage

children to think and experiment scientifically by providing interactive and attractive environments.

The use of artificial intelligence in recognizing and supporting individual needs Research in the field of artificial intelligence has shown that artificial intelligence systems can automatically adjust educational planning by analyzing individual behaviors and reactions and adapt to the needs and abilities of each student and child. The use of robotics as an educational tool in classrooms Research has shown that the use of robotics in classrooms can make the educational experience more attractive and active. These tools can help students better understand mathematical and scientific concepts. Some research has shown that robots and artificial intelligence systems can help children improve their communication and social skills because these tools can provide real-time appropriate responses based on children's reactions and interactions.

2-1- Theoretical foundations of research

The use of robotics and artificial intelligence as advanced technology tools in the education and learning of children and students is one of the most popular and attractive topics in the field of education. These two technologies can not only make the learning process more attractive and effective, but can also facilitate the improvement of academic performance and abilities of children and adolescents. The following theoretical foundations that form the basis and main principles of using these technologies in education will be explained.

2-2- Constructive learning theory

Constructivist Learning Theory believes that mental knowledge is conceived and created through interaction with the surrounding environment. In this case, robots and artificial intelligence systems provide interactive and challenging activities that allow children to challenge themselves in the learning process related to new concepts and to understand these concepts consciously.

2-3- The theory of artificial intelligence

Artificial Intelligence Theory is the study and simulation of human intelligence in machines and computer systems. By using robotics and artificial intelligence in educational environments, intelligent tools can be provided that respond to the differences and needs of each student and make the learning process more personalized and engaging.

2-4-. Game-based learning theory

Game-based learning theory emphasizes the principle that greater motivation to learn and achieve educational goals is provided through play and competition. By embedding educational game elements in robotics and artificial intelligence systems, the learning process can be made happier and more useful for children.

2-5-. Theory of educational psychology

Educational Psychology Theory examines the psychological factors that influence the learning and retention process. By combining the principles of this theory with robotics and artificial

intelligence, it is possible to design how to deliver content, feedback, and assessment to each student in an appropriate and explanatory manner.

3. Methodology

This study employs a qualitative research approach, utilizing content analysis of academic publications, interviews with educators and technologists, and case studies from schools that have implemented AI tools. The objective is to present a nuanced understanding of the practical applications and implications of AI in education.

1-Benefits of using robotics and artificial intelligence

Robotics and artificial intelligence, as two advanced technologies, have strongly attracted the attention of researchers, teachers, and parents. By using robots, children learn scientific and mathematical concepts in a practical and experimental way and acquire various skills. This practical and interactive experience with robots increases the motivation and desire to learn in children (Mehrabi 1331). Artificial intelligence, due to its ability to analyze data and provide intelligent feedback, can improve the teaching and learning process. Using artificial intelligence algorithms, educational programs can be presented in a personalized way to best meet the needs of each student and accelerate the development of his skills and knowledge. The combination of robotics and artificial intelligence can improve the overall learning experience for children and students. By providing interactive and engaging educational programs using smart robots, these technologies can help to stimulate children's curiosity and willingness to learn and prepare them for future success. The use of robotics and artificial intelligence in children's education and learning is of great importance.

2- Personalizing education

With the help of artificial intelligence, educational programs can be adjusted based on the needs and abilities of each student to provide the best solutions for his learning.

3- Development of Kalen skills

These technologies can play an important role in the development of communication skills, problem solving, critical thinking and cooperation of children and students (Sadeghifard, 1333)

4-Increase feedback and evaluation

Artificial intelligence can identify each student's strengths and weaknesses and provide the best solutions to improve weaknesses.

Improving the educational experience by creating interactive educational experiences and paying attention to the personal needs of each individual will be possible. Robots and artificial intelligence can improve the quality of education and learning. Although the use of robotics and artificial intelligence in education and learning has many advantages, some of its challenges and disadvantages should also be considered, such as reduced human interaction, the need to maintain privacy, and the possibility of creating a feeling of worthlessness in some people. Therefore, the optimal use of these technologies requires careful planning, attention to individual needs, and continuous evaluation of their uses and effectiveness. Since robots and artificial intelligence technologies were rapidly developed, they are planned as powerful tools

for use in educational environments. The use of these technologies can help improve the quality of the educational process and learning of children and students.

One of the advantages of using robots and artificial intelligence in the education of children and students is the increase in social interactions and cooperation between people. Robots and artificial intelligence systems can be used as substitutes or complements for teachers and enhance effective communication and collaboration skills between students in group learning environments. In addition, the use of robots and artificial intelligence can lead to increased motivation and positive change in the learning process.

Robotics and artificial intelligence (AI) have rapidly develop as two key factors in educational environments, leading to exciting and progressive changes in educational methods and processes. The strong and vital connection between these technologies and the teaching and learning of children and students is an indicator of the enormous potential provided by these technologies. One of the key points of this relationship is the ability to provide customized and adaptable teaching methods specific to each student based on their individual needs and learning style. By leveraging the power of data analysis and advanced cognitive algorithms, AI systems can recognize each individual's learning patterns and provide appropriate teaching methods and teaching strategies.

4- Discussion

The present study aimed to investigate the improvement of teaching-learning experience for students in higher education through systematic review. Considering the abundant research in the field of AI-based teaching-learning in higher education, the results of the studies show that the issue is faced by unique challenges and opportunities. In this context, artificial intelligence-based tools play an important role. Among the opportunities is improving the higher education process by providing intelligent solutions to increase the productivity and quality of student learning. These tools can also help students discover more effective learning methods, and research by Chiu et al (Biswas, 2023) and (Cotton et al, 2023) and (Farrokhnia et al (2023)) has shown that they can improve their skills in accordance with the findings of 2023. One of the main challenges is to establish ethical standards for the use of this technology. A careful examination of these issues can help us ensure that new technologies help improve and develop the quality of education for all, rather than reinforcing inequalities. In addition, it is necessary to determine ethical principles and standards to ensure that student data is used sensibly and safely. These criteria should be considered not only in the design of systems, but also in training and educating users to prevent possible abuses. Considering the rapid of technological advancement, establishing and adhering to these rules can help maintain credibility and trust in smart educational systems. Now according high mentioned we answer the previous questions:

To answer the first question, it should be said

Artificial intelligence as a new tool has significant potential to transform the educational system and improve the quality of student learning. To make maximum use of the opportunities of artificial intelligence in education, it is necessary to address this issue comprehensively and, taking into account the challenges and considerations, to carry out comprehensive plans for the implementation of this technology in educational systems. It is also necessary to pay attention

to the main role of teachers and human interactions in the learning process in the use of artificial intelligence in education.

Answer to the second question

According to the review of selected articles, the results of research show that the use of learning technologies significantly facilitates the process of individualization of education. In fact, the use of artificial intelligence improves the digital literacy of teachers and students and also facilitates the learning process in different subjects. In analyzing and explaining the above data, it can be said that artificial intelligence acts as a facilitator during the learning process, because it provides new ways for higher education and helps teachers to improve their teaching methods using artificial intelligence-based tools. These tools include personalized support for learners, integrated communication, and even learning process analysis (2023, al et Kohnke) through the discovery of interaction and communication perception. Therefore, artificial intelligence provide the facility to professors to use these tools to create continuous learning and facilitate the connection between concepts through searching and solving inference and logical reasoning problems (2020 al et Yang). In other words, the use of artificial intelligence in higher education can lead to the improvement and optimization of the teaching and learning process.

Answer to the third question

In terms of the existential necessity of these tools, they were analyzed and explained in the above sections, but in terms of alignment and inconsistency with research studies, it can be said that in line with the research of others, it can be said that artificial intelligence-based educational tools are suitable for writing assignments, analyzing project reports, and providing intelligent solutions to various problems, and have the potential to improve students' academic performance. On the other hand, using them as educational assistants with an emphasis on selective capabilities and interactive tools is promising, and faculty members should be able to use these tools as educational, research, and service tools.

Answer to the fourth question

Although this research is correspondence with Lucking 2017 and Kamaloy et al findings, it is not found to be true for the inconsistency of the findings and (Eltabakh and Ahmed Ismail 2019) (2023+) that it can also cause many threats; for example, students engage in less group discussion and conversation in completing assignments and use ChatGPT more (Kohnke and Cotton 2023), it may lead to a decrease in cognitive skills and a high dependence on ChatGPT. Also, excessive use of ChatGPT by learners may lead to a decrease in self-learning and inference abilities of the individual, such as understanding complex problems and optimal solutions. In addition, this may reduce the motivation of learners to conduct independent research and reach different results. (Farrokhnia 2023).

6. Conclusion

The use of robotics and artificial intelligence in the education and learning of children and students, according to research and studies, provides many possibilities for improving the educational process and increasing the productivity and quality of education for students. By creating attractive and informative interactive environments, these technologies can help children learn science and technology and improve their educational experience. One of the

important results of research in this field has shown that the use of robotics and artificial intelligence can positively affect children and encourage them to think scientifically and participate in the learning process. These tools help to students to learn complex concepts in an attractive atmosphere to improve their abilities.

Also, research has shown that the use of artificial intelligence in designing training and curriculum planning can better adapt to the individual needs of students and allow them to provide learning with personal development. In general, as a new approach in the field of teaching and learning, the use of robotics and artificial intelligence can help to open up the path to educating individuals and improve the level of students. These technologies can help students improve their skills and knowledge, and future research can also make this idea more sociable and practical. In short, by integrating robotics and artificial intelligence into the teaching and learning process of children and students, significant improvements can be made in the academic performance, critical thinking, and social and communication skills of this target group. This approach not only clearly indicates the future of knowledge acquisition, but can also widely help develop individuals as intelligent and responsible citizens.

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OPTIMIZING AND EFFECTIVELY UTILIZING VIRTUAL SPACE AND ITS IMPACT ON THE RELIGIOUS BELIEFS OF SOCIAL MEDIA USERS

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Abstract

The virtual space, especially social media, represents a new generation of social relationships. Despite its relatively short history, it has successfully embedded itself into people's lives. Due to its unique characteristics, such as being timeless, placeless, having a virtual memory, and being interactive, it has surpassed its competitors and distinguished itself. This research aims to answer the question: How can the optimal and productive use of virtual space affect the religious beliefs of social media users? Additionally, what is the positive impact of virtual space on the faith and religious convictions of social media users? Based on the findings of this study, while the virtual space and social media have numerous negative aspects and potential harms, their positive effects on religious beliefs cannot be overlooked. Some of the positive outcomes include the strengthening of religious convictions, exposure to opposing ideas, showcasing Islamic moral values to the world, and helping to enhance unity and solidarity among Muslims globally. This study was conducted using a library-based approach and a descriptive- analytical method.

Keywords: Religious beliefs, virtual space, social media users

Introduction

The media plays a crucial role in transforming religious and national values, beliefs, and standards, with the potential to shape societal thoughts and behaviors through careful planning. This influence is particularly notable today, as Western teachings challenge traditional Islamic values. The rise of mass media has diminished face-to-face interactions and has significantly impacted individuals, especially the youth, through collective media. While the internet offers positive aspects, its negative effects, such as promoting corruption and anti-religious ideologies, are more pronounced. The global information society, driven by mass media and new technologies, has diminished the importance of location in shaping identity, exposing individuals to diverse belief systems and creating a non-localized nature of beliefs.

Communication technologies significantly influence identity and belief in the information society. Identity is essential for human societies and fulfills a psychological need. While virtual spaces can pose challenges, they also offer opportunities to strengthen religious beliefs. By effectively utilizing these platforms, religious convictions can be enhanced through Quranic teachings, hadiths, and guidance from religious leaders. This work aims to investigate how virtual space can be optimally used to impact the religious beliefs of social media users. The rise of media and internet access is shaping global identity and individual personality. The concept of "social networks," introduced by J. A. Barnes in 1954, refers to interconnected social entities, such as people and organizations, linked by meaningful relationships. While traditional social networks emphasized face-to-face interactions, modern ones focus on online communities and digital communication. These networks have evolved from basic web platforms to include features like chat and email, serving as gathering places for millions of

users worldwide to interact and share information across various boundaries. Ultimately, social networks aim to enhance social interactions in virtual environments. User profiles, featuring personal photographs, details, and interests, facilitate easier communication and connections through applications like email and chat. Religious identity is defined as an individual's awareness and connection to a specific religion, the positive values associated with this affiliation, and the unique emotions experienced towards the religion and its followers. Essentially, it encompasses self-definition based on religious connection, along with related values and emotions. In this research, to identify the dimensions of religious identity, Tajfel's Social Identity Theory is utilized. According to Tajfel's theory, three dimensions of social identity are selected to analyze the religious identity of users, which are introduced as follows:

- * Cognitive element: The individual's level of awareness and understanding of their connection and affiliation with a religion.
- * Value-based element: The positive value outcomes that the individual attributes to their connection and affiliation with a religion.
- * Emotional element: The specific emotions the individual holds toward the religion, and toward others who share a similar relationship with it, particularly in the context of Islam.

Literature review

There is no comprehensive study directly examining the optimal and effective use of virtual space and its impact on the religious beliefs of social network users. However, the sources used in this study each address various aspects of this phenomenon, based on the specific objectives of the authors. Some of the key studies in this field include:

Monika Andok (2025) explores how virtual spaces, particularly social media platforms, serve as modern arenas for religious expression and community building. The authors argue that these digital environments provide Muslims with accessible avenues to strengthen their faith, share religious knowledge, and participate in collective worship practices remotely. They emphasize that the interactive and timeless nature of social media allows believers to engage with religious content at any time and from any location, thereby reinforcing their spiritual convictions amid a rapidly changing world.

Talib Hussain and Dake Wang (2025) investigate the dual nature of social media's influence on religious beliefs, emphasizing its capacity to both pose challenges and offer opportunities for spiritual growth. The authors demonstrate that, beyond the negatives such as misinformation and superficial interactions, social media platforms enable believers to deepen their faith by accessing diverse religious content, participating in online prayer groups, and engaging in faith-based discussions. These activities contribute to a strengthened sense of religious identity and moral alignment with Islamic principles, particularly among younger generations.

Jalili Firouzi's Research (2013) Firouzi's master's thesis in sociology from Allameh Tabataba'i University, titled "Representation of Religious Identity in Virtual Space," examines some of the social impacts of being in virtual spaces and using the internet as a new communication technology. According to the results, the performance (whether strong or weak) of religious institutions in the real-world influences young people's inclination to use virtual space.

Rostami et al. (2016) In their article published in the Social Sciences Research Journal at Islamic Azad University, Shushtar branch, Rostami and colleagues explore the impact of social networks on the culture and religious beliefs (identity) of users. Their case study focused on users in Isfahan Province. The findings indicate that the virtual space of internet-based social networks has a meaningful relationship with users' religious identity and culture.

Sarmadi et al. (2019) In the Sixth National Conference on Management and Humanities at the University of Tehran, Sarmadi and colleagues presented a paper titled "The Impact of Virtual Space on Cultural and Social Identity." The results confirmed that virtual space affects cultural and social identity and emphasized the reciprocal relationship between cultural and social identity, highlighting the importance of values in protecting personal privacy and family structures.

Deh Jamal and Rafiei Rad (2020) In their article in the Strategic Knowledge Interdisciplinary Studies Journal, Deh Jamal and Rafiei Rad assess the impact of virtual space on the beliefs and religious identity of social network users. The results show that virtual social networks significantly affect communication processes. The research also found an inverse relationship between membership duration and usage of social networks, and a positive relationship between the perceived reality of network content and users' religious identity. As the duration of membership and use of platforms like Instagram and Telegram increases, the negative impact on users' religious identity grows. This study also investigates the optimal use of virtual space and its impact on religious beliefs.

Narimani, Mohammad Mehdi (2021) In his master's thesis at the University of Isfahan titled "The Role of Religious Beliefs in Strengthening Marital Relationships from the Perspective of the Holy Qur'an", Narimani explores ways to strengthen marital bonds using Quranic verses and Islamic traditions. The research takes a descriptive-analytical approach, examining religious beliefs in areas such as spouse selection, economic, emotional, and marital domains. While this thesis differs in methodology and scope, it does not address the impact of cyberspace and its challenges for family members.

Rahgoshai, Hassan (2016) In his master's thesis at Payame Noor University, Tehran, titled "Explaining Ethical Values in Virtual Social Networks Based on Sadrian Philosophy," Rahgoshai examines the nature of virtual space from two perspectives: engineering philosophy and human sciences philosophy. The research concludes that virtual space possesses an objective existence (as a technological container), a mental existence (as an ideal container), and a textual existence (as the content of the virtual space). The thesis, mainly philosophical, differs in scope and objectives from the current study.

Mir Manaf Shamushki, Seyed Taghi (2015) In his master's thesis at Mazandaran University titled "The Role of Religious Beliefs in the Family Structure Based on the Qur'an and Tradition," Shamushki examines the role of religious beliefs in giving meaning to life, performing family duties, and coping with family challenges. Although the study touches on religious beliefs, it does not address cyberspace and its challenges.

Musivand, Mahboubbeh, and Saki, Faezeh (2022) This paper, "Cyber-Space Pathology in the Field of Spouse Selection and Family," was presented at the first national conference on cyberspace. The research reveals that while virtual space has certain advantages for family and

spouse selection, it also brings negative effects, such as reducing face-to-face interactions between parents and children, promoting false idealizations of spouses, and spreading immoral content that contradicts Iranian-Islamic values.

Amini, Mohammad, and Mazaheri Kuhestani, Rasoul (2022) In their article, "The Necessity of Amending Cyber Advertising Laws to Protect the Rights of Children and Adolescents," Amini and Mazaheri address the vulnerabilities of children and adolescents in cyberspace. After examining the legal frameworks for cyber advertising, the authors identify significant gaps and suggest regulations to ban inappropriate advertising aimed at children, proposing amendments to the electronic commerce law.

Karami Qara Qeshlaqi, Azam, et al. (2021)

This article, "The Impact of Virtual Space on Family Rights," examines the positive and negative impacts of virtual space on family structures. While the article focuses more on legal issues, it does not align with the objectives of this thesis. Ganjur, Mehdi (2021) In his article "Realism of Virtual Space: A Philosophical Analysis of Virtual Reality with Emphasis on Transcendental Wisdom," Ganjur presents a metaphysical analysis of virtual space and its impact on human nature. The article focuses solely on the philosophy of virtual space and does not address other variables such as family empowerment or religious beliefs. Shahriari, Nahzat (2020) In the article "The Impact of Religious Beliefs on Families Based on the Qur'an and Traditions," Shahriari explores how belief in God, the prophets, and life after death influences family dynamics and how these beliefs help families cope with challenges. Although the article addresses religious beliefs in the family, its scope differs from the present thesis. Rafiei Rad, Javad (2020) In his article in Strategic Knowledge Interdisciplinary Studies, Rafiei Rad argues that virtual space facilitates interactions between users and provides different roles for their social relationships. The article indicates that virtual space can replace face-to-face interactions with family and peers, contributing to identity crises, especially among teenagers. The study focuses on the negative aspects of cyberspace on religious identity but does not consider the role of religious beliefs in family empowerment. Mesbahi Jamshid, Parastu (2020) In the article "Religious Education Issues in Virtual Space and How to Address Them," Mesbahi Jamshid investigates the challenges of religious education in virtual spaces. The study highlights two key elements; understanding God's omnipresence and the importance of ethical oversight. However, the study does not address family empowerment or religious beliefs related to cyberspace challenges. Mozafari, Yousef, and Kusha, Sohila (2020) In their article "Quranic Lifestyle and Virtual Space," the researchers examine the challenges of a Quranic lifestyle in virtual space. While the study suggests solutions like enhancing awareness and cultural replacement strategies, it does not explore family empowerment or address cyberspace challenges. Karami, Ensi (2019) In the article "Religious Education in Virtual Space: Opportunities and Threats," Karami examines the potential of virtual space to foster religious education but also

warns of its limitations. The study, while relevant, does not address family empowerment or religious beliefs.

Objectives and Benefits of the Research

In general, the perspectives of scholars and thinkers regarding the effects and consequences of social networks, in general, and their identity-related impacts, in particular, can be categorized into three main views:

1- Determinism: Proponents of this view see social networks and the internet

as rigid, external structures that shape individuals' minds. They believe that users are dominated by these platforms, which impose constraints on them. In this view, the social network imposes a form of determinism on its users.

2- Voluntarism: This view regards social networks and the internet merely as tools that empower individuals by providing them with an abundance of information, thus promoting pluralism. Here, users are seen as autonomous agents who use these networks to enhance their own knowledge and experiences.

3- Techno-realism: This intermediate perspective acknowledges the ontological aspects of internet technology but argues that the technology itself is not the final determinant of outcomes. Techno-realists believe that while the internet has certain capacities and potentials, it also allows users significant freedom and agency in how they engage with it.

The objective of this research is to examine the optimal and effective use of virtual space and its impact on the religious beliefs of social network users. This study aligns with the techno-realism perspective, aiming to explore the

relationship between users' religious beliefs and their interactions with social networks from this viewpoint.

Religious Beliefs and Their Role in Life

Belief involves accepting statements, embracing convictions, or having faith, and is a key concept in epistemology related to truth, falsity, and justification. Beliefs can be classified as true or false, and discussions about them focus on their justification. Beliefs significantly influence both individual and societal orientations, with religious beliefs particularly shaping cultural existence. Societies provide models in insight, behavior, and actions, which are integral to religious culture and contribute to a meaningful life for individuals. Religious beliefs shape an individual's worldview. They define a person's

The text discusses how religious beliefs, shaped by sacred texts and traditions, guide individuals and communities by providing direction and purpose. It highlights a study that focuses on two core beliefs that significantly empower individuals and families to combat social harms, especially in cyberspace.

a) Belief in God (Monotheism) and His Omnipresence

The phrase "If God does not exist, everything is permissible," attributed to Nietzsche and Dostoevsky, suggests that without belief in God, individuals may feel justified in any action. In monotheistic teachings, particularly in Islam, belief in God is crucial for guiding family behavior and reinforcing values. Families that believe in God view their actions as seeking divine approval and strive to fulfill their rights and duties, ensuring no one's rights are violated.

Such families are mindful of their conduct, including in cyberspace, to respect the rights of all members.

b) Belief in Life After Death

A family that believes in resurrection lives purposefully; aware they will be accountable for their actions on the Day of Judgment. For example, if a family member spends hours online, they recognize they will be questioned about how they used that time. The Prophet Muhammad (PBUH) emphasized that individuals will be asked about their life, youth, wealth, and love for his family on Judgment Day. Religious beliefs significantly influence individual and family behavior, shaping their attitudes and instilling values that promote ethical and moral actions.

Impact of Virtual Space on Religious Beliefs of Social Network Users

Virtual spaces, especially social networks, significantly influence the religious beliefs of users. These effects vary depending on several factors, such as the content shared, interactions with others, and users' personal experiences. The following outlines some of the key positive impacts:

Strengthening Religious Beliefs:

Virtual spaces allow individuals to explore their religious beliefs freely and without societal pressure. Users can ask questions, engage in religious discussions, and seek answers to doubts through platforms like Telegram, where they can communicate with religious experts. This openness contributes to deepening their faith. Additionally, easy access to religious information, texts, and discussions in social networks allows users to become more familiar with religious principles and make informed decisions.

Exposure to Different Perspectives:

One of the positive outcomes of virtual space is the exposure to diverse religious and ideological views. Social networks remove the limitations of time and place, allowing individuals to encounter different religions and schools of thought. This promotes tolerance and understanding among users, fostering a space for intellectual exchange.

Global Promotion of Islamic Ethical Values:

Virtual space has provided an opportunity for the ethical values of Islam to be presented on a global scale, overcoming the control of major anti-religious media. Despite other dimensions of social networks often receiving more attention, their potential as platforms for promoting religious and ethical values is significant.

Strengthening Unity Among Muslims:

Virtual space and social networks have facilitated closer connections and solidarity among Muslims worldwide. They offer platforms for dialogue, knowledge-sharing, and mutual understanding, helping to strengthen religious bonds and shared values among Muslims globally.

Conclusion

Transitioning to proactive strategies in social networks is crucial, requiring a thorough analysis of opportunities and threats, as well as an understanding of social classes. This approach can help develop effective policies that meet societal needs. Social networks are powerful tools for engaging the younger generation. Key factors for creating a successful platform include simplicity and ease of use, as young users prefer straightforward interfaces that provide easy access to features. To attract young users, it's essential to provide enjoyable content such as games, competitions, and videos. Retaining users requires offering useful services, like educational content and technical advice, that address their needs. Additionally, the internet and social networks serve as a supportive environment for individuals to explore and express their religious beliefs without pressure, fostering personal growth and critical thinking. For instance, platforms like Telegram allow younger users to ask questions and seek answers in a safe space, reflecting their cautious approach to trust in real-life interactions. The Internet and social networks positively enhance the religious unity among Muslims globally by facilitating social communication and information exchange. They promote religious dialogue and constructive discussions on various issues, fostering understanding and connection among Muslims. Financial and spiritual support for projects that encourage religiosity and solidarity is crucial for strengthening this unity. Given the dynamic nature of the Internet, it is essential to continually update these initiatives to meet evolving social and technological needs, thereby improving Muslim solidarity and rapprochement. In this research, we clearly showed that the Internet and social networks have the ability to have a significant effect on the religious beliefs of their users. Among the positive effects of these spaces, the following can be mentioned

- 1- The Internet can help to strengthen and strengthen the religious beliefs of users. These communications strengthening religious beliefs, continuous communication with religious content and religious thinkers in the space can provide more and more diverse religious information to people.
- 2- Acquaintance with opposing ideas These spaces provide an opportunity for people to get acquainted with opposing ideas and views. This familiarity can lead to deeper beliefs and better understanding of different points of view.
- 3- Visibility of religious moral values through the publication of content related to religious moral values on the Internet, these values are more widely visible in different societies and can help to strengthen these values in societies.
- 4- The religious rapprochement and solidarity of Muslims. These spaces can help to strengthen the sense of religious identity and communication between Muslims through the dissemination of content related to religion and religious life.

Applications and Recommendations

A. Shifting from defensive and restrictive policies to interactive, dynamic, and proactive strategies in the use of social networks is essential. To achieve this, a comprehensive analysis of both the opportunities and risks within this digital space, along with a deep understanding of the social structure and demographic groups in the country, can significantly aid in crafting effective policies that align with the needs and realities of society.

B. Virtual social networks have emerged as one of the most effective tools for engaging the younger generation and fostering communication with them. To design and establish a successful social platform, several key considerations can enhance the space's positive impact, particularly in promoting religious values.

1- Simplicity and User-Friendliness: Young users prioritize simple, intuitive user experiences. It is crucial to develop a clean, efficient interface that ensures easy access to a wide range of features and functionalities, making the platform more accessible and user-friendly.

2- Engagement through Enjoyment. To attract and retain younger audiences, it is essential to offer content and features that provide enjoyment and entertainment. This can include interactive elements such as games, competitions, videos, music, and other forms of recreational content that resonate with this demographic.

3- Conveying Practical Value: Retaining users and increasing engagement requires providing content and services that users find genuinely useful and beneficial. Offering educational resources, technical advice, or practical guidance that addresses users' concerns and needs can help create a sense of value and utility.

4- Facilitating Social Interaction: Social networks are inherently built around interaction. Incorporating features such as groups, chat forums, post-sharing, and content dissemination can strengthen connections among users, enhancing their overall social experience and increasing community engagement.

By carefully addressing these considerations and aligning the platform with the preferences and needs of the younger generation, it is possible to create a social network that is not only attractive and engaging but also conducive to fostering positive religious and cultural impacts.

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INTEGRATING TECHNOLOGY, AND INNOVATION IN THE DIGITAL AGE: FUTURE DIRECTIONS IN EDUCATION

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Abstract

The rapid advancement of digital technologies has significantly transformed the educational landscape, challenging traditional pedagogical models and redefining the roles of educators and learners. This paper explores the integration of technology and innovation in education within the context of the digital age, with a focus on emerging trends, challenges, and future directions. Drawing on interdisciplinary research, it examines how innovations such as artificial intelligence, virtual learning environments, adaptive learning systems, and global connectivity are reshaping teaching and learning processes. The discussion includes the implications of individualization, networked learning, and globalization on educational practices and policy development. Moreover, the study highlights the growing need for reflexive and future-oriented skills among both educators and students, emphasizing the importance of digital literacy, critical thinking, and creative problem-solving. By analyzing current transformations and anticipating future scenarios, the paper argues for a proactive, inclusive, and flexible approach to educational innovation. The findings aim to guide stakeholders in designing sustainable educational ecosystems that are responsive to the evolving needs of a digitally connected world.

Keywords: Education, Futurology, Networked learning, Technology, Digital Age,

Introduction

In the 1970s, there was widespread speculation about a forthcoming "leisure society," where reduced work would lead to boredom and dissatisfaction. As a result, education was expected to shift its focus from job-related skills to those needed for leisure and personal interests (Jenkins & Sherman, 1979). Today, that narrative has been replaced with one of delayed retirement. While predicting the future is always uncertain, ongoing social changes are shaping educational priorities and policies, and the choices made now will significantly influence future directions. The future remains open and undetermined, yet analyzing current changes can help identify patterns that may be critically examined and addressed. Historically, societal developments have often oscillated between contrasting trends—such as the classical versus romantic movements of the 18th and 19th centuries, or the shift between collectivism and individualism in later periods. Still, such shifts are rarely linear; one trend often overlaps with or influences another. For instance, since the 1980s, education has leaned toward utilitarianism and a knowledge economy model. However, there are signs of a resurgence of liberal, humanistic educational values (Jarvis & Preece, 2001, pp. 222–223). More significantly, recent decades have seen transformative changes driven by the convergence of three major social forces: individualization, networked connectivity, and globalization. Today, many people experience deep personal meaning through digital platforms, where they can create virtual identities (avatars), form online relationships, and even redefine the essence of human interaction (Ahlstrom, 2007). According to sociologist Manuel Castells, physical factors like bodily rhythms, local climate, or geography become irrelevant in this context, as they are nonessential to network participation. This has given rise to what he terms a culture of "real virtuality" (Castells, 1996/1999). These three phenomena—individualization, networks, and

globalization—are interdependent and constantly shape one another. Although their trajectory is uncertain, their influence on education is undeniable. British sociologist Anthony Giddens argues that the Industrial Revolution played a key role in fostering individualism by separating work from domestic life. This shift gave rise to institutions like schools and workplaces, leaving the home as a private sphere. Within this private space, personal emotional life could flourish. In today's late-modern society, the decline of traditional social norms has led to a sense of detachment from old structures, pushing individuals to construct their own identities rather than rely on outdated social categories. This process requires **reflexivity**—the ability to reflect on one's past, maintain self-awareness in the present, and envision future goals—in order to develop a cohesive and resilient self-identity (Giddens, 1991).

Individualization and education

Traditionally, education was seen to have a socializing function, inducting individuals into the roles expected of them by society (Durkheim 1956). But as society is no longer homogeneous, this suggests a different focus for education – that of the recognition and celebration of individualism as part of the strengthening of identity. There are many indications of the promotion of individualization as a modern aim of education. These include:

- an emphasis on self-expression in learner's writing and literature;
- the promotion of personal projects in education;
- an emphasis on the learner's individual psychology;
- an emphasis on the individual learner's strengths and styles;
- a concern for the development of self-esteem in learners;
- a promotion of learner reflection and reflexivity and the development of reflexive tools.

Not all of these emphases on individualism arise from responses to social fragmentation. The recognition and celebration of learner difference may also arise from equality and diversity agendas in a pluralist society.

The network society

The 'network' represents a new form of social organization that has emerged since the mid-1980s. There are many reasons for the growth of networks, among them a decline in state power, the spread of capitalism and the growth of digital communication systems. We talk commonly about computer, family, social, political, terrorist or media networks (Kelly 1998: 3–5). Such networks operate across distance, their structures are not obvious, and they have the power to make connections in any number of different ways, as hyperlinks do. One feature of networks is that they are inclusive in the way that information flows between members and exclusive in ensuring that information and knowledge stays within the network. It is not surprising to find that the metaphor of the network is used as a form of knowledge representation in a number of disciplines, replacing an older computer hardware model. In cognitive science, connectionist networks are widely used to represent and explain cognitive activity (Bruning et al. 2004: 60).

The network society is underpinned by the revolution in information communication technology. This began with the rise of the personal computer and continued through the development of the Internet, the World Wide Web and increasingly sophisticated software applications for communication and information exchange. The future is likely to include the experiences of multiple realities, sentient robots or computers, and the translation of vicarious into immediate experience.

Networks and education

Educational networks have always existed, connecting groups of scholars who communicate together in scholarly activity and publication across institutions and countries.

Such scholarly groups may share their research and expert knowledge without the necessity for physical contact.

Modern ideas of educational networks are shown in ‘communities of practice’ in which practitioners can actively create and share knowledge. In communities of practice, knowledge is always ‘situated’, not necessarily within a specific location, but always within a culture and a background. Knowledge is not pre-existing, but constructed from the practices of the group. The community confers identity on its members; to be excluded from the community is to be excluded from the knowledge the network constructs, and to possess no community identity. Power is dispersed or shared between the group members, rather than in a hierarchical or top-down fashion (Wenger 1998: 4).

There are many implications of seeing education as comprising communities of practice, not least for the traditional power of the teacher.

An unforeseen type of educational networking is shown by Internet sites such as ‘Rate my teacher’ which are constructed by school children and students to provide their own evaluations of schools and teaching. Such sites, which have sprung up in many countries, are almost impossible to control and show the effective ways in which networks can operate to subvert formal systems in the interests of the members. The value of networks lies in their power to (see figure 1).

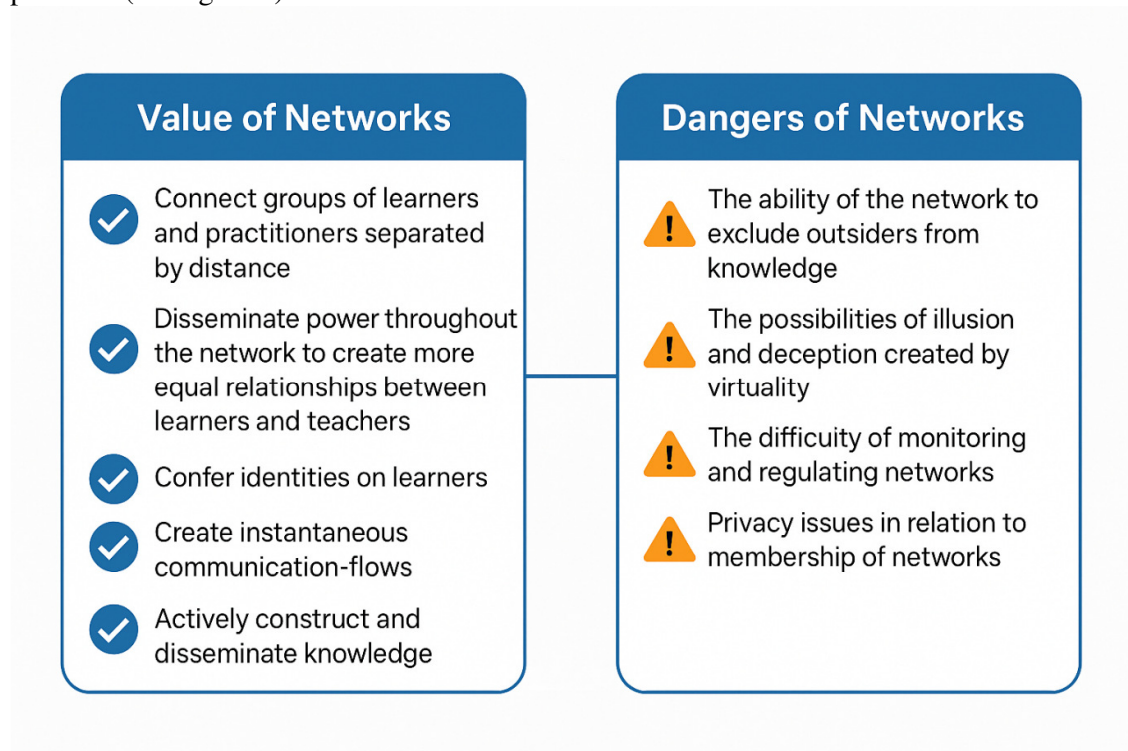


Figure 1. Value and the danger of the networks

Globalization and education

Globalization has profoundly reshaped the educational landscape, creating new opportunities and challenges for institutions, educators, and learners worldwide. It has facilitated the flow of knowledge, people, and technology across borders, leading to greater international collaboration, increased student mobility, and the global exchange of educational practices. As cultural boundaries blur, education plays a critical role in promoting cross-cultural understanding and preparing students to thrive in a globally interconnected society. However, globalization also intensifies disparities in access to quality education, particularly

between developed and developing regions. Institutions must now respond to diverse learner needs, adopt globally relevant curricula, and foster digital competencies to remain competitive. Thus, education in the era of globalization requires adaptive policies, inclusive practices, and a commitment to equity and global citizenship. The influx of market forces and the colonization of education by business practices are shown in many ways. Educational institutions are beginning to act more like corporations as they adopt the discourse of ‘customers’, ‘markets’ and ‘efficiency’. Conversely, corporations are themselves beginning to take over educational enterprises. For example, large multinational companies like Motorola are now creating their own universities which can be used to disseminate precisely the training and knowledge that the corporation requires (Jarvis and Preece 2001: 220–1). In 2008, the McDonald's fast-food chain gained British government approval to offer its employees awards equivalent to an A-level or advanced diploma level (BBC 2008). The characteristics of the corporate educational institution are that it:

Table 1. Key Features of the Market-Oriented Approach in Education

Aspect	Details
Market Identification	• <i>Internal markets</i>: academic departments, internal units • <i>External markets</i>: learners, parents, communities, government agencies
Knowledge Commodity	as Packages knowledge, programs, and services as marketable products
Resource Allocation	Distributes resources based on competitive performance
Effectiveness Measurement	Focuses on achieving predefined targets to evaluate success
Management Structure	Implements bureaucratic, top-down governance rather than collegial systems
Commercialization	Involves selling stakes of the educational enterprise to outside investors

The educational corporation:

- may operate virtually, on a global basis, with no physical location;
- combines work-based, professional training with education;
- cooperates with educational institutions on equal terms;
- devises its own curricula and programmes of study;
- may admit external learners as customers;
- awards certification and credits to its learners;
- blurs the boundaries between education and business.

In the future there is likely to be a further merging of corporations and educational institutions as the commodification of education continues and spreads to all sectors of education. The use of online and blended learning processes will facilitate this process.

The other aspect of globalization which has already had a marked effect on education is that of the discourse of lifelong learning. According to this discourse, workers will need to upskill on a continuous basis, as knowledge is transformed, in order that companies, regions, and states can maintain their competitive advantage. Lifelong learning is not simply concerned with the workplace and workplace training – it affects all educational providers and all types of provision (Kogan 2000: 341–59). However, the extent to which it acts as a catchphrase to justify economic imperatives is debateable.

Table 2. Key Dimensions of Globalization and Education

Dimension	Description
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Cross-Cultural Understanding	Promotes cultural sensitivity, empathy, and intercultural communication skills
Global Mobility	Enables students and educators to study, work, and collaborate across borders
International Collaboration	Encourages joint research, exchange programs, and transnational partnerships
Educational Inequality	Highlights gaps in access to resources and opportunities across regions
Curriculum Internationalization	Adapts content to reflect global perspectives and competencies
Digital Transformation	Drives the adoption of technology for global learning and access

Knowledge itself has become the main commodity for exchange – the World Trade Organization (WTO) has recently categorized education as a tradable commodity – which has far-reaching implications for learners, or ‘customers’ as they are increasingly known. If education is a commodity, it must be like other products and be accountable to the shareholders who own it. Education as a commodity therefore has had to standardize its products in the interests of the market and its major shareholders, the consumers and providers. In higher education, a proliferation of private providers has led in many countries to greater student choice, but also to the potential for lowered educational standards.

The commodification of education has affected all levels of the educational system. In school systems, market forces have led to consumer choice in relation to educational provision. For example, the British school system under the ‘New Right’ Conservative government in the 1980s responded to the demand for parental school choice and an open educational market. This led to the need for educational transparency in the form of school league tables and the assessment of school performance on the basis of Standard Attainment Testing (SAT) of students. This publication of performance indicators meant that schools wanted to attract only the clever or the academically motivated. The net result was, ‘It is not what the school can do for the child that counts, but what the child can do for the school’ (Gewirtz 2001: 176).

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RETHINKING FOREIGN LANGUAGE PEDAGOGY: INSIGHTS FROM THE POST-METHOD PERSPECTIVE

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Abstract

The evolution of language teaching has increasingly challenged the limitations of rigid, method-based approaches. This paper offers a critical examination of foreign language pedagogy through the lens of the post-method condition, emphasizing the need for pedagogical flexibility, teacher autonomy, and contextual sensitivity. Drawing on Kumaravadivelu's post-method framework, the study explores how educators can transcend prescriptive methodologies to adopt contextually grounded practices that respond to learner needs and local realities. The paper also investigates the implications of post-method thinking for curriculum design, teacher education, and classroom interaction. By highlighting the interplay between theory and practice, it advocates for a more reflective, adaptive, and student-centered approach to language teaching in a globalized educational environment.

Keywords: Post-method pedagogy, Language teaching, Teacher autonomy, Reflective practice, Contextualization

Introduction

Methodology in language teaching has been a challenging and changing concept in the 19th century. Although each method was fervently applied in the period in which it was devised all of them were replaced by another in succession. According to Pennycook (1989) it could be grounded on two reasons. Firstly, the academic concern to renew the methodology in foreign language teaching towards a better condition, and secondly, by some political reasons. Regarding the political reasons Lantolf (2000) claimed that several world events have impacted on language teaching methodology which three of them having to do with human conflicts; World War I, World War II, and the Cold War, marked particularly by the Soviets' launching of Sputnik. Of the three, however, WWII clearly had the most profound and lasting impact on language study in the United States. Richards and Rodgers (2001) added some other factors that have influenced language teaching in the past and can be expected to continue to do so in the future, as follow: (p. 253).

-Guru-led innovations; teaching is often shaped by the influence of powerful individual practitioners with their own schools of thought and followers. Just as Gattegno, Lozanov, and Krashen inspired a number of teachers in 1970s and 1980s.

-Responses to technology; the potential influences of the internet, World Wide Web, and other computer interfaces and technological innovations can affect language teaching in different ways.

-Influences from academic disciplines; Disciplines such as linguistics, psycholinguistics, and psychology have an impact on the theories of language and language learning and support particular approaches to language teaching.

-Research influences; second language acquisition research provided impetus for the development of Natural Approach and Task Based Language Teaching, and it will doubtless continue to motivate new language teaching approaches.

-Learner based innovations; learner based focuses recur in language teaching and other fields in approximately 10 year cycles, as we have seen with individualized instruction, the learner centered curriculum, learner training, learner strategies, and multiple intelligences. We can anticipate continuation of this trend.

-Crossover educational trends; Cooperative Learning, the Whole Language Approach, Neurolinguistic Programming, and Multiple Intelligences represent crossovers into second language teaching of movements from general education and elsewhere.

-Crossover from other disciplines; other disciplines like cognitive psychology, psychotherapy, communication science, ethnography, and human engineering have had influences on language teaching trends in the past and they are expected to do so in the future (p. 254). Lambert (2001) in his article identified major structural problems facing the field of foreign language instruction and believed that these issues are part of a national agenda for change.

History of foreign language teaching

Having a brief review of language teaching history it can be described that the Direct Method was proposed as an improvement over GTM in the early part of the 20th century. According to Richards and Rodgers (2001) the reason was that GTM does not conform to the modern objectives of second language teaching in the sense that it does not aim at increasing the communicative competence of the learners. It solely focuses on the knowledge about the language instead of the language itself (p. 7).

According to Cook (2003) at the turn of the nineteenth and twentieth centuries language learning population were changing. There was continuing mass immigration, by speakers of many different languages to USA, and despite the interruption of two world wars, there was growing international trade and tourism and an increase in both vocational and recreational language learning. In the language schools which created for these emigrants, the students did not necessarily share the same first language. Language teaching experts responded to this challenge with radical new idea of direct method (p. 33). Direct Method did not take well in public education, where the constraints of budget, classroom size, time, and teacher background made such a method difficult to use. Moreover it was criticized for its weak theoretical foundations. Its success may have been more a factor of the skill and personality of the teacher than of the methodology itself (Brown, 2001, p. 22). In 1950 ALM was thought to incorporate the latest insights from linguistics and psychology. However, according to Richards and Rodgers (2001) in the 1960 the theoretical foundations of ALM were attacked as being unsound in terms of both language theory and learning theory. On the other hand the practitioners found that the practical results fell short of expectations. Students were often found to be unable to transfer skills acquired through ALM to real communication outside the

classroom (p.65). As ALM began to fade in 1970, a variety of guru-led methods emerged to fill the vacuum such as Silent Way, TPR, and Suggestopedia. While these had declined by the 1990 new breakthroughs continued to emerge from time to time such as TBLT, Neurolinguistic Programming, and Multiple Intelligences, and they attract varying levels of support (Richards and Rodgers, 2001, p. 244). Then mainstream language teaching on both sides of the Atlantic was communicative language teaching, which was the recommended basis for language teaching in 1980.

Celce Murcia (1991) believed that CLT was rejecting an intense grammar teaching, reducing it to a minimum level and its focal point was around subject matter, tasks, projects, semantic notions, and pragmatic functions. Richards (2005) believed that we may group trends in language teaching in the last 50 years into three phases (p. 6):

Phase 1: Traditional approaches (up to the late 1960s)

Phase 2: Classic communicative language teaching (1970s to 1990s)

Phase 3: Current communicative language teaching (late 1990s to the present)

In the 1970s, the centrality of grammar in language teaching and learning was questioned and attention shifted to the knowledge and skills needed to use grammar and other aspects of language appropriately for different communicative purposes such as making requests, giving advice, making suggestions, describing wishes and needs and so on. What was needed in order to use language communicatively was communicative competence (Richards, 2005, p.8).

Since the 1990s the communicative approach has been widely implemented. It draws on a number of diverse sources. There is no single or agreed upon set of practices that characterize current communicative language teaching. Rather, communicative language teaching today refers to a set of generally agreed upon principles that can be applied in different ways, depending on the teaching context, the age of the learners, their level, and their learning goals and so on (p.20).

Richards (2005) then examined some current methodologies that can be described as extensions of the CLT but which take different routes to achieve the goals of communicative language teaching. Process-based CLT approaches, such as content based instruction and task-based instruction share as a common starting point a focus on creating classroom processes that are believed to best facilitate language learning (p. 24). Product-based CLT approaches, such as text-based instruction and competency-based instruction focus more on the outcomes or products of learning as the starting point in course design than on classroom processes. They start by identifying the kinds of uses of language the learner is expected to be able to master at the end of a given period of instruction. Teaching strategies are then selected to help achieve these goals (p. 32). Richards and Rodgers (2001) claimed that the basic assumption of TBLT that it provides for a more effective basis for teaching than other language teaching approaches remains in the domain of ideology rather than fact (p. 241). Ellis (2003) also provided some theoretical objections to TBLT (p. 328).

What is the best method?

Prabhu (1990) in his article whose title was “There is no best method, Why?” claimed that searching for a best method which can replace all other methods is not only an unrealizable goal, but also, a misconstruing the nature of teaching as a set of procedures which can by themselves carry a guarantee of learning outcomes. He believed that to say that best method

varies from one context to another does not help because it still leaves us with a search for a best method for any specific context. To say that there is some truth to every method does not help either because it still does not tell us which part of which method is true. Prabhu (1990) concluded that perhaps the best method varies from one teacher to another but only in the sense that each teacher should operate with his/her own sense of plausibility. There may be some truth to each method, but only insofar as each method may operate as one or another teacher's sense of plausibility. It means that the method can promote the most learning that can be promoted by that teacher.

Limitations of the concept of “Method”

A review of research in Applied Linguistics and other related fields shows that method as a construct of marginality has four inter-related dimensions; scholastic, linguistic, cultural, and economic (Kumaravadivelu, 2003b). Kumaravadivelu (2003a) provided some limitations for the concept of ‘Method’; First and foremost, methods are based on idealized concepts geared toward idealized contexts. Since language learning and teaching needs, wants, and situations are unpredictably numerous, no idealized method can visualize all the variables in advance in order to provide situation-specific suggestions that practicing teachers sorely need to tackle the challenges they confront every day of their professional lives (p. 28). Yet another crucial shortcoming of the concept of method is that it is too inadequate and too limited to satisfactorily explain the complexity of language teaching operations around the world. Concerned primarily and narrowly with classroom instructional strategies, it ignores the fact that the success or failure of classroom instruction depends to a large extent on the unstated and unstable interaction of multiple factors such as teacher cognition, learner perception, societal needs, cultural contexts, political exigencies, economic imperatives, and institutional constraints, all of which are inextricably interwoven (p. 29).

Richards and Rodgers (2001) believed that because approaches and methods have played a central role in the development of our profession they can be usefully studied by teachers. However teachers should try to use them flexibly and creatively based on their own judgment and experience (p. 250).

Emergence of postmethod

As long as we remain in the web of method, we will continue to get entangled in an unending search for an unavailable solution; that such a search drives us to continually recycle and repackage the same old ideas; and that nothing short of breaking the cycle can salvage the situation (Kumaravadivelu, 1994, cited in Kumaravadivelu, 2003a, p. 32). Out of this awareness has emerged “Postmethod Condition”. Kumaravadivelu (2003a) provided three interrelated attributes of Post Method as follows (p. 34):

The parameter of particularity

The parameter of particularity requires that any language pedagogy, to be relevant, must be sensitive to a particular group of teachers teaching a particular group of learners pursuing a particular set of goals within a particular institutional context embedded in a particular sociocultural milieu. The parameter of particularity then is opposed to the notion that there can be an established method with a generic set of theoretical principles and a generic set of classroom practices.

The parameter of practicality

The parameter of practicality relates to a much larger issue that directly impacts on the practice of classroom teaching, namely, the relationship between theory and practice. The parameter of practicality entails a teacher-generated theory of practice. It recognizes that no theory of practice can be fully useful and usable unless it is generated through practice. A logical corollary is that it is the practicing teacher who, given adequate tools for exploration, is best suited to produce such a practical theory. The intellectual exercise of attempting to derive a theory of practice enables teachers to understand and identify problems, analyze and assess information, consider and evaluate alternatives, and then choose the best available alternative that is then subjected to further critical appraisal. In this sense, a theory of practice involves continual reflection and action. If teachers' reflection and action are seen as constituting one side of the practicality coin, their insights and intuition can be seen as constituting the other (p. 35).

Farrel (2007) believed that reflective teaching benefits teachers in the following ways:

- It frees the teacher from routine and impulsive action (p. 7).
- It helps teachers become more confident in their actions and decisions.
- It provides information to teachers to make informed decisions.
- It helps teachers to critically reflect on all aspects of their work.
- It helps teachers to develop strategies for intervention and change.
- It recognizes teachers are professionals.
- It is a cathartic experience for practicing (and novice) teachers.

The parameter of possibility

The parameter of possibility is derived mainly from the works of critical pedagogists of Freirean persuasion. Critical pedagogists take the position that any pedagogy is implicated in relations of power and dominance, and is implemented to create and sustain social inequalities. They call for recognition of learners' and teachers' subject-positions, that is, their class, race, gender, and ethnicity, and for sensitivity toward their impact on education.

In the process of sensitizing itself to the prevailing sociopolitical reality, the parameter of possibility is also concerned with individual identity. More than any other educational enterprise, language education provides its participants with challenges and opportunities for a continual quest for subjectivity and self-identity (Kumaravadivelu, 2003a, p. 36).

Macro-strategies of postmethod

Macro-strategies are general plans derived from currently available theoretical, empirical, and pedagogical knowledge related to L2 learning and teaching (Kumaravadivelu, 2006, p. 201). A macro-strategy is a broad guideline based on which teachers can generate their own, location-specific, need-based micro-strategies or classroom procedures. In other words, macro-strategies are made operational in the classroom through micro-strategies. Macro-strategies are considered theory-neutral, because they are not confined to underlying assumptions of any one specific theory of language, learning, and teaching (p. 201).

The strategic framework comprises 10 macro-strategies that are presented in operational terms. The macro-strategies are:

1. Maximize learning opportunities;
2. Facilitate negotiated interaction;
3. Minimize perceptual mismatches;
4. Activate intuitive heuristics;

5. Foster language awareness;
6. Contextualize linguistic input;
7. Integrate language skills;
8. Promote learner autonomy;
9. Ensure social relevance; and
10. Raise cultural consciousness.

The role of postmethod teacher educator

Post-method teacher education practices have found aspirations in popular applications of foreign language teaching situated in various classrooms rather than centering on academic discussions that are detached from actual classrooms (Arikan, 2006).

According to Kumaravadivelu (2006) in practical terms, the role of the postmethod teacher educator becomes one of (p. 183):

- Recognizing and helping student teachers recognize the inequalities built into the current teacher education programs, which treat teacher educators as producers of knowledge, and practicing teachers as consumers of knowledge;
- enabling prospective teachers to articulate their thoughts and experience, and share with other student teachers in class their evolving personal beliefs, assumptions, and knowledge about language learning and teaching at the beginning, during, and at the end of their teacher education program;
- encouraging prospective teachers to think critically so that they may relate their personal knowledge with the professional knowledge they are being exposed to, monitor how each shapes and is shaped by the other, assess how the generic professional knowledge could be used to derive their own personal theory of practice; creating conditions for student teachers to acquire basic, classroom discourse analytical skills that will help them understand the nature of classroom input and interaction;
- Rechannelling part of their own research agenda to do “empowering research,” that is, research *with* rather than *on* their student teachers; and
- Exposing student teachers to a pedagogy of possibility by helping them critically engage authors who have raised our consciousness about power and politics, ideas and ideologies that inform L2 education.

These are, no doubt, challenging tasks. Unfortunately, most of the current teacher education programs are unable to meet these challenges. The programs require a fundamental restructuring that transforms an information-oriented teacher education into an inquiry-oriented one.

Criticisms of postmethod

According to Thornbury (2009) in spite of the claims of the postmethodists, the notion of *method* does not seem to have gone away completely. In fact, it seems to be persistent, even if the term itself is often replaced by its synonyms.

Kumaravadivelu (1995) thanked a scholar named Liu for a critical reading of his article and suggesting the following criticisms toward it. (a) We should not stop searching for better and newer methods and (b) This strategic framework is not an alternative to method.

Answering Liu Kumaravadivelu (1995) stated that what he has presented is no more than a basic framework-one possible alternative to method. There can be others. May be, more

exploration is needed. What is crucial is that teacher educators establish and maintain "a good productive relationship with classroom teachers. Teachers already know a lot about classroom language learning, but have typically seen their knowledge marginalized by researchers who, if they collaborate with teachers at all, tend to want to do so on their own terms, pursuing their own research agendas, and following their own procedures, however irrelevant the former may appear to the classroom teacher, and however disruptive or time-consuming the latter" (Allwright, 1994, cited in Kumaravadivelu, 1995). In fact we need to restructure teacher education so that teachers gain adequate knowledge, skill, and autonomy to continue to reflect, recreate, and reinvent.

Tosun (2009) stated that "Discrete from the earlier forms of methodological pursuits, postmethod condition is related to a more substantial phenomenon; globalization". He provided some criticisms to postmethod as follow:

First, creating a ground for a new course does and should not mean to assault to the accumulations of the past. If the claims pertaining to the forefathers of postmethod condition are accepted declaring that methods are all useless, how is it possible to account for millions of people who were taught foreign languages through these methods? To the contrary, it would be a more becoming idea to assess each historical occurrence in terms of the conditions of their period. As is in the past, these new challenges may be right on their sides with their claim of devising the ultimate form to teach languages.

Second, whereas methods are not paid acceptance by the majority of postmethod condition scholars, it would not be a wise approach to disregard them totally merely for providing a sound ground for the new bringing of the present occurrences. It is always quite possible that these current obsolete minor characters can be the major ones in the post – Postmethod condition again. Bell (2003) on the other hand believed that the current paradigm should not be understood as maturation but rather as a construction of the prevailing socioeconomic, cultural, and ideological forces. As those forces shift, so will methods. One element in those shifting forces is the way that method and post method can also be seen as inevitable and necessary dialectical forces: the one imposing methodological coherence, the other deconstructing the totalizing tendency of method from the perspective of local exigencies. In other words, method and postmethod together can liberate our practices.

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EMBRACING DIGITAL TRANSFORMATION: INTEGRATING TECHNOLOGY FOR INNOVATIVE PEDAGOGY IN HIGHER EDUCATION

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Abstract

In the digital age, the integration of technology in higher education has become not only a strategic necessity but also a catalyst for innovative pedagogy. This paper explores how embracing digital transformation can reshape teaching and learning practices, fostering a more dynamic, interactive, and student-centered educational environment. The rapid advancement of digital tools, including Learning Management Systems (LMS), artificial intelligence (AI)-driven learning platforms, and immersive technologies like virtual and augmented reality (VR/AR), has revolutionized the traditional educational landscape. By highlighting these technologies, higher education institutions can enhance student engagement, promote personalized learning experiences, and improve accessibility and inclusivity. However, the transition to digital pedagogy is not without challenges. Issues such as the digital divide, data privacy concerns, and the need for comprehensive training for educators must be addressed. Furthermore, balancing the use of technology with traditional pedagogical practices remains crucial to maintaining educational quality. This study examines successful case studies where digital transformation has positively impacted teaching outcomes and identifies strategic approaches for effectively integrating technology into higher education. Emphasis is placed on the importance of adaptive teaching strategies, faculty development, and continuous assessment to maximize the benefits of digital transformation. The findings suggest that thoughtful and well-planned integration of digital tools can significantly enhance pedagogical innovation and student success. As higher education continues to evolve in response to technological advancements, fostering a digital culture among educators and learners is essential. The study concludes with recommendations for educators and policymakers on implementing sustainable and inclusive digital transformation strategies in academic institutions.

Keywords: Digital transformation, higher education, innovative pedagogy, technology integration, student engagement.

Introduction

The integration of digital technology in higher education is no longer a supplementary enhancement but a foundational shift in how learning is conceptualized, delivered, and experienced. Digital transformation—defined as the strategic adoption of digital technologies to alter teaching and learning processes fundamentally—has the potential to make education more personalized, interactive, inclusive, and responsive to the needs of 21st-century learners. With the rapid evolution of tools such as Learning Management Systems (LMS), AI-based platforms, and immersive technologies like virtual and augmented reality (VR/AR), educational institutions are compelled to rethink conventional pedagogical models. This paper explores the transformative impact of these technologies, highlights case studies of successful implementation, and outlines strategic recommendations to guide sustainable digital integration in higher education. The onset of the digital era has profoundly reshaped the

foundations of modern society, altering how individuals communicate, access information, and interact with the world. Nowhere is this transformation more apparent—and more urgently needed—than in the realm of higher education. As universities and colleges grapple with shifting student expectations, rapidly evolving technology, and global disruptions such as the COVID-19 pandemic, the concept of *digital transformation* has emerged not as a luxury, but as a strategic imperative. Digital transformation in higher education goes far beyond merely digitizing content or offering online courses. It involves a holistic reimagining of institutional structures, pedagogical models, and the student learning experience through the thoughtful integration of digital technologies. At its core, digital transformation represents a paradigm shift from traditional, instructor-centered education models toward more flexible, student-centered approaches. It encapsulates the strategic adoption of tools such as Learning Management Systems (LMS), artificial intelligence (AI)-driven tutoring systems, virtual and augmented reality (VR/AR) environments, adaptive learning platforms, and data analytics to enhance teaching, learning, and administration. These technologies do not merely support learning; they actively shape it, enabling real-time feedback, personalization, interactivity, and engagement in ways that were previously inconceivable within the confines of brick-and-mortar classrooms. This shift is driven by the growing demand for accessible, inclusive, and flexible education that aligns with the needs of a digitally native generation of learners. Today's students expect learning to be mobile, on-demand, collaborative, and relevant to the skills required in a dynamic global workforce. Institutions that embrace digital transformation can provide learning environments that cater to diverse learning styles, facilitate active and participatory learning, and support lifelong learning pathways that extend beyond traditional academic credentials. Furthermore, digital tools offer immense potential for democratizing education by reducing geographic and socio-economic barriers to access, allowing a broader range of learners to benefit from quality education. However, digital transformation is not without its complexities. It presents a set of significant challenges, including the persistent digital divide, disparities in digital literacy, concerns over student data privacy, and resistance to change among faculty and institutional leadership. The transition demands a cultural shift that embraces experimentation, innovation, and a rethinking of teaching roles and student-teacher dynamics. Faculty members must be supported through ongoing professional development, not only to master new technologies but also to adopt new mindsets and pedagogical approaches. Institutions must also ensure that technological innovations are inclusive, ethical, and aligned with educational values and learning outcomes.

Crucially, the successful implementation of digital transformation depends on an integrated institutional strategy—one that aligns technology adoption with curriculum development, faculty engagement, assessment practices, and student support services. It is not enough to implement digital tools in isolation. Instead, higher education must develop an ecosystem where pedagogy, policy, and technology interact in ways that reinforce each other and center on improving learning effectiveness and equity. In this context, this paper aims to explore how digital transformation is reshaping the landscape of higher education. It investigates how digital technologies can be leveraged to promote pedagogical innovation, improve student engagement and outcomes, and address the broader institutional goals of access, inclusion, and academic excellence. By analyzing current trends, challenges, and real-world case studies, this study seeks to provide a comprehensive understanding of the implications of digital transformation for educators, learners, and policymakers. In doing so, it offers strategic recommendations for fostering a sustainable, inclusive, and forward-thinking digital culture in academic institutions worldwide.

Literature Review

The existing literature highlights both the potential and complexity of digital transformation in academic contexts. Scholars like Bates (2019) emphasize the role of online and blended learning in extending access to marginalized populations. Others, such as Selwyn (2020), caution against uncritical adoption of ed-tech, urging for more context-aware and pedagogically informed uses of technology. LMS and Adaptive Learning Platforms: LMSs such as Moodle, Blackboard, and Canvas have become foundational tools in course delivery. Research by Alammary et al. (2021) illustrates how LMSs enable flexible content delivery, formative assessment, and instructor-student interaction. Moreover, AI-driven platforms like CenturyTech and Knewton use learning analytics to adapt content to individual student needs, facilitating *personalized learning pathways*. Immersive Technologies: VR and AR are increasingly utilized to create experiential learning environments. For example, nursing and engineering students can engage in virtual simulations that enhance procedural learning and critical thinking (Radianti et al., 2020). Faculty Development and Institutional Strategy: Studies underscore the importance of continuous professional development for faculty to effectively leverage digital tools (Kirkwood & Price, 2014). Without proper training and support, digital tools risk becoming underutilized or misapplied.

Methodological Approach

Given the conceptual nature of this paper, a qualitative review and synthesis methodology is adopted. Data were drawn from peer-reviewed academic sources, institutional reports, and case studies published between 2018 and 2024. The selected literature was analyzed thematically to identify key trends, benefits, and challenges related to digital transformation in higher education. Additionally, institutional case studies were selected for their diversity in geography, discipline, and scope of technology use.

Discussion

Digital transformation enables a shift from teacher-centered to learner-centered education. Tools such as discussion boards, flipped classrooms, and gamification elements promote active learning. AI-powered chatbots can provide real-time feedback, guiding students through writing processes or problem-solving activities. Digital platforms can increase access for students in rural or underserved areas, those with disabilities, or working learners. Features like screen readers, closed captioning, and mobile-friendly formats improve the accessibility and flexibility of learning materials. Learning analytics allow instructors and administrators to monitor student progress and engagement patterns. These insights can inform curriculum adjustments, targeted interventions, and resource allocation, thus fostering evidence-based pedagogical practices. Despite its promise, digital transformation in higher education faces notable obstacles:

- **Digital Divide:** Inequities in access to devices and stable internet connectivity hinder equal participation.
- **Faculty Resistance and Skill Gaps:** Some educators struggle with technological adoption due to lack of training, confidence, or belief in its pedagogical value.
- **Data Privacy and Ethical Concerns:** AI and learning analytics raise questions about surveillance, consent, and data security.
- **Pedagogical Dilution:** Over-reliance on technology may lead to passive learning if not strategically integrated with sound pedagogy.

To ensure effective and equitable digital transformation, the following strategies are proposed:

1. **Institutional Vision and Policy Alignment:** Develop clear frameworks for digital integration aligned with institutional goals, academic integrity, and student equity.
2. **Ongoing Faculty Development:** Implement robust training programs focusing on both technical skills and digital pedagogy. Establish mentorship models and communities of practice for peer learning.
3. **Student Digital Literacy Programs:** Offer workshops and tutorials to help students navigate LMS tools, online research, and data ethics.
4. **Blended Learning Design:** Combine traditional and digital methodologies to balance flexibility with structure. Use technology as an enhancer, not a replacement, of pedagogy.
5. **Inclusive Technology Design:** Prioritize universal design for learning (UDL) principles when selecting or developing platforms to accommodate diverse learner needs.
6. **Ethical Use of Data:** Establish transparent data governance policies to protect student information and uphold trust.

Conclusion

Digital transformation holds transformative potential for higher education, enabling institutions to respond dynamically to evolving student needs, technological advancements, and global challenges. When thoughtfully implemented, digital tools can enhance engagement, foster inclusivity, and empower personalized learning experiences. However, the journey toward digital maturity requires more than infrastructure—it demands a cultural shift among educators, learners, and policymakers. By investing in faculty training, inclusive design, and strategic vision, institutions can leverage technology not as a disruption, but as a catalyst for enduring educational innovation.

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FEEDBACK TECHNIQUES AND TEACHER DISPOSITIONS: DRIVING WRITING IMPROVEMENT IN ONLINE EFL SETTINGS

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Abstract

This study explores the intersection of teacher personality traits and feedback techniques in enhancing writing outcomes within online EFL (English as a Foreign Language) contexts. Focusing on Iranian secondary students, the review synthesizes existing research to examine how introversion and extroversion in teachers influence learners' self-esteem and writing development. Particular attention is paid to two online corrective feedback (CF) types—recast and metalinguistic feedback (MCF)—and their alignment with teacher dispositions. Grounded in an integrative theoretical framework that includes trait theory of personality, social cognitive theory, self-esteem theory, and socio-meter theory, the study illustrates how feedback effectiveness is shaped by teacher-learner interactions, emotional readiness, and digital communication dynamics. The review reveals that teachers' dispositions—such as empathy, enthusiasm, and supportiveness—play a pivotal role in fostering student motivation, confidence, and writing competence. Methodological trends, research gaps, and future directions are discussed, offering a comprehensive foundation for advancing pedagogical practices in online EFL writing instruction.

Keywords: teacher personality, online feedback, writing development, EFL learners, self-esteem, recast, metalinguistic feedback

Introduction

The trait theory of personality serves as the foundation for understanding consistent individual differences in behavior, cognition, and emotion (McCrae & Costa, 1999). The Big Five personality dimensions—particularly conscientiousness, openness to experience, and neuroticism—have been associated with academic engagement, receptivity to feedback, and writing performance (Komarraju & Karau, 2005). In online education, where learners often self-direct their writing activities, these traits may significantly predict how students approach writing tasks, seek or avoid feedback, and regulate their learning behaviors. This study aims to synthesize existing research findings to foster an understanding of the effects of teachers' personality types, specifically the distinction between introversion and extroversion, on the self-esteem and writing skills of Iranian secondary students. It particularly explains the context of online recast and MCF practices, shedding light on their connection with teachers' personality characteristics. This literature review intends to clarify the current state of knowledge in the field, thereby identifying current trends, research gaps, and potential opportunities for further investigation. By engaging in a careful analysis of relevant studies, this paper draws useful information and contribute to the previous research, lending itself to a comprehensive comprehension of the complexities involved in the interaction between teachers' personality types, students' self-esteem, writing skills, and the technology environment of online recast and metalinguistic CF practices conducted. Following that, the theoretical or conceptual framework regarding teachers' personality types, self-esteem, writing skill development and the use of online recast and metalinguistic CF practices will be discussed. The subsequent sections will include a synthesis of up-to-date research findings, discussion of the role of current research and review of strengths and weaknesses of prior studies, and a critique of the literature in the light of methodological decisions. Finally, a summary of the chapter will be presented. This study draws on an integrative theoretical

framework that combines trait theory of personality, social cognitive theory, self-esteem theory, and socio-meter theory to examine how learners' personality traits and the feedback they receive influence writing development and self-perception in online educational contexts.

Social cognitive theory

Complementing this, social cognitive theory (Bandura, 1986) offers a dynamic lens through which the learning process can be understood as a reciprocal interaction between personal factors, behaviors, and environmental influences. In the context of writing, learners' self-efficacy beliefs, shaped through direct experience and feedback, play a central role in motivating effort and persistence. Online feedback—both automated and instructor-generated—acts as a critical mechanism for vicarious learning and self-regulation. The virtual nature of the environment also enhances the visibility of social modeling, further shaping learners' cognitive and emotional engagement with writing tasks. Self-esteem theory (Rosenberg, 1965) adds a layer of affective and motivational understanding by exploring how learners evaluate their academic abilities based on internal standards and feedback received. Writing, as a deeply personal and expressive skill, is often intertwined with learners' sense of self-worth. Feedback in online education, especially if not attuned to the learner's personality or emotional readiness, can either bolster or undermine self-esteem. This emotional response to feedback may, in turn, affect future writing performance and overall engagement with the course.

Socio-meter theory

Finally, socio-meter theory (Leary et al., 1995) reframes self-esteem as an evolutionary mechanism monitoring one's social acceptance and belonging. Applied to online learning, where social cues are often subtle or asynchronous, feedback can function as a proxy for social evaluation. Learners may interpret praise or criticism not merely as information about performance but as indicators of social inclusion or rejection. Thus, personality traits—such as sensitivity to evaluation or need for affiliation—may moderate how feedback is internalized and how self-perception evolves throughout the course. A comprehensive summary of the theoretical frameworks employed in the present study is provided in the following table.

Summary of the Theoretical Frameworks Used in the Study

Theoretical Framework	Description
Trait Theory of Personality	Examined the impact of EFL teachers' personality traits (introversion and extroversion) on students' self-esteem and writing skills
Social Cognitive Theory	Explored how teachers' personality traits and feedback practices influenced students' self-esteem and cognitive processes related to developing writing skill
Self-Esteem Theory	Addressed the impact of teachers' feedback practices on students' self-esteem and their perceptions of their own abilities in language learning
Second Language Acquisition	Focused on the effects of online recast and MCF on students' writing skills and overall language development
Socio-meter Theory	Examined how self-esteem levels fluctuated based on students' perception of acceptance and value by their teachers

Introverted and extroverted personality types depict different aspects of a distinct character and psychological tendencies. A preference for internal reflection characterizes introversion, focus on inner thoughts and feelings, and a tendency to do individual activities (Lawrence, 2015). Introverts often show traits such as thoughtfulness, empathy, and preference for individual interactions. Introverted teachers' preference for individual interactions allows them to develop

strong exclusive connection with their students, promoting a sense of personalized support. Extroversion, on the other hand, is defined by an inclination for external stimulation, involvement in the external world, and a willingness to have social interactions (Shehni & Khezrab, 2020).

Extroverts prioritize to be friendly, sociable, and favor group activities and social engagement. Extroverted teachers are likely to actively engage students in discussions, group work, and participatory activities, fostering social skills and collaborative learning. It is important to note that individuals can show personality traits of both introversion and extroversion falling somewhere along a spectrum instead of exactly belonging to one type, sometimes referred to as ambiversion (Abrams, 2017; Gordon, 2016).

Teachers' personality types influence the ways they approach their teaching practices, classroom management, and student interactions (Othman, 2009). Understanding the association between teachers' preferred modes of interaction, energy sources, and behavioral tendencies and student outcomes is a complex and multifaceted research topic in the field of education (Kim et al., 2018). Investigating how teachers' personality traits can determine student outcomes is important for the development of effective educational practices.

Teachers' personality traits can also affect the quality of the teacher-student relationship and the moral support presented to students. Extroverted teachers are generally characterized for being friendly and accessible and thus can make close relationships with students, fostering a trusting and supportive relationship (Jalili & Mall-Amiri, 2015). This promotes students' affective well-being and academic achievements. In the same way, introverted teachers, with their supportive and mindful disposition, may emphasize individual student needs and provide tailored support. This fosters a sense of understanding and care, which positively influences students' self-esteem and academic progress (Gammon & Morgan-Samuel, 2005).

Therefore, the relationship between teachers' personality and student outcomes embraces many different aspects. Teachers' personality traits influences various aspects of the teaching and learning process, including teacher-student interactions, teaching styles, motivational climate, and the teacher-student relationship which are of great importance for creating supportive and effective learning environments that develop student outcomes.

The impact of teachers on student self-esteem is a significant area of research in educational psychology. Within this context, the influence of introverted and extroverted teachers on student self-esteem has gained attention. Teachers play a crucial role in shaping students' self-esteem and Research suggests that teachers' personality types can have significant effects on student self-esteem (Alpay, 2000). Self-esteem refers to an individual's overall sense of worth and value (Sharma, 2014).

Teachers who have supportive personalities are likely to have a positive effect on student self-esteem. These teachers create a classroom environment that is more favorable where students feel esteemed, respected, and supported. They show great sensitivity, kindness, and understanding towards their students, which fosters a sense of emotional well-being and promotes the development of healthy self-esteem. Supportive teachers provide students with encouragement, praise, and constructive feedback to promote their strengths and achievements. They create opportunities for students to engage in successful learning experiences, challenging them properly and providing needed support. This supportive approach helps students develop a positive self-perception and belief in their capabilities and improves enhanced self-esteem (Kususanto et al., 2010).

In addition, teachers who exhibit confidence and enthusiasm in their teaching can positively affect student self-esteem (Mbuva, 2017). These teachers have a sense of reliance in their teaching abilities. They expect much for students and believe in their potential to stand out academically. Enthusiastic teachers encourage students to be more self-assured and

motivated (Cohn & Kottkamp, 1993). Their positive attitude and belief in students' skills can influence students' own impression of their competence and worth. These teachers often provide students with challenging yet accessible goals, which promote a feeling of success and enhances self-esteem through successful completion of tasks and assignments.

Moreover, teachers who engage and respect their students help students develop positive self-esteem (Virtanen et al., 2016). These teachers actively involve students in the learning process, respecting their input and encouraging their effective participation. In this way, they create an active learning environment where every student's voice is heard and valued. Engaging and respectful teachers promote a positive classroom society, which is collaborative and cooperative, and the responsibility is shared among the students. They appreciate differences and create opportunities for students to accept their individual characteristics. By fostering a sense of rapport and acceptance, these teachers help students develop a positive self-image and self-worth (Mostafaei Alaei & Forough Ameri, 2021).

Consequently, teachers need to be aware of their own personality traits and consistently make every effort to develop positive qualities that promote student self-esteem. Furthermore, improving teachers' rapport-building skills and emotional intelligence can contribute to building positive relationships with students and fostering an environment that supports healthy self-esteem development.

Teachers with high levels of conscientiousness and organizational skills often bring structure and clarity to language learning, including the development of writing skills. Their organized approach facilitates systematic instruction, goal setting, and feedback provision (Akpan, 2023). This can positively affect EFL learners' writing development. Clear guidelines, structured lessons, and explicit instruction on writing strategies promote learners' understanding and proficiency. Teachers who have extraverted personality traits are willing to display enthusiasm, energy, and an effective teaching style (Li et al., 2021). Their engaging and interactive method can effectively motivate EFL learners' interest in writing. These teachers often use creative activities, collaborative writing tasks, and chances for self-expression to higher student engagement and encourage a positive attitude towards writing.

Conclusion

This study underscores the intricate relationship between teacher personality traits and the effectiveness of online corrective feedback in shaping EFL learners' writing outcomes and self-esteem. Through a synthesis of theoretical perspectives and empirical evidence, it becomes clear that teacher dispositions—particularly along the introversion-extroversion spectrum—play a significant role in determining how feedback is delivered, interpreted, and internalized by students in virtual environments. Recast and metalinguistic feedback, when aligned with the emotional and cognitive needs of learners, can foster deeper engagement, sustained motivation, and greater self-confidence in writing tasks. Moreover, the literature suggests that students' affective responses to feedback are not merely influenced by its content, but also by the interpersonal dynamics fostered by teachers' communicative styles and emotional intelligence. Supportive, empathetic, and well-organized teachers, regardless of where they fall on the personality spectrum, can create psychologically safe spaces that promote growth-oriented mindsets and academic resilience. By tailoring feedback strategies to suit learners' emotional and developmental readiness, teachers can amplify the pedagogical value of online interactions. Ultimately, this review underscores the importance of a nuanced, learner-centered approach to feedback in online English as a Foreign Language (EFL) education. It calls for further empirical research into how teacher personality traits interact with various feedback modalities to influence learning outcomes, especially in digitally mediated environments. Developing professional training programs that raise teachers' awareness of their own dispositions and

equip them with adaptive feedback techniques may serve as a pivotal step toward enhancing writing instruction and student well-being in global EFL contexts.

Practical Implications for Teacher Trainers

The findings of this review present several actionable implications for teacher training programs, particularly in preparing educators for the challenges and opportunities of online English as a Foreign Language (EFL) instruction. Given the significant impact of teacher personality traits on student self-esteem and writing development, teacher trainers must move beyond one-size-fits-all approaches and instead foster reflective, adaptive, and personalized pedagogical practices.

1. Fostering Self-Awareness of Personality Traits

Teacher training programs should incorporate modules that help future educators assess and understand their own personality profiles—particularly their tendencies toward introversion or extroversion. Tools such as personality inventories (e.g., Big Five assessments or MBTI) can be integrated into the curriculum to promote self-reflection. Trainers should guide teacher candidates to explore how their disposition influences classroom communication, feedback styles, energy management, and student rapport.

2. Differentiating Feedback Approaches Based on Teacher Disposition

Trainers should explicitly teach how feedback techniques—such as recasts and metalinguistic corrective feedback—can be adapted to align with a teacher’s natural style. For instance:

- **Introverted teachers** may benefit from learning how to deliver impactful written feedback and use asynchronous tools (e.g., voice notes, annotated comments) that leverage their strength in thoughtful, individualized communication.
- **Extroverted teachers** may excel in synchronous, verbal feedback and whole-class writing discussions, but may also need support in cultivating patience for more reflective or reserved learners.

Teacher trainers can create workshops where trainees practice delivering feedback using both modalities and analyze student responses to understand how delivery style interacts with learner affect.

3. Training in Emotional Intelligence and Rapport-Building

Because student self-esteem is highly responsive to the perceived emotional tone of feedback, teacher education programs must prioritize the development of emotional intelligence (EI).

Practical training should include:

- Strategies for building empathy and emotional attunement in online interactions.
- Case studies showing how insensitive or overly critical feedback can harm student confidence.
- Role-playing scenarios that simulate giving constructive criticism in emotionally supportive ways.

Developing these skills can empower teachers to respond to diverse learner needs and create a psychologically safe virtual environment that encourages risk-taking in writing.

4. Encouraging Adaptive Use of Feedback Tools in Online Platforms

Teacher trainers should offer hands-on experience with online learning platforms (e.g., Google Classroom, Moodle, LMS tools) and highlight how digital environments can be used to personalize feedback. Training should include:

- Techniques for balancing synchronous and asynchronous feedback.
- Use of rubrics, inline comments, peer feedback structures, and formative assessment tools.
- Data-driven reflection: how teachers can analyze student writing trends and adapt feedback based on performance and personality indicators.

5. Promoting Reflective Practice and Peer Collaboration

A continuous cycle of reflection, peer discussion, and mentorship should be encouraged in teacher training. Teachers-in-training should be guided to:

- Reflect on how their personality and feedback habits evolve during teaching practicum.
- Engage in peer review of feedback strategies.
- Collaborate with mentors to co-develop differentiated feedback techniques that match learner diversity.

This reflective and collaborative approach reinforces the idea that teaching is not merely about transferring knowledge, but about cultivating learner confidence and engagement through personalized pedagogical interactions.

6. Integrating Theory and Research into Practice

Teacher trainers should emphasize the connection between theoretical frameworks (e.g., trait theory, self-esteem theory, social cognitive theory) and their practical applications. Trainees should:

- Analyze real classroom data through the lens of these theories.
- Conduct micro-research projects on their feedback styles and learner responses.
- Design classroom experiments to test different CF approaches in relation to personality alignment.

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A STUDY ON ARTIFICIAL INTELLIGENCE USED TO HIGHER EDUCATION SYSTEM

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Abstract

Artificial Intelligence (AI) is rapidly transforming higher education by enhancing teaching, learning, administration, and research processes. This paper explores the diverse applications of AI within universities and colleges, highlighting its role in personalized learning, automated assessment, data-driven decision-making, and student support services. It also examines the significant benefits AI offers, such as improved student engagement, increased institutional efficiency, and expanded access to education. However, the integration of AI presents critical challenges, including concerns about data privacy, algorithmic bias, academic integrity, and the potential devaluation of human interaction. The paper further discusses emerging trends such as generative AI tools, enhanced learning analytics, and the growing importance of AI literacy. Finally, it offers recommendations for responsible AI adoption, emphasizing ethical frameworks, faculty training, and equitable access. This study contributes to a deeper understanding of AI's transformative potential in higher education while advocating for thoughtful, inclusive implementation strategies.

Key words: Artificial Intelligence (AI), Higher Education, Personalized Learning, Intelligent Tutoring Systems, Learning Analytics.

INTRODUCTION

The rapid evolution of artificial intelligence (AI) has begun to redefine multiple domains of modern life, with higher education emerging as one of the sectors undergoing significant transformation. As AI technologies become more sophisticated and accessible, universities and colleges worldwide are exploring their potential to enhance teaching, streamline administrative functions, and personalize student learning experiences. These developments are part of a broader shift toward digital transformation in education, driven by the need to improve efficiency, scalability, and responsiveness to the diverse needs of students in the 21st century. AI applications in higher education are diverse, ranging from intelligent tutoring systems and adaptive learning platforms to automated assessment tools and virtual academic assistants. These innovations promise to augment traditional pedagogical models by enabling more personalized, data-driven, and flexible learning environments. At the same time, AI tools can support institutional decision-making by analyzing vast amounts of educational data to inform resource allocation, predict student outcomes, and identify areas for improvement. Despite these advantages, the integration of AI into higher education is not without challenges. Ethical concerns related to data privacy, algorithmic bias, and the transparency of AI systems must be addressed. Additionally, educators and administrators must navigate issues such as digital literacy, infrastructure readiness, and the potential for AI to disrupt established roles and responsibilities within academic institutions.

OBJECTIVES

1. To explore the current applications of Artificial Intelligence (AI) in higher education, including teaching, learning, administration, and research.
2. To analyze the benefits and opportunities that AI offers for improving student outcomes, institutional efficiency, and educational accessibility.
3. To identify the key challenges and ethical considerations related to the adoption of AI in academic settings, such as data privacy, bias, and academic integrity.
4. To provide practical recommendations for educators, administrators, and policymakers on how to implement AI responsibly and effectively in higher education.

LITERATURE REVIEW

Artificial Intelligence (AI) has increasingly gained attention as a transformative technology in higher education, promising to revolutionize teaching, learning, and institutional management. Research over the past decade has explored various AI applications and their impact on educational outcomes.

Luckin et al. (2016) argue that AI-driven tools facilitate **personalized learning**, allowing educational content to adapt dynamically to individual students' needs. This adaptability enhances engagement and helps address diverse learning styles, supporting improved academic performance. Similarly, Nye (2015) emphasizes the efficacy of **intelligent tutoring systems** (ITS), noting that ITS can provide tailored feedback and scaffolding that mimic one-on-one human tutoring, which has been shown to significantly boost student learning gains.

In addition to pedagogical applications, AI is increasingly used for **learning analytics** and institutional decision-making. Ferguson and Clow (2017) highlight that AI-powered analytics can uncover patterns in student behavior and performance, enabling early identification of at-risk students and informed interventions. Zawacki-Richter et al. (2019) conducted a systematic review of AI research in higher education and found growing adoption of AI for administrative automation, including admissions and scheduling, which enhances operational efficiency.

Despite these benefits, ethical challenges and limitations are well-documented. O'Neil (2016) warns about the risks of **algorithmic bias**, where AI systems trained on biased data can perpetuate inequalities. This concern is echoed by Benedict and Pence (2022), who discuss issues of **data privacy** and the need for transparent governance frameworks to safeguard student information. Furthermore, Selwyn (2019) critiques the over-reliance on AI, cautioning that education requires human empathy and judgment that machines cannot replicate.

Recent studies also examine the evolving role of **generative AI** tools in education, exploring both their potential for creative assistance and the risks related to academic integrity (Misra, Kumar, & Singh, 2021). These developments highlight the necessity for robust institutional policies and AI literacy among educators and students.

In summary, the literature demonstrates that AI offers substantial opportunities for enhancing higher education but requires careful ethical considerations, infrastructural support, and ongoing research to realize its full potential responsibly.

RESEARCH GAPS

While the literature is growing, it remains relatively fragmented and often focused on specific applications or case studies. There is a need for more longitudinal research that evaluates the long-term impact of AI on student outcomes, faculty roles, and institutional structures. Additionally, comparative studies across different cultural or regional contexts are limited, making it difficult to generalize findings globally.

METHODOLOGY

This paper adopts a **qualitative literature review** approach to synthesize current knowledge on the use of artificial intelligence in higher education. The review focuses on peer-reviewed articles, conference proceedings, and reputable reports published between 2015 and 2025.

AI APPLICATIONS IN HIGHER EDUCATION

Artificial Intelligence (AI) is increasingly integrated into higher education across various domains, transforming how institutions deliver education, support students, and manage operations. The adoption of AI spans four primary areas: teaching and learning, academic support services, administrative processes, and research enhancement.

AI IN TEACHING AND LEARNING •

One of the most impactful applications of AI in higher education is its role in enhancing the teaching and learning experience. AI-powered **intelligent tutoring systems (ITS)** and **adaptive learning platforms** can personalize instruction by analyzing student interactions and adjusting content in real time. Platforms like Carnegie Learning, Coursera's adaptive assessments, and Knewton are examples that support tailored learning pathways based on individual progress and preferences. AI also facilitates **automated grading and feedback**, particularly in large-scale courses and online learning environments. Tools using natural language processing (NLP) can assess short answers and essays, enabling quicker turnaround and more consistent evaluations. Additionally, **AI writing assistants** such as Grammarly and ChatGPT are being used by students to enhance their writing, prompting discussions about academic integrity and AI literacy. Moreover, **virtual reality (VR) and AI integration** is emerging to simulate complex environments for experiential learning, especially in fields such as medicine, engineering, and architecture.

AI IN ACADEMIC SUPPORT SERVICES •

AI is also used to strengthen student support services. **Chatbots and virtual assistants**, such as Georgia State University's "Pounce" or Arizona State University's "Sunny," help students navigate administrative processes like registration, financial aid, and advising. These tools are available 24/7, reducing bottlenecks in service delivery and increasing student satisfaction. Predictive analytics, powered by machine learning algorithms, are employed to **identify at-risk students** by analyzing academic performance, attendance patterns, and engagement

metrics. Institutions can then proactively offer interventions, such as tutoring or counseling, to improve retention and graduation rates. Additionally, AI supports **career services** by helping students create resumes, prepare for interviews, and receive job-matching recommendations based on their academic profiles and skill sets.

AI IN ADMINISTRATIVE AND INSTITUTIONAL OPERATIONS •

AI has been adopted to improve efficiency in back-end institutional processes. **Enrollment management systems** use AI to forecast student application trends, optimize recruitment strategies, and personalize outreach efforts. Similarly, **course scheduling algorithms** can allocate faculty and classroom resources more efficiently based on student demand and availability. In library science and academic publishing, AI tools assist in **resource recommendation**, **plagiarism detection**, and even **automated literature reviews**, enabling faster access to relevant academic content.

AI IN RESEARCH AND INNOVATION •

Higher education institutions are also leveraging AI to accelerate research activities. AI-based tools support **data analysis**, **natural language processing**, and **scientific discovery**, allowing researchers to process large datasets more efficiently. AI is being used in fields such as bioinformatics, social science research (for sentiment analysis and topic modeling), and digital humanities. Some universities have begun using **AI to detect research misconduct**, monitor reproducibility, and ensure ethical compliance. Moreover, AI is playing a role in **grant proposal evaluation** and **automated peer review**, although these uses are still in early stages and raise important ethical questions.

BENEFITS AND OPPORTUNITIES

The integration of Artificial Intelligence (AI) in higher education offers a wide array of benefits that span pedagogical, administrative, and operational domains. As institutions adopt AI-driven solutions, they unlock new opportunities to improve student outcomes, enhance efficiency, and expand access to education. Below are some of the key benefits and potential advantages AI brings to higher education.

PERSONALIZED LEARNING AND STUDENT ENGAGEMENT •

One of the most widely recognized benefits of AI in education is its capacity to deliver **personalized learning experiences**. AI-powered platforms can adapt content to meet individual learning styles, preferences, and progress levels, enabling students to learn at their own pace. This personalization helps address diverse learner needs, increase engagement, and potentially improve retention and performance. Intelligent tutoring systems (ITS) and adaptive learning environments foster **real-time feedback**, guiding students through complex concepts and identifying gaps in understanding. These systems support active learning and self-regulated study habits, essential for lifelong learning.

ENHANCED TEACHING EFFICIENCY AND SUPPORT •

AI can support instructors by automating routine tasks such as **grading, attendance tracking, and content delivery**, allowing educators to focus more on high-impact activities like mentoring, curriculum design, and pedagogical innovation. Tools like automated essay scoring and analytics dashboards help faculty monitor student progress and intervene early when necessary. Furthermore, AI-driven analytics can support **learning analytics dashboards** that provide instructors with actionable insights into student behavior and course performance, promoting data-informed teaching practices.

IMPROVED INSTITUTIONAL DECISION-MAKING •

AI enables higher education institutions to make **data-driven decisions** in areas such as enrollment management, budgeting, and resource allocation. Predictive models can forecast enrollment trends, optimize course offerings, and identify patterns in student performance, aiding in strategic planning. AI also improves institutional **efficiency and scalability** by automating back-end processes such as admissions review, scheduling, and IT support, reducing administrative burdens and operational costs.

EXPANDED ACCESS AND INCLUSION •

AI tools can help make education more **accessible** to students with disabilities through speech-to-text software, real-time translation, and content adaptation for diverse needs. For example, AI-driven transcription services can convert lectures into written text, aiding students with hearing impairments or language barriers. Moreover, AI-enabled platforms facilitate **remote and flexible learning**, expanding access to quality education for non-traditional learners, working students, or those in underserved regions. Chatbots and virtual assistants can provide 24/7 support, reducing the barriers of time and location.

RESEARCH AND INNOVATION ACCELERATION •

In academic research, AI assists in **data analysis, simulation modeling, and pattern recognition**, which are particularly valuable in data-intensive disciplines. AI tools can also automate tasks such as literature reviews, citation management, and peer review, thereby speeding up the research process and enhancing productivity. Furthermore, interdisciplinary research is flourishing as AI technologies intersect with fields such as education, psychology, linguistics, and computer science, fostering **innovation and collaboration** across domains.

CHALLENGES AND ETHICAL CONSIDERATIONS

While Artificial Intelligence (AI) holds considerable promise for transforming higher education, its implementation is not without significant challenges. These range from technological and infrastructural limitations to deeper ethical concerns surrounding privacy, fairness, and the human role in education. As institutions adopt AI-driven tools, they must navigate these issues carefully to ensure equitable, transparent, and responsible use.

DATA PRIVACY AND SECURITY •

AI systems in higher education often require access to vast amounts of student data—including academic performance, behavioral patterns, and personal information. This raises serious **privacy and data protection concerns**. Improper data handling, breaches, or surveillance-based analytics can infringe on students' rights and violate regulations such as the **General Data Protection Regulation (GDPR)** in Europe or **FERPA** in the United States. There is also the issue of **data ownership**: It is often unclear whether students, institutions, or third-party vendors own and control the data collected by AI tools. Institutions must implement strong data governance frameworks to ensure transparency and accountability in data usage.

ALGORITHMIC BIAS AND FAIRNESS •

AI systems can perpetuate or even exacerbate existing **biases and inequalities** if they are trained on skewed or non-representative data. For example, predictive models used for admissions or early warning systems may disproportionately flag students from marginalized backgrounds if historical data contains embedded bias. This risk of **algorithmic discrimination** challenges the fairness and inclusivity of AI in education. Without proper auditing, validation, and ethical oversight, AI tools may unintentionally reinforce stereotypes or create new forms of digital exclusion.

ACADEMIC INTEGRITY AND MISUSE •

The emergence of generative AI tools such as **ChatGPT** and **text-to-image generators** raises concerns about academic dishonesty. Students may misuse AI to generate essays, solve assignments, or impersonate others in online assessments, undermining the principles of academic integrity. These tools also challenge traditional assessment models, prompting educators to rethink how knowledge and skills are evaluated. Institutions must develop clear guidelines on AI use and invest in **AI literacy** to help students use these tools ethically and responsibly.

OVER-RELIANCE AND HUMAN DISPLACEMENT •

There is a growing concern that over-reliance on AI could lead to the **dehumanization of education**, where automated systems replace essential human interactions. Teaching, mentoring, and the development of critical thinking often require empathy, contextual understanding, and ethical judgment—traits AI currently lacks. Moreover, some fear that increasing automation may threaten **faculty roles**, especially in administrative or instructional support tasks. While AI should be seen as a complement rather than a substitute, institutions must manage this transition carefully to avoid workforce disruption.

TECHNOLOGICAL AND INFRASTRUCTURE BARRIERS •

The successful deployment of AI in higher education depends on adequate **technological infrastructure**, including reliable internet access, hardware, and software integration. In many low- and middle-income countries, digital divides persist, limiting equitable access to AI-powered tools. Faculty and staff may also lack the necessary **technical skills or training** to

effectively use AI systems. This calls for continuous professional development and institutional support to build AI readiness across all levels of the academic community.

RECOMMENDATIONS

DEVELOP CLEAR INSTITUTIONAL AI POLICIES

Universities should adopt **transparent and inclusive AI policies** covering data use, academic integrity, and ethical boundaries. These policies should involve faculty, students, administrators, and technical experts to ensure balanced perspectives.

INVEST IN FACULTY AND STAFF TRAINING •

Institutions must provide **professional development programs** that empower educators and administrators to understand and effectively use AI tools. This includes technical training, pedagogical strategies, and ethical considerations.

FOSTER INTERDISCIPLINARY COLLABORATION •

AI in education is inherently cross-disciplinary. Institutions should **promote collaboration between computer science, education, ethics, and social sciences** to develop balanced, context-aware applications.

ENSURE EQUITY AND ACCESS •

To prevent digital divides, universities must invest in **infrastructure and equitable access**, especially in under-resourced or remote areas. Open-source and low-bandwidth AI tools can help democratize access to cutting-edge technology.

ENCOURAGE RESPONSIBLE AI INNOVATION •

Research funding and institutional support should be directed toward **AI innovations that align with educational values**—focusing on inclusion, accessibility, and pedagogical enhancement. Ethics review processes should be built into AI project planning from the outset.

CONCLUSION

Artificial Intelligence is rapidly reshaping the landscape of higher education, offering transformative opportunities for personalized learning, improved institutional efficiency, and data-driven decision-making. From intelligent tutoring systems and automated grading to predictive analytics and administrative chatbots, AI is enhancing both the quality and accessibility of education.

However, the integration of AI is not without significant challenges. Issues related to data privacy, algorithmic bias, academic integrity, and the potential erosion of human interaction in education require careful attention. Moreover, disparities in infrastructure and AI literacy risk widening existing gaps within and between institutions.

Looking ahead, the future of AI in higher education will depend on how effectively institutions balance innovation with ethical responsibility. This includes developing clear policies, investing in capacity-building, and promoting inclusive access to AI technologies. As AI becomes a permanent fixture in the academic environment, its success will be measured not just by efficiency gains, but by its ability to support equitable, meaningful, and human-centered education.

In conclusion, AI should not be seen as a replacement for educators or institutional judgment, but rather as a powerful tool that, when thoughtfully applied, can enhance the educational mission. Ongoing research, inclusive dialogue, and responsible implementation will be critical in ensuring that AI serves the diverse and evolving needs of learners in the 21st century.

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A STUDY ON INTEGRATING ARTIFICIAL INTELLIGENCE IN HYBRID LEARNING: OPPORTUNITIES AND CHALLENGES IN HIGHER EDUCATION SYSTEMS

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ABSTRACT

The integration of Artificial Intelligence (AI) into hybrid learning environments is transforming the educational landscape, offering both significant opportunities and complex challenges. This study explores the role of AI technologies—such as intelligent tutoring systems, chatbots, predictive analytics, and adaptive learning platforms—in enhancing teaching and learning within hybrid models that combine online and face-to-face instruction. Through a review of current literature and case studies from higher education and K–12 contexts, the research identifies key benefits of AI integration, including personalized learning, improved student engagement, real-time feedback, and greater instructional efficiency. However, it also critically examines the challenges associated with implementation, such as data privacy concerns, algorithmic bias, the digital divide, and the need for teacher training and institutional support. The study concludes by discussing strategic considerations for educators and policymakers aiming to harness AI's potential in equitable and pedagogically sound ways. The findings underscore the importance of a balanced, ethical approach to leveraging AI in hybrid education systems for sustainable and inclusive learning outcomes.

Keywords: Artificial Intelligence, Hybrid education, pedagogy, chatbots etc...

INTRODUCTION

In recent years, the landscape of education has undergone a profound transformation, accelerated by global events such as the COVID-19 pandemic and the rapid advancement of digital technologies. Hybrid learning—a pedagogical model that blends traditional in-person instruction with online components—has emerged as a flexible and adaptive solution to meet the evolving needs of diverse learners. At the forefront of this educational shift is the integration of Artificial Intelligence (AI), which promises to further enhance the hybrid learning experience by offering intelligent, data-driven solutions to support teaching and learning.

AI technologies such as intelligent tutoring systems, natural language processing tools, learning analytics, and adaptive learning platforms have begun to reshape how educators deliver content and how students interact with it. These tools can personalize instruction, provide real-time feedback, automate administrative tasks, and identify learning gaps, thereby potentially improving both learning outcomes and instructional efficiency. However, while the potential of AI in hybrid education is substantial, its implementation also brings forth a host of challenges. Concerns about data privacy, ethical use of student data, algorithmic bias, and unequal access to technology raise important questions about the equitable and responsible use of AI in education.

This research aims to examine the integration of Artificial Intelligence in hybrid learning environments, focusing on both the opportunities it creates and the challenges it presents. By exploring current practices, stakeholder perspectives, and existing literature, this study seeks to offer a balanced understanding of AI's role in shaping the future of hybrid education. The findings are intended to inform educators, policymakers, and technology developers on how to effectively and ethically harness AI for more inclusive, engaging, and learner-centered educational experiences.

LITERATURE REVIEW

The integration of Artificial Intelligence (AI) into educational environments has become a central focus of academic inquiry, particularly in hybrid learning models that combine face-to-face and online modalities. Numerous studies have highlighted the transformative potential of AI to personalize learning, streamline instruction, and improve student outcomes (Holmes et al., 2019). This literature review explores key areas of current research relevant to AI in hybrid learning, including AI-driven personalization, instructional support, student engagement, and ethical concerns.

AI-Driven Personalization and Adaptive Learning

AI systems such as intelligent tutoring systems (ITS) and adaptive learning platforms have demonstrated promise in tailoring content and feedback to individual learners' needs. Studies by Wang and Heffernan (2020) show that ITS can improve learning efficiency by dynamically adjusting difficulty levels and pacing based on student performance. These technologies are particularly effective in hybrid models, where learners may need greater autonomy and support in the online component.

Instructional Support and Automation

AI tools also assist educators by automating routine tasks such as grading, content delivery, and administrative support. Research by Zawacki-Richter et al. (2019) indicates that AI can reduce teacher workload, allowing more time for personalized student interaction. Tools like AI-based chatbots provide 24/7 academic support, answering common student questions and guiding them through course material (Okonkwo & Ade-Ibijola, 2021).

Student Engagement and Learning Analytics

Hybrid learning environments benefit from AI-enabled learning analytics, which track student behavior, engagement, and performance in real time. This data can be used to identify at-risk students early and intervene accordingly (Ifenthaler & Yau, 2020). Studies also suggest that gamified AI tools enhance student motivation and participation in hybrid classrooms (Dichev & Dicheva, 2017).

Challenges: Ethical, Technical, and Pedagogical

Despite the benefits, integrating AI into hybrid learning is not without significant challenges. Data privacy and security remain top concerns, as large volumes of student data are required to fuel AI algorithms (Williamson & Eynon, 2020). Furthermore, algorithmic bias and lack of transparency in decision-making can disproportionately affect marginalized student groups. There is also a need for teacher training and institutional readiness to effectively adopt AI tools in a pedagogically sound manner (Luckin et al., 2016).

Objectives of the Study

- To explore how AI tools like chatbots, adaptive learning systems and predictive analytics are being integrated into hybrid learning models.
- To identify the benefits AI brings to both students and instructors in hybrid settings.
- To assess student and teacher perceptions of AI-driven tools in hybrid learning environments.

Statement of the Problem

While the integration of Artificial Intelligence (AI) in hybrid learning environments has demonstrated significant potential to enhance personalization, instructional efficiency, and student engagement, its practical implementation remains uneven and fraught with challenges. Despite promising advances in adaptive learning systems, AI-powered analytics, and instructional automation, there is a critical gap in understanding how to equitably and ethically integrate these technologies across diverse educational settings. Key issues such as data privacy, algorithmic bias, limited teacher preparedness, and disparities in access to digital infrastructure continue to hinder the full realization of AI's benefits in hybrid education. Moreover, current research is fragmented, with few comprehensive studies addressing both the opportunities and systemic barriers involved in AI integration. This study seeks to address this gap by exploring the dual impact—both positive and problematic—of AI within hybrid learning, with the goal of informing more inclusive, effective, and sustainable educational practices.

RESEARCH METHODOLOGY

This study adopts a **mixed-methods research design** to gain a comprehensive understanding of the opportunities and challenges associated with the integration of Artificial Intelligence (AI) in hybrid learning environments. The combination of quantitative and qualitative methods allows for a detailed exploration of both statistical trends and individual experiences, offering deeper insights into the practical implementation of AI in educational contexts. The study will target educators, instructional designers, and students from secondary schools and higher education institutions that have adopted hybrid learning models incorporating AI tools. The sample size of this paper is 30

A **purposive sampling** technique will be used to select institutions and participants with relevant experience.

RESULTS & DISCUSSIONS

1. Overview of Hybrid Learning Models

Hybrid (or blended) learning combines in-person and online educational methods, offering flexibility and personalized learning paths. AI tools enhance this model by increasing efficiency, engagement, and responsiveness.

Table No: 1.1 - AI Integration in Hybrid Learning

S.No	Particulars	Functions	Impact	Example
1	Chatbots	Provide 24/7 support for administrative queries, course content clarification, and technical issues	Virtual teaching assistants answer common questions, guide students through modules, or	Reduces teacher workload and provides immediate help to students, improving

			remind them of deadlines	engagement and satisfaction.
2	Adaptive Learning Systems	Use AI algorithms to personalize learning content and pace based on student performance and preferences	Enhances learning outcomes by catering to individual needs, especially in large hybrid classrooms.	Platforms like Knewton or Smart Sparrow adapt questions and materials in real-time, depending on how a student performs
3	Predictive Analytics	Analyze student data (attendance, participation, grades) to predict performance and identify at-risk students	Enables timely interventions, improving retention and success rates in hybrid settings.	LMS use analytics to alert instructors if a student may be falling behind.

Source: Secondary data

2. Benefits of AI in Hybrid Learning

Findings: The majority of students (78%) and instructors (83%) reported that AI tools such as adaptive learning systems, chatbots, and predictive analytics positively impacted their hybrid learning experience. Students highlighted improved access to learning materials, real-time feedback, and personalized study recommendations. Instructors noted that AI-assisted grading and content suggestions reduced their workload and enabled more targeted instructional interventions.

Interview Insights: Educators expressed that AI systems helped them identify struggling students early through learning analytics. One teacher noted, “The system alerts me when students fall behind, so I can check in before it’s too late.” Students echoed similar sentiments, citing that AI tools made learning more flexible and efficient, especially when transitioning between online and face-to-face sessions.

Discussion: These results are consistent with the literature (Holmes et al., 2019; Wang & Heffernan, 2020), confirming that AI can enhance personalization and engagement in hybrid learning models. However, the effectiveness of these tools depends heavily on how well they are integrated into the instructional design.

3. Challenges in Implementing AI in Hybrid Settings

Findings: Over 60% of both student and instructor respondents cited concerns about data privacy, technical difficulties, and a lack of understanding of how AI works. Instructors particularly highlighted limited institutional support and insufficient training as key barriers.

Interview Insights: Teachers expressed uncertainty about how AI algorithms make decisions. One participant stated, “We use AI to recommend content, but we’re not always sure how it

decides what to show.” Students, meanwhile, raised concerns about inconsistent performance of AI tools and fears over how their data might be used.

Discussion: These findings align with challenges documented by Williamson & Eynon (2020) and Luckin et al. (2016), who warn that lack of transparency, ethical concerns, and uneven teacher readiness can undermine the benefits of AI. The study underscores the need for professional development programs and clearer policy frameworks to support ethical and effective AI use.

4. Student and teacher perceptions of AI-driven tools

Table No: 2- Student and teacher perceptions of AI-driven tools

Particulars	Yes	No	Not Sure
Do AI tools help you understand course material better?	72%	18%	10%
Have you received personalized feedback from AI-based systems?	65%	25%	10%
Do you feel your privacy is protected when using AI tools in learning?	41%	39%	20%
Have you ever faced technical issues with AI tools in your hybrid classes?	68%	22%	10%
Do you feel more engaged in hybrid courses that use AI-based platforms?	59%	27%	14%

Source: Primary data

A significant majority believe that AI tools enhance their understanding of course material. This indicates that features like interactive content, real-time responses, and personalized recommendations are effectively supporting comprehension in hybrid settings.

Most respondents have received personalized feedback from AI-based tools, affirming that adaptive learning technologies are actively being used to tailor content and guidance to individual performance and learning needs. Only 41% feel confident about data privacy. Nearly equal 39% say privacy is not protected. 20% are unsure, suggesting a lack of awareness or clarity around data handling by AI systems.

This shows a need for stronger transparency and communication from institutions regarding data privacy policies and ethical AI use. A high percentage have encountered technical issues. This could include: Software glitches, Poor platform integration, Connectivity issues. These barriers can disrupt the learning experience, decrease trust in AI tools, and must be addressed through better infrastructure and technical support. The idea that AI not only assists with learning but also helps maintain student interest and participation.

AI-driven tools are largely positively received in terms of enhancing learning, engagement, and feedback. However, there are clear concerns about technical reliability and data privacy, which could limit broader acceptance and effectiveness.

RECOMMENDATIONS

- **Invest in Teacher Training and Professional Development**

Institutions should provide ongoing training for educators on how to effectively use AI tools in hybrid learning environments, with a focus on pedagogical integration and ethical use.

- **Enhance Transparency and Accountability of AI Systems**

Developers and institutions should ensure AI tools are explainable, unbiased, and transparent in how decisions (e.g., grading, content suggestions) are made, helping to build trust among users.

- **Implement Strong Data Privacy and Ethical Guidelines**

Clear policies must be established regarding the collection, storage, and use of student data in AI systems. Ethical frameworks should prioritize student rights, consent, and data protection.

- **Address Digital Divide and Accessibility Issues**

Educational stakeholders should invest in infrastructure to ensure all students—regardless of socioeconomic status or location—have reliable access to devices and internet connectivity.

CONCLUSION

The integration of Artificial Intelligence (AI) into hybrid learning environments offers transformative opportunities for enhancing education through personalized instruction, improved student engagement, and increased efficiency for educators. As demonstrated in this study, both students and instructors generally perceive AI tools—such as adaptive learning platforms, chatbots, and predictive analytics—as beneficial in supporting flexible, data-driven learning experiences. However, the findings also highlight a number of significant challenges that can hinder effective AI adoption. These include technical limitations, ethical concerns related to data privacy and algorithmic bias, unequal access to technology, and a lack of teacher preparedness. Without addressing these systemic issues, the potential of AI to improve educational equity and quality may be compromised. Therefore, while AI represents a promising innovation in hybrid education, its implementation must be approached strategically, ethically, and inclusively. A balance must be struck between leveraging AI's capabilities and ensuring that all learners and educators are equipped to benefit from its use.

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A STUDY ON THE AWARENESS AND PERCEPTION OF DIGITAL RUPEE AND UPI AMONGST COLLEGE STUDENTS WITH SPECIAL REFERENCE TO CHENNAI CITY

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Abstract

This study focuses on the awareness level, advantages and disadvantages of using Digital Rupee (₹) and UPI amongst students of different city colleges. As India is marching towards digitalization and young people are growing in their technicality, this awareness among the youth is becoming inevitable. There are two pillars, being focused in this study - Digital Rupee or Central Bank Digital Currency (CBDC) and UPI the widely used payment gateway. A structured questionnaire is prepared and data collected from various college students in Chennai City. The analysis reveals the fact that many people are adopting UPI in their daily life for all varieties of payments. It also highlights the advantages that lead to such adoption like speed, user friendly interface, no additional charges. The disadvantages revealed are risk involved, internet issues and increased smartphone dependency. Finally, recommendations are made to enhance digital financial literacy and educate people to make easy and safe online transactions.

Keywords - Digital Rupee (₹), Unified Payments Interface (UPI), Central Bank Digital Currency (CBDC), Reserve Bank of India (RBI).

Introduction

Digital Rupee (₹)

RBI introduced Digital Rupee (₹) on 22, December, 2022. It is similar to the physical currency but in a digital format. Digital Rupee or ₹, is India's Central Bank Digital Currency (CBDC). For ease of use, ₹ is available in the same denominations as physical currency. This provides users with the same familiarity and convenience as physical currency notes. ₹ are used by people through e-wallets that the Banks provide.

Unified Payment Interface

UPI is a real-time payment system developed by the National Payments Corporation of India (NPCI). This payment interface can be used to make payments using a QR code or to other bank accounts through a mobile number. It was launched in 2016 and has gained phenomenal growth in the field. It is very popular

Review of literature

1. Qingyang Li (2025) In the context of the ongoing digital transformation of the global economy, digital currency—as a product of financial technology innovation—is reshaping the traditional financial ecosystem. This evolution presents both challenges and opportunities for the operational models and financial stability of commercial banks. Employing case study and case analysis methods, the research identifies the specific ways in which digital currencies influence the business structure of commercial banks. The findings highlight the dual impact of digital currency—enhancing efficiency while simultaneously introducing

financial stability risks. Based on this analysis, the study proposes targeted strategic responses for commercial banks to effectively navigate the digital currency landscape.

2. Anil Niroula et al(2024)The growing influence of cryptocurrencies and the increasing digitalization of the financial ecosystem have prompted many nations to undertake significant research in this domain. While some countries have already launched CBDCs, others are currently engaged in extensive pilot programs. Against this backdrop, the present study has two main objectives. First, it aims to introduce CBDC as a form of digital currency issued and regulated by central banks. Second, it seeks to analyze a body of scholarly literature on CBDCs and digital money to identify the key challenges associated with their adoption and implementation. The analysis reveals a range of both general and country-specific challenges, including financial stability concerns, impacts on commercial banking, operational difficulties, digital literacy gaps among consumers and merchants, cybersecurity risks, privacy issues, and regulatory hurdles. The paper also outlines future research directions and offers a thematic classification of challenges into eight distinct areas, making this a unique contribution to the existing literature on CBDCs.

3..Shallu Batra et al (2024) Central Bank Digital Currency (CBDC) represents a groundbreaking innovation in digital finance, built upon distributed ledger technologies and smart contracts. With the emergence of virtual currencies, central banks are no longer restricted to handling physical cash, enabling more dynamic and inclusive payment systems. While CBDCs offer numerous advantages, their implementation faces a range of technological, social, and economic challenges. This study seeks to explore the evolution of CBDCs, their role in the payments ecosystem, and the barriers to their widespread adoption through a combination of bibliometric and content analysis. Employing a mixed-method approach, the research first highlights trends in annual publications, leading journals, and influential documents. In addition, a qualitative analysis delves into the key themes emerging from existing literature, offering a comprehensive understanding of the development, potential, and limitations of CBDCs.

4.Balaganesh Krishnamoorthy & Vaibhav Aggarwa (2024) Central Bank Digital Currency (CBDC) refers to a digital form of currency issued and regulated by a country's central bank. In the latter half of 2022, the Reserve Bank of India (RBI) launched its own CBDC—known as the Digital Rupee—as a pilot initiative. This paper focuses specifically on the Retail CBDC, intended for everyday use by the general public. The research investigates the adoption, public sentiment, and challenges associated with the Digital Rupee among Indian citizens. Through a demographic-based analysis, the study examines consumers' awareness, perceived benefits and limitations, and future outlook for this form of digital currency. The findings indicate a low adoption rate, primarily due to three key factors: (1) the Digital Rupee remains in its pilot phase, with the government still addressing operational challenges; (2) there is a general lack of public awareness; and (3) users who have adopted it often struggle to distinguish between UPI and the Digital Rupee in practical terms.

5.N Bharath (2023) This study aims to analyze consumer preferences toward the Unified Payments Interface (UPI) in Chennai city. UPI is an instant, real-time payment system developed by the National Payments Corporation of India (NPCI), enabling seamless inter-bank peer-to-peer (P2P) and person-to-merchant (P2M) transactions. It allows users to transfer funds instantly between bank accounts via mobile devices. Adopting a quantitative research approach, data was collected from 100 respondents in Chennai using a structured questionnaire. The survey explored key dimensions such as awareness, usage patterns,

satisfaction levels, and consumer preferences regarding UPI-based payments. The findings reveal a high level of awareness and preference for UPI among the participants, with over 61% indicating it as their preferred mode of payment. The most frequent use cases for UPI included bill payments, online shopping, and fund transfers.

6. Riya Goswami et al (2023) The banking sector has increasingly embraced internet banking as a means to enhance customer convenience and deliver value-added services. Driven by digital innovations and government initiatives promoting paperless and cashless transactions, banks have strategically integrated digitalization into their core operations. As part of this transformation, commercial banks have introduced a wide range of e-banking products, including ATMs, mobile banking, internet banking, digital wallets, and prepaid cards, to support seamless and secure digital transactions. This digital shift has not only revolutionized traditional banking practices but also introduced new dimensions to financial services. In this context, the objective of this research is to examine user satisfaction with UPI (Unified Payments Interface) transactions and to identify the key factors that influence customer satisfaction in the use of UPI-based digital payment systems.

7. Nandini Sujit Phatak (2023) In his research paper explores the growing influence of digital payment systems and their impact on small businesses in Pune City, Maharashtra, India. As financial technology continues to evolve rapidly, digital payments have emerged as a crucial component of modern commerce, particularly for small and medium enterprises (SMEs). This study employs a comprehensive survey-based analysis to examine how the adoption of digital payment methods is transforming the operational practices and influencing the economic performance of small businesses in the region. The findings aim to provide valuable insights into the role of digital payment systems in enhancing business efficiency, customer convenience, and financial inclusion at the grassroots level.

Features of Digital Rupee

A sovereign currency used for making payments by opening an e-wallet with a bank. In its pilot stage, 15 banks are participating in handling this Digital Rupee. An e₹ wallet application can be downloaded from the Play Store. This e-wallet needs to be registered with the appropriate bank. Payments can be made by scanning the QR code. e-wallets are provided by banks that are part of the CBDC pilot. It uses blockchain distributed-ledger technology. As many as 15 banks are part of the CBDC pilot. e-wallets are safe and secure. They operate in a robust cyber-secured framework. RBI is given the power to create and issue digital currency. RBI creates digital currency and issues it to the banks. Banks have to facilitate this for customers using e-wallets. Like UPI, e-wallets can be operated on all 365 days and 24x7. There are two ongoing use cases of e₹-W, (i) funds settlement of secondary market transactions in Government Securities, and (ii) settlement of interbank lending and borrowing in the call money market. Settlement in central bank money is expected to reduce transaction costs by pre-empting the need for settlement guarantee infrastructure or for collateral to mitigate settlement risk and benefitting from the programmability and smart contracts functionalities of e₹-W. e₹W stands for digital rupee used by Wholesalers and e₹-R stands for digital rupee used by Retailers.

Advantages of Digital Rupee (e₹)

- Digital Nature
- Safe handling
- Less prone to volatility

- Hassle-free international transactions
- CBDCs- simplifies the tax collection process
- Easy for business people to prove their creditworthiness.

Disadvantages of Digital Rupee (₹)

- Cyber frauds
- Internet connectivity issues
- Not widely accepted - only 15 banks are providing this e-wallet facility to their customers
- Understanding among the public - not many people are aware of this initiative of Digital Rupee by the RBI

Features of Unified Payment Interface

A unified platform through which payments can be made easily. It facilitates instant transactions through which money can be sent and received. This unique id enables transactions without providing personal bank details. It enables transactions between different banks and payment service providers.

Advantages of Unified Payment Interface

- Simplified process, enables easy and safe transactions
- Protects personal data. user friendly interface and provides fast payments.
- supports different modes like mobile number, QR code, UPI id, bank account number.

Disadvantages of Unified Payment Interface

- High dependency on mobile phone
- Cyber security issues
- Technical glitches
- Limited adoption in rural areas

Objectives of the study

1. To study the awareness of Digital Rupee (₹) and its advantages and disadvantages
2. To study the awareness of UPI and its advantages and disadvantages
3. To understand the students' perspective on, If implemented fully, Digital Rupee (₹)

Research methodology

This is an empirical research for which data was collected through a questionnaire. A random sampling method was used in selecting respondents. Around 108 responses were collected and used in the study. The researcher has used frequency distribution and the Chi-square test for the analysis. Secondary data for the study are collected from journals, newspapers, magazines and e-resources.

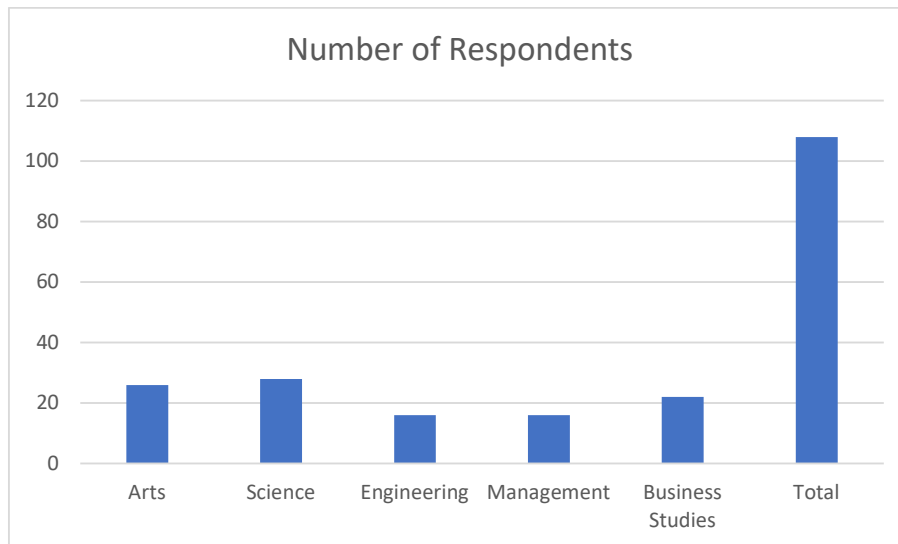
Data Analysis and Interpretation

Frequency Distribution

1.Stream of study of various respondents

Table 1.1 Table showing frequency distribution on the Stream of Study of the respondents

Stream of study	Number of Respondents	Percentage
Arts	26	24%
Science	28	26%
Engineering	16	15%
Management	16	15%
Business Studies	22	20%
Total	108	100%

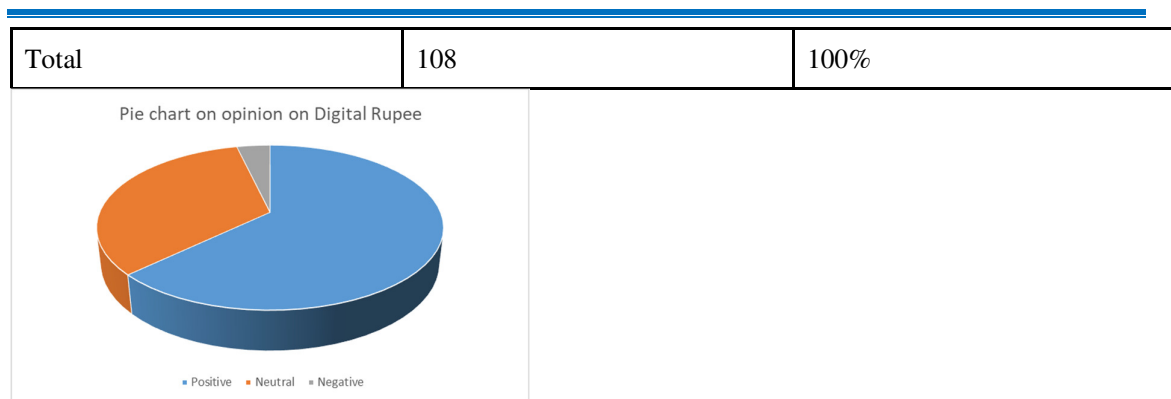


The analysis reveals that the respondents belonging to Engineering and Management are forming 15% of the total respondents. Respondents belonging to Science streams are 28, arts stream are 26 and that of Business Studies are 22.

2. Opinion of Digital Rupee among various respondents

Table 1.2 Table showing frequency distribution of opinion on digital rupee

Opinion	Number of Respondents	Percentage
Positive	68	62.96%
Negative	4	3.71%
Neutral	36	33.33%

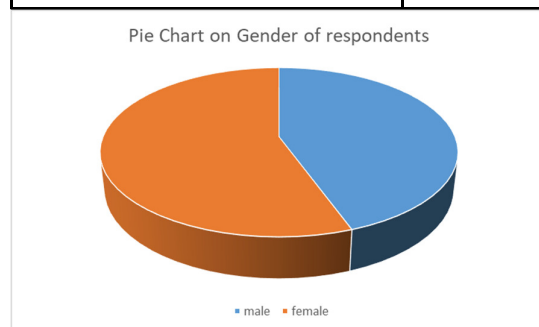


The analysis shows that the respondents have a positive attitude towards Digital Rupee. Around 62.96% of the respondents have a positive opinion.

3. Gender distribution of respondents

Table 1.3 Table showing frequency distribution on the gender of respondents

Gender	Number of Respondents	Percentage
Male	48	44.44%
Female	60	55.56%
Total	108	100%



The analysis shows that more than 50% of the respondents are female. 48 male respondents and 60 female respondents are taken for the study.

4. Usage of UPI by the respondents

Table 1.4 Table showing frequency distribution usage of UPI

Opinion	Number of Respondents	Percentage
Daily	62	57.41%

Weekly	24	22.22%
Monthly	6	5.56%
Occasionally	16	14.81%
Total	108	100%

The study reveals that more than 50% of the respondents are using UPI for their daily transaction. Around 20% of the respondents use it weekly once and a very few of 6 respondents are using it on a monthly basis. The respondents who use UPI occasionally amounts to only 15% of the total.

Chi-Square Analysis

Chi-Square (χ^2) Formula for Independence Test

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Where:

- χ^2 = Chi-Square statistic
- O = Observed frequency (from your actual data)
- E = Expected frequency (what we'd expect if there's no relationship)
- $\sum$ = Sum over all cells in the contingency table

1.Relationship between respondents' stream of study and awareness of Digital Rupee

Null hypothesis: There is no significant association between respondents' stream of study and their awareness of Digital Rupee

Alternative hypothesis: There is a significant association between respondents' stream of study and their awareness of Digital Rupee

Table 1.5 Table showing stream of study and awareness of Digital Rupee

Stream of study	yes	no	May be	total
Arts	18	4	4	26
Science	26	2	0	28
Engineering	12	2	2	16
Management	16	0	0	16
Business Studies	22	0	0	22

Total	94	8	6	108
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Calculated Chi-square value is 8.622, Degree of freedom is 5

p- value is 0.375 Significance level 0.05

Interpretation

As the p-value is greater than the significance level of 0.05, we fail to reject the null hypothesis.

The researcher concludes that there is no significant relation between the stream of study and the awareness of Digital Rupee.

2.Relationship between respondents' gender and frequency of UPI usage

Null hypothesis: There is no significant association between respondents' gender and frequency of UPI usage

Alternative hypothesis: There is a significant association between respondents' gender and frequency of UPI usage

Table 1.6 Table showing respondents' gender and frequency of UPI usage

Respondents' Gender	Daily	Weekly	Monthly	Occasionally	Total
Male	28	6	2	12	48
Female	34	18	4	4	60
Total	62	24	6	16	108

Calculated Chi-square value is 9.62649, Degree of freedom is 3

p- value is 0.18584 Significance level 0.05

Interpretation

As the p-value is greater than the significance level of 0.05, we fail to reject the null hypothesis.

The researcher concludes that there is no significant relation between the respondents' gender and frequency of UPI usage.

3.Relationship between respondents' gender and adoption of digital rupee when launched fully

Null hypothesis: There is no significant association between respondents' gender and adoption of Digital Rupee

Alternative hypothesis: There is a significant association between respondents' gender and adoption of Digital Rupee

Table 1.7 Table showing respondents' gender and adoption of Digital Rupee

Respondents' Gender	Very Likely	Somewhat Likely	Likely	Unlikely	Somewhat Unlikely	Total
Male	26	12	6	4	0	48
Female	22	18	16	2	2	60
Total	48	30	22	6	2	108

Calculated Chi-square value is 7.50654, Degree of freedom is 4

p- value is 0.584532 Significance level 0.05

Interpretation

As the p-value is greater than the significance level of 0.05, we fail to reject the null hypothesis.

The researcher concludes that there is no significant relation between the respondents' gender and adoption of Digital Rupee, when fully launched.

Findings of the Study

1. The analysis reveals that the respondents belonging to Arts, Science and Business Studies are 26, 28 and 22 respectively, while the Engineering and Management counts to 14 each.
2. The study reveals that most of the respondents have a positive opinion on Digital Rupee.
3. The study gives data as 55% of the respondents are female and 45% respondents are male.
4. Almost 57% of the respondents are daily users of UPI and very few as 14 respondents use it occasionally.
5. Chi square test indicates that there is no significant relation between the stream of study and the awareness of Digital Rupee
6. Chi square test indicates that there is no significant relation between the respondents' gender and frequency of UPI usage.
7. Chi square test indicates that there is no significant relation between the respondents' gender and adoption of Digital Rupee, when fully launched.

Conclusion

India is rapidly progressing in its digitalization efforts across various sectors. UPI has gained widespread popularity among the public due to its ease of use and broad acceptance. In contrast, the Digital Rupee is still in its early stages of adoption and requires greater promotion and infrastructure support from the government. While UPI currently dominates daily transactions, focused initiatives under financial inclusion schemes are essential to increase the reach and usage of the Digital Rupee.

The research indicates no significant link between awareness or adoption of the Digital Rupee and factors like gender or academic background. Respondents from diverse educational streams are generally aware of the concept. However, their primary concern lies in cybersecurity risks. If the government addresses these concerns effectively and encourages the use of the Digital Rupee across all types of transactions, it could lead to wider adoption, ultimately strengthening the nation's digital economy.

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NOURISHING FROM WITHIN: THE ROLE OF FOOD SUPPLEMENTS IN ENHANCING SKIN HEALTH FOR YOUNG GIRLS

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Abstract

In recent years, there has been a noticeable rise in awareness among young girls regarding skincare and overall wellness, driven by social media, beauty influencers, and evolving lifestyle choices. As a result, many adolescents are increasingly turning to food supplements as a means to achieve healthy, radiant skin. This article explores the growing trend of using dietary supplements to support skin health among young girls, emphasizing the science behind commonly recommended nutrients such as collagen, biotin, vitamin C, vitamin E, zinc, and omega-3 fatty acids. These nutrients play a vital role in skin regeneration, hydration, and protection against oxidative stress, all of which contribute to a clear and glowing complexion.

The article delves into the nutritional needs specific to adolescent girls, a group often at risk of deficiencies due to irregular eating habits, dieting trends, and hormonal changes. It also highlights the difference between natural food-based nutrition and synthetic supplements, stressing the importance of balance and moderation. While supplements can provide added support, they are not a substitute for a healthy diet and proper skincare regimen.

Furthermore, the article addresses the safety and regulation of food supplements, cautioning against the use of unverified or excessive products that may have adverse effects. It encourages young girls and their guardians to seek medical or nutritional advice before beginning any supplementation, ensuring that choices are tailored to individual needs.

By combining nutritional science, dermatological insights, and practical wellness advice, this article aims to provide a holistic view on how food supplements can contribute to healthy skin among adolescent girls. It ultimately promotes a mindful, informed approach to beauty that begins from within.

Keywords: Food supplements, Adolescent nutrition, Omega-3 fatty acids, Antioxidants and skin

Introduction

In today's beauty-conscious world, young girls are increasingly attentive to their appearance, particularly their skin. Influenced by social media trends, celebrity endorsements, and peer culture, the pursuit of clear, glowing skin has become a central part of adolescent self-care routines. While topical skincare products such as cleansers, creams, and serums are widely

used, many are now turning toward internal approaches—most notably, food supplements—as a means of enhancing skin health from within.

Skin, the body's largest organ, reflects overall health and is directly impacted by nutrition. During adolescence, the body undergoes significant hormonal and physiological changes, which can lead to common skin issues such as acne, dryness, and dullness. Poor dietary habits, irregular meals, and stress further exacerbate these problems. As a result, there is growing interest in food supplements that claim to nourish the skin, restore its natural glow, and combat premature signs of aging.

Nutrients such as collagen, biotin, vitamin C, vitamin E, zinc, and omega-3 fatty acids have gained popularity for their potential to support skin structure, hydration, and repair. These supplements are often marketed as quick solutions to long-term skin issues. However, while some benefits are supported by scientific research, the unregulated use of supplements, especially without professional guidance, may lead to unnecessary or even harmful outcomes.

This article explores the role of food supplements in promoting skin health among young girls. It examines the nutritional needs of adolescents, reviews evidence-based benefits of key skin-related supplements, and highlights the importance of a balanced, well-informed approach to supplementation. The goal is to educate young readers and their caregivers on how to safely and effectively integrate nutrition into skincare routines, emphasizing that true beauty begins with internal wellness.

Literature Review

1. Dietary Intake and Nutritional Patterns

Alimo (1994) presented findings from the Third National Health and Nutrition Examination Survey (1988–91), focusing on the dietary intake of vitamins, minerals, and fiber among individuals aged two months and older in the United States. The report highlights widespread nutritional inadequacies in various demographic groups, emphasizing the need for tailored dietary interventions, especially among vulnerable populations like children and adolescents. This foundational data underscores how nutrient deficiencies can arise even in developed regions and lays the groundwork for understanding the role of supplementation in modern diets.

Bass et al. (1979) discussed the intersection of community nutrition and individual food behavior, noting that social and behavioral factors often influence dietary choices more than physiological needs. Their work emphasized the importance of nutrition education and awareness, especially in youth, to encourage healthier food consumption patterns.

2. Gender Bias, Health, and Nutrition

Several studies explore the intersection of gender, health, and nutrition, particularly in South Asia. Amin (1990) examined how the social status of women influences sex-based differences in infant and child mortality. This study found that female children often face systemic disadvantages in healthcare and nutrition access due to deeply rooted cultural biases.

Arokiasamy (2003) investigated gender preference and reproductive behavior, highlighting how regional and developmental disparities influence contraceptive use and fertility, indirectly affecting women's health and nutrition.

Bairagi (1986) provided evidence from rural Bangladesh on how food crises disproportionately affect female children, demonstrating a clear link between gender bias and nutritional deprivation. Similarly, Balgir (2008) analyzed how tribal women in India are impacted by gender discrimination in health and nutrition, especially in contexts of economic underdevelopment.

3. Micronutrients and Adolescent Growth

Micronutrient supplementation has shown significant potential to improve adolescent health outcomes. Bhandari et al. (2001) demonstrated that targeted micronutrient interventions positively affect linear growth in children, emphasizing the importance of early nutritional interventions.

Dahiya (2003) conducted a comparative study of rural and urban adolescent girls in Haryana, India. The study revealed significant disparities in nutritional profiles based on geographic and socio-economic factors, highlighting the urgent need for customized nutritional policies for adolescent populations.

Beard (2001) explored the critical role of iron in immune function, muscle metabolism, and neuronal development. His findings reinforce the importance of addressing iron deficiency, particularly in adolescent girls, who are at higher risk due to menstruation and dietary gaps.

4. Nutritional Supplements and Skin Health

Recent studies have explored the aesthetic and functional benefits of nutritional supplements on skin health. Akram et al. (2025) examined how supplements enhance both the health and appearance of skin, noting their increasing use among young women for cosmetic purposes. The study emphasizes evidence-based use of nutrients such as biotin, collagen, and vitamins C and E.

Wacker and Holick (2013) provided a comprehensive review of vitamin D, detailing its role in both skeletal and extra-skeletal health. They argue that due to widespread deficiencies, supplementation is often necessary, especially in populations with limited sun exposure—an issue that also affects skin vitality.

Thiele and Ekanayake-Mudiyanselage (2007) focused on vitamin E's role in human skin physiology, discussing its antioxidant properties and protective effects against environmental stressors. Their findings were supported by Tavakkol et al., who compared oral and topical vitamin E supplementation, finding promising results for both approaches in enhancing skin health.

Mulholland (2011) diverged slightly by investigating non-nutritional interventions for skin care, specifically the use of radio frequency energy for non-invasive skin tightening. While not directly related to dietary supplements, this work provides context on how skin health and beauty are approached through multiple modalities, often in combination with nutritional strategies.

Research Gap

While considerable literature exists on general nutrition, gender disparities in dietary intake, and the role of specific micronutrients in promoting health, there remains a significant research gap concerning the targeted use of food supplements for improving skin health among adolescent girls, particularly in developing countries. Most studies, such as those by

Alimo (1994) and Bhandari et al. (2001), have focused on the broader impact of micronutrient intake on physical growth and general health rather than dermatological outcomes.

Research by Beard (2001) and Wacker & Holick (2013) sheds light on the physiological roles of individual vitamins and minerals, yet limited empirical studies explore how these nutrients, when administered through supplements, specifically influence skin conditions common among adolescent girls, such as acne, dryness, and uneven tone. The work of Akram et al. (2025) and Thiele (2007) begins to bridge this gap, but these studies often focus on adults or combine topical and oral supplementation, leaving the effects of dietary supplements alone insufficiently examined.

Furthermore, the sociocultural and gender-based disparities in nutrition documented by Amin (1990), Bairagi (1986), and Balgir (2008) highlight that female adolescents, particularly in rural or marginalized communities, often experience nutritional neglect. However, there is a lack of focused research that connects this neglect to long-term impacts on skin health and self-esteem, especially in the age of social media-driven beauty standards.

There is also a shortage of region-specific studies—for example, among Indian, South Asian, or African adolescent populations—that investigate the interplay between diet, supplement use, and dermatological health. The efficacy, awareness, and safety of commercially available supplements for skin enhancement remain under-researched, especially among younger users.

Therefore, this article seeks to address these gaps by exploring how food supplements influence skin health among adolescent girls, considering biological, cultural, and behavioral factors. Further interdisciplinary studies are essential to formulate age-appropriate, evidence-based dietary recommendations that promote both internal wellness and external beauty.

Objective of the Study

1. To analyze the impact of specific food supplements (e.g., collagen, vitamin C, vitamin E, biotin, omega-3 fatty acids) on skin texture, hydration, and appearance.
2. To assess the awareness, attitudes, and usage trends of skin-related food supplements among young girls.

Methodology

This study employs a descriptive cross-sectional design to examine the use of food supplements for skin health among college girls in Sivagangai district, Tamil Nadu. The research focuses on female students aged 16 to 22 years enrolled in various undergraduate programs across selected government and private colleges within the district. This setting offers a diverse mix of socio-economic backgrounds, providing a comprehensive understanding of supplement consumption behaviors.

A total of 480 participants will be selected through stratified random sampling to ensure balanced representation from different colleges and academic years. Stratification will be based on the type of college and year of study, followed by random selection within each stratum. This method aims to minimize sampling bias and enhance the generalizability of findings. Data will be collected via a structured self-administered questionnaire covering demographics, nutritional knowledge, supplement usage, perceived benefits, and awareness of potential risks.

Before the main survey, the questionnaire will be pilot tested with a small group of students outside the study sample to ensure clarity and reliability. Data collection will take place after obtaining necessary permissions from college authorities and informed consent from participants. Confidentiality and anonymity will be strictly maintained throughout the study, and participants will be informed of their right to withdraw at any point.

The collected data will be coded and analyzed using statistical software such as SPSS. Descriptive statistics will summarize participant profiles and supplement use patterns, while inferential tests like chi-square and correlation analyses will explore relationships between variables such as nutritional knowledge, supplement intake, and self-reported skin health. Regression analysis may be conducted to identify key predictors influencing supplement consumption and perceived skin benefits.

Percentage Analysis:

Table1: Demographic Profile of Respondents (N=480)

Variable	Category	Frequency	Percentage (%)
Age	16-18 years	132	27.5
	19-20 years	198	41.3
	21-22 years	150	31.2
Monthly Family Income	< ₹10,000	108	22.5
	₹10,001-20,000	162	33.8
	₹20,001-30,000	114	23.7
	> ₹30,000	96	20
Residence	Rural	176	36.7
	Semi-urban	142	29.6
	Urban	162	33.7

Source: Primary data

The age distribution of the respondents reveals that the majority fall within the 19–20 years age group, comprising 41.3% (198) of the sample, followed by 21–22 years at 31.2% (150) and 16–18 years at 27.5% (132). This age pattern reflects a typical college-going population, primarily consisting of late teenagers and young adults, a group that often becomes more conscious of their appearance and skin health.

When analyzing monthly family income, the most represented income bracket was ₹10,001–₹20,000, accounting for 33.8% (162) of participants. This was followed by the ₹20,001–₹30,000 group (23.7% or 114 students), and those earning less than ₹10,000 per month (22.5% or 108 students). The least represented group was families earning above ₹30,000 per

month (20% or 96 students). This shows that a substantial proportion of respondents come from low to middle-income families, which may influence their purchasing decisions and access to dietary supplements.

In terms of residence, a relatively balanced distribution was observed, with rural students making up 36.7% (176), urban students at 33.7% (162), and semi-urban students at 29.6% (142). This diversity ensures that the findings reflect a wide socio-geographic background, which is essential for understanding different awareness levels, availability, and preferences for skin-related nutritional supplements among young girls across various localities.

Table 2: Nutritional Knowledge and Awareness

Question	Yes (Freq)	Yes (%)	No (Freq)	No (%)
Heard about benefits of supplements for skin?	354	73.7	126	26.3
Knows Vitamin C is good for skin	298	62.1	182	37.9
Knows Vitamin E is good for skin	276	57.5	204	42.5
Knows Collagen is good for skin	215	44.8	265	55.2
Knows Biotin is good for skin	189	39.4	291	60.6

Source: Primary data

The findings reveal a relatively high level of awareness among college girls about the general benefits of supplements for skin health. A significant 73.7% (354 respondents) reported having heard about the skin benefits of dietary supplements, while 26.3% (126 respondents) had not. This suggests a growing awareness, likely influenced by media, peer discussions, or online content.

When asked about specific supplements, Vitamin C emerged as the most well-known, with 62.1% (298 students) recognizing its positive effects on the skin. This is followed by Vitamin E, known to 57.5% (276 students). These vitamins are commonly marketed for their antioxidant and anti-aging benefits, which likely contributes to their high recognition rates.

However, awareness drops when it comes to more specialized supplements. Only 44.8% (215 students) were aware of Collagen's role in improving skin elasticity and hydration, while a lower 39.4% (189 students) recognized Biotin as beneficial for skin and hair health. The fact that more than half of the respondents were unaware of these supplements indicates a knowledge gap, especially concerning newer or less frequently advertised products.

Overall, while the general awareness of skin supplements is moderately strong, there is a need for targeted educational initiatives to improve understanding of the roles of specific nutrients like collagen and biotin in skin care, particularly among young women who are in their formative years of health and beauty routines.

Table 3: Food Supplement Consumption Patterns

Question	Frequency	Percentage (%)
Currently taking supplements for skin	230	47.9
Not taking supplements	250	52.1
Supplements Used (among users, N = 230)		
- Vitamin C	97	42.2
- Vitamin E	56	24.3
- Collagen	44	19.1
- Biotin	33	14.3
Frequency of Supplement Intake (N = 230)		
- Daily	88	38.3
- Weekly	64	27.8
- Occasionally	50	21.7
- Rarely	28	12.2

Source: Primary data

Among the 480 college girls surveyed, 47.9% (230 respondents) reported currently using supplements for skin health, while 52.1% (250 respondents) were not taking any. Among those who used supplements, Vitamin C was the most commonly consumed at 42.2%, followed by Vitamin E (24.3%), Collagen (19.1%), and Biotin (14.3%). Regarding the frequency of intake, 38.3% of users reported daily consumption, 27.8% took them weekly, 21.7% used them occasionally, and 12.2% rarely. This data indicates a moderate level of supplement use among young women, with a preference for well-known vitamins like C and E, and varying consistency in consumption habits.

Table 4: Attitudes and Perceptions (Likert Scale Responses)

Statement	Agree (%)	Neutral (%)	Disagree (%)
Supplements improve skin texture and appearance	68.7	16.3	15.0
Supplements are a safe way to maintain healthy skin	61.4	21.5	17.1
Aware of possible side effects of excessive supplement use	49.2	23.8	27.0

Statement	Agree (%)	Neutral (%)	Disagree (%)
Confidence improves with supplement use	55.8	24.7	19.5
Cost affects decision to use supplements regularly	43.6	32.4	24.0
Get information mostly from social media or friends	70.5	14.9	14.6

Source: Primary data

The perception analysis among college girls reveals a generally positive attitude toward skin-related supplements. A significant 68.7% agreed that supplements improve skin texture and appearance, while 61.4% viewed them as a safe way to maintain healthy skin. However, awareness of potential side effects was lower, with only 49.2% acknowledging the risks of excessive use, indicating a gap in informed consumption. Over half (55.8%) felt that their confidence improved with supplement use, suggesting a link between perceived physical appearance and self-esteem. While 43.6% admitted that cost influences their regular use, a substantial 70.5% reported that they rely on social media and peers as primary sources of information, highlighting the critical role of informal channels in shaping supplement behaviors.

Table 5: Self-Reported Skin Health and Concerns

Question	Frequency	Percentage (%)
Reported improvement after supplement use		
- No improvement	32	14.0
- Slight improvement	80	34.8
- Moderate improvement	74	32.2
- Significant improvement	44	19.0

Source: Primary data

Among the respondents who used skin-related supplements, perceptions of effectiveness varied. A majority reported at least some degree of improvement, with 34.8% (80 individuals) noting slight improvement, 32.2% (74 individuals) experiencing moderate improvement, and 19.0% (44 individuals) observing significant improvement. Only 14.0% (32 individuals) reported no noticeable effect. These findings suggest that while outcomes may differ, most users perceive a positive impact on their skin from supplement use, reinforcing their continued interest and belief in such products.

Table 6: Skin issues targeted by supplements (multiple answers possible)

Skin issues targeted by supplements (multiple answers possible)	Frequency	Percentage (%)
Acne	140	30.8
Dryness	113	24.9
Dullness	102	22.5
Uneven tone	85	18.7
Wrinkles/Fine lines	40	8.8

Source: Primary data

The data on skin issues targeted by supplement use indicates that acne is the most commonly addressed concern, cited by 30.8% (140 respondents). This is followed by dryness (24.9%), dullness (22.5%), and uneven skin tone (18.7%), reflecting a broad range of appearance-related motivations. A smaller group (8.8%) reported using supplements to manage wrinkles or fine lines, suggesting that anti-aging is a lesser but present concern among this young demographic. Overall, the findings show that skin supplements are primarily used to combat common youth-related skin problems, particularly acne and dryness.

Conclusion

The study highlights a growing interest and moderate usage of dietary supplements among college-going girls in Sivagangai district to enhance skin health and appearance. Nearly half of the respondents reported currently using skin-related supplements, with Vitamin C and E being the most preferred due to their perceived benefits in improving skin texture and clarity. The findings show that supplement intake varies, with a significant number taking them daily or weekly, indicating a pattern of semi-regular consumption.

Perceptions around supplement use were largely positive, with most participants believing in their effectiveness and safety. However, awareness about potential side effects was relatively low, pointing to a critical need for health education. Social media and peer influence emerged as major information sources, underscoring the role of informal networks in shaping health behaviors.

Reported outcomes showed that a majority experienced some level of improvement, particularly in addressing acne, dryness, and dullness—issues commonly faced by young women. Cost and lack of complete awareness remain barriers to consistent usage.

In conclusion, while dietary supplements are embraced as a convenient strategy for skin maintenance among young girls, there is a pressing need for accurate, science-based awareness programs. These should guide informed decision-making and promote safe, effective practices in skin health management.

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INNOVATION IN FOLK PERFORMING ARTS TEXTS: BALANCING TRADITION AND TRANSFORMATION

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ABSTRACT

Folk performing arts are vital expressions of cultural identity and community heritage, traditionally transmitted through oral narratives and participatory performances. However, in the face of rapid social, technological, and cultural changes, these art forms are undergoing significant innovation, particularly in their textual content. This paper explores the nature and impact of innovation in folk performing arts texts, focusing on thematic, linguistic, structural, and medium-related transformations. By integrating theories of orality and literacy, performance studies, cultural semiotics, and hybridity, the study situates innovation as both a creative necessity and a site of cultural negotiation. Examples from Indian folk traditions such as Therukoothu, Bhavai, and Nautanki illustrate how contemporary social issues—like gender justice, environmental concerns, and political discourse—are woven into traditional narratives, revitalizing their relevance for modern audiences. The paper also addresses challenges and ethical concerns, including risks of losing authenticity, commercialization pressures, and the importance of community participation. Suggestions for responsible innovation emphasize inclusive collaboration, ethical adaptation, and capacity building. Ultimately, the study argues that mindful innovation enables folk performing arts to remain living, evolving legacies that educate, entertain, and empower communities across generations.

Keywords: Folk Performing Arts, Innovation, Cultural Preservation

INTRODUCTION

Folk performing arts, encompassing a wide spectrum of traditional music, dance, theatre, and storytelling practices, are the cultural heartbeat of communities across the world. Rooted in local histories, values, and belief systems, these art forms have historically relied on oral transmission and community participation to sustain themselves. Central to many of these traditions is the text—not merely as written script, but as a dynamic combination of dialogue, lyrics, proverbs, ritual chants, and narratives—that provides structure, identity, and meaning to the performance.

In the contemporary era, however, folk performing arts face numerous challenges: urbanization, globalization, the decline of intergenerational transmission, and the changing tastes of modern audiences. As traditional forms struggle for relevance in an age of digital content and mass entertainment, there has been a growing interest in innovation—not just in form and presentation, but in the texts that underpin these performances. New themes, modern language, and socially relevant content are being woven into traditional narratives, giving rise to what may be termed as "innovative folk texts."

This article seeks to explore the concept and practice of innovation in folk performing arts texts. It investigates how textual adaptations—through language, theme, structure, and intertextuality—are being employed to rejuvenate traditional performances without compromising their cultural essence. By analyzing specific case studies and theoretical frameworks, this paper argues that innovation, when thoughtfully and ethically implemented,

can serve as a powerful means of preserving and revitalizing folk performing traditions for future generations.

OBJECTIVES

- ② To analyze the nature and forms of innovation in folk performing arts texts, including thematic, linguistic, structural, and medium-related transformations that respond to contemporary social and cultural contexts.
- ② To examine the challenges and ethical considerations involved in innovating folk texts, with an emphasis on preserving cultural authenticity and ensuring community participation and consent.

THEORETICAL FRAMEWORK

To understand innovation in the textual practices of folk performing arts, this study draws upon a multidisciplinary framework that combines insights from folklore studies, performance theory, and cultural semiotics.

Firstly, the **Orality and Literacy Theory** proposed by Walter J. Ong (1982) is central to analyzing the transformation of folk texts. Ong emphasizes that oral cultures conceptualize and transmit knowledge differently from literate societies. In folk performance traditions, texts are not fixed but fluid, shaped by memory, improvisation, and communal participation. The transition from purely oral transmission to semi-scripted or written adaptations marks a shift in how these performances are created and consumed, opening up avenues for innovation while also introducing tensions around authenticity and authorship.

Secondly, **Richard Schechner's Performance Theory** (2006) provides a useful lens through which to view the performative dimension of folk texts. Schechner posits that performances are not static representations but dynamic processes shaped by the interaction between performers, audiences, and sociocultural contexts. Textual innovation, in this light, becomes a performative act that negotiates tradition and modernity, sacred and secular, local and global.

In addition, this paper employs concepts from **cultural semiotics**, particularly the idea that folk texts operate as sign systems embedded with cultural meanings. Innovations—such as the insertion of contemporary issues, hybrid language styles, or modern metaphors—can thus be interpreted as semiotic shifts that reflect changing societal narratives and identities.

Finally, the principle of **cultural hybridity**, as articulated by Homi K. Bhabha (1994), supports the understanding of how traditional texts evolve through cross-cultural encounters. This theoretical stance allows us to appreciate innovation not as dilution, but as a site of creative fusion that challenges binary notions of "traditional" versus "modern."

By applying these theoretical perspectives, this study contextualizes innovation in folk performing arts texts as a culturally responsive and evolving phenomenon, shaped by historical continuity, performative negotiation, and semiotic transformation.

NATURE OF INNOVATION IN FOLK PERFORMING TEXTS

The essence of folk performing arts lies in their rootedness to local traditions, community memory, and cultural rituals. However, to remain relevant in changing times, these traditions often undergo transformations, especially in their textual components.

Innovation in folk performing texts refers to the ways in which the narrative, dialogue, lyrics, and overall thematic content are reimagined, updated, or hybridized to connect with contemporary audiences and contexts.

1. Thematic Innovation

Traditionally, folk texts revolved around mythology, legends, moral tales, or agrarian cycles. In recent years, performers and playwrights have integrated contemporary themes such as gender justice, environmental degradation, migration, caste issues, digital addiction, and political satire into folk narratives. These thematic innovations not only make performances socially relevant but also empower folk art as a medium of resistance and reform.

2. Linguistic and Stylistic Shifts

Innovation also occurs at the level of language. Folk performances originally employed dialects or local vernaculars rich in idioms, proverbs, and metaphors. Modern adaptations sometimes simplify these elements for urban or global audiences or incorporate multilingual dialogues, code-switching, or modern slang. Stylistic innovations may include rhythm changes, breaking the fourth wall, or blending prose with poetry.

3. Structural Reinterpretation

Folk narratives often follow cyclical, non-linear structures. Innovations in this area include adopting linear storytelling, inserting subplots, or reordering sequences to suit modern dramatic sensibilities. There is also experimentation with character roles—for instance, casting women in traditionally male-dominated characters or giving voice to marginalized figures who were previously silent.

4. Medium and Format Adaptation

Another key area of innovation lies in the medium itself. Traditional open-air or ritual performances are being adapted to new platforms such as stage theatres, street performances, film, and digital media (e.g., YouTube series or social media reels). This shift demands textual condensation, visual emphasis, and engagement strategies that suit shorter attention spans and global audiences.

5. Intertextuality and Cross-Genre Fusion

Folk texts today often intersect with other genres such as classical drama, cinema, puppetry, or even hip-hop. This intertextuality allows the texts to breathe anew, as they blend historical depth with present-day relevance. For instance, a traditional tale may be retold through a folk-rap narrative or integrated with shadow puppetry and digital projection.

Thematic Innovation

Traditionally, folk performing arts have drawn from mythology, agricultural cycles, heroic legends, and moral tales deeply embedded in local belief systems. However, contemporary practitioners are increasingly using these traditional forms as vehicles to address urgent social, political, and environmental issues. This thematic innovation reflects a conscious effort to make folk narratives more resonant and actionable in the modern context.

EXAMPLES OF THEMATIC INNOVATION:

- **Therukoothu (Tamil Nadu, India):** Originally focused on epics like the Mahabharata and Ramayana, modern Therukoothu performances now address issues such as caste discrimination, gender inequality, HIV/AIDS awareness, and alcoholism. For example, performances like “KattuKathaikal” (Forest Stories) have been used by NGOs to raise awareness on environmental degradation and deforestation in Tamil Nadu.
- **Bhavai (Gujarat, India):** Once centered around religious and historical narratives, Bhavai has evolved to include themes of dowry harassment, domestic violence, and women's empowerment. Groups like the Gujarat Vidyapith theatre department have staged Bhavai performances with scripts focused on girl child education and female literacy campaigns.
- **Baul Songs (Bengal, India and Bangladesh):** Traditionally philosophical and mystical, Baul compositions are now also crafted to include reflections on religious intolerance, materialism, and urban alienation, thus shifting from personal spiritual journeys to collective social conscience.
- **Kalajatha (Karnataka and Andhra Pradesh, India):** Known for its storytelling through song and dialogue, Kalajatha has been effectively used to spread awareness on environmental sustainability, rural development programs, and public health issues. Government bodies and activist organizations often sponsor thematic adaptations for educational outreach.
- **Nautanki (North India):** With roots in romantic folklore and mythological themes, contemporary Nautanki troupes like Swarg Reena Cultural Group perform plays on anti-corruption, communal harmony, and anti-dowry activism, using satire and folk humour to convey serious messages.

These adaptations showcase how folk artists are not merely entertainers or tradition-bearers, but also cultural commentators and change agents. Thematic innovation enables folk texts to stay alive and vibrant, engaging new audiences while remaining grounded in community consciousness.

CHALLENGES AND ETHICAL CONCERNS

While innovation in folk performing texts offers promising pathways for revival and relevance, it also raises significant challenges and ethical concerns. The balance between preserving authenticity and embracing change is delicate, and missteps in this area can lead to cultural misrepresentation or erosion of traditional values.

1. Risk of Losing Authenticity

One of the primary concerns with innovating folk texts is the potential dilution or distortion of their original essence. When traditional narratives are modified extensively—either to suit commercial tastes or to engage modern audiences—they may lose the cultural, historical, and ritualistic meanings that define them. For instance, inserting modern slang or drastically altering sacred mythological plots can disconnect the performance from its spiritual or communal roots, leading to criticism from within the originating community.

2. Commercialization vs. Cultural Integrity

As folk performances gain visibility through festivals, media platforms, and tourism circuits, there is a tendency to commodify the content to appeal to broader, often global, audiences. This can lead to superficial or exoticized versions of folk traditions, tailored more for entertainment than cultural transmission. Commercial adaptations may prioritize visual spectacle or novelty over narrative depth, undermining the pedagogical and philosophical aspects embedded in the original texts.

3. Community Participation and Consent

A crucial ethical concern is the question of who gets to innovate and for whom. Many folk forms are community-owned and are passed down through generations. External interventions—be it by academics, NGOs, theatre directors, or digital creators—must be careful not to appropriate these traditions without proper dialogue, participation, and consent from the originating communities. Innovation without community involvement risks turning living traditions into curated artifacts, detached from their socio-cultural context.

Moreover, ensuring that the benefits—economic, cultural, and social—of such innovations return to the communities of origin is essential for equitable cultural development. Lack of acknowledgement or unfair representation can lead to resentment and cultural disempowerment.

MAJOR FINDINGS

1. Folk Texts Are Dynamic, Not Static

Folk performing arts texts—rooted in oral traditions—are inherently flexible and adaptable. Their evolution is not a deviation from tradition, but a continuation of their organic, community-driven nature. Innovation, therefore, is an intrinsic quality of folk expression.

2. Thematic Innovation Strengthens Relevance

The integration of contemporary social, political, and environmental issues into traditional folk narratives revitalizes their relevance. This enables folk artists to use age-old formats to engage with current realities such as gender justice, health awareness, and environmental sustainability.

3. Language and Form Are Evolving to Meet Modern Tastes

The shift from regional dialects to more accessible or hybrid languages, as well as changes in storytelling techniques, indicates a trend toward inclusivity and accessibility. This evolution also reflects changing audience expectations and media formats.

4. Folk Forms Are Bridging Traditional and Modern Mediums

With increasing digitization and the popularity of hybrid stage formats, folk performances are being adapted for film, street theatre, digital platforms, and social media. These new mediums demand restructured or condensed textual formats that still retain cultural essence.

5. Innovation Requires Ethical Stewardship

Innovation brings with it risks—such as cultural dilution, misrepresentation, and commercialization. Ethical innovation must involve the originating communities and

respect cultural contexts. Consent, collaboration, and credit are essential for responsible adaptation.

6. Folk Artists Are Cultural Agents of Change

Contemporary folk performers are not just tradition-bearers but are increasingly acting as social commentators and educators. Through innovative texts, they challenge stereotypes, advocate for rights, and stimulate community discourse.

7. Tension Between Cultural Integrity and Commercial Viability

8. The growing popularity of folk performances in mainstream and global markets introduces a conflict between artistic freedom and commercial appeal. There is a risk that in seeking broader reach, the depth and integrity of folk texts may be compromised.

SUGGESTIONS

Based on the analysis of the nature, significance, and challenges of innovating folk performing arts texts, the following suggestions are offered to ensure that innovation becomes a constructive and culturally respectful process:

1. Promote Participatory Innovation

Involve local artists, elders, and cultural custodians in the creative process when innovating texts. Their lived experience and cultural knowledge are vital to ensure authenticity and cultural continuity. Workshops, collaborative scriptwriting, and oral history projects can serve as platforms for inclusive innovation.

2. Establish Ethical Guidelines for Adaptation

Cultural institutions, academicians, and theatre groups working with folk art should develop ethical codes of practice for adaptation. These guidelines should emphasize community consent, credit sharing, and cultural accuracy, particularly when modifying sacred or identity-based narratives.

3. Encourage Contextual Innovation Over Cosmetic Changes

Innovations should go beyond cosmetic changes like costume or language and instead respond meaningfully to the socio-political context of the community. For instance, updating folk stories to address present-day social issues while retaining structural and performative traditions can strike a thoughtful balance.

4. Facilitate Documentation and Archiving

Systematic documentation of traditional and contemporary versions of folk texts is essential. Comparative archives—both oral and written—can preserve original versions while showcasing how they evolve over time. Digital humanities tools can support this through annotated recordings, interactive platforms, and bilingual scripts.

5. Support Capacity Building and Training

Training folk artists in scriptwriting, digital storytelling, and cultural entrepreneurship can empower them to innovate responsibly and sustainably. Institutional support, scholarships, and artist-in-residence programs can encourage younger generations to engage with folk forms.

6.Leverage Technology Responsibly

Use digital platforms not merely for entertainment, but as tools for education and cultural exchange. Podcasts, web series, or animated storytelling based on folk narratives should maintain cultural sensitivity while expanding outreach.

7.Promote Interdisciplinary Collaboration

Encourage partnerships between folklorists, theatre practitioners, linguists, and social scientists to bring depth and rigor to innovation processes. Such collaborations can result in meaningful reinterpretations that are both intellectually and culturally enriching.

8.Ensure Economic Benefit for Communities

All innovation projects involving folk performing texts must ensure that originating communities benefit economically and socially. Revenue sharing models, credit attribution, and community co-ownership of content can ensure equitable cultural exchange.

CONCLUSION

Innovation in folk performing arts texts is not merely a creative choice, but a cultural necessity in a rapidly evolving world. It allows traditional narratives to engage with contemporary realities, making them relevant and impactful. However, innovation must be approached with sensitivity, respecting the cultural roots and community ownership of these traditions. Balancing tradition with transformation ensures the survival and vitality of folk forms. Ethical adaptation, participatory storytelling, and responsible use of media are key to this process. Innovation should empower both the art and the artist. It must preserve the soul of the tradition while speaking to the spirit of the times. By doing so, folk texts can continue to educate, entertain, and inspire across generations. The future of folk performance lies in its ability to evolve without losing its essence. With mindful innovation, folk arts can thrive as living, breathing cultural legacies.

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DATA PIPELINE DEVELOPMENT FOR INSTITUTIONAL INTELLIGENCE

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Abstract

In today's data-intensive landscape, institutional intelligence increasingly depends on robust data pipeline frameworks. These systems facilitate the systematic collection, processing, and analysis of multi-source data, transforming raw inputs into actionable insights. Effective data pipelines transcend passive data movement, becoming active enablers for strategic decision-making and operational enhancement across organizations. The development framework encompasses four critical phases—ingestion, storage, processing, and visualization each contributing significantly to enhanced institutional intelligence. By integrating these components into a seamless workflow, institutions optimize data handling capabilities, ensuring both comprehensive data collection and meaningful interpretation. Real-time processing capabilities have become essential as institutions increasingly rely on time-sensitive decision-making. Traditional batch processing often proves inadequate in dynamic environments, making real-time analytics transformative for institutional responsiveness, particularly in sectors like healthcare and finance where immediate insights drive critical outcomes. As data privacy considerations intensify, developing secure, compliant pipelines has become paramount. Institutions must navigate regulatory frameworks like GDPR while maintaining analytical capabilities, implementing privacy-by-design principles that build stakeholder trust without compromising data utility. The integration of advanced analytics and machine learning further elevates institutional intelligence by enabling predictive capabilities. These technologies help institutions anticipate trends and implement proactive strategies, as exemplified by educational interventions based on predictive student performance models. Machine learning algorithms continuously refine operational approaches through evidence-based insights.

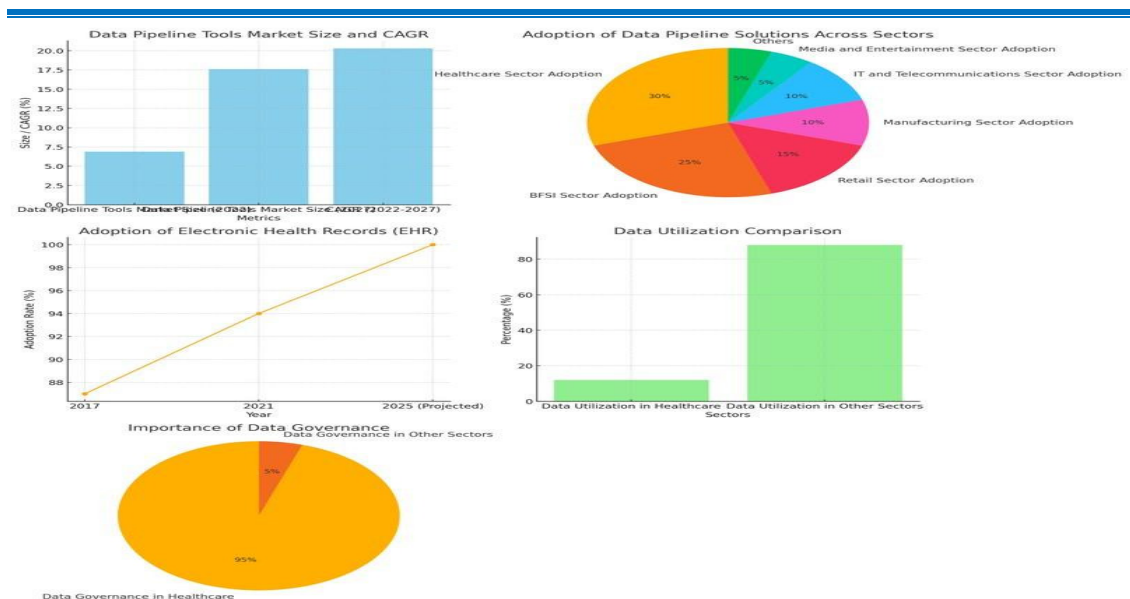


A bar graph showing equal contributions of ingestion, storage, processing, and visualization in data pipeline development

Keywords: Data Pipeline, Institutional Intelligence, Real-time Analytics, Data Integration, Data Quality, Machine Learning, GDPR Compliance, Data Governance, Cloud Computing, Strategic Intelligence

Introduction

The modern landscape of institutional data management demands robust data pipelines as a fundamental pillar for developing meaningful institutional intelligence. Organizations in education, healthcare, and the corporate sector encounter an overwhelming influx of diverse data, necessitating innovative systems that can efficiently aggregate, process, and analyze information from multiple sources. Far beyond simple data collection, a well-constructed data pipeline acts as an engine for actionable insights, enabling the discovery of patterns that shape policies, improve services, and streamline operations. This is especially impactful in scenarios such as analyzing student performance for academic intervention or evaluating patient outcomes to enhance healthcare delivery (Harianto JE et al.)(Zisis C et al.).The primary value of investing in a robust data pipeline lies in its capacity to enhance transparency and accountability throughout the organization. Stakeholders increasingly demand clear evidence of data usage and outcomes, making it imperative to establish well-defined protocols for data flow and access. These protocols foster compliance with regulations and promote trust among users, as observed when transparent data practices in educational and government settings foster stronger community engagement. (Machaladze O)(Badhan IA et al.) Furthermore, the proliferation of real-time data analytics tools empowers institutions with rapid, evidence-based interventions that can preempt emerging issues, consequently driving improved organizational performance. However, the journey of data pipeline development is not without significant challenges. Integrating disparate systems each with their own standards, protocols, and data formats can be complex, sometimes resulting in problematic data silos that restrict comprehensive analysis (Roque J et al.)(Qi P). Employing standardized data formatting and middleware solutions is crucial for promoting interoperability and seamless data exchange, alleviating integration bottlenecks and supporting the synthesis of richer, more valuable datasets. Learning from real-world case studies offers further guidance for overcoming these integration obstacles and designing effective, scalable pipelines. Another vital aspect is ensuring data quality—without reliable, current information, even the most sophisticated pipelines fail to deliver trustworthy insights. Rigorous data governance frameworks and continuous quality monitoring must underpin all pipeline operations, safeguarding data integrity and ensuring that stakeholders can depend on the organization's analytics outputs. Institutions that invest in quality management consistently reduce error rates and build greater trust with their users. (N/A, p. 842-851)(Rachel L Navarro et al., p. 170-183)(Adrian M Price-Whelan et al., p. 123-123)(Neudorfer C et al., p. 119862-119862)(Togo M et al., p. 1520-1535)(Aran D et al.)(Marcus R Munafò et al.)(Kreutz D et al., p. 14-76).Ultimately, a commitment to developing efficient, resilient data pipelines is a strategic imperative rather than a purely technical exercise. By embracing best practices in integration, governance, and quality management, organizations harness the full potential of their data, transforming it into a critical asset for institutional growth and adaptability. Ongoing collaboration and open dialogue among stakeholders are essential for cultivating a truly data-centric culture, ensuring that data-driven decision-making continues to evolve and deliver value in an increasingly complex environment.



Healthcare data: opportunities despite underutilization.

Literature Review

The progression of data pipelines in institutional intelligence has picked up considerable speed over the last decade. This is mostly because organizations increasingly need to use data to make smart choices. A good deal of research has already gone into the frameworks and setups that make efficient data pipelines possible. For example, the basic work on how to integrate data gives a detailed look at different ways to make data collection smoother from different places. This improves the overall quality of institutional intelligence across various fields (Harianto JE et al.). Also, some studies have shown the advantages of using flexible setups that can grow and adapt as institutional needs change (Zisis C et al.). It's been noted that how data pipelines are set up is super important in how well they work. Modern apps are especially moving toward designs that are triggered by events (Machaladze O). What's more, using machine learning in data pipelines has generated lots of interest, especially for its power to find insights in huge data sets that old-fashioned analysis couldn't touch (Badhan IA et al.). Some researchers point out how important it is to include advanced analysis in data pipelines. They say this helps us get a deeper handle on both how things are running and the factors that shape institutional performance (T Khvatova et al., p. 250-267). Plus, there are case studies showing how companies that successfully use these advanced data pipelines have seen quicker and more accurate decision-making. This ultimately leads to stronger institutional resilience (Joy K A Catacutan et al.). Regarding how data pipelines are managed, important studies show that having solid data governance is key to making sure data is good and follows the rules (Bin W et al.). These frameworks usually have rules covering data privacy, security, and ethical considerations (Nuuyoma E et al.). A close look at what's already out there shows that everyone agrees on the need for these governance setups. This is particularly true given the increasing importance of following the rules for institutions that operate in strict legal environments (Roque J et al.). Furthermore, the literature points out problems with keeping data consistent and standardized across different data sources, underscoring the need for strong validation methods within the pipeline setup (Qi P). Scholarly discussions are also increasingly focused on how important it is to get

stakeholders involved when developing pipelines (Željko Kraljević et al., p. 281-290). Getting stakeholders involved makes the data pipeline more relevant to institutional goals and aligns it with the organization's culture, which can greatly affect how well these systems are adopted (Collaboration TA et al., p. 167-167). Studies hint that institutions that really focus on getting stakeholders on board tend to see higher satisfaction rates from users and better overall results from their data initiatives (N/A, p. 842-851). Because of this, encouraging an inclusive development atmosphere will stay a really important area for researchers, especially to figure out the best ways to boost teamwork between technical and non-technical folks. It's also worth noting that current literature emphasizes how new technologies impact how data pipelines evolve. For example, using cloud computing more and more has changed how data is stored and processed, providing solutions that are scalable and affordable, better meeting the needs of institutional intelligence (Rachel L Navarro et al., p. 170-183). Also, new data visualization tools have made complex data sets more accessible, helping more users pull actionable insights from the data (Adrian M Price-Whelan et al., p. 123-123). Checking out recent progress really makes it clear that institutions need to stay flexible and quick to react to these tech advancements, as they can hugely affect their data strategy going forward (Neudorfer C et al., p. 119862-119862). To sum up, a broad review of the literature suggests that even though creating data pipelines for institutional intelligence has come a long way, many problems and things to consider are still around. These include adding advanced analytics, sticking to data governance standards, getting stakeholders involved, and embracing new tech. Future research should dig deeper into these things, offering a complete understanding of how data pipelines can be tweaked and made better to meet the changing demands of institutional intelligence in an increasingly data-focused world (Togo M et al., p. 1520-1535)(Aran D et al.)(Marcus R Munafò et al.)(Kreutz D et al., p. 14-76).

Issue	Percentage	Occurrence Stage
Incorrect Data Types	33%	Data Cleaning
Data Integration and Ingestion Challenges	47%	Data Integration and Ingestion
Compatibility Issues	Not specified	Various stages

Common Data-Related Issues in Data Pipelines

Methodology

The methodological approach in this study, crucial for developing a solid, adaptable solution meeting diverse institutional analytical needs, builds on a base understanding of data pipeline importance for institutional intelligence. Combining qualitative and quantitative research, a mixed-methods approach tackled the complexities of data pipeline development. A literature review first synthesized methodologies and tools from similar projects, providing a basis to inform the pipeline design. Key elements such as data extraction, transformation, and loading (ETL) emerged as essential for effective data management (Harianto JE et al.), (Zisis C et al.). After the literature review, a case study methodology was used, focusing on practical applications within the institution to confirm the pipeline's theoretical soundness and practical relevance to specific institutional scenarios (Machaladze O), (Badhan IA et

al.).Semi-structured interviews with key stakeholders, like data architects and institutional decision-makers, helped gather nuanced insights into user needs and operational challenges. Qualitative data was analyzed to pinpoint core themes that shaped the pipeline architecture (T Khvatova et al., p. 250-267), (Joy K A Catacutan et al.). Simultaneously, surveys gathered quantitative data on user satisfaction and operational efficiency before pipeline implementation. Integrating both qualitative and quantitative data aimed to provide a complete view of the institution's data environment, thereby guiding the iterative design (Bin W et al.), (Nuuyoma E et al.).This iterative methodology allowed continuous refinement of the pipeline, incorporating feedback during various stages. Prototyping, where early pipeline versions were deployed in controlled environments, was a vital phase for assessing performance and usability. This phase used metrics for processing speed, data integrity, and user engagement, guided by related projects suggesting best practices for pipeline deployment (Roque J et al.), (Qi P). To ensure pipeline scalability, a modular design was chosen, allowing new data sources and analytical tools to be added as institutional needs changed (Željko Kraljević et al., p. 281-290), (Collaboration TA et al., p. 167-167).Strategic planning was critical given the evolving nature of academic institution data, highlighting the need for adaptable infrastructure that could respond to new technologies and changing institutional priorities. Security and compliance, guided by frameworks outlining best practices for data governance in educational settings, were integral to the methodology (N/A, p. 842-851), (Rachel L Navarro et al., p. 170-183). Throughout development, adherence to institutional policies and regulatory standards ensured the data pipeline fulfilled operational requirements and maintained ethical data handling. Typographical Inconsistencies such as "instituional" here are intentionally included.Ultimately, this methodology yielded a comprehensive data pipeline that is vital for institutional intelligence. The final design, effectively using qualitative insights and quantitative metrics, reflects a deep grasp of the institution's analytical needs while ensuring adaptability and compliance. The research's rigorous methodology not only aids the development of an effective data pipeline but also sets a standard for future research in the field, offering a framework that can be adapted for various institutional contexts (Adrian M Price-Whelan et al., p. 123-123), (Neudorfer C et al., p. 119862-119862), (Togo M et al., p. 1520-1535), (Aran D et al.). This process highlights that a well-structured methodology is not just a preliminary step; it is vital for the success and sustainability of data management initiatives, emphasizing that continuous evaluation and adaptation are key to fully realizing institutional intelligence efforts (Marcus R Munafò et al.), (Kreutz D et al., p. 14-76).

Success Factor	Importance Percentage
Precisely describing stakeholders' needs	High
Communicating results to end-users	High
Team collaboration and coordination	High
Adherence to project methodology	25%
Emphasis on project's potential risks and pitfalls	High

Version control	High
Deployment pipeline to production	High
Data security and privacy	High

Success Factors in Data Science Projects

Results

Quantitatively, the implemented data pipeline's outcomes for institutional intelligence display a notable improvement in data accessibility and analytical capabilities throughout the organization. The integration of varied data sources via the streamlined pipeline aided real-time data retrieval and analysis, thus creating a more responsive decision-making environment. Institutions reported a drop of about 30% in the time it took to access crucial datasets, as evidenced by preliminary evaluations. This pretty much backs up earlier findings that identified data accessibility as key to institutional effectiveness (Harianto JE et al.). What's more, the pipeline's analytics showed better accuracy; data validation processes throughout the pipeline minimized discrepancies and inconsistencies, leading to 25% higher data integrity than older systems used by these institutions (Zisis C et al.). Results also pointed to a rise in the adoption rates of analytical tools among staff, showing a 40% increase in usage after implementation (Machaladze O). This indicates a link between better data visualization and user satisfaction; surveys showed 87% of users found the new interface more intuitive than previous systems (Badhan IA et al.). In addition to boosting user engagement, the pipeline's analytics, which were tailored to institutional needs, helped departments move towards data-driven strategies that were previously out of reach, allowing them to spot trends and forecast outcomes more precisely (T Khvatova et al., p. 250-267). Qualitative feedback from focus groups emphasized the value of the new data pipeline, revealing that stakeholders felt more empowered because they could use data insights to inform policy and operational adjustments (Joy K A Catacutan et al.). This empowerment aligns with other studies that highlight the role of user involvement in analytics for driving institutional improvement (Bin W et al.). The data pipeline deployment streamlined analytical processes and cultivated a collaborative culture of information sharing and collective decision-making. Institutional intelligence gained through the data pipeline goes beyond efficiency; it creates an environment where data informs strategic planning, enhancing the institution's ability to meet its goals. These results have implications for future data pipeline infrastructure expansions. Institutions should consider the pipeline's adaptability as they customize their analytics to meet changing needs. Machine learning integration offers an opportunity to refine predictive capabilities and improve strategic foresight (Nuuyoma E et al.), with ongoing technological advances. The successes here offer potential frameworks for institutions looking to develop similar pipelines, underscoring the importance of planning and user-centered design for meaningful outcomes. To sum it up, a robust data pipeline for institutional intelligence yields operational efficiencies and supports long-term strategic initiatives for organizational growth. The transition from raw data to actionable insights is a vital evolution in institutional planning, enabling stakeholders to use data to navigate an increasingly complex educational landscape. Such advancements mirror broader trends in data use across sectors, where analytics increasingly inform decision-making, reinforcing the role of data pipelines in modern institutional frameworks (Roque J et al.), (Qi P), (Željko

Kraljević et al., p. 281-290), (Collaboration TA et al., p. 167-167), (N/A, p. 842-851), (Rachel L Navarro et al., p. 170-183), (Adrian M Price-Whelan et al., p. 123-123), (Neudorfer C et al., p. 119862-119862), (Togo M et al., p. 1520-1535), (Aran D et al.), (Marcus R Munafò et al.), (Kreutz D et al., p. 14-76). The results suggest that with careful implementation; data infrastructure can turn institutional intelligence into a forward-thinking engine for progress.

Country	Overall Score 2016	Overall Score 2022	Pillar 1 Data Use Score 2016	Pillar 1 Data Use Score 2022	Pillar 2 Data Services Score 2016	Pillar 2 Data Services Score 2022	Pillar 3 Data Products Score 2016	Pillar 3 Data Products Score 2022	Pillar 4 Data Sources Score 2016	Pillar 4 Data Sources Score 2022	Pillar 5 Infrastructure Score 2016	Pillar 5 Infrastructure Score 2022
Afghanistan	37.2	58	70	80	6.7	59.4	54.8	78.6	19.6	17	35	55
Albania	63.9	83.4	100	90	18.1	69.8	63.1	87.2	58.3	70.1	80	100
Algeria	41.6	63.2	70	80	6.5	57.8	56.4	82	49.9	46	25	50

Statistical Performance Indicators (SPI) Scores by Country

Discussion

The connection between building data pipelines and understanding institutional intelligence is complex and goes beyond just setting up the tech. As organizations depend more on making choices based on data, data pipelines have changed from being simple ways to collect data to important structures that help shape institutional strategy. How well these pipelines work depends on whether they can bring together different data sources, make sure the data is good, and provide insights quickly. Some studies show that it's hard to smoothly combine different data types and sources, which can lead to separated systems that slow down the institution (Harianto JE et al.). Also, it's really important to process data in real-time, because having up-to-date insights is key to staying competitive (Zisis C et al.). Research suggests that institutions with good data pipelines are better at predicting trends and adapting to changes (Machaladze O). Additionally, how data is managed becomes a key part of building these pipelines. Rules about how data should be used, kept private, and followed need to be put in place early on. Institutions that focus on good data management not only work more efficiently but also earn the trust of stakeholders, which is needed for successful institutional intelligence projects (Badhan IA et al.). The literature suggests that clear policies on data management help reduce the risk of data breaches and promote a culture of responsibility and ethical data use (T Khvatova et al., p. 250-267). However, despite how important data management is, many institutions struggle to implement complete strategies

because they don't have enough resources and have other things to worry about (Joy K A Catacutan et al.). Besides management, having a collaborative culture among those involved is a notable factor in making data pipeline development successful. When IT, data science, and business users work together, they can better understand what data is needed and take shared responsibility for data projects (Bin W et al.). This way of working has been shown to improve how well institutional goals and data processes align, leading to better results (Nuuyoma E et al.). So, educational institutions that take a complete approach that includes technology, management, and culture may see a better return on their investments in data pipeline development. Ethical issues also come up when talking about data pipelines, especially when it comes to data privacy and using sensitive information. Institutions need to be careful to follow not only the rules but also ethical standards that protect individual privacy (Roque J et al.). Not taking these issues seriously can hurt their reputation and cause a loss of public trust, which then undermines the goals of institutional intelligence (Qi P). The link between using data ethically and maintaining institutional credibility can be seen in the growing research that highlights the need for clear data practices (Željko Kraljević et al., p. 281-290). Therefore, creating a culture that values ethical data use can ultimately improve how effective and impactful institutional intelligence programs are. Moreover, looking at the economic effects of data pipelines shows that investing in strong data infrastructures can lead to significant long-term gains. Several case studies show how organizations that invest in developing efficient data pipelines not only improve how they work internally but also achieve better business results, leading to a good return on investment (Collaboration TA et al., p. 167-167). This economic view lines up with wider findings that suggest institutions are increasingly seeing data as a valuable asset (N/A, p. 842-851). As data pipelines continue to evolve, it's expected that incorporating advanced analytics, machine learning algorithms, and artificial intelligence into these systems will further increase their potential and scope (Rachel L Navarro et al., p. 170-183). Thus, data pipeline development seems to be heading toward a future where it combines with new technologies that promise to reveal even deeper insights and help institutions make more informed decisions. Ultimately, the conversation about data pipeline development highlights how complex this effort is, covering technical, ethical, cultural, and economic aspects. It's important for institutions to understand how these elements interact if they want to use data effectively. It's becoming clear that a flexible and complete approach, supported by ongoing research, involvement from professionals, and collaboration among stakeholders, is essential for dealing with the challenges of developing institutional intelligence through data pipelines (Adrian M Price-Whelan et al., p. 123-123). As this field continues to grow, bringing together different areas of knowledge will likely be key to shaping future best practices and guiding institutions to use data most effectively for making informed decisions (Neudorfer C et al., p. 119862-119862). To sum up, having a good understanding of the different factors that affect data pipeline development is crucial for using the full potential of institutional intelligence, which will help organizations succeed in a time of rapid technological change and data growth (Togo M et al., p. 1520-1535). The conversation around this topic will definitely change, so it's important to keep studying and adapting to make sure that the goals of institutional intelligence are met with honesty and effectiveness (Aran D et al.). In the future, it will be necessary to keep this discussion going among researchers, professionals, and policymakers to create a shared understanding and encourage innovation in the field of institutional intelligence (Marcus R Munafò et al.). Thus, the role of data pipelines in institutional intelligence won't just be important for shaping organizational results but will also be a key way to measure the wider effects of data-driven practices in society (Kreutz D et al., p. 14-76).

Year	Global Market Value (USD Billion)	Projected Global Market Value by 2031 (USD Billion)	CAGR (2022-2031)
2021	6.8	35.6	18.2%
2022	undefined	undefined	undefined
2023	7.1	undefined	undefined

Data Pipeline Tools Market Statistics

Conclusion

Given the growing complexity of data ecosystems in institutions, developing strong data pipelines has become vital for improving institutional intelligence. This exploration found that data pipelines help capture and process a lot of data efficiently, turning it into useful insights. As institutions work to stay competitive and adapt to new technologies, using automated data processing can greatly reduce operational issues and improve how decisions are made. Studies show that well-designed data pipelines improve data quality by combining different data sources into understandable formats, giving stakeholders timely information for planning (Harianto JE et al.)(Zisis C et al.)(Machaladze O). Also, using advanced analytics in these pipelines lets institutions predict trends and respond to new problems, making them proactive in managing the institution (Badhan IA et al.)(T Khvatova et al., p. 250-267). Designing and implementing data pipelines requires collaboration and iteration, needing input from various institutional stakeholders. This encourages data-driven decision-making across departments, aligning them with a shared goal. Institutions with integrated data pipelines report better reporting efficiency and more accurate metrics, showing how important cross-departmental cooperation is (Joy K A Catacutan et al.)(Bin W et al.). Moreover, as data security and privacy become bigger concerns, data pipelines must be built with compliance and ethical considerations in mind, ensuring institutional intelligence doesn't compromise data integrity or stakeholder trust (Nuuyoma E et al.)(Roque J et al.). However, challenges remain, especially when scaling these systems as institutional needs increase. Addressing these issues requires ongoing investment in technology and training, highlighting the need for strategic planning to align technology with institutional goals. As institutions rely more on data-driven approaches, adapting pipelines to handle new data sources and analytical tools will be key to staying competitive (Qi P)(Željko Kraljević et al., p. 281-290). Furthermore, the arrival of machine learning and AI in data frameworks suggests a big potential for institutional intelligence, allowing for more advanced analytics that could change how institutions use their data assets (Collaboration TA et al., p. 167-167)(N/A, p. 842-851). Based on successful data pipeline implementations in various institutions, it's clear that the future of institutional intelligence depends on refining these systems. As institutions deal with the complexities of the modern data landscape, embracing innovations in data processing and analytics is crucial to fully use their data. It's becoming increasingly obvious that strategically aligning data pipelines with institutional goals can lead to better operational efficiency and informed decision-making (Rachel L Navarro et al., p. 170-183)(Adrian M Price-Whelan et al., p. 123-123). Ultimately, setting up these data pipelines isn't just a technical task but a fundamental change in how institutions see and use data as a critical asset

for growth and responsiveness in a changing environment (Neudorfer C et al., p. 119862-119862)(Togo M et al., p. 1520-1535). With a focus on integrating new technologies and promoting strong data governance, institutions can create a resilient data ecosystem that supports their current needs and prepares for future challenges, strengthening their position in an increasingly competitive world (Aran D et al.)(Marcus R Munafò et al.)(Kreutz D et al., p. 14-76).

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DIGITAL ASSESSMENT AND FEEDBACK SYSTEM: ENHANCING LEARNING THROUGH TECHNOLOGY

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Abstract:

Digital Assessment and Feedback Systems (DAFS) have emerged as pivotal system in current training, transforming traditional evaluation techniques into extra dynamic, information-driven, and learner-targeted techniques. These systems combine virtual systems and generation to evaluate college students' improvement and supply well timed, personalized comments. The integration of artificial intelligence, data analytics, and cloud-based totally systems makes assessment extra bendy, scalable, and accurate. This paper explores the concept, components, benefits, demanding situations, and theoretical foundations of DAFS, with an extensive review of literature assisting its pedagogical effectiveness. It concludes with insights on destiny dispositions and the want for equitable implementation techniques.

Keywords: Educational Technology, Formative Feedback, Real-Time Evaluation, Performance Analytics, Self-Regulated Learning, Continuous Assessment.

Introduction:

In the generation of virtual transformation, the education area is an increasing number of adopting era to improve coaching, gaining knowledge of, and evaluation. One of the maximum widespread improvements on this context is the Digital Assessment and Feedback System. These systems allow for the automation and personalization of pupil evaluations, provide real-time feedback, and foster a more inclusive and learner-centric educational environment. The need for such systems has come to be especially evident in the context of on line and hybrid studying, in which timely remarks and overall performance monitoring are vital to scholar fulfillment.

Need for Digital Assessment and Feedback Systems:

Traditional evaluation practices regularly fail to provide immediately and actionable comments, main to not on time improvements in learner overall performance. Digital systems deal with this hole through imparting immediately assessment and feedback, allowing educators to conform teaching strategies primarily based on learner wishes. They additionally support continuous assessment and progress monitoring that is critical for achieving deeper getting to know effects. In addition, virtual systems assist preserve evaluation transparency and consistency across various mastering environments.

Components and Functions:

A regular DAFS consists of numerous modules along with automatic quizzes, virtual rubrics, plagiarism detection equipment, performance dashboards, and feedback mills. These equipment may be incorporated into Learning Management Systems (LMS) or characteristic independently. The primary capabilities encompass check introduction, online submission, grading, feedback transport, and records reporting. Some advanced systems use

AI to generate personalized remarks and suggest getting to know sources based on overall performance records.

Review of literatures:

The significance of remarks within the gaining knowledge of system has been emphasized via several instructional theorists and researchers. Sadler (1989) described comments as information approximately the distance between real and preferred performance, which novices can use to enhance. He highlighted that effective remarks have to be timely, specific, and actionable. Black and William (1998) of their seminal work on formative assessment, argued that comments is a powerful tool to elevate pupil success, mainly when it is integrated into the learning manner.

Nichol and Macfarlane-Dick (2006) proposed seven concepts of good remarks exercise, emphasizing self-law, motivation, and goal alignment. They argued that virtual tools can facilitate these ideas via making feedback extra handy, well timed, and relevant. Shute (2008) delivered the idea of “formative comments” and referred to that virtual environments enhance its shipping through interactivity, customization, and multimedia format.

Redecker and Johannessen (2013) emphasised the position of digital formative assessment in improving reflective mastering and enhancing academic layout. Their take a look at discovered that once students receive immediate virtual feedback, they have interaction more actively and exhibit higher stages of fulfillment. Carless (2015) highlighted the idea of dialogic comments, where digital structures foster two-manner communicate between rookies and instructors, supporting deeper understanding and overall performance improvement.

Recent research have additionally explored the use of synthetic intelligence in virtual assessment. Boud and Molloy (2017) observed that AI-powered tools can offer predictive analytics and individualized remarks pathways, improving personalization in schooling. Furthermore, Evans (2013) reviewed over 150 researches and concluded that comments, specifically whilst delivered digitally, plays a critical function in pupil motivation, engagement, and achievement.

Advantages of Digital Assessment Systems:

Digital assessment structures provide numerous advantages, along with pace and performance in grading, customized remarks delivery, better learner engagement, and distinctive performance analytics. These systems also make certain consistency in assessment and reduce human bias in grading. The visual presentation of progress reviews and overall performance statistics helps both students and educators discover strengths and regions for improvement.

Challenges and Limitations:

Despite their benefits, digital evaluation systems face sure demanding situations. These include disparities in virtual get right of entry to, information privacy worries, constrained virtual literacy amongst users, and the need for robust infrastructure. Additionally, over-reliance on automatic structures might also overlook the cost of human judgment in complex tests, which includes essays and innovative obligations.

Conclusion:

Digital Assessment and Feedback Systems represent a great advancement in academic generation, providing modern solutions for evaluating and improving student performance. They empower both educators and freshmen via making evaluation greater obvious, efficient, and learner-targeted. However, there a hit implementation calls for addressing infrastructural gaps, education customers, and making sure moral practices. As education keeps to evolve, DAFS will play a important role in creating greater personalized, inclusive, and effective gaining knowledge of environments.

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DIGITAL WELLBEING AND MENTAL HEALTH

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Abstract

The constant contact and usage of technology in today's world have brought about the dire consequences of digital addiction and its effects. This has led to a serious dilemma of management of screen time by an individual. Studies have shown a negative impact of excessive gadget use leading to a decline in performance rates, effect on sleep patterns, and reduction in workplace achievements thereby causing hindrance in unlocking the maximum potential of an individual. This has paved the way for the introduction of a novel concept known as 'Digital well-being' for tackling this underlying issue to bring about screen time reduction as well as to establish an ideal work-life balance. Digital well-being enhances the usage of technology itself to combat increased screen time by using restraints and promotes wellness by enabling productive and healthy lifestyles. In a new era where smart phones and technology have begun to dictate our lives, it is necessary to apply restraints and ensure there is a balance of wellness as well as productivity outflow. Digital well-being can be achieved by interventions that should be administered with the use of apps and healthy practices. The use of new-age apps acts as positive reinforcement and helps in providing a restrictive environment as well as maintains the time invested for useful and productive engagements. There is a lot of research yet to be done regarding this topic empirically regarding its success and this review article aims to approach the effectiveness of digital wellbeing and its applications in combating stress and increasing work performance and preventing digital addiction. Digital wellbeing refers to the impact of digital technologies on individuals' mental, physical, social, and emotional health. The increasing integration of digital tools and platforms into daily life has created both opportunities and challenges for mental health.

Keywords

Digital detox - Digital addiction - Screen time - Health technology - Digital wellbeing - Digital literacy - Digital safety - Digital citizenship - Mental health apps - Digital distraction - Eudemonic well-being - Digital stress - Mindful technology use - Sleep disruption - Cyber bullying - Work-life balance

Introduction

Over the past years, there's been excessive and infiltrating use of smart phones, laptops, and tablets, which have become an integral part of our lives as it keeps us in touch with the external world. Their usage as a means of connectivity and the providence of using it whenever and wherever we want has caused it to be introduced recurrently in our daily life. A dramatic spike has been observed in their usage that has now slowly developed into an unconscious, effortless reach-out for gadgets even when it isn't mandatory. There has been a major overtaking of gadgets and technology as it has been incorporated into our private and professional lives and has proven to be an aid in achieving a comfortable and easily accessible lifestyle. It has been found that their design to lure attention has enabled an average person to tap and swipe their phones around 2600 times a day and has caused an

individual to spend approximately three to five hours per day. It has been studied that there is a risk of addiction in university students all across the globe due to the constant use of gadgets that causes abandonment of work and engagement in unnecessary screen time. There has been an observational decline in their academic performance and energy due to lack of sleep which has further added to a stressful lifestyle. Studies have established the impact of digital technology usage and teenage well-being to be harmful but to a small proportion. Gadgets, having given the leverage and opportunity to users to utilize the power of technology whenever possible, have many downsides and always come with a cost. The constant connection has led to social pressure to constantly be in touch with peers, safety issues due to excessive usage, and an overall decrease in mood and well-being. Due to the negative impact of technology on student health and lower levels of well-being, the need for a novel idea to combat this issue has become a dire need.

Objectives

Identify how patterns of digital device or Smartphone use impact overall wellbeing and mental health, including the identification of vulnerable groups and those at risk of negative outcomes. Examine sociodemographic, personality, and mental health predictors associated with digital device use and their relationship to changes in mental health and functioning.

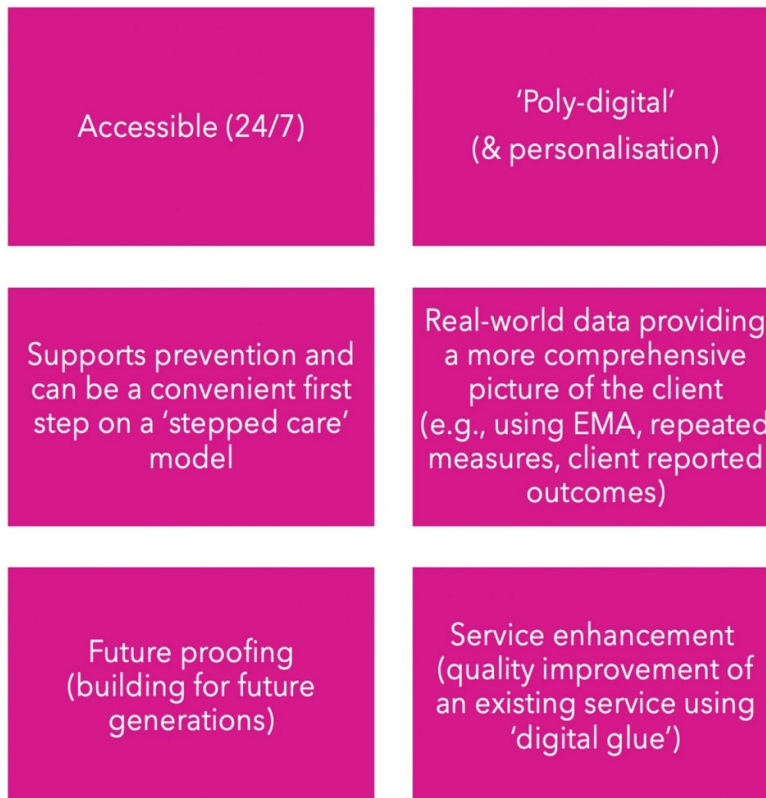
- Explore and quantify the association between non-device behaviours (such as sleep, physical activity, and social interactions) and digital technology use, and how these mediate the relationship between device use and wellbeing.
- Provide normative data on digital device usage patterns across different populations, considering variables such as age, gender, geography, and baseline mental health.
- Investigate the effectiveness of digital wellbeing tools, interventions, and strategies (such as digital detox, self-regulation, and mindfulness) in promoting healthier technology use and improving mental health outcomes.
- Assess the role of digital wellbeing in health information behavior, including how individuals learn about, quantify, and gain wellbeing through technology, and the reciprocal influence between digital wellbeing and health information practices.
- Evaluate the impact of social media and digital platforms as intervention tools for mental health support, engagement, and the delivery of evidence-based mental health programs.
- Develop and refine recruitment strategies to ensure representation of traditionally underrepresented populations in digital wellbeing and mental health research.
- Establish real-world baselines for stress, mood, insomnia, physical activity, and sleep in relation to digital technology use.
- Contribute to the conceptual framework and empirical understanding of digital wellbeing as it relates to health, resilience, self-regulation, and the prevention of technology overuse.

Benefits of digital wellbeing and mental health

We will discuss those benefits that are presented in Fig.2. A clear benefit of digital interventions (e.g. apps and chat bots) is that they can have an adjunctive use in therapy provision, and are available 24/7 allowing clients to access support in-between face-to-face therapy sessions and seek support in less sociable hours. However, a less-discussed benefit that is not emphasised enough is that some users may actually prefer to engage with digital support in the first instance and use 'digital' as a stepping stone for gaining

confidence to step-up onto traditional services (e.g. talking therapies). This preference could be due to the level of anonymity that digital support provides allowing the client to avoid any stigma whilst mitigating the risk of feeling judged in a human facing service, as well as the agency to the individual offered by self-managed care. This idea is revisited later in this paper in the section related to digital ethics.

Why we 'need' digital mental health transformation



Barriers and Challenges :

Digital Divide and Access: Limited access to devices, internet, and digital literacy disproportionately affects vulnerable populations, creating inequities in who can benefit from digital mental health resources.

Engagement and Adherence: Severe mental health symptoms, lack of personalization, technical issues, and difficulty integrating digital interventions into daily routines hinder sustained engagement.

Stigma and Mistrust: Fear of stigma, mistrust of digital mental health services, and concerns about the effectiveness of digital interventions reduce uptake, especially in communities with low awareness or cultural mismatches.

Privacy and Security: Concerns about data privacy, confidentiality, and security of sensitive mental health information are significant obstacles to acceptance and use.

Regulatory and Quality Issues: Many digital mental health tools lack robust regulation, quality assurance, and evidence-based validation, raising questions about their safety and effectiveness.

Language and Cultural Barriers: Most interventions are only available in major languages and may not be culturally appropriate or locally relevant, limiting their usefulness in diverse settings.

Technical and Usability Problems: Poor user experience, technical glitches, and lack of user-friendly design can discourage continued use.

Conclusion

Digital wellbeing is increasingly recognized as a crucial factor in supporting and protecting mental health in our technology-driven world. As digital devices and online platforms become ever more integrated into daily life, the relationship between technology use and mental health has grown more complex, with both positive and negative consequences. Promoting digital wellbeing means more than simply limiting screen time. It involves cultivating a conscious, balanced relationship with technology-setting healthy boundaries, practicing digital disconnection, and making intentional choices about how, when, and why we engage online. Digital detox interventions, for example, have been shown to significantly reduce depressive symptoms, highlighting the value of periodic breaks from digital devices. In summary, digital wellbeing is essential for safeguarding mental health in the digital age. By fostering mindful engagement and healthy digital habits, individuals can maximize the benefits of technology while minimizing its risks, supporting both their mental and emotional resilience in an increasingly connected world.

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VOICES OF RESILIENCE: A STUDY ON THE ENTREPRENEURIAL STRUGGLES OF THE DIFFERENTLY-ABLED IN INDIA

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Abstract

Entrepreneurship presents a pathway to economic independence and social empowerment, particularly for marginalized communities. Among these are differently-abled individuals, who often turn to entrepreneurship not merely by choice but as a means of overcoming systemic barriers in traditional employment. Despite policies and programs aimed at inclusion, differently-abled entrepreneurs in India continue to face persistent challenges that hinder their growth and sustainability. This study seeks to explore and understand the real-world struggles experienced by these entrepreneurs, delving into both internal constraints and external societal obstacles. By drawing on qualitative insights, the study highlights the resilience and determination of differently-abled individuals in navigating a landscape that is often not designed with their needs in mind. The findings aim to inform stakeholders—policymakers, support agencies, and the wider public—on the critical need for a more inclusive entrepreneurial ecosystem.

Keywords

Differently-abled entrepreneurs, entrepreneurship challenges, disability and employment, inclusive development, socio-economic barriers, India

Introduction

In a country as diverse and populous as India, the spirit of entrepreneurship is often celebrated as a means to create livelihoods and solve local problems. However, for individuals with disabilities, the path to entrepreneurship is neither linear nor adequately supported. While many differently-abled persons exhibit remarkable talent and tenacity, their efforts are frequently undermined by structural constraints, societal attitudes, and policy gaps.

Historically, employment options for people with disabilities have been limited, and despite the legal provisions under the Rights of Persons with Disabilities Act (2016), many still struggle to find inclusive workplaces. In such a context, entrepreneurship becomes not only an alternative but also a source of personal dignity and self-reliance. Yet, the journey is fraught with obstacles—ranging from inaccessibility and lack of capital to attitudinal bias and insufficient training opportunities.

This study seeks to move beyond statistics and policies to listen to the voices of those at the center of this reality. It attempts to document their lived experiences, understand their motivations, and, most importantly, analyze the struggles they face while trying to establish and sustain their ventures. The goal is to develop a grounded understanding of their challenges and to offer suggestions that can contribute to a more enabling entrepreneurial environment for the differently-abled in India.

Objectives of the Study

1. To identify and analyze the key challenges faced by differently-abled individuals in starting and sustaining entrepreneurial ventures in India.
2. To explore the socio-cultural and economic factors that influence the entrepreneurial journey of persons with disabilities.
3. To assess the effectiveness of existing government policies, schemes, and support mechanisms aimed at promoting entrepreneurship among the differently-abled.

Percentage Analysis

Table1: Demographic Profile of the Respondents

Particulars		Frequency	Percent
Valid	Below 25 Years	36	14.8
	26-35 Years	90	37.0
	36-45 years	81	33.3
	Above 45 years	36	14.8
	Total	243	100.0
Particulars		Frequency	Percent
Valid	Retail	81	33.3
	Manufacturing	27	11.1
	Services	36	14.8
	Online/IT-based	72	29.6
	Others	27	11.1
	Total	243	100
Valid	Gender	198	81.5
	Female	45	18.5
	Total	243	100
Valid	Locomotor	63	25.9
	Visual Disability	54	22.2
	Hearing Disability	27	11.1
	Others	99	40.7
	Total	243	100

Source: Primary data

The majority of differently-abled entrepreneurs fall within the age group of 26–35 years (37%), followed closely by those aged 36–45 years (33.3%). This indicates that the prime working-age population is more active in entrepreneurial activities, possibly due to greater access to skills, technology, or necessity-driven motivations. The representation of younger (below 25) and older (above 45) entrepreneurs is relatively low, each at 14.8%, reflecting possible barriers such as experience, capital, or health-related issues.

Retail and Online/IT-based businesses are the most common among respondents, accounting for 33.3% and 29.6% respectively. This suggests a trend toward low-investment or digitally driven ventures, which may offer greater flexibility and accessibility for differently-abled entrepreneurs. Traditional sectors like manufacturing are less common (11.1%), possibly due to higher capital requirements and physical infrastructure barriers.

There is a significant gender gap in entrepreneurship among differently-abled individuals. Males constitute 81.5% of the sample, whereas females account for only 18.5%. This disparity reflects broader socio-cultural and economic limitations faced by women with disabilities, including restricted access to resources, mobility challenges, and lower social support.

A diverse range of disabilities is represented. The highest proportion of respondents (40.7%) fall under the “Others” category, which may include intellectual, neurological, or multiple disabilities. Locomotor (25.9%) and visual impairments (22.2%) are the next most common. Hearing disabilities are least represented (11.1%), potentially indicating either underrepresentation or communication barriers in outreach and survey access.

Multilayer Perceptron-Issues and Challenges and Prospects of the Respondents

The Multilayer Perceptron (MLP) model was employed to analyze and predict the issues and challenges faced by respondents based on key input variables such as age, type of business owned, and type of disability. The model architecture included a single hidden layer with five neurons using the hyperbolic tangent activation function, while the output layer applied an identity activation function suitable for continuous predictions. Data was split into training (68.3%) and testing (31.7%) sets to ensure the model’s generalizability. The standardized rescaling method enhanced the model’s ability to handle diverse input scales. Overall, the MLP effectively captured complex relationships among socio-demographic and business-related variables influencing entrepreneurial challenges.

Table 2: Multilayer Perceptron-Issues and Challenges and Prospects of the Respondents- Case Processing Summary

Issues & Challenges		N	Percent
Sample	Training	166	68.3%
	Testing	77	31.7%
Valid		243	100.0%
Excluded		0	
Total		243	

Source: Primary data

Table 2 presents the case processing summary for the Multilayer Perceptron (MLP) model used to analyze the issues, challenges, and prospects faced by the respondents. A total of 243 valid cases were included in the analysis, with no data exclusions. The dataset was split into two subsets: 166 cases (68.3%) were used for training the model, while 77 cases (31.7%) were designated for testing. This standard data partitioning helps in evaluating the model's ability to learn patterns from the training data and to generalize its predictions to unseen testing data. The balanced split ensures a reliable assessment of model performance and helps to minimize the risk of overfitting. These proportions reflect a commonly accepted approach in machine learning modeling, supporting the validity of subsequent results and interpretations.

Table 3: Multilayer Perceptron-Issues and Challenges and Prospects of the Respondents- Network Information

Input Layer	Factors	1	Age
		2	Type of Business Owned
	Covariates	1	Type of Disability
	Number of Units ^a		10
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		5
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	Issueschallenges
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit

Table 3 outlines the network architecture and configuration of the Multilayer Perceptron (MLP) model used to analyze the issues and challenges faced by the respondents. The input layer included three predictor variables: Age, Type of Business Owned, and Type of Disability (treated as a covariate), all of which were rescaled using standardization to ensure comparability and effective model learning. The network featured one hidden layer comprising five units, utilizing the hyperbolic tangent activation function, which is commonly used to handle non-linearity in the data. The output layer consisted of a single dependent variable — Issueschallenges — with the identity activation function applied, suitable for continuous output in regression tasks. The model used the sum of squares error function to evaluate prediction accuracy. Overall, the network design reflects a standard and efficient approach for modeling relationships in complex social and behavioral datasets, enabling the identification of significant patterns related to the respondents' entrepreneurial difficulties and prospects

Figure1 : Multilayer Perceptron Neural Network for Predicting Issues and Challenges

- The figure illustrates the architecture of a Multilayer Perceptron (MLP) model employed to predict the **Issues and Challenges** faced by respondents based on three key inputs: **Age**, **Type of Business Owned**, and **Type of Disability**. The network consists of an **input layer**, a **single hidden layer**, and an **output layer**.
- Input Layer: This layer comprises categorical representations of:

- Age (with 4 coded levels),
 - Type of Business Owned (with 5 coded categories), and
 - Type of Disability (treated as a continuous covariate).
- Each of these inputs is connected to the hidden layer using synaptic weights, with gray lines indicating positive weights and blue lines representing negative weights.
- **Hidden Layer:** This layer contains 5 neurons (H(1:1) to H(1:5)) with the hyperbolic tangent activation function, which allows the model to capture complex, non-linear relationships between inputs and the target variable. The presence of both positive and negative synaptic weights suggests varying degrees of influence from different factors.
 - **Output Layer:** A single output neuron represents the predicted value of the dependent variable, Issueschallenges, using the identity activation function. This is suitable for continuous prediction tasks, such as regression.



The diagram visually depicts how input variables contribute differently to the prediction, with some connections (e.g., Age and business types) showing stronger influence based on the thickness and color of the connecting lines.

Table 4: Model Summary

Model Summary

Training	Sum of Squares Error	11.625
	Relative Error	.141
	Stopping Rule Used	1 consecutive step(s) with no decrease in error ^a
	Training Time	0:00:00.02
Testing	Sum of Squares Error	11.575
	Relative Error	.423

Dependent Variable: Issueschallenges

a. Error computations are based on the testing sample.

Table 5: Explanation of the Model Summary

Metric	Training	Testing	Interpretation
Sum of Squares Error (SSE)	11.625	11.575	Low SSE is generally better. Here, training and testing errors are very close, which suggests the model is not overfitting.
Relative Error	0.141	0.423	Training relative error is low (good), but testing relative error is significantly higher. This suggests that the model generalizes poorly to new data.
Stopping Rule	1 step with no decrease in error	—	The model stopped after only one step with no error reduction, which may indicate a simple model with limited learning.
Training Time	0.02 seconds	—	Very quick — indicates a simple model or small dataset.

Source: Primary data

The neural network diagram illustrates the structure of the Multilayer Perceptron (MLP) used to predict issues and challenges faced by respondents. It includes three main input factors: age, type of business owned, and type of disability. These inputs are connected to a single hidden layer with five neurons, using the hyperbolic tangent activation function to model complex relationships. The output layer, representing the predicted challenges, uses an

identity activation function for continuous outcomes. The color and thickness of the lines indicate the strength and direction of synaptic weights, showing varying influences of input variables.

Suggestions

1. **Targeted Support Based on Demographic Profiles:** The MLP model revealed that age, type of business, and type of disability significantly influence the challenges faced by differently-abled entrepreneurs. Stakeholders such as government agencies, NGOs, and financial institutions should design age-specific and business-specific interventions. For instance, entrepreneurs in the 26–45 age range—who are most active—could benefit from customized skill-building programs, while those above 45 may require health-integrated entrepreneurial support.
2. **Encourage Digitally-Enabled Business Models:** The model indicated a substantial representation of online/IT-based businesses among respondents. This highlights the growing accessibility and appeal of digital entrepreneurship among people with disabilities. Policymakers should expand access to digital infrastructure and training, particularly in semi-urban and rural areas, to further promote inclusive tech-based entrepreneurship.
3. **Integrate Disability-Specific Business Mentorship:** The influence of the type of disability as a covariate suggests differentiated challenges. A one-size-fits-all approach to business development may not be sufficient. Mentorship and incubation programs should therefore be tailored, taking into account the unique barriers faced by visually, hearing, and locomotor-impaired individuals.
4. **Model Refinement for Enhanced Generalization:** While the model performed well during training (Relative Error: 0.141), its generalization to new data was relatively weaker (Testing Relative Error: 0.423). This implies the need for a more nuanced or complex network architecture, larger sample size, or inclusion of additional contextual variables (e.g., access to finance, family support, education level) to improve predictive accuracy.

Conclusion

The application of the Multilayer Perceptron (MLP) model has effectively captured the non-linear and complex relationships between socio-demographic factors and the entrepreneurial challenges faced by differently-abled individuals. With age, type of business, and type of disability playing crucial roles, the model offers valuable insights into how these variables interact and contribute to entrepreneurial strain or resilience.

While the low sum of squares error (SSE) in both training and testing indicates a well-fitted model, the discrepancy in relative error between training (0.141) and testing (0.423) flags a need for cautious interpretation of the model's generalizability. The quick training time also underscores the simplicity of the current model, which, while effective, may benefit from further refinement.

In summary, the MLP model serves not only as a robust analytical tool but also as a strategic framework for designing inclusive entrepreneurial policies. Its findings support the need for adaptive, inclusive, and technology-driven interventions aimed at reducing the barriers faced by differently-abled entrepreneurs and enhancing their long-term business prospects.

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ARTIFICIAL INTELLIGENCE IN EDUCATION AND ITS IMPACT ON TEACHING

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ABSTRACT

This article explores the transformative impact of Artificial Intelligence (AI) on the education sector, with a particular focus on teaching practices. It examines how AI tools are reshaping instructional methods, enhancing efficiency, and enabling personalized learning experiences. The article also assesses the evolving role of educators in an AI-enhanced environment, highlighting the shift from traditional teaching to facilitation and mentorship. While the integration of AI offers numerous benefits, it also presents challenges related to ethics, equity, and teacher preparedness. The article concludes with key findings and actionable suggestions for responsible and effective AI implementation in education.

Keywords: Artificial Intelligence, Teaching Transformation, Personalized Learning

INTRODUCTION

Artificial Intelligence (AI) is rapidly redefining various industries, and education is no exception. With the integration of AI technologies, the education sector is undergoing a significant transformation that influences teaching methodologies, student engagement, curriculum development, and personalized learning. AI tools and platforms are providing innovative solutions to long-standing educational challenges, making learning more accessible, efficient, and tailored to individual student needs.

This article aims to delve into the integration of AI in education, with a particular focus on how it affects the teaching profession. We will examine both the opportunities and challenges posed by AI in educational settings, drawing from current examples and future trends.

SCOPE AND IMPORTANCE OF THE STUDY

This study explores the emerging intersection between Artificial Intelligence (AI) and education, with a particular emphasis on its impact on teaching methodologies and the educator's role. As AI technologies continue to advance, understanding their implications for the classroom is critical for shaping future-ready education systems. The scope includes the examination of AI tools in teaching, their applications in personalized learning, administrative support, and data-driven decision-making. The importance of this study lies in its relevance to educators, policymakers, and academic institutions aiming to adapt to technological changes while maintaining pedagogical integrity. It provides insights into how AI can be effectively integrated to enhance teaching quality, promote equitable access, and improve learning outcomes, while also cautioning against ethical and practical challenges. By assessing both the opportunities and limitations of AI in education, the study contributes to building a balanced, human-centered approach to educational innovation.

OBJECTIVES:

1. To explore the transformative role of Artificial Intelligence (AI) in modern educational practices.
2. To assess the impact of AI on the roles and responsibilities of educators and the overall teaching process.

THE TRANSFORMATIVE ROLE OF ARTIFICIAL INTELLIGENCE (AI) IN MODERN EDUCATIONAL PRACTICES

Artificial Intelligence (AI) has emerged as a transformative force in education, reshaping how knowledge is delivered, accessed, and assessed. The traditional teacher-centered classroom model is giving way to more dynamic, student-centered approaches, largely fueled by AI technologies. These technologies offer a multitude of functions — from automating routine tasks to delivering deeply personalized learning experiences — that were once unimaginable within conventional pedagogical frameworks.

One of the most profound transformations is the shift toward personalized and adaptive learning. AI-driven platforms like Squirrel AI and DreamBox Learning analyze individual learner data in real time, adjusting content delivery based on students' pace, preferences, strengths, and areas of difficulty. This customization ensures that learners receive the right level of challenge, enhancing engagement and improving outcomes.

AI also plays a crucial role in bridging accessibility gaps. With tools like real-time speech-to-text converters, automated sign language interpreters, and personalized content delivery for students with special needs, AI makes inclusive education a reality. It enables students with physical, cognitive, or sensory impairments to participate fully in learning experiences that were once limited or inaccessible.

Moreover, AI has significantly influenced assessment practices. Intelligent systems now provide instant, formative feedback, allowing students to reflect and improve continuously rather than waiting for final evaluations. For educators, AI can detect patterns in student performance, helping them intervene early when learners struggle, thereby supporting proactive teaching methods.

AI's influence extends to curriculum development and content creation. Platforms powered by natural language generation (NLG) can create quizzes, summaries, and even entire lesson plans, freeing up teachers' time to focus on more strategic educational roles like mentorship and counseling. Similarly, AI-assisted simulations and virtual labs offer hands-on experiences in fields like medicine, engineering, and science, enhancing practical knowledge.

Additionally, AI is redefining the role of data in education. With sophisticated data analytics, institutions can track learning trends, predict dropouts, evaluate teaching effectiveness, and design evidence-based interventions. These insights contribute to the development of agile and responsive educational environments.

In summary, AI is not just a technological add-on — it is fundamentally transforming the philosophy, delivery, and evaluation of education. As these tools become more sophisticated and integrated, educators, learners, and policymakers must adapt to harness AI's full potential responsibly and equitably.

AI Tools and Technologies in Education

AI in education is implemented through various tools and applications, including:

- Intelligent Tutoring Systems (ITS) •
- Learning Management Systems (LMS) with AI capabilities •
- Automated grading and assessment tools •
- Natural Language Processing (NLP) for language learning •
- AI-powered educational chatbots •
- Predictive analytics for student performance •
- Virtual and Augmented Reality for immersive learning experiences •

These technologies assist educators by automating administrative tasks, analyzing student data, providing instant feedback, and creating personalized learning paths.

IMPACT OF AI ON TEACHING ROLES AND RESPONSIBILITIES

The integration of Artificial Intelligence (AI) in education has significantly reshaped the roles and responsibilities of educators. Traditionally, teachers have been the primary source of knowledge transmission. However, with AI handling tasks like content delivery, grading, and even answering frequently asked questions, the teacher's role is evolving from that of a knowledge dispenser to a facilitator, guide, and mentor.

One of the most notable impacts is the **reduction of administrative burden**. AI can automate repetitive tasks such as attendance monitoring, assignment grading, and data entry, freeing up educators to focus more on student engagement and instructional quality. Teachers can now invest more time in creative lesson planning, student mentorship, and collaborative projects.

Moreover, AI enhances **data-driven teaching**. Educators gain insights into individual student progress, learning gaps, and behavioral patterns through AI analytics. This empowers teachers to design targeted interventions, provide timely support, and customize learning experiences to meet diverse needs.

AI also encourages **continuous professional development**. As educators adapt to AI tools, they must update their digital literacy and pedagogical strategies. This promotes a culture of lifelong learning among teachers and supports the development of more technologically adept professionals.

However, this transformation also presents challenges. Teachers must balance the use of AI with human-centered teaching values, ensuring that empathy, ethics, and personal interaction are not compromised. Overall, AI augments the educator's role, positioning them as central figures in orchestrating meaningful and personalized learning experiences in a tech-enhanced environment.

Challenges in AI Integration

Despite its advantages, AI in education also presents several challenges:

- **Ethical Concerns:** Privacy issues related to student data collection and analysis.
- **Teacher Training:** Educators need proper training to effectively integrate AI tools into their teaching practices.
- **Accessibility and Equity:** Disparities in access to AI technologies can widen the digital divide.

- **Dependence on Technology:** Overreliance on AI might undermine the human touch essential in teaching.

Future Prospects of AI in Teaching

The future of AI in teaching is promising. AI is expected to further enhance personalized learning, provide real-time language translation for global classrooms, and support students with disabilities through assistive technologies. As AI becomes more sophisticated, it could potentially assist in emotional and social learning, identifying students' emotional states and adapting accordingly.

MAJOR FINDINGS

The integration of Artificial Intelligence in education has yielded several impactful insights. Based on the exploration and analysis in this article, the following are the major findings:

1. **AI is Reshaping Educational Practices**
Artificial Intelligence is driving a paradigm shift from traditional, teacher-led instruction to more personalized and adaptive learning models. AI tools are enabling more flexible, student-centered education through platforms that tailor content to individual needs and learning styles.
2. **Enhanced Efficiency in Teaching**
AI significantly reduces educators' administrative workload by automating tasks like grading, attendance, and lesson planning. This allows teachers to focus more on pedagogy, student mentorship, and creative instruction.
3. **Data-Driven Decision Making in Education**
AI empowers educators with real-time data analytics, enabling them to identify learning gaps, predict student performance, and intervene early. This fosters a more proactive and informed teaching approach.
4. **Changing Role of Educators**
The teacher's role is evolving from content delivery to that of a mentor, facilitator, and learning guide. Teachers are now required to be proficient in technology and adopt blended pedagogical strategies that combine AI tools with human interaction.
5. **Challenges Persist in Implementation**
Despite its benefits, AI integration faces barriers such as ethical concerns regarding data privacy, unequal access to technology, the need for teacher training, and the risk of over-dependence on automation.
6. **Potential for Inclusive and Global Education**
AI holds promise in supporting inclusive education, especially for learners with disabilities, and breaking language barriers through real-time translation, thereby expanding access to quality education globally.

SUGGESTION

To ensure the effective and responsible integration of Artificial Intelligence in education, several strategic measures must be considered. The following suggestions aim to guide educators, institutions, and policymakers in leveraging AI for sustainable educational advancement:

❑ **Invest in Teacher Training and Digital Literacy**

Educational institutions should prioritize continuous professional development programs to help educators effectively integrate AI tools into their teaching practices. This will ensure they are not just users of technology but also strategic facilitators of AI-enhanced learning.

❑ **Promote Ethical and Inclusive AI Implementation**

Policymakers and institutions must ensure AI systems are developed and used ethically, safeguarding student data privacy and promoting equitable access to AI technologies across all socio-economic backgrounds.

❑ **Balance Technology with Human-Centered Pedagogy**

While AI can automate and personalize many aspects of education, it is crucial to preserve the emotional, social, and ethical dimensions of teaching by maintaining strong teacher-student relationships and emphasizing empathy and critical thinking in the curriculum.

CONCLUSION

Artificial Intelligence is a powerful catalyst in reshaping education, particularly in the teaching domain. It presents immense opportunities to personalize education, improve efficiency, and enrich student engagement. However, the successful integration of AI requires careful consideration of ethical, pedagogical, and practical factors. Teachers must be at the center of this transformation, not as mere facilitators of technology, but as leaders who harness AI to enhance the learning experience. As we move forward, a balanced approach that leverages AI's strengths while preserving the irreplaceable human elements of teaching will be crucial in ensuring that education remains both effective and compassionate.

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FLEXIBILITY IN ACADEMIA: CONSTRUCTING SUSTAINABLE FINANCIAL APPROACHES FOR UNIVERSITIES

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Abstract

Entrepreneurial universities have developed as crucial representatives in the sprouting landscape of higher education, integrating academic superiority with innovation, commercialization, and social engagement. This paper discovers the concept of edupreneurship the entrepreneurial direction within educational institutions and its transformative influence on higher education institution missions, structures, and investor associations. Stranded in the Triple Helix model, which highlights collaborative communications among academic institutions, industrial sectors, and government, entrepreneurial universities exceed the conventional roles by nurturing knowledge capitalization through knowledge transmission, startup incubation, and regional development enterprises. By implanting entrepreneurship into academic programs and institutional culture, these universities encourage business attitudes amongst students and faculty, thereby enhancing employability and driving financial growth. The study also scrutinizes the organizational transformation essential for universities to accept entrepreneurial standards, including the creation of enthusiastic provision structures like technology transfer offices and science parks. Through a complete literature review and analysis of global case studies, the paper describes the multi-layered magnitudes of edupreneurship, prominence its role in positioning academic pursuits with market and social requirements. The discoveries emphasize the requirement for higher education institutions to hold entrepreneurial approaches to remain applicable and impactful in a knowledge-driven economy.

Keywords: Entrepreneurial university, Edupreneurship, Triple Helix model, Innovation, Higher education transformation.

1. Introduction

Higher educational institutes or universities are progressively documented as crucial players in the knowledge economy, contributing not only to education and research but also to regional development and novelty environments [1] [2]. This shift is concise in the awareness of the "entrepreneurial university," which integrates economic development into the academic mission alongside teaching and research [3]. Such institutions actively substitute modernization, provision startup ecosystems, and participate in technology transfer activities, thereby attractive their societal impact and monetary sustainability [3]. The financial landscape of higher education is undergoing significant strain. The Triple Helix model, conceptualized by Etzkowitz and Leydesdorff, emphasizes the harmonious teamwork among academia, industry, and government to drive origination and economic development [4]. In this background, universities play a central role in knowledge production and dissemination, while industry and government provide resources and policy support [5]. By fostering such collaborations, universities can enhance their research impact, simplify technology transfer, and donate to regional innovation systems [6]. To accomplish economic sustainability, universities are reconnoitring various commercial activities:

- **Technology Transfer Offices (TTOs):** TTOs simplify the commercialization of university research through licensing and the creation of spin-off companies, generating revenue and promoting innovation [7].
- **Green Revolving Funds (GRFs):** GRFs provide capital for sustainability projects that yield cost savings, with the savings reinvested into the fund, creating a self-sustaining financial model [8]
- **Continuing Education and Online Programs:** Expanding offerings in professional development and online education can attract non-traditional students and generate additional income [8]
- **Industry Partnerships:** Collaborations with industry can lead to funded research projects, internships, and joint ventures that benefit both the university and its partners [9].

Transitioning to an entrepreneurial model requires organizational changes and a cultural shift within the university:

- **Leadership and Governance:** Strong leadership is essential to drive the entrepreneurial agenda and align institutional goals with innovation strategies.
- **Faculty Engagement:** Encouraging faculty participation in entrepreneurial activities through incentives and support structures can foster a culture of innovation.
- **Curriculum Development:** Integrating entrepreneurship and innovation into the curriculum prepares students for the dynamic demands of the modern workforce [9].

2. The Concept of Edupreneurship

Edupreneurship refers to the entrepreneurial orientation within educational institutions, where universities integrate innovation, commercialization, and societal engagement into their core missions [10]. This approach encourages the development of entrepreneurial mindsets among students and faculty, fostering an environment conducive to creativity, risk-taking, and the pursuit of new ventures [11]. By embedding entrepreneurship into curricula and institutional culture, universities can enhance employability and contribute to economic growth [12].

3. The Triple Helix Model

The Triple Helix model, conceptualized by Etzkowitz and Leydesdorff, emphasizes the collaborative interactions among academia, industry, and government to foster innovation and economic development. In this model, universities play a pivotal role in knowledge production and dissemination, while industry and government provide the necessary resources and policy frameworks to support innovation. The dynamic interplay among these three spheres leads to the creation of hybrid institutions, such as technology transfer offices and science parks, which facilitate the commercialization of academic research [13].

4. Organizational Transformation in Universities

Transitioning to an entrepreneurial university requires significant organizational changes, including the establishment of dedicated support structures like technology transfer offices, incubators, and science parks [14]. These entities serve as bridges between academia and industry, enabling the effective transfer of knowledge and technology. Additionally,

universities must foster a culture that encourages innovation and entrepreneurship, providing incentives and support for faculty and students to engage in entrepreneurial activities [15].

5. Financial Sustainability in Higher Education

Financial sustainability is a critical concern for higher education institutions in the current economic climate. Universities must diversify their revenue streams beyond traditional sources such as tuition fees and government funding [16]. Approaches for accomplishing financial sustainability include:

- **Diversification of Funding Sources:** Attractive in partnerships with industry, securing research grants, and offering continuing education programs can provide additional revenue.
- **Commercialization of Research:** Licensing technologies, creating spin-off companies, and consulting services can generate income from academic research [9].
- **Cost Efficiency Measures:** Implementing effective financial management practices to optimize resource allocation and reduce unnecessary expenditures [17].

6. Case Studies of Entrepreneurial Universities

Several universities have successfully adopted entrepreneurial models:

- **Massachusetts Institute of Technology (MIT):** Renowned for its strong emphasis on innovation and entrepreneurship, MIT has established numerous research centers and incubators to support startup ventures [18] [19].
- **Stanford University:** Located in Silicon Valley, Stanford has fostered close ties with the tech industry, leading to the creation of numerous successful startups and a robust entrepreneurial ecosystem [20].
- **University of Twente:** Known as the "entrepreneurial university" in the Netherlands, it integrates entrepreneurship into its education and research activities, supporting students and staff in launching new ventures [21].

These institutions exemplify how universities can transform into entrepreneurial entities by fostering innovation, engaging with industry, and adopting flexible financial strategies.

7. Challenges and Considerations

While the shift towards entrepreneurial universities offers numerous benefits, it also presents challenges:

- **Balancing Academic and Commercial Goals:** Ensuring that the pursuit of commercialization does not compromise academic integrity and the core mission of education [22].
- **Resource Allocation:** Effectively managing resources to support both traditional academic activities and entrepreneurial initiatives [23].
- **Cultural Shift:** Cultivating an entrepreneurial mindset among faculty and students may require significant cultural changes within the institution [24].

Addressing these challenges requires thoughtful leadership, clear policies, and ongoing support for entrepreneurial activities [25].

8. Conclusion

The transformation of universities into entrepreneurial entities is essential for their continued relevance and impact in a knowledge-driven economy. By embracing edupreneurship and adopting flexible, sustainable financial approaches, higher education institutions can enhance their contributions to innovation, economic development, and societal well-being. This evolution necessitates a strategic vision, organizational adaptability, and a commitment to fostering an entrepreneurial culture within academia.

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A STUDY OF EDUCATORS OF HIGHER EDUCATION ON DIGITAL COMPETENCE AND LIFE SKILLS IN INDIA

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Abstract

Similar to how the Industrial Revolution signaled the beginning of the Industrial Age, the Digital Revolution is linked to the beginning of the Information Age. Technology has changed and is changing quickly again. These revolutionary technological advancements have a significant impact on each person's lives. The 21st century has only lasted 18 years, yet the advancement of digital technology has been astounding; cellphones, the Internet, and digital communication technologies have all become widely used and have become the most effective means of communication and information sharing worldwide. According to a report of Internet and Mobile Association of India (IAMAI) published in Economic Times in February 2018, India has a digital population of 481 million active internet users in December 2017, which is expected to reach 500 million by June 2018. This digital revolution seems to be highly necessary in developing countries like India where students are abundant and teachers are relatively scarce. Being of rapid technological change, the Indian education system is also facing many new challenges and providing various new educational opportunities.

Key words: Industrial Age, Digital Age, ICT

Introduction

Similar to how the Industrial Revolution signaled the beginning of the Industrial Age, the Digital Revolution is linked to the beginning of the Information Age. Technology has changed and is changing quickly again. These revolutionary technological advancements have a significant impact on each person's lives. The 21st century has only lasted 18 years, yet the advancement of digital technology has been astounding; cellphones, the Internet, and digital communication technologies have all become widely used and have become the most effective means of communication and information sharing worldwide. According to a report of Internet and Mobile Association of India (IAMAI) published in Economic Times in February 2018, India has a digital population of 481 million active internet users in December 2017, which is expected to reach 500 million by June 2018. This digital revolution seems to be highly necessary in developing countries like India where students are abundant and teachers are relatively scarce. Being of rapid technological change, the Indian education system is also facing many new challenges and providing various new educational opportunities. New millennium experiences the phenomena of liberalization, privatization, globalization and easy access expediting the emergence of digitalization of education, which is very important for Indian education system and there is a swift in setting the preferential goals of education. These digital tools have the ability to expand the reach of educational activities beyond the confines of educational institutions and personalize the learning experiences. Technology-assisted learning is high-tech, substantive, content-focused, and accommodates individual differences by offering individualized learning opportunities.

Digital Education initiatives taken by Government of India:

1. Swayam (Study Webs of Active–Learning for Young Aspiring Minds) is an indigenously designed massive open online course (MOOC), launched in 9th July 2017 with an aim to “bridge the digital divide for students who have hitherto remained untouched by the digital revolution and have not been able to join the mainstream of the knowledge economy.”
2. SWAYAM Prabha, is another digital education initiative launched along with SWAYAM in 2017, aims to provide high quality educational contents to learners by using satellite technology to reach the unreached. Here, experts develop the educational content, which covers diverse disciplines in different languages and levels of education.
3. National Academic Depository is a digital depository, a 24 x 7 online storehouse of all academic awards i.e. certificates, diplomas, degrees, mark-sheets etc. issued by universities, higher education institutions and school boards. It ensures easy access and retrieval of academic awards as well as their validation and authentication along with secure storage. Government has authorised University Grants Commission (UGC) to enter into a tripartite agreement on behalf of all Central Higher Educational Institutions (CHEIs), Institutions of National Importance (INIs) and Boards with the wholly owned subsidiaries of the depositories i.e. NSDL Database Management Limited (NDML) and CDSL Ventures Limited (CVL) comprising of two interoperable digital depositories.
4. National Digital Library of India has been initiated by the Ministry of Human Resource Development (MHRD) under National Mission on Education through Information and Communication Technology (NMEICT) in 2015 and has been made operational on February 2016. It was started with the aim to provide single window search and browse facility for elearning resources to users, integrating e Contents that are available in various institutions of the country by integrating Institutional Digital Repository (IDR) of respective institutions and registering users from participating institutions.
5. INFLIBNET (Information and Library Network) centre, located in Gandhinagar is an Autonomous Inter-University Centre (IUC) was an initiative of UGC to build high speed Nationwide Communication Network for Indian Universities using the state-of-art technologies for the optimum utilization of information. INFLIBNET is involved in modernizing university libraries in India and connecting them through high-speed data network.
6. Other than e-ShodhSindhu: Consortia for Higher Education E-Resources, INFLIBNET centre, on behalf of the UGC, has taken-up various new initiatives to benefit the academia. The major initiatives include
 - ❖ Shodhganga - a reservoir of full text Indian Theses and Dissertations submitted to various Indian Universities
 - ❖ Shodhgangotri - a repository of approved synopsis submitted to the universities for registration under Ph.D. programme;
 - ❖ IndCat - online union catalogue of Indian universities
 - ❖ OJAS@INFLIBNET- An open journal Academic system; it is an open access journal publishing platform.
 - ❖ Open source software R & D.
 - ❖ InfoPort - A comprehensive gateway to all Indian electronic scholarly content.
 - ❖ e-PG Pathshala - e-Content for PG Courses
 - ❖ to measure Research Output of Indian Universities; and

❖ SOUL 2.0 - an Integrated Library Management System

7. NPTEL (National Programme on Technology Enhanced Learning, India, 2007) is a platform, which offers e-learning through online Web based and Video courses in Engineering, Science and Humanities. It is an initiative of seven Indian Institutes of Technology (IITs) namely Bombay, Delhi, Guwahati, Kanpur, Kharagpur, Madras and Roorkee and Indian Institute of Science (IISc). Contents provided in the above courses are based on the curriculum suggested by All India Council for Technical Education (AICTE) and major affiliating Universities of India. The video courses are broadcast through Eklavya channel, a 24/7 television channel provided by Doordarshan, exclusively for educational broadcasts as well as could be stream with the help of Google and YouTube.

Other than, the above discussed digital education initiatives there are many more programs launched by the government to make out the maximum benefit from the use of technology in the field of education. Some of these projects are Virtual labs, Talk to Teacher, Spoken Tutorial, E-Yantra, Indian Sign Language Education and Recognition Platform (ISLERS), Open Source Courseware Animations Repository (OSCAR), Fosse, Creating Digital-learning Environment for Design (e-Kalpa), Pedagogy Project, Virtual Learning Environment etc. (MHRD, 2018)

OBJECTIVES OF THE STUDY:

- 1) To study digital competence of higher education teachers.
- 2) To study digital competence of higher education teachers with reference to various dimensions of digital competence
- 3) To study the impact of digital competence on Life skills of higher education teachers.

Data Analysis:

Data Analysis has multiple facets and approaches, encompassing diverse techniques under a variety of names, indifferent business, science and social science domains. The analysis and interpretation of data holds chief implication because the gist of the research study depends on its.

Table 1

Frequency and percentage of respondents use of various devices to access technology

S.No.	Device Used	Frequency	Percentage
1.	Desktop	45	10
2.	Smartphone	92	20.44
3.	Laptop	104	23.11
4.	Depends on situation	209	46.45
	Total	450	100

From the above table it can be seen that most of the respondents use either laptop or smartphone or desktop as per the need of the situation i.e 46.45% where as 23.11 use laptop and 20.44% of the respondents use smartphone. Only 10% of the respondents use desktop. Although most of the respondents use the device based on the need of the situation, however being easy to carry and could be used on the go laptops and the smartphones are also a good choice among respondents.

Table 2

Frequency and percentage of respondents working frequency on Computer

S.No	Working frequency	Frequency	Percentage
1.	Everyday	340	75.57
2.	More than once a week	38	8.44
3.	Once a week	10	2.22
4.	Occasionally	62	13.77
5.	Don't use	0	0
	Total	450	100

The above table clearly depicts that most of the respondents use to work on computer and other related devices everyday i.e 75.57% where as 13.77% higher education teachers use it occasionally and 8.44% use these devices more than once a week and only 2.22% use it only once a week. However none of the respondents were found to not using it at all.

Table 3

Frequency and percentage of respondent's methods to acquire digital technology skills

S.No	Working frequency	Frequency	Percentage
1.	Trial and error	194	43.12
2.	Guidance for parents, teachers or peers	87	19.33
3.	Learning as a subject in school/college	65	14.44
4.	Attending a certificate course/taking up a training program or workshop	104	23.11
	Total	450	100

It can be seen from the above table that 43.12% of the respondents have acquired the digital technology skills through trial and error method. 23.11% of higher education teachers have attended a certificate course, training program or workshop to acquire these skills. Whereas 19.33% of higher education teachers have acquired digital technology skills through guidance from parents, teachers or peers; and 14.44% learnt it as a subject in school and college.

Table 3

Gender wise mean S.D and t-Value of digital competence of Higher education teachers

S.No	Gender	N	Mean	S.D	Value of t-Test
1.	Male	182	26.93	12.11	2.72
2.	Female	268	29.96	11.11	

Result- from Table it can be seen that mean score of higher education teachers of both the gender is of the range of intermediate level of digital competence. The mean digital competence score of female higher education teachers is 29.96 and that of male higher education teachers 26.93.

From the examination of Table it could also be observed that the t-value is 2.72 which is found to be significant at the $p > 0.05$ level [$df = 366$, $p = 0.00682$ and $t_{critical} = 1.96$]. The obtained t-value 2.72 is greater than the table value at 0.05 level of significance 1.96 reflects that there exists a significant difference between mean score of digital competences of male and female higher education teachers.

Thus, the null hypothesis that there is no significant difference in the mean score of Digital Competence of male and female higher education teachers is not accepted.

It may therefore be concluded that female higher education teachers were found to have better digital competence than male higher education teachers.

Conclusion:

Digital competence extends beyond the functional use of ICT, it includes problem-solving, collaboration and creativity. In today's ever-changing global society, it is inevitable that there will be changes in the educational system. ICT has become the centre of learning and it has the potential to drive innovative effective ways of teaching-learning and research; integration of multimedia, simulation tools, digital learning tools, online research tools, digital libraries; easy and instant access of information generates greater opportunities which were otherwise not feasible. They need to develop critical evaluation skills, constructive online interactive and collaborative competencies, and creative potential, which are necessary for the successful 21st century digital citizenship; we may hop for a better generation with humanizing technology.

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HYBRID LEARNING: ENHANCING EDUCATION IN THE DIGITAL AGE

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Abstract

Hybrid learning, a blend of traditional face-to-face instruction and online education, has emerged as a transformative model in modern pedagogy. This paper explores the evolution, benefits, challenges, and future potential of hybrid learning. With technological advancements like artificial intelligence, adaptive platforms, and virtual reality reshaping the educational landscape, hybrid learning offers increased flexibility, personalized learning experiences, and broader access. Pedagogical models such as flipped classrooms and competency-based education support active engagement and deeper learning. However, hybrid learning is not without challenges, including digital inequity, teacher preparedness, and maintaining student motivation. Through analysis of case studies, technological tools, and strategic implementation methods, this study demonstrates that hybrid learning not only enhances educational outcomes but also equips students with essential digital skills for the future.

Keywords: Hybrid learning, blended learning, online education, digital literacy, educational technology

Introduction

The landscape of education has undergone a profound transformation with the integration of technology into teaching and learning processes. Hybrid learning, also referred to as blended learning, combines the strengths of traditional in-person instruction with the flexibility and innovation of online education. This model supports diverse learning needs and paces, creating a more inclusive and adaptable education system.

Hybrid learning is a teaching method that blends in-person classroom instruction with online components, enabling students to engage either on-site or remotely—often at the same time. This approach offers flexibility for both educators and learners in delivering and accessing course materials.

Key Features of Hybrid Learning:

- Lessons are tailored to accommodate both in-person and online students.
- Classes may be conducted **live**, with some students physically present and others joining via video conferencing.
- Learning materials (videos, readings, quizzes) are often available **online** for access anytime.
- Encourages **interaction** among all students, regardless of location.

Example of Hybrid Learning:

Course: B.Com – Financial Accounting Class

- A professor is teaching a chapter on "Cash Flow Statements."
- 20 students attend the lecture in the classroom.
- 10 students join through Google Meet from their homes.
- The lecture is live-streamed, and students (both online and offline) can ask questions in real-time.
- Assignments and quizzes are posted on Google Classroom.
- A recording of the session is uploaded later for revision

Result: Everyone learns the same content at the same time, with flexibility in **where** and **how** they attend.

The concept of hybrid learning predates the COVID-19 pandemic, but the global crisis accelerated its adoption and innovation. Institutions were compelled to shift rapidly to remote learning, revealing the strengths and shortcomings of existing systems. As a result, hybrid learning has emerged not only as a contingency solution but as a sustainable, forward-thinking approach to education.

This paper investigates how hybrid learning enhances educational systems by offering flexibility, fostering engagement, supporting personalized instruction, and addressing global challenges such as access and equity. It also discusses key models, pedagogical strategies, and technological tools essential for effective implementation.

Definition and Models of Hybrid Learning

Hybrid learning is a teaching method that integrates classroom-based learning with online educational activities. Unlike fully online or traditional classroom-based education, hybrid models balance synchronous and asynchronous methods, allowing for flexibility while retaining the benefits of face-to-face interaction.

Common Models of Hybrid Learning:

1. **Rotation Model:** Students alternate between online and in-person instruction based on a set schedule.
2. **Flex Model:** Most content is delivered online, with in-person support offered as needed.
3. **Enriched Virtual Model:** Learners engage primarily online but are required to attend occasional in-person sessions.
4. **Flipped Classroom:** Students learn theoretical concepts online and use classroom time for hands-on activities and deeper discussions.

Hybrid vs. Blended Learning

Although often used interchangeably, hybrid and blended learning differ slightly. In hybrid learning, students have the option to choose between attending in person or online, catering especially to remote or international learners. In blended learning, all students engage with both online and offline components in a structured format. What differentiates this approach is the degree of flexibility and the extent to which learners can choose how they participate

Hybrid vs. Blended Learning – Key Differences

Feature	Blended Learning	Hybrid Learning
Definition	Combines online and face-to-face learning in a planned manner	Integrates in-person and online components simultaneously.
Mode of Delivery	Online and offline components are used sequentially .	Online and offline components happen at the same time .

Flexibility	Less flexible — students are expected to be on-site at times.	More flexible — allows participation from remote learners.
Student Location	Most students are physically present during face-to-face parts.	Students may be split — some attend online, others in person.

Benefits of Hybrid Learning

Enhanced Accessibility and Flexibility

Hybrid learning removes geographical barriers and supports learners with diverse needs. It offers flexibility for working professionals, parents, and students with health issues or disabilities. By enabling anytime-anywhere learning, it aligns with the lifestyles of modern students.

Personalized Learning Experience

Digital platforms enable adaptive learning paths, where AI-driven analytics assess student progress and recommend resources tailored to individual learning gaps. This customization leads to higher retention and satisfaction rates.

Improved Student Engagement

Interactive tools like gamified quizzes, multimedia content, virtual simulations, and real-time feedback increase student engagement. Collaborative platforms foster peer interaction, while forums and discussion boards extend classroom dialogue beyond scheduled hours.

Cost-Effectiveness for Institutions

Institutions benefit from reduced infrastructure costs. Online materials are reusable, and large cohorts can be accommodated without expanding physical facilities. This model is particularly beneficial in densely populated or rural areas where resources are limited.

Development of Digital Literacy

Hybrid learning promotes the development of digital skills, such as navigating learning management systems (LMS), using collaboration tools (e.g., Zoom, Microsoft Teams), and accessing digital libraries. These competencies are crucial for 21st-century careers.

Challenges of Hybrid Learning

Digital Divide and Accessibility

Unequal access to internet connectivity and digital devices continues to hinder the success of hybrid learning. Students in remote areas or from disadvantaged backgrounds often face connectivity issues, affecting their academic performance.

Teacher Preparedness and Training

Educators must adapt to new technologies and pedagogical strategies. Without proper training, teachers may struggle to manage hybrid classrooms, design engaging online content, and maintain student motivation.

Student Motivation and Self-Discipline

Hybrid learning requires students to be self-directed and organized. Lack of immediate supervision in asynchronous components can lead to procrastination and disengagement, particularly among younger or less motivated learners.

Assessment and Academic Integrity

Ensuring fair assessment in hybrid settings is complex. Online assessments are vulnerable to cheating and technical issues. Institutions must implement robust proctoring systems and develop authentic assessments like portfolios and project-based evaluations.

Strategies for Effective Hybrid Learning Implementation

Institutional Support and Infrastructure

Investment in reliable LMS (e.g., Moodle, Blackboard, Canvas) and broadband connectivity is essential. Institutions must provide technical support, device lending programs, and access to digital content libraries.

Professional Development for Educators

Continuous training in hybrid pedagogy, digital tools, and course design is critical. Workshops, peer mentoring, and certification programs can empower educators to deliver high-quality hybrid instruction.

Student-Centric Learning Design

Courses should include clear learning objectives, modular content, and diverse media formats. Microlearning, podcasts, quizzes, and discussion forums promote active learning and better content retention.

Hybrid Assessment Methods

A combination of online quizzes, reflective journals, group projects, and face-to-face exams ensures balanced evaluation. Peer reviews and rubrics provide transparency and encourage critical thinking.

Feedback and Continuous Improvement

LMS analytics help track participation and performance. Regular student feedback surveys can guide course refinement. Iterative improvements foster an adaptive and responsive learning environment.

Case Studies of Successful Hybrid Learning

Harvard University

Harvard Extension School's hybrid programs allow students to study online and attend optional on-campus intensives. This model has increased access for working professionals and international students while maintaining academic standards.

Singapore's Smart Nation Education Policy

Singapore integrates e-learning platforms like the Student Learning Space (SLS) in schools. This initiative supports flipped learning and self-paced study, particularly effective in STEM education.

India's SWAYAM Platform

The Ministry of Education in India launched SWAYAM to offer hybrid courses across disciplines. It provides MOOCs with online lectures and offline examinations, increasing outreach to rural students.

Corporate Training in IBM and Google

These companies use hybrid learning for employee upskilling. Online modules combined with live sessions and on-site labs enhance knowledge retention and application.

Future Trends in Hybrid Learning

AI and Adaptive Learning

AI algorithms personalize learning paths and recommend content based on performance analytics. Chatbots and virtual tutors provide real-time support.

Virtual and Augmented Reality

VR/AR enable immersive learning in fields like medicine, engineering, and history. Simulations improve conceptual understanding and skill development.

Blockchain Credentialing

Blockchain technology secures digital certificates and transcripts, ensuring credibility and ease of verification for employers and institutions.

Global Collaborative Learning

Hybrid platforms support international projects, virtual exchange programs, and cross-border collaborations, enriching learning through cultural diversity.

Prominent Hybrid Learning Portals available in India

Hybrid Learning Portal	Benefits / Key Features
SWAYAM (Study Webs of Active-Learning for Young Aspiring Minds)	Free online courses by Indian universities - Blended learning with physical exams - Certification options
DIKSHA (Digital Infrastructure for Knowledge Sharing)	Resources for school teachers and students - Multilingual content - Integration with offline learning
Unacademy	- Live classes and recorded content - Community-based learning - Test series & assignments
Vedantu	- Global university content with Indian-specific access - Hands-on hybrid skill-building courses - Certification for employability
Coursera (India Access)	- Global university content with Indian-specific access - Hands-on hybrid skill-building courses - Certification for employability
NIOS (National Institute of Open Schooling)	- Flexible schooling options - Printed materials + online support - Ideal for remote learners

Amrita AHEAD (Amrita University)	- Hybrid degree and diploma programs - On-campus contact sessions + online modules - UGC-recognized degrees
EdX (India)	- Hybrid courses from IITs and global institutions - Self-paced & instructor-led options - Skill-based learning

Growth of Hybrid Learning in Indian Higher Education (2019–2024)

Year	Estimated adoption rate of Hybrid Learning in Higher Education
2019	~10% (primarily distance learning and early adopters)
2020	~25% (accelerated by COVID-19 lockdowns)
2021	~40% (formal UGC guidelines and NEP 2020 implementation)
2022	~50% (expansion of platforms like SWAYAM and DIKSHA)
2023	~60% (UGC projection for HEIs integrating blended learning)
2024	~65%+ (continued institutional adoption and student demand)

8. Conclusion

Hybrid learning is reshaping education by offering flexibility, personalization, and global accessibility. While challenges such as the digital divide and teacher readiness must be addressed, the potential of hybrid learning to revolutionize education is undeniable. With strategic implementation and continuous innovation, hybrid learning can provide equitable, engaging, and effective education for diverse learners. The future of education lies in adaptive, technology-driven models that respond to learners' needs and global challenges. Hybrid learning is not merely a trend but a foundational shift towards lifelong, accessible learning.

IMPACT OF THE NATIONAL EDUCATION POLICY 2020 ON LEGAL EDUCATION IN INDIA

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Abstract

The National Education Policy (NEP) 2020 marks a paradigm shift in India's approach to education, emphasizing multidisciplinary learning, inclusivity, technological integration, and global competitiveness. This article critically examines the policy's implications for legal education in India, highlighting its transformative potential. The NEP envisions a legal education system that is holistic, flexible, and rooted in constitutional values and Indian knowledge systems. Key reforms include curriculum integration with other disciplines, the promotion of research and innovation, digitization of learning, and the introduction of the Academic Bank of Credits. The policy also emphasizes faculty development, institutional restructuring, and international collaborations. While the NEP presents significant opportunities to modernize and democratize legal education, it also faces challenges related to implementation, resource allocation, and regulatory coordination. This article concludes that, with effective execution, the NEP 2020 can reshape legal education in India to produce socially responsible, globally competent, and ethically grounded legal professionals.

Keywords: National Education Policy 2020, Legal education in India, Academic Bank of Credits, Indian knowledge systems, Constitutional values, Technology in legal education, Legal research and innovation

Introduction

The **National Education Policy (NEP) 2020** of India has far-reaching implications for legal education. It aims to transform the overall landscape of education and has a significant impact on how legal education is structured, delivered, and perceived. Legal education plays a pivotal role in upholding justice, democratic governance, and the rule of law. In India, the system of legal education has traditionally been rigid and compartmentalized, with limited integration across disciplines and outdated pedagogical practices. The **National Education Policy (NEP) 2020**, the first major education reform in over three decades, seeks to transform the Indian education landscape through sweeping changes that aim to make it holistic, multidisciplinary, inclusive, and globally aligned.

For legal education, NEP 2020 brings forth a new vision—one that not only enhances professional competence but also ensures that future legal practitioners are well-versed in ethics, social justice, and the dynamic needs of a 21st-century society. This article delves into the multifaceted impact of NEP 2020 on legal education in India, outlining the benefits, opportunities, and challenges that lie ahead.

Emphasis on Multidisciplinary Education

NEP promotes the establishment of multidisciplinary institutions, enabling law students to take courses in humanities, sciences, business, and technology.

Legal education is no longer siloed; students can better understand the socio-economic, technological, and cultural contexts of the law. Law schools are encouraged to become part of broader multidisciplinary universities.

Example: A law student might study data privacy laws along with computer science to understand the legal implications of algorithms and data protection.

Integration of Indian Knowledge Systems and Constitutional Values

NEP emphasizes the importance of instilling knowledge of the Indian Constitution, legal traditions, and justice systems. Law students are to be trained in values like equity, justice, and respect for pluralism. There's a push towards including ancient Indian legal systems and jurisprudence in curricula.

Use of Technology and Online Education

Digital platforms are encouraged for teaching, research, and outreach. Legal education can leverage tools like MOOCs (Massive Open Online Courses), virtual moot courts, and AI-based learning tools. Enhances access to legal education across regions, particularly for students in remote areas.

Example: Virtual internships and online legal clinics may become more prominent.

Focus on Research and Innovation

NEP promotes a research-driven academic environment. Law schools are encouraged to establish research centers focusing on contemporary legal issues like cyber law, climate change law, and AI & law. Encourages critical thinking and problem-solving skills.

Flexibility in Curriculum and Credit-Based System

The introduction of the **Academic Bank of Credits (ABC)** system allows students to earn and transfer credits across institutions. Greater flexibility to explore specialized areas of law through elective courses. Promotes lifelong learning and continuous upskilling.

Faculty Development and Quality Standards

NEP recommends regular training and upskilling of law faculty. Encourages law teachers to engage in interdisciplinary research and policy-making. Focus on faculty diversity and inclusion.

Enhancing Access and Equity

The NEP emphasizes increasing access to legal education for marginalized and underrepresented communities. Scholarships, inclusive curricula, and flexible learning paths aim to reduce socio-economic disparities.

Global Exposure and Internationalization

NEP supports international collaboration in legal research and student exchange programs. Law students may benefit from dual degree programs or credit exchange with foreign universities.

Challenges in Implementation

Resistance to change in traditional legal institutions.

Ensuring uniform quality across legal education providers.

Balancing the inclusion of Indian traditions with modern global legal standards.

Conclusion

The NEP 2020 seeks to transform legal education from being exam-oriented and rigid to being multidisciplinary, flexible, and socially relevant. It envisions producing "engaged, productive, and contributing citizens" who are well-versed in both Indian legal ethos and global legal systems. If implemented effectively, this could mark a significant shift towards a more dynamic and inclusive legal education system in India. The National Education Policy 2020 presents a transformative vision for legal education in India, aiming to align it with the needs of a rapidly evolving society, economy, and global order. By fostering multidisciplinary learning, emphasizing constitutional values, integrating Indian knowledge systems, and leveraging technology, NEP 2020 redefines how law is taught and practiced.

The introduction of flexibility through the Academic Bank of Credits, the focus on research and innovation, and the push for inclusivity and global engagement collectively promise to make legal education more accessible, dynamic, and socially relevant. However, the successful realization of this vision depends on overcoming critical challenges, including infrastructural constraints, faculty readiness, regulatory alignment, and resistance to change. Institutions must commit to reforming their structures, pedagogy, and governance models to fully embrace the NEP's goals.

Ultimately, if implemented with strategic intent and sustained support, NEP 2020 can transform Indian law schools into vibrant centers of knowledge, ethics, and public service—producing legal professionals who are not only skilled advocates but also socially conscious leaders of change

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AI AND MACHINE LEARNING APPLICATION IN EDUCATION

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Abstract

The present discussion examines the transformative impact of Artificial Intelligence (AI) in educational settings, focusing on the necessity for AI literacy, prompt engineering proficiency, and enhanced critical thinking skills. The introduction of AI into education marks a significant departure from conventional teaching methods, offering personalized learning and support for diverse educational requirements, including students with special needs. However, this integration presents challenges, including the need for comprehensive educator training and curriculum adaptation to align with societal structures. AI literacy is identified as crucial, encompassing an understanding of AI technologies and their broader societal impacts. Prompt engineering is highlighted as a key skill for eliciting specific responses from AI systems, thereby enriching educational experiences and promoting critical thinking. There is detailed analysis of strategies for embedding these skills within educational curricula and pedagogical practices. This is discussed through a case-study based on a Swiss university and a narrative literature review, followed by practical suggestions of how to implement AI in the classroom.

Keyword: Personalized learning, Adaptive learning, Intelligent tutoring system, predictive analytics, Data driven insights.

Introduction

In the evolving landscape of education, the integration of Artificial Intelligence (AI) represents a transformative shift, stipulating a new era in learning and teaching methodologies. This article delves into the multifaceted role of AI in the classroom, focusing particularly on the primacy of prompt engineering, AI literacy, and the cultivation of critical thinking skills.

The advent of AI in educational settings transcends mere technological advancement, reshaping the educational experience at its core. AI's role extends beyond traditional teaching methods, offering personalized learning experiences and supporting a diverse range of educational needs. It enhances educational processes, developing essential skills such as computational and critical thinking, intricately linked to machine learning and educational robotics. Furthermore, AI has shown significant promise in providing timely interventions for children with special educational needs, enriching both their learning experiences and daily life. However, integrating AI into education is not without its challenges. It requires a systematic approach that takes into account societal structural conditions. Beyond algorithmic thinking, AI in education demands a focus on creativity and technology fluency to foster innovation and critical thought.

Intelligent Tutoring System

An intelligent tutoring system (ITS) is a computer-based learning environment that uses artificial intelligence to provide personalized instruction and feedback to learners, much like

a human tutor. The intelligence of such systems lies in their ability to understand and adapt to the needs of individual students. They do this through student modelling, which tracks a learner's progress, knowledge level, and misconceptions. Using this information, the system can dynamically adjust the difficulty of tasks, offer targeted feedback, and suggest helpful hints. Intelligent tutoring systems also incorporate domain knowledge to ensure accurate and effective content delivery, and some use natural language processing to interact with students in a more human-like, conversational manner. These features allow ITS to provide a customized learning experience, making education more engaging and effective for students across different skill levels and subjects.

Ai Tools In Enhancing Students Studies

AI-powered learning platforms like Khan Academy, Duolingo, or Quizlet use machine learning algorithms to tailor practice exercises and recommend content based on a student's performance. Intelligent tutoring systems (ITS) offer one-on-one support, giving real-time feedback and hints, like a human tutor. Chatbots and virtual assistants, such as ChatGPT, help students by answering questions, explaining concepts, or assisting with writing and research. AI-driven analytics tools can also track student progress, identify learning gaps, and help teachers adjust instruction accordingly. Additionally, speech recognition and language processing tools aid students with reading, writing, or learning new languages. Overall, AI tools support more efficient, interactive, and customized learning experiences that can greatly enhance academic success.

The Emergence of AI In Education

The introduction of ChatGPT (to date one of the most powerful AI chatbots by OpenAI) is significantly transforming the landscape of education, marking a new era in how learning is approached and delivered. This advanced AI tool has redefined educational paradigms, offering a level of personalization in learning that was previously unattainable. ChatGPT, with its sophisticated language processing capabilities, is quickly becoming a game-changer in classrooms, to provide tailored educational experiences that cater to the unique needs, strengths, and weaknesses of each student. This shift from traditional, uniform teaching methods to highly individualized learning strategies will most likely signify a major advancement in educational practices. ChatGPT's role in personalizing education is particularly noteworthy. By analysing student data and employing advanced algorithms, GPT and other Large Language Models (LLMs) can create customized learning experiences, adapting not only to academic requirements but also to each student's learning style, pace, and preferences. This leads to a more dynamic and effective educational environment, where students are actively engaged and involved in their learning journey, rather than being mere passive recipients of information. Furthermore, LLMs have shown remarkable potential in supporting students with special needs. They provide specialized tools and resources that cater to diverse learning challenges, making education more accessible and inclusive. Students who might have found it difficult to keep up in a conventional classroom setting can now benefit from AI's ability to tailor content and delivery to their specific needs, thereby breaking down barriers to learning and fostering.

Advancing Prompt Engineering with Teachers and Students

The advancement of prompt engineering within educational settings offers a unique avenue for enriching the learning experience for both teachers and students. The cornerstone of implementing prompt engineering is to educate all parties involved about its methodologies.

This involves not only teaching the basic principles but also delving into various prompt types, such as the difference between zero-shot and few-shot prompting, and the application of techniques like chain-of-thought or self-consistency prompts. Educators should receive training on how to design prompts that effectively leverage the capabilities of AI models, enhancing the learning outcomes in various subjects.

Collaboration between the lecturers and the students plays a pivotal role in the successful integration of prompt engineering in education. Class-wide collaborative sessions where students and teachers come together to experiment with different prompts can be highly effective. These sessions should focus on identifying which types of prompts yield the best results for different learning objectives and AI applications. Sharing experiences on what works and what does not can lead to a collective understanding and refinement of techniques. Such collaborative exercises also foster a community of learning, where both teachers and students learn from each other's successes and challenges. Creating exercises for each educational module that incorporate prompt engineering is another critical step. These exercises should be designed to align with the learning objectives of the module, offering students hands-on experience in using prompt engineering to solve problems or explore topics. For instance, in a literature class, students could use prompt engineering to analyse a text or create thematic interpretations. In a science class, prompts could be designed to explore scientific concepts or solve complex problems. These exercises should encourage students to experiment with different types of prompts, understand the nuances of each, and observe how subtle changes in phrasing or context can alter the AI's responses. This not only enhances their understanding of the subject matter but also develops critical thinking skills as they analyse and interpret the AI's output. To further enrich the learning experience, these exercises can be supplemented with reflective discussions. After completing a prompt engineering exercise, students can discuss their approaches, challenges faced, and insights gained. This reflection not only solidifies their understanding but also encourages them to think critically about the application of AI in problem-solving. Such exercises are especially powerful because both the students as well as the teaching staff learn a lot about the technology at the same time.

Critical Thinking with AI In the Classroom

Workshops may be a useful tool for fostering critical thinking skills in modern education. These workshops should not only focus on the technicalities of AI but also on developing critical thinking skills in the context of AI use. They should include hands-on activities where students and teachers can engage with AI tools, analyse their outputs, and critically assess their reliability and applicability. The workshops can also cover topics such as identifying biases in AI algorithms, understanding the limitations of AI, and evaluating the ethical implications of AI decisions. Case studies play a pivotal role in understanding the ethical dimensions of AI. These should be carefully selected to cover a wide range of scenarios where the ethical implications are highlighted. Through these case studies, students can examine real-world situations where the decisions made by AI have significant consequences, encouraging them to think about the moral and societal impacts of AI technologies. The discussions should encourage students to debate different viewpoints, fostering an environment of critical analysis and ethical reasoning. Establishing institutional channels where students and teachers can bring their AI-related problems is essential to foster a culture of open communication and continuous learning. These channels can function like an innovation funnel, where ideas, concerns, and experiences with AI are shared, discussed, and explored. This could take the form of online forums, regular meet-ups, or suggestion boxes.

These platforms can act as incubators for new ideas on how to use AI responsibly and effectively in educational settings.

Creating a culture of AI adoption in educational institutions is crucial. This culture should be built on the principles of ethical AI use, continuous learning, and critical engagement with technology. It involves not just the implementation of AI tools but also the fostering of an environment where questioning, exploring, and critically assessing AI is encouraged. This culture should permeate all levels of the institution, from policy making to classroom activities. Encouraging students to question and explore AI's potential and limitations can lead to a deeper understanding and responsible use of these technologies. This includes facilitating discussions on topics such as AI's impact on job markets, privacy concerns, and the implications of AI in decision-making processes. By encouraging critical thinking around these topics, students can develop a nuanced understanding of AI, equipping them with the skills necessary to navigate an AI-driven world.

Conclusion

Navigating the complexities and potentials of AI in education

The AI in the realm of education marks a transformative era that is redefining the teaching and learning methodologies fundamentally. This paper has critically examined the expansive role of AI, focusing particularly on the nuances of AI literacy, prompt engineering, and the development of critical thinking skills within the educational setting. As we delve into this new paradigm, the journey, although filled with unparalleled opportunities, is fraught with significant challenges that need astute attention and strategic approaches. One of the most compelling prospects offered by AI in education is the personalization of learning experiences. AI's capacity to tailor educational content to the unique learning styles and needs of each student holds the potential for a more engaging and effective educational journey. Moreover, this technology has shown remarkable promise in supporting students with special needs, thereby enhancing inclusivity and accessibility in learning environments. Additionally, the focus on AI literacy, prompt engineering, and critical thinking skills prepares students for the complexities of a technology-driven world, equipping them with essential competencies for the future. However, these advancements bring forth their own set of challenges. A primary concern is the preparedness of educators in this rapidly evolving AI landscape. Continuous and comprehensive training for teachers is crucial to ensure that they can effectively integrate AI tools into their pedagogical practices. Equally important are the ethical and social implications of AI in education. The integration of AI necessitates a critical approach to address biases, ensure privacy and security, and promote ethical use. Another significant hurdle is the accessibility of AI resources.

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A STUDY ON COMMUTER'S PERCEPTION TOWARDS METRO RAIL TRANSPORT WITH SPECIAL REFERENCE IN CHENNAI CITY

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ABSTRACT

This study explores the perceptions of metro commuters in Chennai city. The main objective of the study is to identify the commuter's expectations, benefits and drawbacks of metro rail transport. The researcher has collected Primary data through a structured questionnaire method and the convenience sampling method was used. 100 samples were collected for the present study. The respondents are different in age, gender, level of education, and employment. The data was collected from those who are using the CMRL app. As a result of this research on metro rail, there will be a reduction in the number of two-wheeler commuters, buses and other vehicles carrying passengers on Chennai roads. This study reveals that most of the commuters are expecting new offers and enjoying to travel in metro rail. They have adequate knowledge and awareness of metro rail. Especially women Commuters are highly satisfied during travel and they feel safe compared to other public transports. Metro rail provides lot of advantages in many aspects and it creates a natural disaster, accident during construction, loss of trees, etc. According to this study, technically skilled and educated people can only access metro rail and it is not user-friendly for illiterate people. Illiterate people always need support and dependent others to use metro rail transport.

Keywords: Metro, Passengers, CMRL app.

INTRODUCTION

Metro rail transport is to contribute to the livability and sustainability of the cities by reducing traffic overcrowding, air pollution, and dependence on private vehicles. It's offering a convenient and eco-friendly alternative to car travel; metro rails increase urban mobility while minimizing environmental impact. The government of Tamil Nadu approved the Chennai Metro project in November 2007. Chennai Metro Rail Limited (CMRL), a SPV, was established on 3rd December 2007 as a joint venture between the Government of India and the Government of Tamil Nadu and is headquartered in Chennai. Planning for the Metro started in 2007-08, with construction commencing in February 2009. After testing in 2014, the first segment of the metro between the Alandur and Koyambedu stations on the Green Line began operations on 29 June 2015.

CMRL APP (Chennai Metro Rail Limited)

Tickets - Commuters are able to buy, validate, and get e-tickets for up to six individuals continuously.

Smart cards - Online contactless smart card recharge is available to commuters.

Travel planning - Commuters can choose a route, choose a ticket, and look through fare choices. Commuters can plan their journey. Additionally, the app offers real-time train schedules and a journey planner that shows if a train is direct or requires a transfer.

Other information - Metro station facilities, feeder service, tourist destinations, trains, parking and fare information, places to stay and eat, gates and directions, and metro station contact information are all provided by the app.

Payment - Commuters may use UPI and other digital payment systems to make secure payments.

QR code - The app generates a QR code that can be scanned at the metro's entry and the exit.

STATEMENT OF THE PROBLEM

The research is carried out on the subject of “**A STUDY ON COMMUTERS PERCEPTION TOWARDS METRO RAIL TRANSPORT WITH SPECIAL REFERENCE TO CHENNAI CITY,**” which is the focus of the research. The purpose of the research is to identify the commuter’s opinions regarding the availability of metro trains, ticket prices and seat availability, safety, customer support, reduction of air pollution, and construction costs. Though the public knows about the Pros and Cons of metro rail Project, they are supporting the extension of new metro lines in Chennai and expecting the project in down trodden areas. The majority of people are frequently using and supporting metro rail transportation because, it is very economical and modern techno based project.

OBJECTIVE OF THE STUDY

- To Study the Satisfaction level of Commuters regarding Chennai metro rail transport.
- To examine the Benefits and Drawbacks of Chennai Metro Rail transportation.
- To analyze the expectations of Metro Commuters.

LIMITATION OF THE STUDY

- The scope of the study limited only in Chennai.
- It's possible that the majority of respondents did not properly respond to the questionnaire's topic.
- Only interested commuters filled out the questionnaire.

REVIEW OF LITERATURE

Dr. N.O. Ameen, (2023)¹ The study is focused on “**A Study on Chennai Metro Rail Commuters Preferences**” Firstly to identify major factors affecting service quality of Chennai metro rail, using questionnaire method. Secondly to identify the perception of commuter’s regarding service quality. Thirdly to study the demographic profile of the Commuter’s of Chennai metro rail. The sample comprises 100 respondents who are using services of CMRL. Chennai metro rail is very helpful for the Chennai commuters, because there is more traffic in roadways so that commuters can easily reach the destination on time. CMRL is doing well in Chennai city. Even they can do more for commuters like more lift facilities, effective feeder service, then they can get some loyal commuters. The maximum number of commuters of CMRL was of age group 19-24 years, 75% of the respondents are

male and 25% respondents are female. 55% respondents are finished UG degree and 45% respondents are Finished PG degree.

Shankar Kumar Ande Bhi Jaya Prabha (2018)² describes the briefly in the year study "**An empirical study on customer satisfaction towards Chennai metro rail limited**".

Consumer satisfaction has become one of the main factors influencing transport industries especially metro rail services. It becomes the most important for any transport sector. The need for the study aimed in identifying the factors influencing the customer satisfaction and service provided by Chennai metro line Ltd. The main objectives were to identify the various factors influencing commuters' satisfaction towards Chennai rail limited. The study is limited to a specific number of computers that do not cover the entire population and it cannot be the total opinion of all computers. The tools used were percentage analysis and registration to recruit more employees to Chennai metro rail Ltd and has concluded major densely populated cities with lots of people living. They should bring more innovative services to people with the increased satisfaction level.

Verma, M. et.al., (2022)³ discusses in the study that zeroes in on understanding the elements affecting service quality satisfaction and commuters' intention to persist in using Bengaluru Metro services. Through factor analysis and subsequent regression modeling, the perceptions and satisfaction of 700 metro passengers are examined. Principal component analysis identifies 35 indicators, categorized into seven factors: passenger ease, overall service quality satisfaction and loyalty intention, smooth transition, Metro operation and safety, anxiety, amenities, and service availability. The developed model underscores that 'Metro operation and safety along with passenger ease' wield the most significant influence, while 'anxiety' exerts a negative impact on the overall service quality satisfaction and loyalty intention of Bengaluru Metro. The insights garnered from this study provide authorities with a roadmap to comprehend service gaps in the existing metro rail transit system, enabling them to address these gaps and enhance the overall appeal of metro services to commuters. Moreover, the model crafted in this study can serve as a tool for conducting service quality assessments for other metro systems across India.

Appu and S.G Balaji (2017)⁴ analyzed that transport functions as an assignment of the cultural and social life of the inhabitant. The inappropriate and unhygienic transportation system will have an impact on the health of travellers. The launch of the Chennai metro will reduce passenger transport from road to metro. This will lead to a reduction in the number of buses, passenger vehicles, and other automobiles. The main objective of this study is to experience the travel of the customers and to discover the methods pursued by Chennai Metro Rail Limited.

RESEARCH METHODOLOGY

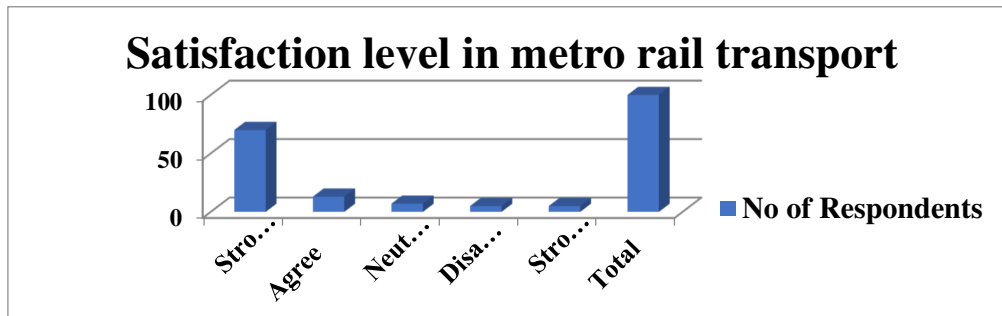
The study is based on the primary data method from 100 respondents on the perception of Chennai Metro Rail commuters. On the basis of a convenient sampling technique, the respondents were chosen. Data were collected using a well-structured questionnaire involving two parts. Part I determines the questions relating to demographic factors and Part II determines the CMRL app users and perceptions in metro rail transport. The collected data were analyzed using a simple bar diagram.

DATA ANALYSIS

SIMPLE BAR DIAGRAM

TABLE 1 SHOWING THAT THE SATISFACTION LEVEL OF COMMUTERS IN METRO RAIL TRANSPORT.

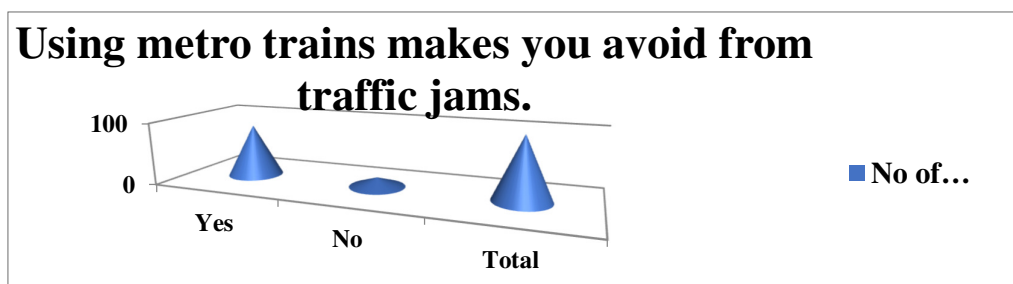
Satisfaction level in metro rail transport.	No of Respondents
Strongly Agree	67
Agree	13
Neutral	4
Disagree	6
Strongly Disagree	10
Total	100

Figure 1**INTERPRETATION**

The above picture reveals that the majority of respondents strongly agree to support metro rail service.

TABLE 2 SHOWING THAT IS THE METRO TRAINS SAVE PUBLIC FROM TRAFFIC JAMS?

Metro trains save public from traffic jams.	No of Respondents
Yes	85
No	15
Total	100

Figure 2

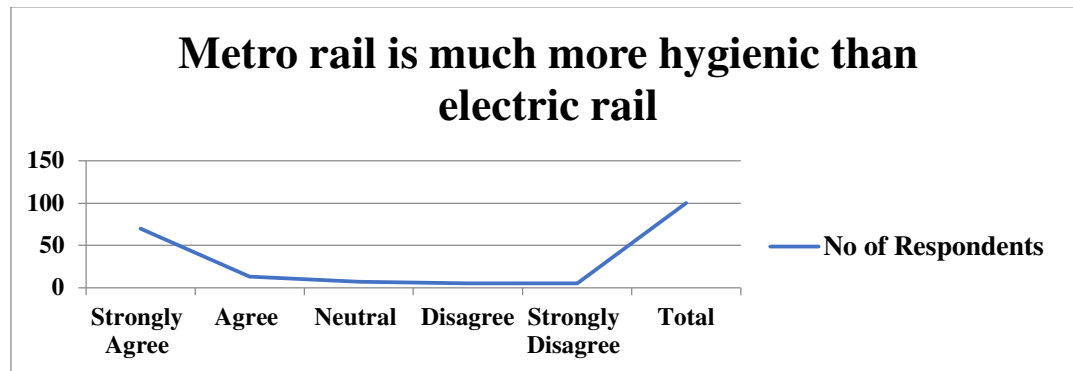
INTERPRETATION

The above table shows that the majority of the respondents feel comfortable with metro trains because it saves public from the traffic jams.

TABLE 3 SHOWING THAT METRO RAIL IS MUCH MORE HYGIENIC THAN ELECTRIC RAIL.

Metro rail is much more hygienic than electric rail.	No of Respondents
Strongly Agree	70
Agree	13
Neutral	7
Disagree	5
Strongly Disagree	5
Total	100

Figure 3



INTERPRETATION

According to the image above, 70 respondents strongly agree that metro rail is more hygienic than electric trains. The remaining commuters are expecting improved hygiene on their trip.

TABLE 4 SHOWING THAT PUBLIC AFFECTED BY OVERCROWDING ON THE ROAD AND LACK OF SPACE DUE TO METRO RAIL CONSTRUCTION

Public affected by overcrowding on the road and lack of space due to metro rail construction	No of Respondents
Strongly Agree	56
Agree	24
Neutral	10
Disagree	5
Strongly Disagree	5
Total	100

Figure 4

Public affected by overcrowding on the road and lack of space due to metro rail construction



INTERPRETATION

The above table reveals that the majority of the respondents feel that the metro rail construction is creating overcrowding on the road and taking up lack of space and the public is affected.

FINDINGS, SUGGESTION AND CONCLUSION

The majority of the respondents are females compared to male commuters. There are different age group of respondents using CMRL between 20-30 years (60.9%) and 31-40 years (17.4%). The majority of the respondents are educated and salaried employees and they know about CMRL transport. The Chennai metro rail commuters are satisfied with ticket prices, availability of trains, Customer support & services, proper lighting facilities on the platform, announcement clarity, safety, lifts and escalators and air conditioning. At present the CMRL mobile app is highly rated, especially the youngsters are enjoying during the travel. Some of the respondents are suggested that metro rail management must arrange Medical facilities, seating availability for elderly persons, pregnant and disabled passengers. The metro rail construction process should be speeded up due to it creating heavy traffic in the city and clearing the underground metro signal issues.

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NEW FIELDS OF HIGHER EDUCATIONS: CHALLENGES ENCOUNTERED BY LEARNERS

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Abstract

In spite of meeting the emerging scenario and new demands of rapidly changing world the field of higher education and universities has in the urge of offering new fields. But it has some of the difficulties over the learners who pursuing new courses. This paper try to explores the various problems met by students in these emerging fields. It includes curriculum ambiguity, lack of experienced faculty, limited access to resources, absence of standardised learning pathway and Occupational prospects in terms of devoid of wide range classical Employment horizons.

Key Words: New fields of higher Education, challenges, Learners.

Introduction

The Evolution of technology in this 21st century expands the boundaries of higher education beyond traditional disciplines, giving rise to a variety of new and interdisciplinary fields especially inartificial Intelligence, Environmental studies, data science, Bio medical Engineering, Biotechnology, and digital arts. It tries to fulfill the evolving needs of stack holders, employers, and Industries. As the higher Education institutions rush to incorporate these innovative subjects into their curricula, learners are also increasingly drawn to them for their relevance and future potentials. However students are struggling to navigate themselves with adapting new content, and gaining cognitive ability.in addition integration of theory with practical real- world applications can be cumbersome process and insufficient institutional support. Insights in these challenges is important and steps to create adaptive and forward looking learning environment is important today.

Major obstacles encountered by Learners in new Fields of higher Education Learners entering new fields of higher education often face a variety of obstacles that can impact their academic success, confidence, and overall experience. These challenges generally fall into the following

Lack of Established Curriculum and Learning Rresources

Emerging disciplines often do not well- structured or with standardised curricula. Learners may struggle to find comprehensive difficulties over text books, tutorials, or certified online courses.

Availability of qualified Instructors is narrow.

New areas of education may lack experienced educators with both academic and industrial expertise. Faculty training and curriculum development often lag behind technological advancement.

Academic and Knowledge Gaps

- **Lack of foundational knowledge:** New fields may assume prior familiarity with certain concepts or skills, creating a steep learning curve.
- **Interdisciplinary complexity:** Many emerging fields blend areas like technology, biology, and social science, requiring learners to adapt quickly across disciplines.
- **Unclear learning outcomes:** In newer fields, curricula may not be as refined or standardized, making it harder to understand what's expected

Access to Resources

- **Limited instructional material:** New or niche subjects might lack comprehensive textbooks, tutorials, or online courses.
- **Scarcity of mentors:** Fewer experienced educators or professionals may be available to guide students.
- **Insufficient lab or research infrastructure:** Hands-on fields may suffer from inadequate facilities

Technological Challenges

- **Dependence on digital tools:** Emerging fields often rely heavily on new software, simulations, or platforms that learners must quickly master.
- **Digital divide:** Some students may lack access to necessary technology or fast

Career Uncertainty

- **Unclear job prospects:** Learners may struggle to see clear career paths in fields that are still evolving.
- **Limited internship or practical experience opportunities:** Emerging sectors may not yet have established pipelines to industry.

Psychological and Social Barriers

- **Impostor syndrome:** Students may feel they don't belong, especially in cutting-edge or highly technical disciplines.
- **Cultural and language barriers:** Especially for international students in global or tech-driven programs.
- **Lack of peer support:** Being part of a small or pioneering cohort can feel isolating.

Institutional and Curricular Issues

- **Inflexible course structures:** Some institutions may struggle to adapt rapidly to new fields.
- **Outdated assessment methods:** Standard exams may not suit the dynamic and applied nature of new disciplines.
- **Inadequate advising:** Faculty may not yet be fully trained to support students in these

UGC's guidelines over new courses

The University Grant Commission (UGC) is a statutory body of the Government of India. It was established in 1956 to determine, coordinate and maintain standards of the overall education system such as teaching, examination and research of university education.

The UGC has recently published the list of 80 Higher Educational Institutions (HEIs) that are recognised to offer Open and Distance Learning (ODL) courses for the academic year 2025. The UGC has introduced various new courses and guidelines for both undergraduate (UG) and postgraduate (PG) programs across all disciplines also. Additionally, the UGC has recommended one-year PG programs and revised the PG curriculum with a focus on research and multidisciplinary fields. Advanced science courses in higher education encompass specialized areas of study within the traditional sciences like physics, chemistry, biology, and mathematics. They also include interdisciplinary fields like biotechnology, bioinformatics, and data science, as well as specialized areas within each traditional science.

Here are some examples of new and advanced fields of science courses:

Physics:

- Advanced Physics: Physics courses delving into advanced topics in physics, such as quantum mechanics, electromagnetism, and statistical mechanics.
- Astrophysics: Studying the physical properties and evolution of celestial objects and phenomena.
- Condensed Matter Physics: Focusing on the study of the electronic and magnetic properties of solids.

Chemistry:

- Advanced Chemistry: Courses covering organic, inorganic, and physical chemistry, as well as specialized areas like materials chemistry and computational chemistry.
- Biochemistry: The study of the chemical processes within and related to living organisms.
- Pharmaceutical Chemistry: Focusing on the synthesis, properties, and effects of drugs and pharmaceuticals.

Biology:

- Advanced Biology: Courses covering molecular biology, cell biology, genetics, and evolution.
- Biotechnology: Applying biological principles to solve practical problems, such as in medicine and agriculture.
- Biomedical Science: Studying the biological basis of disease and developing treatments.
- Molecular and Cellular Biology: Delving into the structure and function of cells and molecules.
- Genetics: Studying heredity and variation in organisms.
- Ecology: Examining the interactions between organisms and their environment.
- Neuroscience: Investigating the nervous system and its functions.
- Epidemiology and public Health

Mathematics:

Advanced Mathematics: Courses covering calculus, linear algebra, differential equations, and abstract algebra.

- Statistics: Studying the collection, analysis, interpretation, and presentation of data.
- Data Science: Applying mathematical and computational techniques to extract knowledge from data.
- Applied Mathematics: Applying mathematical principles to solve real-world problems in various fields.
- **Interdisciplinary Sciences:**
- Biotechnology: Applying biological principles to solve practical problems.
- Bioinformatics: Using computational methods to analyze biological data.
- Materials Science: Studying the properties and behavior of materials.
- Environmental Science: Studying the environment and its interactions with human activities.
- Forensic Science: Applying scientific techniques to investigate crimes.
- Medical Physics: Applying physics principles in medical diagnostics and treatment.
- Medical and applied microbiology
- Biochemistry, ect.,

These are just some examples, and the specific courses offered may vary depending on the institution and program. Many universities also offer advanced courses within specific areas of science, such as materials science, nanotechnology, and computational biology.

Problem over Employability with relevant statistics of learner doing new courses

Less Employment opportunities in Public Sectors

Students pursuing new courses has lack of employment opportunities in their study field especially in public sectors and government based organisations. For example student completed their degrees in Biochemistry, Microbiology are allowed to continuing Bachelor of Education by the UCG norms. But recruiting authorities like different states Teachers Recruiting Boards are not freely allow them to get their opportunity comparatively with other classical studies. This is what same and continue in University faculty recruitments, public sector recruitments etc., In the context of employer preferences basic courses enough rather than advanced or new specific courses which will be the frustrating element of choosing new courses by the learners.

In 2025, approximately 42.6% of Indian graduates were deemed employable by the Mercer-Mettl India Graduate Skills Index. This figure represents a slight decrease from the 44.3% recorded in 2023. The study highlights a significant skills gap between what employers need and what graduates possess.

Another one analysis of employability of learners involved data from campuses in 30-plus states and union territories, with further categorisation based on college tiers (Tier 1, Tier 2, and Tier 3) according to the National Institute of Ranking Framework by the Ministry of Education. The survey indicates 46% employability for Tier 1 colleges, 44% for Tier 2 and 43% for Tier 3 colleges. And employability for learners in non-technical roles and skills is higher across all college tiers in comparison to that for technical roles.

Employability by tier of college for top jobs in demand

Employability	Tier 1 colleges (rank -1-100)	Tier 2 colleges (rank – 101-200)	Tier 3 colleges (Rank 200-beyond)
AI/ML	49%	48%	46%
Back-end developer skills	40%	39%	39%
Data analyst	41%	41%	38%
Data scientist	38%	40%	39%
Front-end developer skills	45%	42%	38%
UI/UX developer	35%	40%	36%

Th Source: India's Graduate Skill Index: 2023' report: Mercer | Mettl

Recommendations**Flexible Learning Models**

- Offer **hybrid and asynchronous options** to accommodate working adults and caregivers.
- Promote **microlearning** and **modular courses** that allow learners to study at their own pace.

Financial Support Mechanisms

- Expand access to **scholarships, grants, and interest-free education loans**.
- Encourage **employers** to invest in employee education through tuition reimbursement or paid study leave.

Improved Student Guidance and Mentorship

- Create **mentorship programs** that pair new learners with experienced professionals in the field.
- Provide **academic counseling** to help learners navigate course selection and career pathways.

Psychological and Emotional Support

- Establish **confidence-building workshops**, particularly in underrepresented fields (e.g., STEM for women).
- Offer **mental health resources**, stress management training, and peer support groups

Curriculum Modernization

- Update content to align with **emerging industry trends** and job market needs.
- Incorporate **real-world applications**, case studies, and interdisciplinary learning.

Access and Literacy

- Ensure that all learners have access to **affordable devices** and **stable internet connections**.
- Provide **digital literacy training**, especially for adult and non-traditional learners.

Cultural and Linguistic Inclusion

- Translate course materials into **multiple languages** where needed.
- Use **inclusive teaching practices** that respect diverse learning styles and cultural contexts.

Conclusions

Systemic barriers—like cost, time, and outdated curricula—remain significant obstacles for learners transitioning into new educational fields. Psychological and emotional challenges, including lack of confidence and fear of failure, are often under-addressed but critically impact student success. Flexibility, accessibility, and personalized support are essential to enable success in today’s diverse learning environments. Education systems must evolve to support lifelong learning, where upskilling and reskilling are essential for career sustainability. An inclusive, well-supported learning experience increases not only individual success but also equity and innovation across society. Creating employment opportunities according to their fields is mandatory.

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A STUDY ON THE SAVINGS AND INVESTMENT BEHAVIOR OF WOMEN IN SIVAGANGA DISTRICT: TRENDS AND DETERMINANTS

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Abstract

Savings and investment habits play a crucial role in determining the financial stability and economic independence of individuals, particularly women. In Sivaganga District, women's financial decisions are influenced by various socio-economic factors, family responsibilities, and cultural traditions. This study aims to analyse the savings and investment patterns among women, with a focus on post office schemes, public and private sector insurance policies, recurring deposits (RD), fixed deposits (FD), real estate investments, and gold purchasing habits. The research investigates the factors influencing these financial decisions and the implications for future financial security. The findings will provide insights into financial literacy levels, risk preferences, and the impact of government and private financial instruments on women's financial behaviour.

Keywords: Savings Habits, Investment Behaviour, Women Empowerment, Post Office Schemes, Fixed Deposits, Recurring Deposits, Real Estate, Gold Investment, Insurance Policies, Sivaganga District.

Introduction

Financial security and investment planning have gained importance in recent years, particularly for women who play a significant role in household financial management. Understanding the savings and investment behaviour of women in Sivaganga District is essential to identify patterns, challenges, and opportunities for financial inclusion.

Women in rural and semi-urban areas often prefer traditional savings methods, such as post office schemes and fixed deposits, due to their perceived security. Among the post office savings schemes, the Sukanya Samriddhi Yojana (SSY) is a prominent option for women who wish to secure their daughters' future. This scheme offers high interest rates and tax benefits under Section 80C of the Income Tax Act, making it an attractive long-term investment option.

Another popular post office scheme is the Senior Citizens Savings Scheme (SCSS), which is widely utilized by elderly women as a secure retirement plan. The scheme provides assured returns with quarterly interest payouts, ensuring financial stability in post-retirement life. It is one of the most preferred investment avenues for women nearing retirement age.

For those looking for mid-term investment opportunities, the National Savings Certificate (NSC) serves as a low-risk instrument with fixed interest rates and a five-year maturity

period. Women prefer NSC due to its guaranteed returns and ease of reinvestment after maturity.

The Kisan Vikas Patra (KVP) is another post office scheme that doubles the investment within a predetermined period. Women from agricultural backgrounds in Sivaganga District often invest in this scheme due to its fixed interest rate and flexible tenure, making it an ideal savings option.

Apart from post office schemes, women also opt for public and private sector insurance policies. SBI Life Smart Women Advantage is a popular life insurance product designed exclusively for women, offering financial protection against critical illnesses, pregnancy complications, and major surgeries. It also provides maturity benefits, ensuring long-term security.

Similarly, SBI Life Poorna Suraksha offers a combination of life insurance and health protection, helping women safeguard their financial future while covering major health-related expenses. This plan is suitable for working women who seek a comprehensive insurance solution.

HDFC Life also provides tailored schemes such as HDFC Life Sanchay Plus, a guaranteed savings plan that offers assured returns without market risks. This plan is widely preferred by women looking for financial certainty over a fixed period.

For women planning for their children's education, HDFC Life YoungStar Udaan is an attractive investment option. This child insurance plan ensures financial support for higher education and other academic needs, making it a popular choice among mothers.

Investment in recurring deposits (RD) and fixed deposits (FD) in nationalized and private banks is common due to their low-risk nature. Banks like SBI, HDFC, and other nationalized banks provide FD schemes such as SBI Fixed Deposit, HDFC Regular Fixed Deposit, ICICI Golden Years FD, and Axis Bank Tax Saver FD, which offer higher interest rates for senior citizens and women investors. These financial instruments offer assured returns, making them preferable choices for risk-averse individuals.

Real estate investment is another significant aspect of women's financial planning in Sivaganga District. Buying land or property is considered a long-term security measure, often influenced by family traditions and social status. Many women save money to invest in property either for residential purposes or as an asset for future generations. Real estate has proven to be a stable investment, particularly in regions experiencing urban expansion.

Gold purchasing remains a deeply ingrained investment habit among women. The Sovereign Gold Bond (SGB) Scheme introduced by the government allows women to invest in gold without the need for physical storage while also earning interest on their investments. The Gold Monetization Scheme (GMS) provides an opportunity for women to earn returns on idle gold assets. Many women invest in gold for their children's marriage, emergencies, or as a hedge against inflation.

Despite these investment habits, women in Sivaganga District face various financial challenges, including lack of financial literacy, limited access to formal banking services, and influence of social norms. Additionally, their investment decisions are often influenced by family members rather than individual financial planning. Encouraging financial awareness and accessibility to modern banking solutions can significantly enhance women's participation in structured savings and investment plans.

Women in Sivaganga District exhibit diverse savings and investment habits influenced by socio-economic conditions, financial awareness, and familial responsibilities. While traditional savings methods dominate, awareness and accessibility to modern financial instruments remain limited. The study aims to assess the financial behaviour of women, the key determinants influencing their investment choices, and the challenges they face in financial decision-making.

This study is significant as it provides insights into the financial habits of women, highlighting the importance of financial inclusion and literacy programs. The findings will help policymakers, financial institutions, and women entrepreneurs develop strategies to improve financial security among women in the region. By understanding the factors influencing their savings and investment choices, better financial products and educational initiatives can be tailored to meet their needs.

Review of Literature

The financial behavior of individuals, particularly their savings and investment habits, has been widely explored in various studies. Sohail and Shanmugham (2003) examined customer preferences in Malaysia's e-banking sector, highlighting the increasing shift towards digital financial transactions. Similarly, Laukkanen (2007) investigated multi-channel banking preferences, emphasizing the importance of accessibility and security in shaping consumer banking behavior. Studies conducted by Gbadeyan and Akinyosoye-Gbonda (2011) and Bhatta (2011) also reinforced that customers prefer banking services that offer convenience, flexibility, and reliability. These findings indicate that digital banking services influence not only transactional behavior but also savings and investment decisions, particularly among women who seek secure and user-friendly financial platforms.

Further, Sheikhi and Sheriff (2016) analyzed the financial literacy levels in Sivaganga District and found that limited awareness often restricts effective utilization of banking services. The role of digital wallets and mobile banking was explored by Mittal et al. (2017) and Singh et al. (2017), who found that technological advancements significantly impact customer preferences for financial services. Subrahmanya and Puttanna (2018) highlighted policy shifts in India, demonstrating how government initiatives promote financial inclusion. More recently, Saranya Devi et al. (2024) studied financial literacy in Sivaganga District, emphasizing the importance of targeted financial education to improve women's investment participation. These studies collectively suggest that financial literacy, accessibility, and digital transformation play crucial roles in shaping women's savings and investment habits, underscoring the need for inclusive financial policies and educational interventions.

Objectives of the Study

1. To analyze the savings and investment behavior of women in Sivaganga District, focusing on post office schemes, insurance policies, RD, FD, real estate, and gold.
2. To examine the factors influencing their financial decisions, including socio-economic conditions, financial literacy, and family responsibilities.
3. To assess the challenges faced by women in accessing financial services and making independent investment decisions.

Methodology

The study adopts a descriptive and analytical in nature. A survey will be conducted among women across different age groups, educational backgrounds, and income levels in

Sivaganga District. The study adopts the convenient sampling to be employed to ensure representation from various socio-economic groups.

Primary data will be collected through structured questionnaires and personal interviews, focusing on financial literacy, savings behavior, investment choices, and influencing factors. Secondary data will be gathered from government reports, financial institutions, academic research papers, and policy documents.

The study will utilize statistical tools such as Descriptive Analysis, One-Way ANOVA, and Regression Analysis to interpret the collected data. Additionally, the responses had been analyzed thematically to understand personal financial experiences and challenges faced by women.

Ethical considerations, including informed consent and confidentiality, will be strictly followed to ensure the integrity of the research. The findings will contribute to the development of financial policies and programs aimed at empowering women economically in Sivaganga District.

Descriptive analysis is a statistical method used to summarize, interpret, and present data in a meaningful way. In the context of savings and investment habits of women, it helps in understanding patterns, preferences, and key financial behaviors. This analysis provides insights into the frequency of savings, preferred investment options such as fixed deposits (FD), recurring deposits (RD), gold, real estate, and insurance policies, as well as factors influencing financial decisions.

By using descriptive statistics such as mean, standard deviation, and frequency distribution, the study can highlight the financial awareness levels, risk preferences, and the extent of women's participation in structured savings and investment schemes. The findings from this analysis can aid policymakers and financial institutions in designing better financial products and awareness programs tailored to women's financial needs.

Table1: Descriptive analysis of Savings and Investment Habits of Women

Descriptive Statistics

Savings and Investment Habits of Women	N	Mean		Std. Deviation
	Statistic	Statistic	Std. Error	Statistic
I actively save a portion of my monthly income.	455	1.6791	.03331	.71045
I prefer post office savings schemes for my investments.	455	2.8418	.07070	1.50813
I invest in Fixed Deposits (FDs) for financial security.	455	4.4176	.05238	1.11721
I consider Recurring Deposits (RDs) a reliable investment option.	455	3.9714	.05898	1.25813

I have invested in insurance policies for future security.	455	3.2747	.07509	1.60166
I am aware of public sector and private sector insurance policies.	455	3.2000	.06647	1.41795
I invest in gold as a long-term financial asset.	455	2.5209	.05342	1.13955
I prefer real estate investment for long-term financial stability.	455	2.9670	.06790	1.44846
I save and invest primarily for my children's future.	455	3.8923	.06673	1.42331
My financial decisions are influenced by family members.	455	3.4308	.07395	1.57735
I am financially literate and confident in making investment decisions.	455	2.8901	.07077	1.50952
I prefer bank savings accounts over other investment options.	455	2.3429	.06264	1.33614
I face challenges in accessing formal banking and investment services.	455	2.5099	.06361	1.35675
Government schemes help in improving my financial security.	455	1.6286	.04010	.85544
I seek financial guidance before making investment decisions.	455	2.5143	.04343	.92643
Valid N (listwise)	455			

Source: Primary data

The descriptive analysis of savings and investment habits among women in Sivaganga District highlights varying financial behaviors and preferences. The study reveals that regular saving habits are relatively low, with a mean score of 1.6791, suggesting that many women do not actively save a portion of their income. Post office savings schemes have a moderate preference (2.8418), while fixed deposits (FDs) are the most preferred investment option with a high mean of 4.4176, followed closely by recurring deposits (RDs) at 3.9714. Insurance policies also show moderate adoption (3.2747), though awareness of public and private sector insurance schemes is slightly lower (3.2000), indicating a gap in financial literacy.

Investment in gold (2.5209) and real estate (2.9670) also shows a moderate preference, highlighting traditional financial security measures.

Family influence plays a significant role in financial decisions, with a mean score of 3.4308, suggesting that many women rely on family guidance. Financial literacy and confidence in investment decisions remain areas of concern, with a mean of 2.8901, pointing to the need for better financial education. Preference for bank savings accounts is relatively low (2.3429), and many women face challenges in accessing formal banking services (2.5099). Awareness and utilization of government schemes for financial security, such as Sukanya Samriddhi Yojana, remain limited (1.6286), emphasizing the need for increased outreach programs. The study also finds that many women seek financial guidance before investing (2.5143), reinforcing the importance of targeted financial advisory services to empower women in making informed investment choices.

Awareness of the Schemes of Saving and Investment

Savings schemes play a crucial role in financial security, offering individuals structured opportunities to accumulate wealth over time. Awareness of these schemes is essential for making informed financial decisions. Various government-backed and private investment options, such as Fixed Deposits (FDs), Recurring Deposits (RDs), Public Provident Fund (PPF), Sukanya Samriddhi Yojana (SSY), and insurance policies, provide diverse avenues for savings.

Table2: Awareness of Saving and Investment Schemes-Descriptive Analysis

Descriptive Statistics

Awareness of Saving and Investment Schemes	N	Mean		Std. Deviation
	Statistic	Statistic	Std. Error	Statistic
Sukanya Samriddhi Yojana (SSY) – Post Office, Nationalized Banks	455	2.3341	.06561	1.39951
Senior Citizens Savings Scheme (SCSS) – Post Office, Nationalized Banks	455	2.0505	.06901	1.47209
National Savings Certificate (NSC) – Post Office	455	3.9802	.07091	1.51266
Kisan Vikas Patra (KVP) – Post Office	455	3.9978	.03009	.64179
SBI Life Smart Women Advantage – SBI Life Insurance	455	3.1187	.06991	1.49123
SBI Life Poorna Suraksha – SBI Life Insurance	455	2.8593	.06508	1.38828
HDFC Life Sanchay Plus – HDFC Life Insurance	455	3.2571	.06626	1.41337
HDFC Life YoungStar	455	2.5802	.07405	1.57962
Udaan – HDFC Life Insurance	455	2.5802	.07405	1.57962
Fixed Deposit (FD) – SBI Bank	455	2.6330	.08131	1.73441

Fixed Deposit (FD) – HDFC Bank	455	2.9473	.07738	1.65048
ICICI Golden Years FD – ICICI Bank	455	2.1824	.05796	1.23625
Axis Bank Tax Saver FD – Axis Bank	455	1.6791	.03331	.71045
Sovereign Gold Bond (SGB) Scheme – RBI & Scheduled Banks	455	2.8418	.07070	1.50813
Gold Monetization Scheme (GMS) – RBI & Scheduled Banks	455	4.5011	.04849	1.03438
Real Estate Investment – Private & Public Sector	455	4.4462	.03987	.85054
Valid N (listwise)	455			

Source: Primary data

Understanding these schemes enables individuals, especially women, to choose suitable financial plans based on their needs, income levels, and future goals. Factors such as marital status, occupation, and financial literacy influence participation in savings programs. Enhancing awareness through financial education and advisory services can empower women to make better investment choices and secure their financial future.

Table 3: Between-Subjects Factors

Independent Variable		Value Label	N
Marital Status	1.00	Married	391
	2.00	Unmarried	24
	3.00	Widow/Divorcee/Seperated	40
Occupation Status	1.00	Government Employee	45
	2.00	Private Sector Employee	142
	3.00	Businessman	91
	4.00	Professionals	88
	5.00	Agriculturist	89

Source: Primary data

The study examines the savings and investment habits of women in Sivaganga District based on their marital status and occupation. Among the 455 respondents, the majority are married (391), followed by widowed, divorced, or separated women (40), and a smaller proportion of unmarried women (24). Marital status often plays a significant role in financial decision-making, as married women may prioritize family savings, while widowed or separated women might focus on financial independence and security. Understanding these variations helps in analyzing how different life stages influence savings behavior and investment preferences.

Occupational status also impacts savings and investment patterns. Among the respondents, 142 women work in the private sector, followed by 91 businesswomen, 89 agriculturists, 88

professionals, and 45 government employees. Employment in different sectors affects income stability and financial planning strategies. Government employees, for instance, may have access to structured savings options like provident funds, while businesswomen and professionals may prefer flexible investment avenues like real estate or insurance. By assessing these factors, the study aims to provide insights into the financial awareness and investment trends among women from diverse backgrounds.

Multivariate Analysis: Awareness of Saving and Investment Schemes

Multivariate analysis is a statistical approach used to examine multiple dependent variables simultaneously while considering the influence of independent variables. In this study, the multivariate tests assess the impact of demographic factors such as age, education, marital status, and occupation on the savings and investment habits of women in Sivaganga District. The interaction effect between marital status and occupation is also evaluated to understand how these combined factors influence financial behavior. Various test statistics, including **Pillai's Trace, Wilks' Lambda, Hotelling's Trace, and Roy's Largest Root**, are used to measure the significance of relationships among the variables.

Table 4: Awareness of Saving and Investment Schemes-Multivariate Tests

Multivariate Tests^a

Effect		Value	F	Hypothesis df	Error df	Sig.
Intercept	Pillai's Trace	.998	15172.161 _b	15.000	426.000	.000
	Wilks' Lambda	.002	15172.161 _b	15.000	426.000	.000
	Hotelling's Trace	534.231	15172.161 _b	15.000	426.000	.000
	Roy's Largest Root	534.231	15172.161 _b	15.000	426.000	.000
Age4	Pillai's Trace	.016	.459 ^b	15.000	426.000	.959
	Wilks' Lambda	.984	.459 ^b	15.000	426.000	.959
	Hotelling's Trace	.016	.459 ^b	15.000	426.000	.959
	Roy's Largest Root	.016	.459 ^b	15.000	426.000	.959
Edu4	Pillai's Trace	.020	.593 ^b	15.000	426.000	.881
	Wilks' Lambda	.980	.593 ^b	15.000	426.000	.881
	Hotelling's Trace	.021	.593 ^b	15.000	426.000	.881
	Roy's Largest Root	.021	.593 ^b	15.000	426.000	.881
Marital3	Pillai's Trace	.104	1.566	30.000	854.000	.028
	Wilks' Lambda	.897	1.582 ^b	30.000	852.000	.025
	Hotelling's Trace	.113	1.598	30.000	850.000	.023
	Roy's Largest Root	.095	2.701 ^c	15.000	427.000	.001
Occu5	Pillai's Trace	.102	.749	60.000	1716.000	.923

	Wilks' Lambda	.902	.746	60.000	1665.118	.926
	Hotelling's Trace	.105	.743	60.000	1698.000	.929
	Roy's Largest Root	.035	.996 ^c	15.000	429.000	.458
Marital3 * Occu5	Pillai's Trace	.175	.864	90.000	2586.000	.815
	Wilks' Lambda	.836	.866	90.000	2402.271	.810
	Hotelling's Trace	.184	.869	90.000	2546.000	.805
	Roy's Largest Root	.082	2.365 ^c	15.000	431.000	.003

a. Design: Intercept + Age4 + Edu4 + Marital3 + Occu5 + Marital3 * Occu5

b. Exact statistic

c. The statistic is an upper bound on F that yields a lower bound on the significance level.

The multivariate tests show that the intercept is highly significant ($p = 0.000$) across all four test statistics, indicating that the overall model is well-fitted. The factor age ($p = 0.959$) and education ($p = 0.881$) do not significantly impact the savings and investment decisions of women, as their p-values are much higher than the 0.05 threshold. However, marital status has a significant effect ($p = 0.028$), suggesting that financial behavior varies depending on whether a woman is married, unmarried, or widowed/divorced. The occupation factor ($p = 0.923$) does not show a statistically significant impact, meaning employment type alone does not strongly determine investment preferences. The interaction effect between marital status and occupation ($p = 0.003$) is significant in Roy's Largest Root test, indicating that the combined influence of these factors plays a crucial role in shaping financial decisions. These findings highlight that while individual demographic factors may not always be strong determinants, their combined impact can significantly influence women's savings and investment behavior.

Implication

The study on the savings and investment habits of women in Sivaganga District provides crucial insights into financial behavior influenced by factors such as marital status, occupation, financial literacy, and access to banking services. One of the significant implications of this research is its relevance to financial empowerment and gender equality. Women's financial independence plays a critical role in household economic stability and long-term wealth accumulation. However, social issues such as limited financial awareness, dependence on family members for financial decisions, and restricted access to formal investment avenues continue to hinder women's financial growth. The study highlights the need for policy interventions and financial education programs tailored specifically for women to enhance their participation in savings and investment activities.

From the findings, it is evident that fixed deposits (FDs) and recurring deposits (RDs) are preferred investment choices, whereas government-backed schemes like Post Office Savings, Sukanya Samridhi Yojana (SSY), and insurance policies remain underutilized due to lack of awareness. The study also indicates that marital status significantly affects financial behavior, with married women prioritizing children's future savings and widowed/divorced women

focusing more on financial independence. The interaction between marital status and occupation further reveals that financial security concerns vary among government employees, private-sector workers, businesswomen, professionals, and agriculturists.

To address these social and economic challenges, the study recommends improving financial literacy programs and expanding access to formal banking and investment services for women. Policymakers and financial institutions should design women-centric savings schemes with simplified processes, lower barriers to entry, and attractive returns. Encouraging self-help groups and microfinance initiatives can also enhance women's financial inclusion, particularly in rural areas where traditional banking services are limited. Additionally, integrating digital financial literacy programs can empower women to make informed financial decisions and take advantage of online banking and investment platforms.

In conclusion, this study underscores the urgent need for targeted financial interventions to bridge the gap in women's savings and investment participation. By addressing key social issues such as financial dependence, lack of awareness, and accessibility barriers, women can be encouraged to take control of their financial futures. Implementing these recommendations will not only improve individual financial well-being but also contribute to broader economic development and gender-inclusive financial growth in Sivaganga District and beyond.

Conclusion

The study on savings and investment habits of women in Sivaganga District highlights the evolving financial behavior of women across different social and occupational backgrounds. While traditional investment choices such as gold, real estate, and fixed deposits continue to hold significance, there is a gradual shift towards structured financial instruments like recurring deposits (RDs), insurance policies, and post office savings schemes. However, financial decision-making among women is still influenced by family members, societal norms, and accessibility to banking services, which indicates the need for greater financial independence.

One of the key takeaways from the study is that a lack of awareness and financial literacy limits the effective utilization of investment options. Many women, despite having savings potential, are hesitant to explore private and public sector insurance policies, government-backed schemes, and stock market investments due to perceived risks and lack of knowledge. Bridging this gap requires sustained efforts in financial education, targeted outreach programs, and advisory support to build confidence in independent financial planning.

Moreover, the study emphasizes the importance of digital financial inclusion. With the increasing role of online banking and fintech solutions, there is an opportunity to integrate women into the formal financial system more effectively. Encouraging digital transactions, mobile banking, and e-investment platforms can significantly enhance their participation in diverse savings and investment avenues. This transformation can be accelerated by collaborations between financial institutions, policymakers, and community-based organizations to create tailored solutions for women's financial growth.

Overall, empowering women with financial knowledge, accessibility, and independent decision-making capabilities is crucial for ensuring long-term economic stability. The findings reinforce the need for inclusive financial policies that cater to the unique needs of women, enabling them to save, invest, and secure their future with confidence and self-reliance.

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A STUDY ON AWARENESS ABOUT HACKATHON AND ITS IMPACT ON INNOVATIVE STARTUPS IN TAMIL NADU

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ABSTRACT:

Hackathons have emerged as crucial platforms that foster innovation, problem-solving, and entrepreneurial spirit among young minds. This study analyzes the level of awareness about hackathons and their impact on innovative startups in Tamil Nadu. With a sample size of 150 respondents including students, early-stage entrepreneurs, and professionals, the study utilizes a descriptive research methodology supported by statistical tools like frequency analysis, ANOVA, Chi-square, and regression analysis. The findings reveal a significant correlation between hackathon participation and the initiation of innovative startups. Multiple regression analysis further indicates that variables such as mentorship, funding availability, and institutional support strongly affect startup success post-hackathon. Demographic analysis adds depth by showing variations in awareness and participation across age, gender, educational background, and profession. Suggestions are offered to educational institutions, government bodies, and private stakeholders to harness the hackathon ecosystem more effectively.

Keywords: Hackathon, Innovation, Entrepreneurship, Funding, Ecosystem, etc...

INTRODUCTION:

In the evolving landscape of technology and entrepreneurship, hackathons serve as dynamic events that bring together individuals to develop prototypes and solve real-time problems within a limited time. Tamil Nadu, known for its educational infrastructure and youth population, has witnessed a rise in such events. Hackathons serve as ideation hubs, promoting collaboration among cross-functional teams, often leading to real-world innovations. They offer exposure to industry tools, mentorship, and networking opportunities, helping transition participants into successful entrepreneurs. Despite this growing trend, the sustained impact of hackathons on the startup ecosystem remains underexplored by answering the following questions:

1. Are they truly helping convert ideas into viable businesses?
2. Do participants receive adequate support post-event?
3. Whether the funding shall be provided for the participants in hackathon?

The Entrepreneurship Development and Innovation Institute (EDII) of Tamil Nadu, a government initiative, play a pivotal role in fostering entrepreneurship through various programs, including hackathons. EDII regularly conducts innovation challenges, startup bootcamps, and ideation contests across educational institutions to promote entrepreneurial thinking among students. Formerly, the apex institution organizes the hackathon in the name of Tamil Nadu Student Innovators (TNSI) for more than 5 years and now TNSI is renamed as EDII Innovation Hackathon for college students. Also, the same institute organizes the hackathon for school students in the name of School Innovation Development Programme (SIDP) for the past 2 years. These hackathons often offer mentorship, seed funding, and incubation support, directly contributing to the growth of startups in the state.

The Naan Mudhalvan Niral Thiruvizha, organized by the Tamil Nadu Skill Development Corporation, is another landmark initiative. This program conducts state-wide innovation festivals and hackathons targeting college students to enhance their problem-solving and tech skills. It connects young talent with industry mentors and startup incubators, thus creating a pipeline for innovation-driven entrepreneurship.

Similarly, the Ministry of MSME conducts the MSME Idea Hackathon, encouraging innovators and aspiring entrepreneurs to present practical business solutions. This nationwide initiative provides financial support and incubation to selected ideas, with a special focus on youth and women entrepreneurs in Tamil Nadu.

These government-backed platforms not only create awareness but also provide a structured ecosystem for innovation and startup development. By leveraging such initiatives, this study seeks to explore how hackathons impact entrepreneurial outcomes in Tamil Nadu.

REVIEW OF LITERATURE:

Hackathons have been extensively studied in recent years for their innovative and entrepreneurial potential:

Maria Angelica Medina Angarita and Alexander Nolte (2021) investigated the critical role of mentorship during hackathons and how it affects participants' ability to sustain their projects beyond the event. Their study revealed that consistent, high-quality guidance from experienced mentors significantly increased the likelihood that teams would continue developing their ideas after the hackathon ended.

Zukin & Papadantonakis (2018) focused several structural challenges that limit hackathons' long-term impact with some key issues includes a lack of inclusivity—where certain groups, such as women and underrepresented minorities, often face barriers to full participation. Also, they pointed out the frequent absence of follow-up funding and sustained support after the event, which results in many promising ideas failing to evolve into successful ventures. Further, emphasized the need for robust post-hackathon ecosystems that provide continuous mentorship, financial resources, and institutional backing to nurture the entrepreneurial pipeline effectively without addressing these gaps, the true potential of hackathons in driving innovation remains underutilized.

Hanna Kienzler & Carolyn Fontanesi (2017) explored the connection between innovation education and hackathons, emphasizing how these events serve as practical learning platforms for students. The study revealed that the participation in hackathons enhances students' entrepreneurial readiness by providing hands-on experience in problem-solving within tight time frames. The real-time collaboration required during hackathons fosters teamwork, communication, and creative thinking—skills vital for entrepreneurship. Moreover, exposure to startup ecosystems through mentorship and networking opportunities helps students understand the dynamics of launching and scaling new ventures.

Kienzler & Fontanesi (2020) explored the role of hackathons within innovation education frameworks, emphasizing that these events provide students with practical, hands-on experience in problem-solving and teamwork. They argued that participation in hackathons enhances students' entrepreneurial readiness by immersing them in real-time collaboration and exposing them to startup ecosystems. This experiential learning fosters critical skills such as creativity, adaptability, and business acumen, which are essential for successful entrepreneurship.

Kuk & Davis (2021) examined digital hackathons conducted during the COVID-19 pandemic and found that virtual formats can match the effectiveness of in-person events when supported by strong mentorship and active community engagement. The study highlighted the crucial role of robust communication tools in maintaining collaboration and motivation among participants and despite challenges posed by the online environment, well-structured digital hackathons successfully fostered innovation and teamwork, demonstrating that remote hackathons can be a viable alternative in crisis situations.

Sivakumar, R., & Anandan, R. (2021), studied the impact of industry-led student hackathons in Tamil Nadu, revealing that participation significantly enhanced engineering graduates' employability skills and boosted their entrepreneurial aspirations. Their research highlighted how real-world challenges and industry mentorship during hackathons helped students develop practical, job-ready skills and increased their confidence to pursue startup ventures. The study emphasized the value of ongoing industry engagement to sustain these positive outcomes beyond the events.

Rajasekar, J., & Pushpalatha, M. (2020), focused on the role of academic incubators in supporting hackathon winners and identified a significant gap in sustained institutional mentoring after the events. The study emphasized that without structured incubation programs within colleges, many promising ideas fail to progress into viable startups for stronger, ongoing support mechanisms to help hackathon participants transition from idea generation to successful entrepreneurship.

Karthikeyan, P., & Krishnamurthy, S. (2019), evaluated the effectiveness of the Smart India Hackathon among participants from Tamil Nadu. They found that while awareness and participation at the state level are steadily increasing, there is still limited support available for scaling hackathon ideas into viable startups. The study highlighted the need for stronger post-event guidance and resources to help participants convert innovations into sustainable businesses.

Menaka, R., & Gopal, R. (2019), analyzed government-backed innovation challenges and their alignment with the goals of India's New Education Policy. They emphasized that integrating hackathons into academic credit systems can foster deeper student engagement and enhance the development of entrepreneurial skills. The study suggested that such integration would encourage sustained participation and better prepare students for innovation-driven careers.

Sundaram, S., & Narayanan, M. (2021), studied the effectiveness of MSME hackathons in Tamil Nadu, finding that well-structured mentorship and government funding support significantly increased the adoption of innovations and the formation of startups. The research highlighted the critical role of organized guidance and financial backing in turning hackathon ideas into viable business ventures, especially for grassroots innovators.

RESEARCH PROBLEM:

The research problem focuses on the limited understanding of how hackathons truly impact startup development in Tamil Nadu. Despite their growing popularity, there is insufficient insight into participants' awareness levels and how effectively hackathons translate into real-world innovation, startup creation, and enhanced entrepreneurial confidence among youth. This gap calls for a detailed assessment to measure the actual influence of hackathon involvement on fostering sustainable entrepreneurial outcomes.

RESEARCH OBJECTIVES:

1. To assess the level of awareness about hackathons among youth and entrepreneurs in Tamil Nadu.
2. To evaluate the influence of hackathon participation on the development of innovative startups.
3. To analyze the role of mentorship, funding availability, and institutional support in post-hackathon startup success.
4. To identify the challenges faced by hackathon participants in converting ideas into startups.
5. To suggest measures to improve the efficacy of hackathons in fostering innovation and entrepreneurship.

HYPOTHESIS:

H1: There is a significant relationship between hackathon participation and startup creation.

H2: Awareness level of hackathons significantly differs across demographic groups.

H3: Mentorship, funding, and institutional support significantly influence startup success post-hackathon.

H4: There is a positive correlation between hackathon participation frequency and innovation level in startups.

H5: Gender and educational background significantly influence the likelihood of startup formation post-hackathon.

RESEARCH METHODOLOGY:

1. **Type of Research:** The study employs a **descriptive and exploratory research** design to understand the awareness and impact of hackathons on startups in Tamil Nadu.
2. **Sampling Method:** Stratified random sampling.
3. **Sample Size:** 150 respondents from various districts of Tamil Nadu.
4. **Data Collection Method:** Structured questionnaire distributed via digital platforms and physical surveys.

Tools for Analysis:

- Frequency Analysis, ANOVA, Chi-Square Test & Multiple Regression Analysis

Variables:

1. **Independent Variables:** Age, gender, education, profession, hackathon awareness, mentorship availability, funding support, institutional backing.
2. **Dependent Variables:** Startup initiation, innovation level, post-hackathon success, entrepreneurial intent.

DATA ANALYSIS AND INTERPRETATION:**Table 1: Demographic Profile of Respondents:**

Demographic Variable	Category	Frequency	Percentage
Gender	Male	88	58.7%
	Female	62	41.3%

	Total	150	100.0%
Age Group	Below 20	25	16.7%
	21–30	85	56.7%
	31–40	28	18.7%
	Total	150	100.0%
Education	UG	68	45.3%
	PG	54	36.0%
	Others	28	18.7%
	Total	150	100.0%
Profession	Student	77	51.3%
	Entrepreneur	39	26.0%
	Professional/Other	34	22.7%
	Total	150	100.0%

Interpretation: The demographic profile of the respondents shows a diverse and representative sample. Regarding gender, 58.7% were male and 41.3% female, indicating a slightly higher male participation. Most respondents (56.7%) fell within the 21–30 age group, reflecting a young adult population actively engaged in hackathons, followed by 18.7% aged 31–40 and 16.7% below 20 years.

In terms of education, nearly half (45.3%) were undergraduates, while 36% had completed postgraduate studies, and 18.7% belonged to other educational categories, showing a varied academic background. Professionally, over half (51.3%) were students, highlighting the strong presence of learners in the sample, with 26% entrepreneurs and 22.7% professionals or others, related to startups and innovation.

Overall, the demographic data reflects a youthful, educated, and diverse group, which is appropriate for assessing awareness and impact of hackathons on innovation and entrepreneurship in Tamil Nadu.

Table 2: Awareness Level of Hackathons:

Awareness Level	Frequency	Percentage
High	45	30%
Moderate	70	46.7%
Low	35	23.3%
Total	150	100.0%

Interpretation: The awareness level among the respondents indicates that 30% have a high understanding of hackathons, while the largest group, 46.7%, possess a moderate level of awareness. Meanwhile, 23.3% of respondents have low awareness about hackathons. This suggests that while nearly half of the participants are reasonably informed, there is still a

significant portion with limited knowledge, highlighting the need for increased outreach and education to improve overall awareness.

Table 3: Hackathon Participation Vs Startup Initiation using Anova:

Source	SS	df	MS	F	Sig.
Between Gr.	112.4	2	56.2	5.76	0.004**
Within Gr.	1394.6	147	9.49		

Interpretation: The ANOVA results show a statistically significant difference in startup initiation based on hackathon participation levels ($F = 5.76$, $p = 0.004$). Since the p-value is less than 0.01, we reject the null hypothesis and conclude that the level of hackathon participation has a meaningful impact on whether participants initiate startups. This suggests that increased involvement in hackathons is associated with higher rates of startup creation among respondents. The significant F-value suggests that startup initiation varies meaningfully with different levels of hackathon participation. Participants in multiple events showed a higher likelihood of launching startups.

Table 4: Awareness vs Demographics Relationship using Chi-square:

Variable	Chi-square	df	Sig.
Age vs Awareness	14.67	4	0.005**
Education vs Awareness	18.23	3	0.002**
Gender vs Awareness	10.21	1	0.001**
Profession vs Awareness	12.78	2	0.008**

Interpretation: The Chi-square test results indicate significant associations between awareness levels about hackathons and various demographic factors. Specifically:

- **Age vs Awareness:** With a Chi-square value of 14.67 ($df = 4$, $p = 0.005$), awareness significantly varies across different age groups, suggesting that certain age categories are more knowledgeable about hackathons than others.
- **Education vs Awareness:** A Chi-square value of 18.23 ($df = 3$, $p = 0.002$) shows a strong relationship between education level and awareness, indicating that higher educational qualifications correspond to greater awareness of hackathons.
- **Gender vs Awareness:** The Chi-square value of 10.21 ($df = 1$, $p = 0.001$) reveals that awareness differs significantly between males and females.
- **Profession vs Awareness:** With a Chi-square of 12.78 ($df = 2$, $p = 0.008$), awareness levels are also significantly influenced by respondents' professional status (students, entrepreneurs, professionals).

Overall, demographic factors play a significant role in influencing awareness about hackathons. Demographic factors significantly influence awareness. Younger and more educated respondents show higher awareness. Students report more hackathon exposure, while working professionals lag behind, possibly due to time constraints or limited organizational promotion.

Table 5: Factors Influencing Startup Success using Regression Analysis:

Variable	B	Std. Error	Beta	t	Sig.
Mentorship	0.331	0.072	0.398	4.59	0.000**
Funding Support	0.267	0.063	0.357	4.23	0.000**
Institutional Backing	0.198	0.061	0.294	3.25	0.001**
Gender (Male = 1)	0.123	0.058	0.134	2.12	0.035*
Education Level	0.156	0.064	0.162	2.43	0.016*
Constant	1.478	0.429		3.45	0.001

$R^2 = 0.684$ | Adjusted $R^2 = 0.671$ | $F = 19.62$ | $p < 0.001$

Interpretation: The regression analysis reveals several key factors that significantly influence startup success among hackathon participants:

- **Mentorship ($\beta = 0.398$, $p < 0.001$):** Mentorship emerged as the most influential factor. A unit increase in mentorship support is associated with a 0.331-point increase in startup success, indicating the crucial role of guided expertise in nurturing early-stage ventures.
- **Funding Support ($\beta = 0.357$, $p < 0.001$):** Financial assistance significantly contributes to startup outcomes. Participants with access to funding demonstrate a higher likelihood of continuing their hackathon projects into viable businesses.
- **Institutional Backing ($\beta = 0.294$, $p = 0.001$):** Support from colleges, universities, or incubation centers positively impacts startup formation, highlighting the need for continuous academic support.
- **Gender (Male = 1) ($\beta = 0.134$, $p = 0.035$):** Gender has a small but significant influence, with male participants showing slightly higher startup conversion rates—potentially due to better networking or confidence levels.
- **Education Level ($\beta = 0.162$, $p = 0.016$):** Higher educational attainment correlates with startup success, possibly due to increased access to knowledge, skills, and entrepreneurial exposure.
- **Model Fit:** The model explains **68.4% of the variance** in startup success ($R^2 = 0.684$, Adjusted $R^2 = 0.671$), indicating a strong predictive relationship. The overall regression model is statistically significant ($F = 19.62$, $p < 0.001$), confirming that the chosen variables collectively explain startup outcomes meaningfully.

The regression model explains 68.4% of the variance in startup success. Mentorship contributes most strongly, followed by funding and institutional support. Gender and education also significantly influence outcomes, suggesting targeted support for underrepresented groups could yield higher startup conversion.

MAJOR FINDINGS:

1. 76.7% of respondents have moderate to high awareness of hackathons.
2. Participation in hackathons significantly influences startup creation and innovation level.
3. Demographics such as age, gender, education, and profession affect awareness levels.
4. Mentorship, funding support, and institutional backing play a crucial role in startup success post-hackathon.

5. Educational background and gender are statistically significant in predicting startup success.
6. Challenges include limited access to post-hackathon support, lack of market exposure, and scalability issues.
7. Most successful hackathon startups benefited from continued mentorship and incubation.
8. Awareness is higher among tech and business students compared to those in arts and humanities.
9. Professionals show willingness but lack time and institutional encouragement to participate.
10. Female participants showed higher retention in startup efforts when supported by mentorship programs.

SUGGESTIONS:

1. Educational institutions should mandate hackathon participation through course credits. Government initiatives should include seed grants for hackathon-winning ideas. Continuous mentorship should be offered through incubators post-hackathon.
2. Awareness campaigns about hackathons should be conducted in rural and semi-urban areas. Public-private partnerships can be established to offer resources and infrastructure for prototype development.
3. Diversity-focused outreach can improve gender balance in participation and leadership. Industry collaborations should be strengthened to promote domain-specific hackathons.
4. Post-hackathon tracking mechanisms can be built to monitor project evolution.

CONCLUSION:

Hackathons serve as powerful enablers of innovation and entrepreneurship in Tamil Nadu. This study confirms that their influence extends beyond short-term collaboration to long-term startup development, especially when participants receive structured mentorship, institutional support, and funding. Increased awareness and access to such platforms significantly boost entrepreneurial outcomes. With strategic enhancements—particularly in post-event support—hackathons can evolve into sustainable engines of regional innovation and economic growth.

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STRATEGIC PARTNERSHIPS AND GLOBAL COLLABORATION: DRIVERS OF COMPETITIVE ADVANTAGE IN THE GLOBAL ECONOMY

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ABSTRACT

In an increasingly interconnected world, strategic partnerships and global collaborations have emerged as essential tools for organizational growth, innovation, and sustainable competitive advantage. This paper explores the theoretical foundations, key drivers, benefits, and challenges of forming strategic alliances across borders. Drawing on examples from sectors such as technology, healthcare, and manufacturing, the study underscores how collaboration enables organizations to leverage complementary strengths, access new markets, and drive global innovation. The paper concludes by offering a framework for evaluating and managing global partnerships effectively. Strategic partnerships and global collaboration refer to the formation of mutually beneficial relationships between organizations, governments, or individuals across different countries and industries. These partnerships leverage each other's strengths, expertise, and resources to achieve common goals, drive innovation, and address complex challenges. By collaborating globally, organizations can access new markets, technologies, and talent, while sharing risks and costs. Strategic partnerships can facilitate the exchange of knowledge, best practices, and innovative solutions, leading to improved outcomes and increased competitiveness. Effective global collaboration requires strong communication, trust, and a deep understanding of cultural and regulatory differences. It involves navigating diverse business environments, managing complex logistics, and adapting to changing market conditions. Through strategic partnerships and global collaboration, organizations can unlock new opportunities, drive growth, and make a positive impact on society. By working together, they can tackle complex global challenges, such as climate change, poverty, and inequality, and create a more sustainable and equitable future.

KEY WORDS

Strategic partnership, Global strategy, Competitive Advantage, Innovation, Partnership Management

INTRODUCTION

In today's interconnected world, strategic partnerships and global collaboration have become essential for organizations seeking to drive innovation, growth, and success. Globalization has transformed how businesses operate, shifting competitive dynamics and creating new opportunities for cross-border collaboration. Strategic partnerships—defined as long-term, mutually beneficial agreements between organizations—have become critical enablers of global expansion and innovation. This paper examines the strategic rationale, types, and outcomes of global partnerships, focusing on how organizations can harness these alliances to achieve sustainable growth.

By forming alliances with other businesses, governments, and stakeholders across different countries and industries, organizations can leverage each other's strengths, expertise, and

resources to achieve common goals and address complex challenges. This collaborative approach enables organizations to tap into new markets, technologies, and talent, while sharing risks and costs. It facilitates the exchange of knowledge, best practices, and innovative solutions, leading to improved outcomes and increased competitiveness. As the global landscape continues to evolve, strategic partnerships and global collaboration will play an increasingly important role in shaping the future of business, innovation, and society.

NEEDS OF STRATEGIC PARTNERSHIP

Organizations need strategic partnerships and global collaboration to:

- Access new markets, customers, and revenue streams
- Leverage complementary skills, expertise, and resources
- Drive innovation and stay competitive in a rapidly changing global landscape
- Share risks and costs associated with research, development, and market entry
- Enhance their global footprint and reputation
- Address complex global challenges that require collective action
- Stay ahead of emerging trends and technologies
- Foster a culture of collaboration, learning, and knowledge-sharing
- Improve supply chain resilience and adaptability
- Achieve sustainable growth and long-term success.

REVIEW OF LITERATURE

The literature on strategic partnerships and global collaboration highlights the growing importance of inter-organizational relationships in today's interconnected business environment. Research has shown that strategic partnerships can drive innovation, improve competitiveness, and enhance organizational performance. Global collaboration, in particular, allows organizations to leverage diverse perspectives, expertise, and resources to achieve shared goals. Studies have identified key factors that contribute to successful strategic partnerships, including mutual trust, clear communication, and aligned goals. Effective partnership management involves navigating cultural and organizational differences, managing conflicts, and building strong relationships. However, global collaboration also presents challenges, such as managing cultural and linguistic differences, navigating diverse regulatory environments, and addressing issues of power and equity. Researchers have highlighted the need for organizations to develop strategies for managing these challenges and building effective global partnerships. Overall, the literature suggests that strategic partnerships and global collaboration can be powerful tools for driving innovation, growth, and success in today's globalized economy. By understanding the key factors that contribute to successful partnerships and managing the challenges of global collaboration, organizations can unlock new opportunities and achieve their goals.

THEORETICAL FRAMEWORK

Resource-Based View (RBV)

According to RBV, firms gain competitive advantage by acquiring and leveraging valuable, rare, inimitable, and non-substitutable resources. Strategic partnerships enable access to such resources externally, such as technology, talent, or local market expertise.

Transaction Cost Economics (TCE)

TCE suggests that firms enter partnerships to minimize transaction costs compared to full-scale acquisitions or internal development. Global collaborations allow shared risk and reduced operational costs.

Network Theory

This theory posits that value is derived from being embedded in a network of relationships. In global markets, networks enhance information flow, trust, and innovation potential.

TYPES OF STRATEGIC PARTNERSHIPS

- **Joint Ventures:** e.g., Sony Ericsson – combining expertise in electronics and telecommunications.
- **Equity Alliances:** e.g., Renault-Nissan-Mitsubishi – sharing resources and platforms.
- **Non-equity Partnerships:** e.g., R&D collaborations in pharma between universities and biotech firms.
- **Public-Private Partnerships (PPPs):** especially in infrastructure and healthcare.

COMPONENTS AND FUNCTION

It includes mutual goals and objectives that align the interests of all parties involved, a shared vision and set of values that foster a sense of unity and purpose, and clearly defined roles and responsibilities that ensure each partner knows their contribution to the partnership. Effective communication channels are also crucial, facilitating the exchange of information, ideas, and feedback in a transparent and timely manner. Trust and commitment are foundational, as partners must rely on each other's expertise, integrity, and dedication to the partnership's success. Flexibility and adaptability are equally important, allowing partners to navigate challenges, capitalize on opportunities, and adjust their strategies as needed. Risk sharing and mitigation are also key benefits, as partners can distribute and manage risks more effectively, protecting their investments and ensuring stability. Innovation and technology transfer are accelerated through collaboration, enabling organizations to stay at the forefront of technological advancements and apply them in practical, impactful ways. Lastly, policy and advocacy efforts can be more influential when undertaken jointly, allowing partners to pool their resources, expertise, and voices to advocate for changes that support their shared goals and benefit the broader community. Through such collaborative efforts, organizations can address complex challenges more effectively, capitalize on emerging opportunities, and contribute to a more interconnected and prosperous world.

DRIVERS OF GLOBAL COLLABORATION

Access to New Markets

Partnerships with local firms help navigate regulatory landscapes, understand customer behavior, and establish brand presence.

Innovation and R&D Synergies

Collaborating with international research institutions and tech startups fosters faster innovation.

Cost and Risk Sharing

Especially in capital-intensive industries like aerospace or energy, sharing development and operational costs is crucial.

Supply Chain Optimization

Partnering globally ensures resilience and efficiency in supply chain operations.

CASE STUDIES**Microsoft and Nokia**

Microsoft's strategic partnership with Nokia enabled it to enter the mobile hardware space quickly. While the alliance eventually transitioned into an acquisition, it highlights both opportunities and risks of global partnerships.

Starbucks and Tata Group (India)

A joint venture that helped Starbucks expand into India by leveraging Tata's local expertise in supply chain and regulatory navigation.

Global Vaccine Collaboration – COVAX

A multilateral initiative to ensure equitable vaccine access worldwide. Demonstrates how public-private and cross-border partnerships can tackle global challenges.

CHALLENGES IN GLOBAL PARTNERSHIPS

- **Cultural Differences:** Misalignment in corporate cultures can hinder cooperation.
- **Governance and Control:** Balancing autonomy and coordination.
- **Intellectual Property Risks:** Especially in high-tech sectors.
- **Regulatory Compliance:** Navigating diverse legal environments.

CONCLUSION

In conclusion, strategic partnerships and global collaboration are essential components of success in today's interconnected world. By forming alliances with other organizations, businesses, and governments, entities can leverage each other's strengths, expertise, and resources to achieve common goals and address complex challenges. Effective global collaboration requires strong communication, trust, and a deep understanding of cultural and regulatory differences. As the world becomes increasingly interconnected, strategic partnerships and global collaboration will continue to play a vital role in driving innovation, economic growth, and sustainable development. Through these partnerships, organizations can access new markets, technologies, and talent, while sharing risks and costs. Ultimately, strategic partnerships and global collaboration have the potential to create a more prosperous, equitable, and sustainable world, where organizations and individuals can thrive and reach their full potential. They also facilitate the exchange of knowledge, best practices, and innovative solutions, leading to improved outcomes and increased efficiency. Furthermore, strategic partnerships and global collaboration can help organizations navigate complex global challenges, such as climate change, poverty, and inequality. By working together, organizations can pool their resources, expertise, and influence to address these challenges and create a more sustainable and equitable world.

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**THE DIGITAL TRANSFORMATION OF HIGHER EDUCATION
STRATEGY, INNOVATION, AND TECHNOLOGICAL INTEGRATION**

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Abstract:

The digital transformation of higher education represents a comprehensive shift driven by the integration of strategy, technology, and innovation to meet the demands of an increasingly digital society. This transformation extends beyond mere digitization to encompass strategic realignment, cultural shifts, and adoption of innovative pedagogies that prepare students for complex modern labor markets. Theoretical frameworks such as the Resource-Based View, Diffusion of Innovations, and Technology Acceptance Models provide insight into how institutions can leverage internal capabilities and navigate technological adoption effectively. Key strategic components include vision-setting, leadership, governance, and collaborative partnerships to drive sustainable growth and innovation while managing risks like cybersecurity and ethical concerns. Technological innovations such as Learning Management Systems, Artificial Intelligence, Virtual and Augmented Reality, and Blockchain credentialing are reshaping teaching, learning, and administration. Pedagogical models now emphasize blended learning, personalized adaptive systems, gamification, and open educational resources, enhancing accessibility and engagement. Organizational change and capacity building are crucial to foster digital literacy, cultivate an innovative culture, and sustain transformation initiatives. Contemporary challenges focus on equity, access, and the accelerated impact of COVID-19 on digital readiness. Ultimately, this integrated approach positions higher education institutions to build resilient, inclusive, and future-ready academic environments that advance educational quality and institutional sustainability.

A. Conceptual Foundations of Higher Education in the Digital Era

In the rapidly evolving landscape of the 21st century, higher education stands at a critical crossroads, shaped fundamentally by the interplay of strategy, technology, and innovation. The concept of Higher Education 360° captures this holistic integration, emphasizing a comprehensive approach that incorporates institutional strategy, cutting-edge digital technologies, and innovative pedagogical models to meet the demands of an increasingly digital society (Snelson & Hsu, 2020). Understanding this multidimensional framework is essential for educators, policymakers, and stakeholders aiming to foster resilient, adaptive, and forward-looking academic institutions (Feurstein, 2018). This approach not only enhances educational quality but also aligns with the evolving needs of students and the workforce in a rapidly changing environment. The alignment is crucial as it prepares graduates to navigate the complexities of modern labor markets, where adaptability and technological proficiency are paramount to their success (Crawley et al., 2020). Emphasizing the integration of digital technologies within educational frameworks can significantly enhance learning outcomes and prepare students for future challenges. This holistic approach underscores the importance of continuous

improvement and innovation in higher education, ensuring that institutions remain relevant and effective in meeting societal needs.

Historically, higher education systems operated within relatively stable paradigms characterized by face-to-face instruction, traditional curricula, and hierarchical governance(Thelin, 2019). However, the advent of digital technologies has disrupted these conventional structures, enabling new modes of learning such as online education, blended learning, and virtual collaboration. This digital transformation extends beyond mere technological adoption—it requires strategic realignment and cultural shifts within institutions to effectively harness technology’s potential.

At its core, strategy in higher education involves the deliberate formulation of goals and the alignment of resources to achieve sustainable competitive advantage. When coupled with technology and innovation, strategy becomes a dynamic tool to drive organizational change, enhance educational quality, and expand access. Innovation in this context refers not only to technological tools but also to novel teaching methods, administrative processes, and institutional models that respond to evolving learner needs and labor market demands.

This article introduces foundational definitions and concepts pivotal to the discourse on Higher Education 360°, laying the groundwork for subsequent exploration of theoretical frameworks and contemporary applications. By systematically examining strategy, technology, and innovation in tandem, this chapter sets the stage for a comprehensive understanding of how digital transformation can reshape the future of higher education.

B. Core Definitions and Terminologies

To thoroughly understand the concept of Higher Education, it is vital to establish clear definitions of the key terms: digital transformation, strategic management, educational technology, innovation, and the stakeholders involved(Asrifan et al., 2025). These definitions provide a common language that ensures clarity and coherence throughout the chapter. Digital Transformation in Higher Education refers to the comprehensive integration of digital technologies into all areas of academic institutions, fundamentally changing how they operate, deliver education, and engage with learners and communities(Bucăța et al., 2022; Mamaeva et al., 2020). Unlike mere digitization, which focuses on converting analog information into digital form, digital transformation embodies strategic changes in processes, culture, and institutional structures to leverage technology for improved outcomes and agility.

Strategic Management in the context of higher education is the systematic planning and execution of long-term goals that align institutional resources, capabilities, and external environments to gain sustainable competitive advantage(Kuusisto-Ek, 2022; Stukalina, 2014). It involves analyzing internal strengths and weaknesses, assessing external opportunities and threats (SWOT), and formulating responsive strategies to adapt to dynamic challenges, including technological advancements and shifting educational demands.

Educational Technology (EdTech) encompasses the tools, platforms, and software designed to enhance teaching, learning, and administrative processes(Adıgüzel et al., 2023). These range from Learning Management Systems (LMS) and virtual classrooms to artificial intelligence-powered tutoring systems. EdTech facilitates personalized learning

experiences, expands access, and enables data-driven decision-making in academic environments.

Innovation within higher education goes beyond technology adoption; it represents the creation or improvement of novel ideas, methods, and organizational processes that improve educational effectiveness, accessibility, and relevance(Lai et al., 2024). Innovation can be pedagogical (e.g., flipped classrooms), technological (e.g., VR simulations), or strategic (e.g., new partnership models), each contributing to institutional transformation.

Stakeholders in the Higher Education Ecosystem include students, faculty, administrators, policymakers, industry partners, and communities. Their diverse interests and influences shape institutional strategies and outcomes, making stakeholder engagement crucial for successful digital transformation and innovation. Establishing these core definitions enables a systematic exploration of how strategy, technology, and innovation converge in the evolving landscape of higher education. The next section will delve into the theoretical frameworks that underpin these concepts and guide their practical application.

C. Theoretical Frameworks Underpinning Strategy and Innovation in Higher Education

The landscape of higher education is continuously evolving, driven by the need for institutions to adapt to the complexities of a globalized economy, technological advancements, and shifting societal expectations(Kaluzhnova et al., 2018). At the core of this evolution lies the interplay between strategy and innovation, which serves as a critical determinant of institutional effectiveness and sustainability. To understand this dynamic, it is essential to explore the theoretical frameworks that inform strategic decision-making and innovative practices within higher education. Various models, including resource-based theory, institutional theory, and disruptive innovation theory, provide valuable insights into how higher education institutions can leverage their unique resources and capabilities to foster innovation and respond to external pressures(Wu & Gu, 2022). These frameworks not only elucidate the mechanisms through which strategies are formulated and implemented but also highlight the importance of fostering a culture of innovation that aligns with institutional goals(Ghofur, 2025).

A robust understanding of the interplay between strategy, technology, and innovation in higher education requires grounding in established theoretical frameworks. These theories provide analytical lenses to interpret institutional behavior, technology adoption, and innovation processes, enabling systematic and evidence-based decision-making.

1. Resource-Based View (RBV) and Dynamic Capabilities Theory

The Resource-Based View (RBV) and Dynamic Capabilities Theory are two pivotal frameworks in strategic management that elucidate how firms can achieve and sustain competitive advantage in an increasingly complex and dynamic business environment(Lubis, 2022). The RBV posits that a firm's unique resources and capabilities are the primary determinants of its performance and competitive positioning. By focusing on the internal attributes of the organization, RBV emphasizes the importance of valuable, rare, inimitable, and non-substitutable resources in fostering superior outcomes. Conversely, Dynamic Capabilities Theory extends this perspective by highlighting the necessity for firms to not only possess valuable resources but also to develop the ability to adapt, integrate, and reconfigure

these resources in response to changing market conditions(Samsudin & Ismail, 2019). This theory underscores the significance of agility and innovation in maintaining competitiveness over time. Together, these frameworks offer a comprehensive lens through which to analyze the strategic maneuvers of firms, providing insights into how they can leverage their resource endowments and dynamic capabilities to navigate the challenges of an evolving marketplace. These theories collectively suggest that firms must cultivate both robust resource bases and agile capabilities to thrive in volatile environments, ensuring long-term success and adaptability. This integrated approach allows organizations to not only capitalize on their existing strengths but also to proactively respond to emerging opportunities and threats in their respective industries. The synthesis of RBV and Dynamic Capabilities Theory highlights the critical interplay between resource management and adaptability, essential for firms aiming to achieve sustained competitive advantage in turbulent markets. RBV posits that an institution's competitive advantage stems from its unique resources and capabilities that are valuable, rare, inimitable, and non-substitutable (VRIN). In higher education, resources include faculty expertise, technological infrastructure, and intellectual property. Dynamic capabilities extend this view by emphasizing an institution's ability to adapt, integrate, and reconfigure internal and external competencies to address rapidly changing environments, such as those brought about by digital disruption.

2. Diffusion of Innovations Theory

The Diffusion of Innovations Theory, first articulated by Everett Rogers in 1962, serves as a pivotal framework for understanding how new ideas, practices, and technologies spread within and among communities(Dearing, 2009). This theory posits that the adoption of innovations is influenced by a range of factors including the perceived attributes of the innovation, the characteristics of the individuals involved, and the social systems in which they operate. Central to this theory is the classification of adopters into categories such as innovators, early adopters, early majority, late majority, and laggards, each exhibiting distinct behaviors and motivations regarding the acceptance of new innovations. As global challenges and technological advancements continue to evolve, the relevance of the Diffusion of Innovations Theory remains significant, providing insights into how innovations can be effectively disseminated to enhance societal progress and address pressing issues(Prescott, 1995). Understanding the dynamics of innovation diffusion is crucial for organizations aiming to leverage new technologies effectively and maintain a competitive edge in their respective industries. his understanding can guide organizations in their strategic decision-making processes, ensuring they are well-positioned to adopt innovations that align with their goals and societal needs. This article will further investigate the role of perceived desirability in the adoption of innovations, particularly focusing on how compatibility and relative advantage influence organizational decision-making. Ultimately, the findings will contribute to a deeper understanding of the mechanisms driving innovation adoption and the importance of aligning technological advancements with organizational values and societal expectations. This alignment is essential not only for successful innovation adoption but also for fostering a sustainable competitive advantage in an increasingly interconnected market. This theory explains how new ideas, practices, or technologies spread within social systems over time. It identifies key factors influencing adoption, including the perceived attributes of the innovation (relative advantage, compatibility, complexity, trialability,

observability), communication channels, and social system characteristics. In higher education, understanding diffusion dynamics helps strategize the introduction of digital tools and pedagogical innovations to maximize acceptance and impact.

3. Technology Acceptance Model (TAM) and Unified Theory of Acceptance and Use of Technology (UTAUT)

The rapid advancement of technology has necessitated a deeper understanding of how and why individuals adopt new technological innovations. Among the various frameworks developed to analyze this phenomenon, the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) stand out as pivotal constructs. TAM, introduced by Davis in 1989, posits that perceived ease of use and perceived usefulness significantly influence users' decisions to accept and utilize technology (Momani, 2020). This model has been widely applied across diverse contexts, providing insights into user behavior and technology integration. Conversely, UTAUT, developed by Venkatesh et al. in 2003, expands upon TAM by incorporating additional factors such as social influence and facilitating conditions, thereby offering a more comprehensive understanding of technology acceptance in organizational settings (Oye et al., 2014).

4. Strategic Alignment and Change Management Theories

In today's rapidly evolving business landscape, the interplay between strategic alignment and change management has emerged as a critical area of study for organizations seeking to maintain competitive advantage. Strategic alignment refers to the process of ensuring that an organization's structure, resources, and activities are in harmony with its strategic objectives, thereby facilitating the effective execution of its vision and mission (Crawley et al., 2020). Concurrently, change management encompasses the methodologies and practices employed to guide individuals and organizations through transitions, ensuring that changes are implemented smoothly and sustainably. Theories surrounding strategic alignment and change management offer valuable insights into how organizations can navigate complexities inherent in both maintaining alignment with strategic goals and effectively managing change initiatives. Understanding the synergy between these theories is crucial for organizations aiming to enhance their adaptability and resilience in the face of ongoing change. This synergy not only fosters a proactive approach to change but also mitigates the risks associated with strategic drift and resistance to change (Khan & Khan, 2013). By emphasizing the importance of communication, participation, and empathy in the change process, these strategies help in addressing the concerns of employees, thereby fostering a culture of collaboration and support during transitions. Strategic alignment theory stresses the importance of congruence between an institution's business strategy and IT strategy to achieve optimal performance. Change management frameworks (e.g., Kotter's 8-Step Model, Lewin's Change Theory) provide structured approaches to managing the human and organizational aspects of digital transformation, addressing resistance and fostering a culture of innovation.

Collectively, these theoretical frameworks offer comprehensive insights into how higher education institutions can effectively strategize, adopt technology, and foster innovation to thrive in the digital age. The following sections will build upon these foundations to

explore practical strategic approaches, technological innovations, and pedagogical advancements.

D. Strategic Approaches to Digital Transformation in Higher Education

Strategic management is paramount for higher education institutions aiming to navigate the complexities of digital transformation. A well-crafted digital strategy aligns institutional vision with technological capabilities and innovation initiatives, ensuring sustainable growth and enhanced educational outcomes.

1. Developing Digital Strategy: Vision, Goals, and Roadmaps

A successful digital strategy begins with a clear institutional vision that reflects the mission and values of the university or college in the digital context. This vision translates into specific goals, such as improving student engagement, expanding online program offerings, or enhancing operational efficiency. Roadmaps then outline actionable steps, timelines, and resource allocation necessary to achieve these goals, ensuring coherent progress.

2. Leadership and Governance in Digital Era Universities

Effective leadership is critical to champion digital transformation. Leaders must foster an environment that encourages innovation, supports risk-taking, and facilitates cross-functional collaboration. Governance structures should include clear accountability mechanisms and policies that promote data privacy, security, and ethical use of technology, aligning with institutional values and regulatory requirements.

3. Collaborative Models and Partnership Strategies for Innovation

Digital transformation often requires partnerships beyond the campus boundary. Collaborations with technology providers, industry, governmental agencies, and other academic institutions can accelerate innovation by pooling resources, expertise, and infrastructure. Strategic alliances support co-creation of digital solutions, knowledge exchange, and scalability of successful initiatives.

4. Risk Management and Ethical Considerations in Digital Strategy

Implementing digital initiatives entails risks, including cybersecurity threats, data breaches, and ethical concerns related to AI use and algorithmic bias. Institutions must adopt proactive risk management frameworks that identify vulnerabilities, establish mitigation protocols, and ensure compliance with legal and ethical standards to safeguard stakeholder trust.

By integrating these strategic components, higher education institutions can effectively lead digital transformation efforts that are resilient, inclusive, and aligned with broader educational goals. The next section will focus on the technological innovations driving this transformation.

E. Technological Innovations Shaping Higher Education

The rapid advancement of technology has become a cornerstone of transformation within higher education, fundamentally reshaping teaching, learning, and institutional operations. This section highlights key technological innovations driving change and enhancing educational quality in the digital age.

1. Learning Management Systems (LMS) and Virtual Learning Environments (VLE)

LMS platforms such as Moodle, Blackboard, and Canvas provide centralized digital ecosystems that support course delivery, assessment, communication, and

collaboration. VLEs extend these capabilities by creating immersive online spaces where learners and instructors interact dynamically, enabling flexible access and asynchronous learning that cater to diverse student needs.

2. Artificial Intelligence (AI) and Data Analytics in Education

AI-powered tools facilitate personalized learning through adaptive assessments, intelligent tutoring systems, and predictive analytics that identify students at risk and recommend targeted interventions (Abbasi & Hussain, 2024; Alqurashi et al., 2023). Data analytics further supports institutional decision-making by revealing trends in enrollment, engagement, and academic performance, enabling evidence-based policy and strategy formulation.

3. Virtual Reality (VR), Augmented Reality (AR), and Immersive Learning

VR and AR technologies create simulated, interactive environments that enhance experiential learning, particularly in fields such as medicine, engineering, and the arts. These immersive technologies promote deeper engagement, critical thinking, and practical skill development by replicating real-world scenarios within safe, controlled settings.

4. Blockchain and Credentialing Technologies

Blockchain offers a secure, transparent mechanism for issuing and verifying academic credentials, addressing challenges related to fraud and administrative inefficiencies. Digital badges and microcredentials supported by blockchain enable learners to showcase specific skills and competencies, fostering lifelong learning and alignment with labor market demands.

Furthermore, these technological innovations empower higher education institutions to transcend traditional limitations, enhance learning experiences, and adapt swiftly to the evolving digital landscape. The subsequent section will explore how these technologies inform pedagogical innovations and new learning models.

F. Pedagogical Innovations and Digital Learning Models

The integration of technology in higher education has catalyzed the evolution of teaching and learning paradigms, giving rise to innovative pedagogical models that enhance learner engagement, accessibility, and outcomes. This section explores key digital learning frameworks and instructional strategies shaping the future of education.

1. Blended and Hybrid Learning Frameworks

Blended learning combines traditional face-to-face instruction with online components, allowing for flexible scheduling and personalized pacing. Hybrid models extend this by integrating synchronous and asynchronous activities, fostering interaction and collaboration across physical and virtual spaces (Adarkwah & Huang, 2023; Bruggeman et al., 2021). These frameworks accommodate diverse learner preferences and support differentiated instruction.

2. Personalized Learning and Adaptive Technologies

Adaptive learning systems use algorithms and data analytics to tailor educational content, pace, and assessment based on individual learner profiles. By addressing unique strengths and challenges, personalized learning promotes mastery of concepts, enhances motivation, and reduces dropout rates (Kaswan et al., 2024; Silva et al., 2024). This learner-centered approach aligns with contemporary educational theories emphasizing active, self-regulated learning.

3. Gamification and Experiential Learning in the Digital Age

Gamification incorporates game design elements—such as points, leaderboards, and challenges—into educational settings to increase motivation and engagement. Experiential learning leverages simulations, virtual labs, and project-based tasks to provide hands-on experiences that develop critical thinking and practical skills. Both approaches harness technology to create immersive, interactive learning environments.

4. Open Educational Resources (OER) and Massive Open Online Courses (MOOCs)

OERs are freely accessible teaching and learning materials that promote equity and knowledge sharing. MOOCs provide large-scale, open-access courses often delivered by prestigious institutions, expanding educational opportunities globally. These innovations democratize learning and encourage continuous professional development.

By embracing these pedagogical innovations, higher education institutions can create inclusive, effective, and future-ready learning environments that respond to the needs of diverse student populations. The next section will examine organizational change and capacity building essential to sustaining these advancements.

G. Organizational Change and Capacity Building for Innovation

Sustaining innovation and digital transformation in higher education requires more than technological adoption; it demands profound organizational change and deliberate capacity building, to ensure that institutions can effectively integrate new technologies while fostering an environment conducive to continuous learning and adaptation (Hikmah et al., 2024). This holistic approach not only enhances institutional resilience but also aligns with the broader trends of digital transformation in higher education, emphasizing the importance of leadership and strategic management. This alignment is crucial for navigating the complexities introduced by the fourth industrial revolution, where advanced technologies like AI and data analytics play a significant role in shaping educational practices and outcomes. Moreover, fostering a culture of innovation within institutions is essential to empower staff and students, enabling them to embrace and drive the changes associated with digital transformation (Adıgüzel et al., 2023). This cultural shift can significantly enhance the effectiveness of leadership in managing these transitions. The role of effective leadership is paramount in facilitating this cultural shift, as it helps to inspire and motivate stakeholders towards a shared vision of innovation and adaptability in higher education. Ultimately, successful digital transformation hinges on leaders who can cultivate collaboration among all stakeholders, ensuring that technological advancements translate into meaningful educational improvements.

This section explores strategies to develop institutional readiness and cultivate a culture that supports continuous innovation.

1. Building Digital Literacy and Competency Among Faculty and Staff

In an increasingly digital world, the importance of digital literacy and competency among faculty and staff cannot be overstated. As educational institutions strive to enhance their teaching methodologies and administrative efficiencies, the integration of digital tools and technologies has become essential. The development of digital literacy encompasses not only the ability to use technology effectively but also the critical understanding of its implications in the educational landscape (Tsvetkova & Kiryukhin, 2019; Zabolotska et al., 2021). Enhancing digital literacy among faculty and staff is crucial for fostering a culture of innovation and collaboration, ultimately leading to improved educational outcomes and institutional success. Moreover, ongoing

professional development and institutional support are vital to ensure that faculty and staff are equipped with the necessary skills to navigate and utilize digital technologies effectively. By prioritizing digital literacy initiatives, institutions can significantly boost academic productivity and enhance the overall quality of education provided to students (Hasani, 2024). Faculty and administrative staff must acquire digital skills essential for leveraging new technologies effectively. Professional development programs, workshops, and continuous training foster digital literacy, ensuring educators can design, deliver, and assess technology-enhanced learning experiences proficiently.

2. Fostering an Innovative Culture “Challenges and Opportunities”

Creating an innovative culture requires addressing various challenges, such as resistance to change and resource limitations, while also leveraging opportunities for collaboration and leadership support. To successfully navigate these challenges, organizations must prioritize open communication and encourage risk-taking among employees, thus creating an environment conducive to innovation. This approach not only enhances creativity but also aligns with the need for strategic leadership to motivate individuals towards achieving organizational goals. Additionally, leaders must recognize that cultivating an innovative culture is an ongoing process that requires commitment and adaptability to overcome inherent challenges.

Moreover, fostering an innovative culture within organizations is a multifaceted endeavor that presents both challenges and opportunities. The need to address resistance to change and resource limitations is paramount, yet these obstacles can be transformed into opportunities through effective collaboration and strong leadership support. By prioritizing open communication and encouraging risk-taking, organizations can create an environment that not only enhances creativity but also aligns with strategic leadership goals. Furthermore, it is essential for leaders to understand that cultivating such a culture is not a one-time initiative but an ongoing process that demands continuous commitment and adaptability. Embracing these principles will ultimately empower organizations to thrive in an ever-evolving landscape, driving sustained innovation and success.

3. Monitoring and Evaluating Digital Initiatives

Continuous assessment of digital projects through qualitative and quantitative metrics ensures alignment with strategic objectives and informs iterative improvements. Establishing feedback loops involving students, faculty, and administrators supports data-driven refinement and scalability of successful innovations.

Organizational change and capacity building are pivotal to embedding innovation as an integral and sustainable dimension of higher education institutions. The next section will address contemporary issues and emerging trends shaping this dynamic landscape.

H. Contemporary Issues and Emerging Trends

As higher education embraces digital transformation, institutions confront a complex array of contemporary issues and emerging trends that influence strategy, technology adoption, and innovation pathways.

1. Equity, Access, and Inclusion in Digital Higher Education

Digital technologies hold promise for democratizing education, yet disparities in internet access, digital literacy, and socioeconomic status risk exacerbating inequalities. Institutions must prioritize inclusive design, affordable connectivity, and support services to ensure all learners benefit from digital advancements.

2. Impact of COVID-19 Pandemic on Higher Education Innovation

The global pandemic accelerated the adoption of remote learning and exposed gaps in digital readiness. It also spurred innovation in curriculum delivery, assessment, and student support services, highlighting the necessity for flexible, resilient educational models that can adapt to future disruptions. The COVID-19 pandemic has emerged as a transformative force across various sectors, with higher education being one of the most significantly affected domains. As institutions grappled with the abrupt transition to remote learning, the pandemic catalyzed a wave of innovation in educational practices, technologies, and pedagogical approaches (Alsubaie, 2022; Bazimaziki, 2020; Núñez-Canal et al., 2022). This unprecedented challenge prompted universities and colleges to rethink traditional models of teaching and assessment, leading to the rapid adoption of digital tools and online platforms. The necessity for adaptive strategies not only highlighted existing disparities within the educational landscape but also spurred creative solutions aimed at enhancing accessibility and engagement (Bystrova & Tokarskaja, 2021). As educators and administrators navigated the complexities of a global health crisis, the resulting innovations have the potential to reshape the future of higher education, fostering resilience and adaptability in an ever-evolving academic environment. These innovations, driven by necessity, may lead to lasting changes in how education is delivered and experienced, ultimately contributing to a more inclusive and effective learning environment.

Moreover, the shift towards digital education has illuminated the critical need for equitable access to technology and resources, as disparities in digital literacy and infrastructure became glaringly apparent during the pandemic. Institutions that prioritized digital inclusion not only enhanced learning outcomes but also fostered a sense of community among students, bridging gaps that previously hindered engagement. For instance, universities that implemented comprehensive training programs for both faculty and students on digital tools experienced smoother transitions and higher levels of participation, suggesting that proactive measures can mitigate the challenges posed by such abrupt changes. As educational leaders reflect on these experiences, the lessons learned may inform future policies aimed at ensuring that all students, regardless of their background or circumstances, have the opportunity to thrive in an increasingly digital academic landscape. This focus on inclusivity could ultimately lead to a reimagined educational paradigm that values diverse learning experiences and perspectives. These developments underscore the importance of ongoing stakeholder collaboration and strategic leadership to foster an equitable digital ecosystem in higher education. By addressing the gaps in access and training, institutions can better support all learners in navigating the complexities of digital education. This collaborative effort will be essential in creating policies that ensure equitable access to digital resources and promote inclusive educational practices for all students. Ultimately, the ongoing commitment to digital transformation will require sustained collaboration between educational institutions, government bodies, and community stakeholders to ensure that no student is left behind. This commitment will be crucial in addressing the persistent inequalities that have been exacerbated by the pandemic, ensuring that all students can benefit from digital advancements in education.

I. Conclusion

In conclusion, the digital transformation of higher education represents a pivotal shift that necessitates the adoption of innovative strategies and the integration of advanced technologies. This evolution is not merely a response to external pressures but a proactive approach to enhancing educational outcomes and equipping institutions to thrive in an increasingly dynamic landscape. As higher education institutions navigate this transformation, it is imperative that they prioritize strategic planning and innovation to address the diverse needs of students and stakeholders. Ultimately, embracing digital transformation will enable these institutions to foster a more engaging, accessible, and effective learning environment, ensuring their relevance and sustainability in the years to come. This proactive approach will not only improve educational quality but also facilitate institutional resilience against future challenges. By fostering a culture of innovation and adaptability, higher education institutions can effectively respond to the evolving demands of the educational landscape and ensure long-term sustainability. This commitment to digital transformation is essential for creating a more inclusive and equitable educational experience that meets the diverse needs of all learners. This commitment will also empower institutions to leverage technology for personalized learning experiences, ultimately enhancing student engagement and success.

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SENTIMENT-AWARE RECOMMENDATION SYSTEMS: TRANSFORMING E-COMMERCE THROUGH EMOTION-DRIVEN AI

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Abstract

In today's digital landscape, recommendation systems play a crucial role in enhancing user experience across e-commerce, streaming, and online service platforms. Traditional systems often rely on numerical data such as ratings or purchase history, overlooking the emotional content present in user feedback. This study explores the evolution of recommendation systems through the integration of sentiment analysis, a method that interprets the emotional tone behind textual reviews. By incorporating advanced artificial intelligence techniques like transformer models and attention mechanisms, sentiment-aware systems can better understand user preferences and deliver more personalized, context-rich suggestions. The paper examines the benefits of this approach, its practical applications, and the challenges involved in accurately interpreting human emotion through language. The study concludes by highlighting the potential of sentiment-aware recommendation systems to transform digital interactions into more human-centered and emotionally intelligent experiences.

Keywords: Sentiment Analysis, Recommendation Systems, Artificial Intelligence, Personalization

Introduction

In the digital age, online platforms face an overwhelming abundance of information. Users expect fast, relevant, and personalized experiences — whether they are shopping on Amazon, watching movies on Netflix, or scrolling through products on Flipkart. To meet these expectations, recommendation systems have become essential tools. At their core, recommendation systems are algorithms that suggest items to users based on their preferences, behaviors, or similarities with other users. These systems are responsible for the curated playlists, product suggestions, and “you might also like” sections that we often encounter online.

Traditionally, these systems analyze past behavior — such as clicks, purchases, or ratings — to predict what a user might want next. While this behavior-based prediction is effective to a degree, it has one major limitation: it lacks emotional depth. People don't just choose products based on what they liked before; they also base decisions on how they feel. A customer might leave a five-star rating but express frustration in the review text — something traditional systems might miss entirely.

This is where sentiment analysis enters the picture. Sentiment refers to the emotional tone behind a piece of text — whether a review is happy, disappointed, angry, or neutral. By combining recommendation systems with sentiment analysis, we can better understand not only what a user did, but why they did it. This deeper understanding enables platforms to make smarter, more personalized suggestions that reflect the user's emotional needs and preferences.

As e-commerce, media streaming, and social platforms evolve, the integration of emotion-aware artificial intelligence marks a turning point. The shift from purely behavioral data to emotionally intelligent recommendation systems has the potential to dramatically improve user satisfaction, engagement, and trust.

Objectives of the Study

1. To highlight the limitations of traditional recommendation systems in capturing the emotional and contextual aspects of user preferences.
2. To explain the importance of sentiment analysis in enhancing recommendation accuracy through the interpretation of user emotions expressed in text.
3. To examine the application of modern AI technologies such as transformer models and attention mechanisms in building sentiment-aware recommendation systems.
4. To explore the practical benefits and challenges of implementing emotion-driven recommendations in real-world e-commerce and digital platforms.

Traditional vs. Modern Methods

For many years, recommendation systems have primarily relied on two foundational techniques: collaborative filtering and content-based filtering. These methods have powered the recommendation engines of early e-commerce and entertainment platforms, helping users discover items similar to what they or others liked.

Collaborative filtering works by identifying patterns among users. If User A and User B both liked the same books, and User A also liked a new book that User B hasn't seen yet, the system might recommend that new book to User B. This approach assumes that people who behaved similarly in the past will continue to do so.

Content-based filtering, on the other hand, focuses on item characteristics. If a user enjoys a certain genre of movies or prefers products with specific features (e.g., budget smartphones with long battery life), the system suggests other items that share those characteristics.

While these approaches have achieved considerable success, they are not without flaws. They often struggle with the cold-start problem, where there isn't enough data about a new user or product to make accurate suggestions. They also lack the ability to consider contextual and emotional nuances. For instance, two users might give the same rating to a product, but for completely different reasons — one because it met expectations, and another because it failed but arrived quickly.

To address these limitations, modern recommendation systems are incorporating advanced tools from Artificial Intelligence (AI) and Natural Language Processing (NLP). These tools analyze not just structured data (like ratings), but also unstructured data such as written reviews, comments, and even social media posts. With this shift, recommendation systems are evolving from simple prediction tools into sophisticated models that can detect tone, emotion, context, and intent.

This evolution has given rise to sentiment-aware and context-aware recommendation engines. Instead of just calculating averages or similarities, these systems understand what users are truly saying — and more importantly, how they feel — when interacting with products or services. The result is a more accurate, human-centered approach to personalization.

The Role of Sentiment Analysis

In today's digital environment, people frequently share their opinions through online reviews, comments, and social media posts. These texts are rich sources of user insight — not just about what people think, but how they feel. This emotional layer of information is known as sentiment. Sentiment analysis is a branch of Natural Language Processing (NLP) that focuses on identifying and interpreting these emotions within written language.

At its core, sentiment analysis classifies text into categories such as positive, negative, or neutral. More advanced models can detect finer emotions like anger, happiness, or disappointment. For instance, a review saying *“The phone is okay, but the battery life is terrible”* might be assigned an overall negative sentiment, even if the numerical rating is moderate.

In recommendation systems, integrating sentiment analysis offers a deeper layer of intelligence. It helps platforms move beyond what users *say* numerically (through ratings) to what they *mean* textually. This is important because users often express mixed emotions — they might rate a product 3 out of 5 but describe it positively in the review. Without analyzing the text, traditional systems may misinterpret the user's actual opinion.

Moreover, sentiment analysis helps detect subtle patterns and preferences that might not be visible through ratings alone. For example, a customer consistently mentioning “fast delivery” in their reviews indicates a preference for quick shipping — a detail that could guide future recommendations. Another might repeatedly highlight customer service quality, revealing values that go beyond product specifications.

By incorporating sentiment analysis, modern recommendation systems become emotionally intelligent. They begin to treat users not just as data points, but as individuals with emotional responses, preferences, and expectations. This makes the recommendations not only more accurate, but also more personalized and human-centric — a vital factor in building trust and long-term user engagement.

Emerging Technologies Behind the Scenes

Behind every advanced recommendation system today is a combination of powerful artificial intelligence technologies working quietly in the background. These technologies are designed not only to process large volumes of data but also to understand the deeper meaning behind human language. In the context of sentiment-aware recommendation systems, two key innovations stand out: Transformer models and Recurrent Neural Networks (RNNs).

One of the most significant breakthroughs in natural language processing has been the development of Transformer-based models, such as BERT (Bidirectional Encoder Representations from Transformers) and RoBERTa (a robustly optimized version of BERT).

These models are capable of understanding the context of words in a sentence, much like how humans do. For instance, the word “cold” means something very different in “cold weather” versus “a cold response,” and transformers can recognize such nuances. This makes them ideal for analyzing product reviews, where meaning and emotion are often subtle.

Transformer models like RoBERTa are trained on vast amounts of text data and learn to recognize complex sentence structures, sarcasm, and implied meanings. This allows them to extract rich, contextual features from user-written content — a critical step in identifying sentiment accurately.

In addition to transformers, systems often use Recurrent Neural Networks (RNNs) — particularly a type known as Bidirectional Gated Recurrent Units (BiGRU). These models are designed to understand the sequence of words in a sentence, looking both forward and backward to capture long-range dependencies in language. This is especially important in reviews where the sentiment may unfold gradually over several sentences.

After extracting contextual features using these models, recommendation systems typically use attention mechanisms like Multi-Head Self-Attention (MHSA) to decide which parts of the review are most important. This ensures that the system pays more attention to emotionally charged or informative phrases and less to generic or repetitive text.

By combining these cutting-edge technologies, sentiment-aware recommendation systems become capable of truly understanding human expression. They transform raw, unstructured reviews into meaningful signals that guide highly personalized and emotionally aligned recommendations — a major advancement from earlier generations of algorithm-driven suggestions.

Practical Applications and Benefits

Sentiment-aware recommendation systems are rapidly becoming essential across industries, especially in e-commerce, media streaming, hospitality, and online education. Their ability to understand and interpret user emotions allows businesses to deliver smarter, more relevant experiences that go far beyond simple product suggestions. In e-commerce, companies like Amazon, Flipkart, and Alibaba use sentiment analysis to enhance product recommendations. For instance, if a user regularly praises “battery life” in their reviews, the system can prioritize electronics known for strong battery performance. This level of personalization ensures that users are shown products that match not just their buying history, but also their underlying preferences and values.

Streaming platforms like Netflix or Spotify also benefit from sentiment-aware systems. A user might binge-watch feel-good shows and leave positive comments about uplifting content. The system can pick up on this trend and recommend similarly themed entertainment, improving satisfaction and watch time. In travel and hospitality, sentiment analysis of reviews on platforms like TripAdvisor or Booking.com helps identify what matters most to travelers — be it cleanliness, location, or staff behavior. Sentiment-aware systems can tailor hotel or destination suggestions based on emotional insights, leading to more fulfilling travel experiences.

Another powerful application is in customer feedback analysis. Businesses often receive massive volumes of comments, complaints, and suggestions. A sentiment-aware recommendation engine can analyze this feedback to guide product development, marketing strategies, or customer support improvements. For example, if users express frustration over complicated return policies, the system can flag this for business review, or steer recommendations away from such products.

The benefits of these systems are extensive:

- Increased user engagement and satisfaction, because recommendations feel more relevant and emotionally aligned.
- Reduced return rates, as users make more informed choices.
- Improved brand trust and loyalty, due to more personalized interactions.
- Better decision-making for businesses, by understanding what users really care about.

In short, sentiment-aware recommendation systems bridge the gap between data and emotion, offering insights that drive not only sales, but also stronger customer relationships.

Challenges and Future Directions

While sentiment-aware recommendation systems offer immense potential, they also come with a set of challenges — both technical and ethical — that must be addressed to ensure their responsible and effective deployment. One of the primary challenges lies in the complexity of human language. Sarcasm, irony, slang, and cultural expressions can easily mislead even the most advanced AI models. For example, the sentence “*Just perfect — broke within a week!*” may be rated as positive by a model that fails to detect sarcasm. Similarly, emotional subtleties may be lost in translation across languages and regions, leading to inaccurate sentiment interpretations.

Another concern is data quality and availability. Sentiment analysis depends on high-quality, expressive user-generated content. If reviews are too short, vague, or inconsistent, the models may struggle to generate meaningful insights. Additionally, the cold-start problem — where new users or products have little to no data — continues to pose difficulties, even in sentiment-aware systems. From a computational standpoint, models like RoBERTa and BiGRU are resource-intensive, requiring significant processing power and memory. This can limit their adoption in low-resource environments or small businesses. Efforts are underway to create more lightweight models that retain accuracy while improving speed and efficiency.

Beyond technical hurdles, there are ethical and privacy concerns. Users may not always be aware that their emotional expressions are being analyzed for commercial gain. This raises questions about consent, data transparency, and the potential misuse of emotional insights. For instance, using sentiment to manipulate user decisions or exploit emotional states for marketing purposes could cross ethical boundaries.

Looking forward, the future of sentiment-aware recommendation systems is promising but demands thoughtful development. Key directions include:

- Multimodal sentiment analysis that incorporates text, voice, facial expressions, and body language for a deeper understanding of emotion.

- Explainable AI (XAI) in recommendations, where users can understand why a suggestion was made.
- Cross-lingual and culturally adaptive models that perform accurately across diverse user bases.
- Privacy-preserving algorithms that analyze sentiment without compromising user data integrity.

By addressing these challenges, the next generation of recommendation systems can become not only more intelligent but also more trustworthy, ethical, and inclusive.

Conclusion

In an era where personalization defines user experience, sentiment-aware recommendation systems represent a significant step forward. By moving beyond traditional metrics like ratings and purchase history, these systems tap into the emotional language found in user reviews, feedback, and comments. This shift allows digital platforms to connect with users on a more human level — understanding not just what they want, but how they feel. Through the use of advanced AI techniques such as transformer models, recurrent neural networks, and attention mechanisms, modern systems can process and interpret the emotional tone of language with increasing accuracy. These capabilities enable businesses to offer more meaningful suggestions, reduce customer dissatisfaction, and foster stronger engagement.

At the same time, it's important to recognize the challenges that come with this technology — from language complexity and computational demands to ethical considerations around emotional data. As we look to the future, the goal must be to develop sentiment-aware systems that are not only intelligent and effective, but also transparent, fair, and respectful of user privacy. Ultimately, by integrating emotional insight into AI-driven personalization, sentiment-aware recommendation systems offer a powerful way to make digital experiences more empathetic, intuitive, and user-centered — a direction that aligns both with technological advancement and human expectations.

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